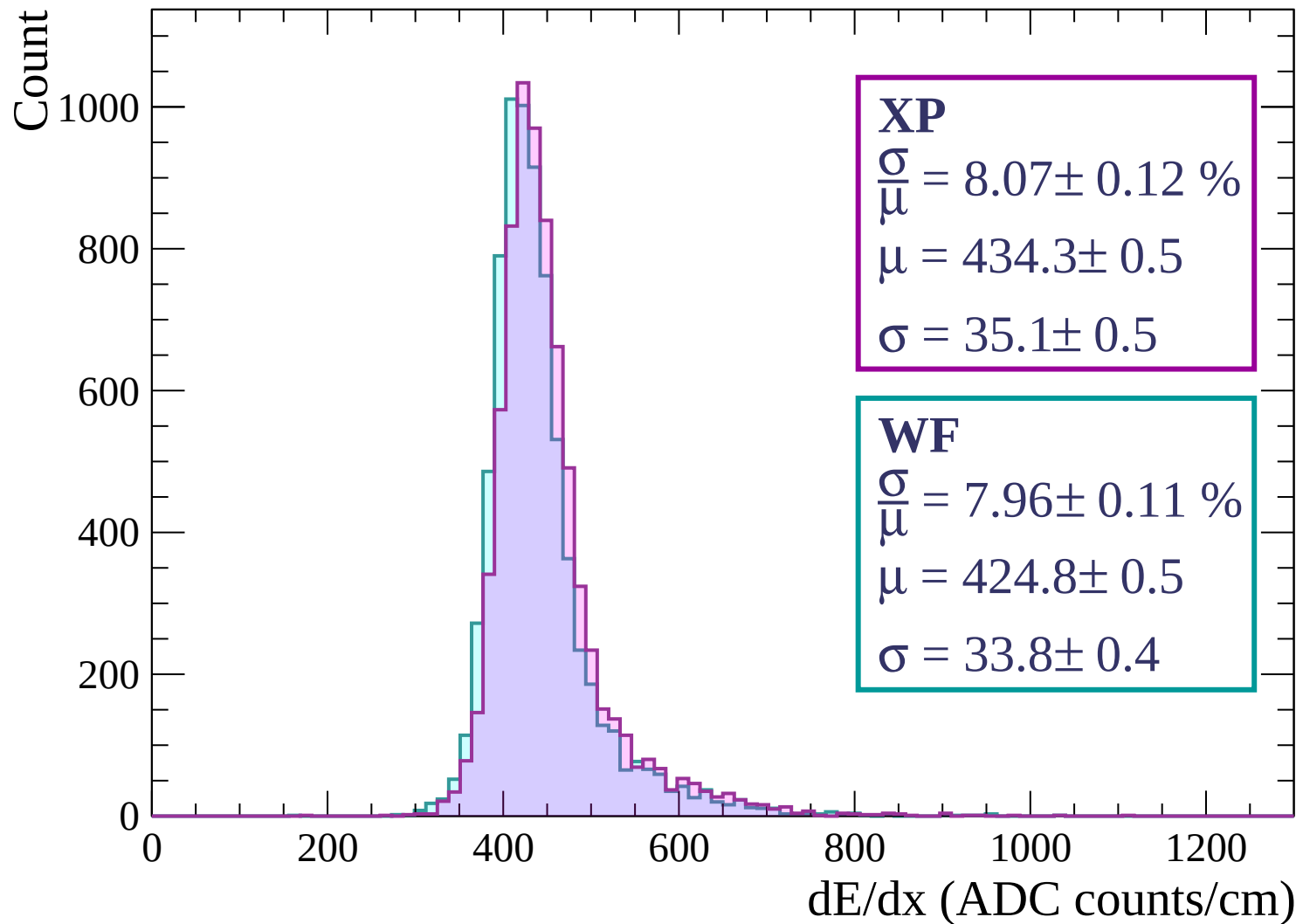
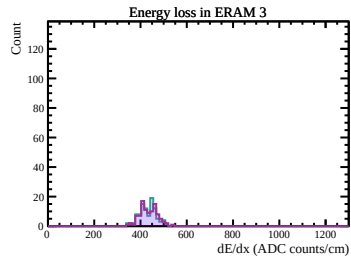
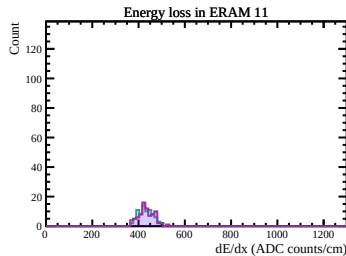
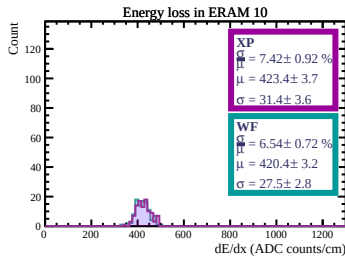
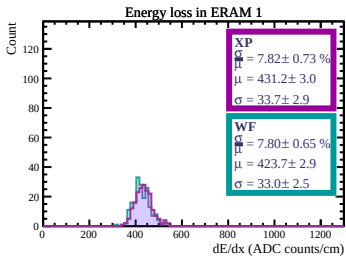
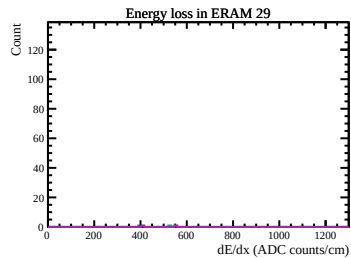
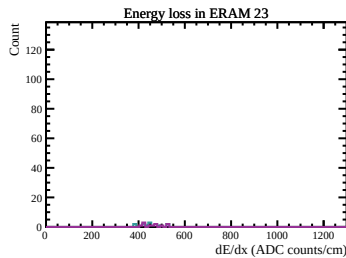
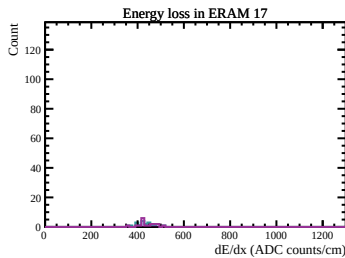
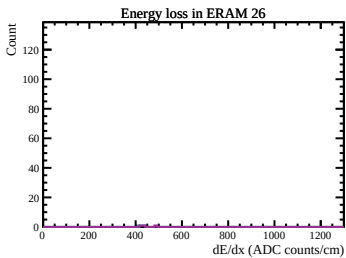
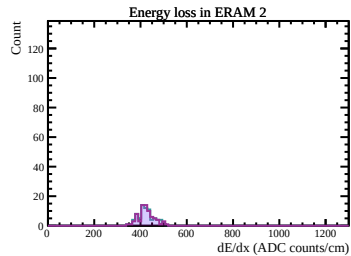
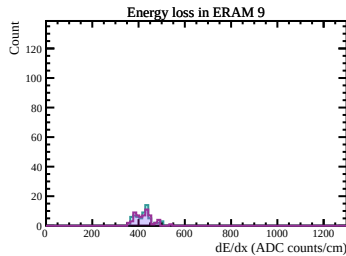
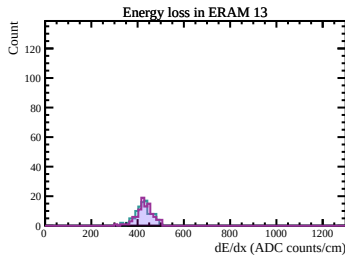
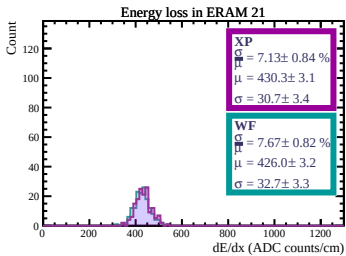
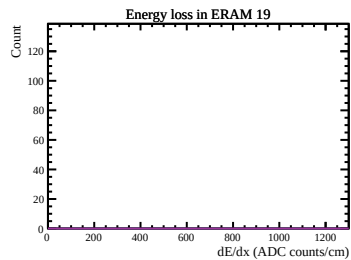
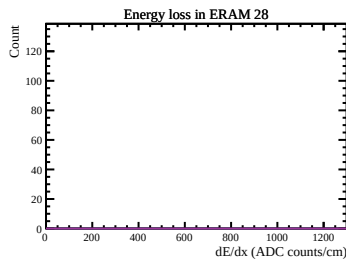
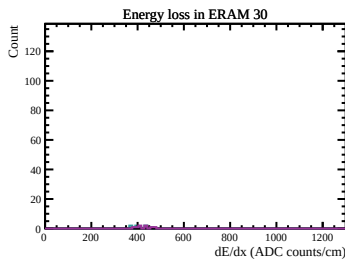
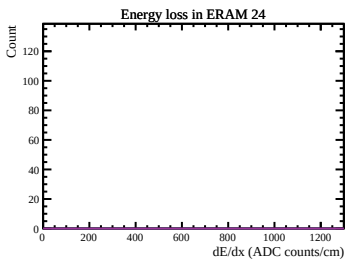
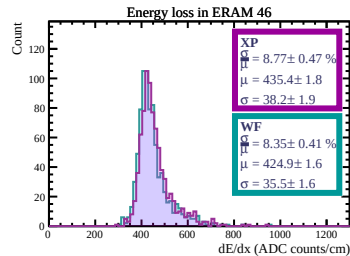
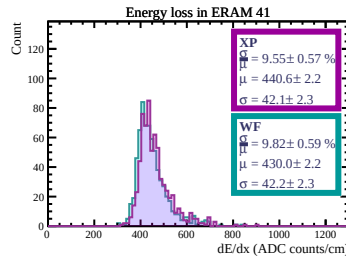
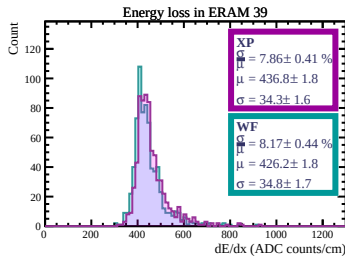
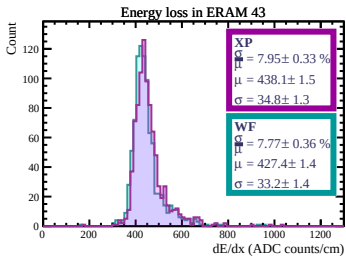
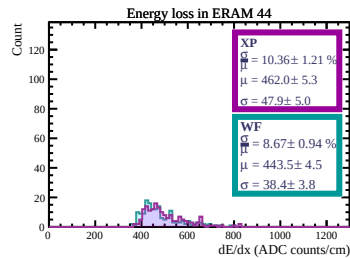
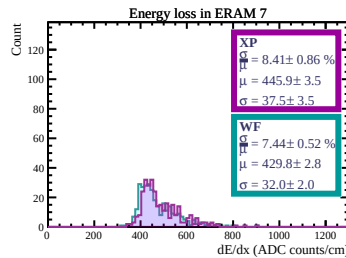
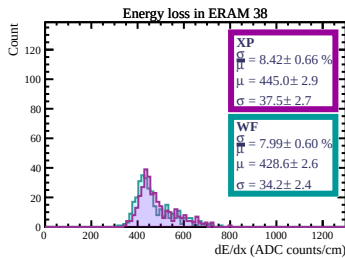
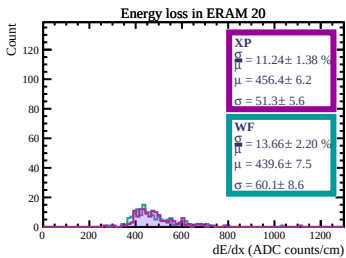
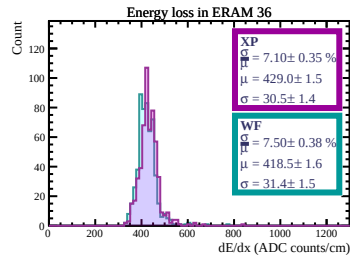
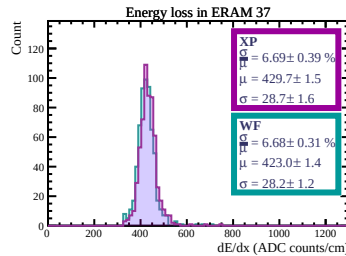
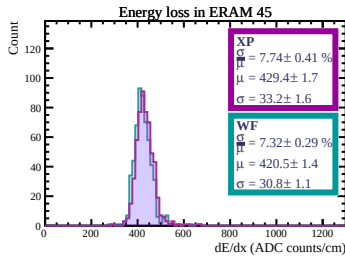
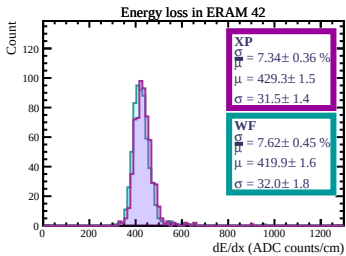
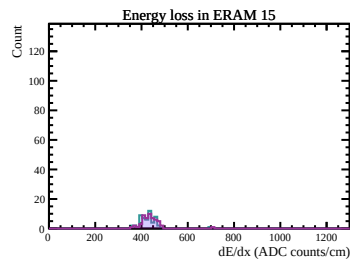
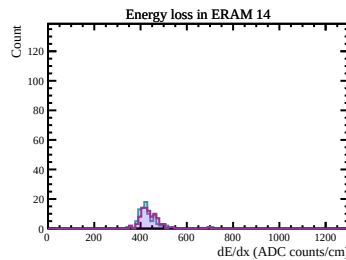
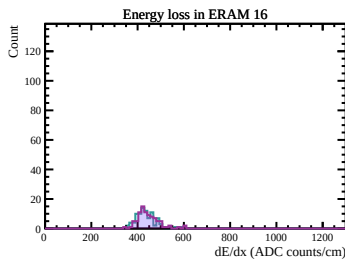
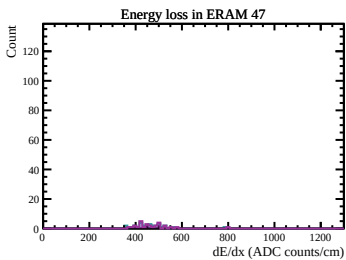
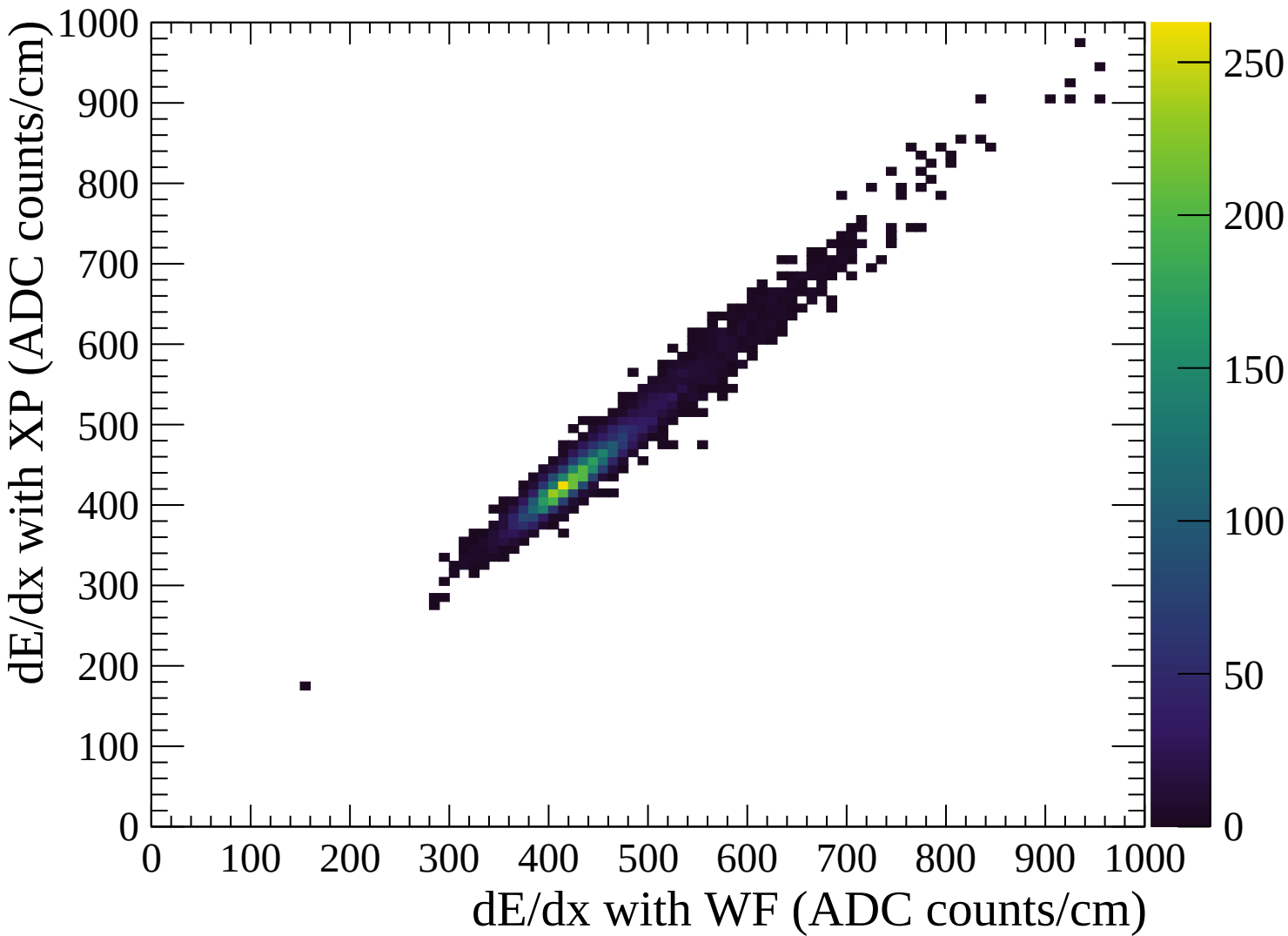
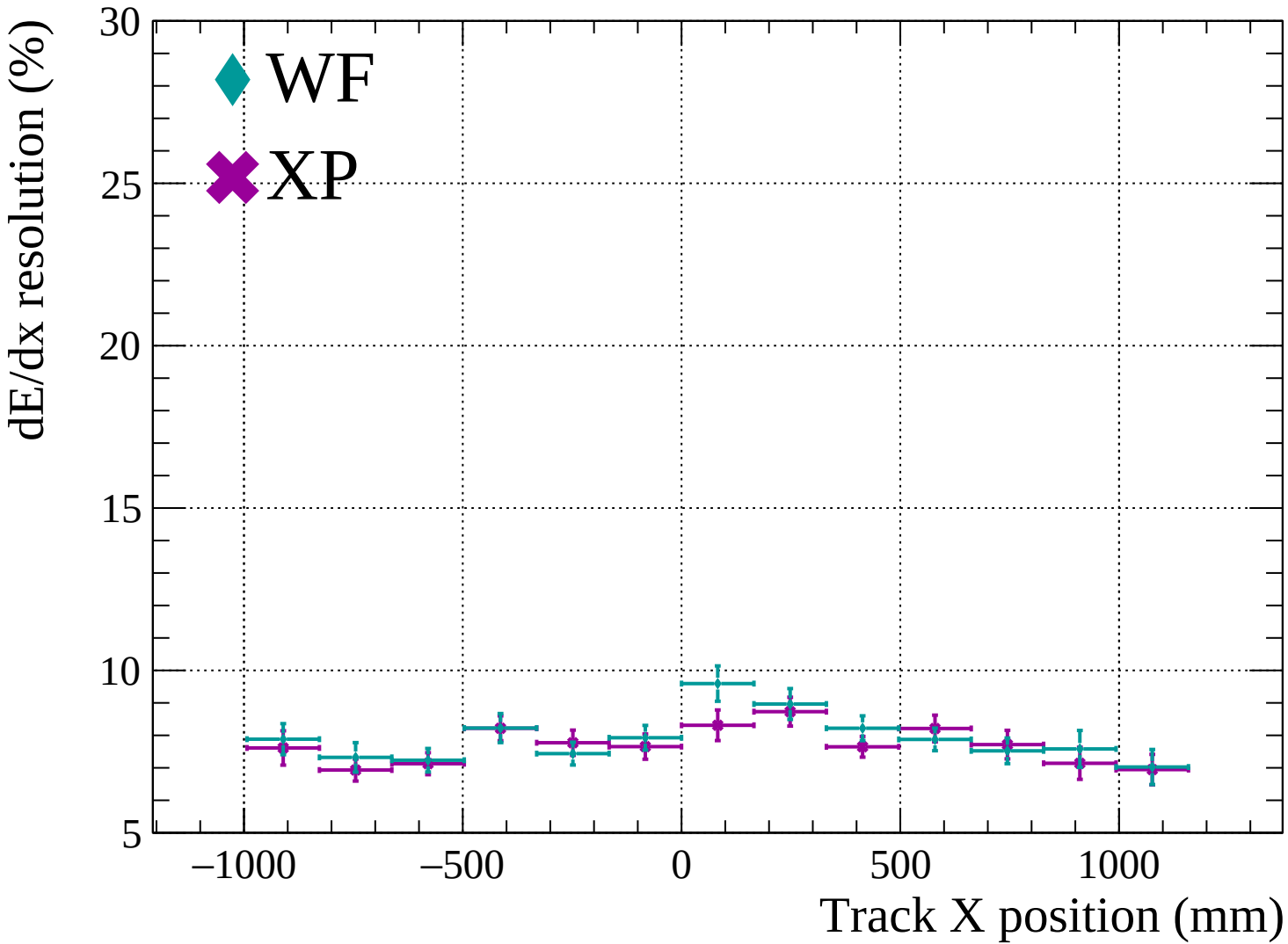


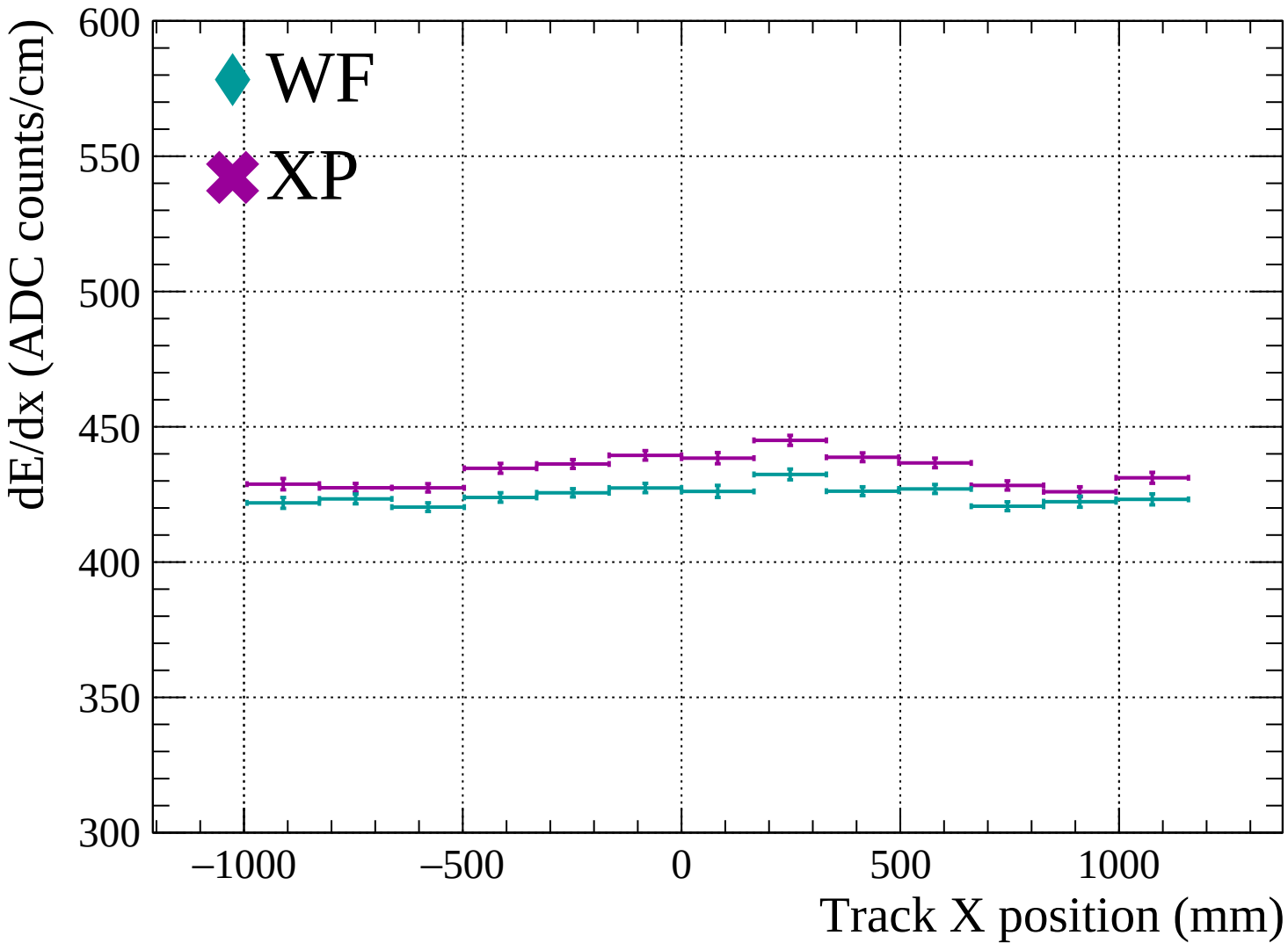


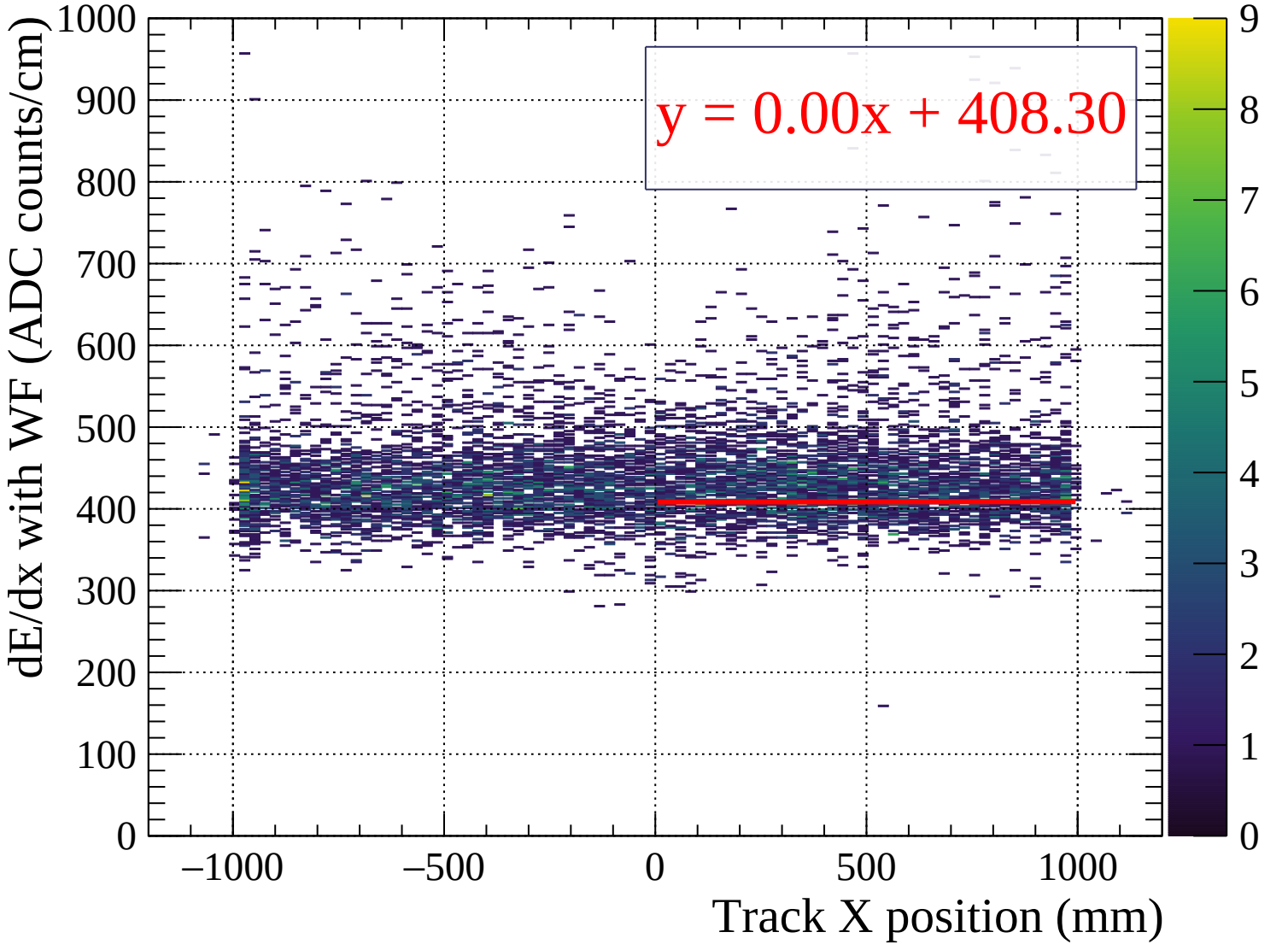


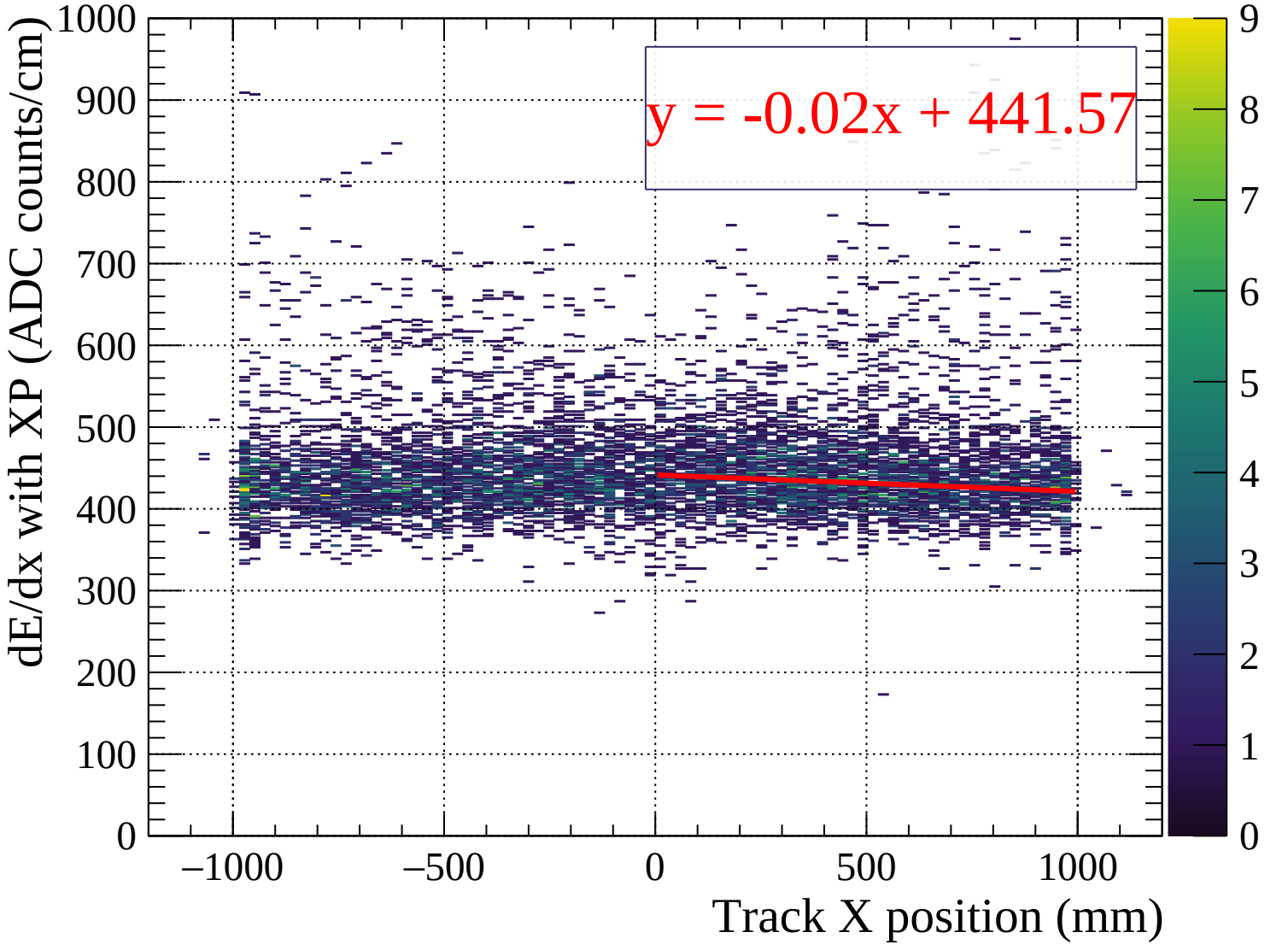


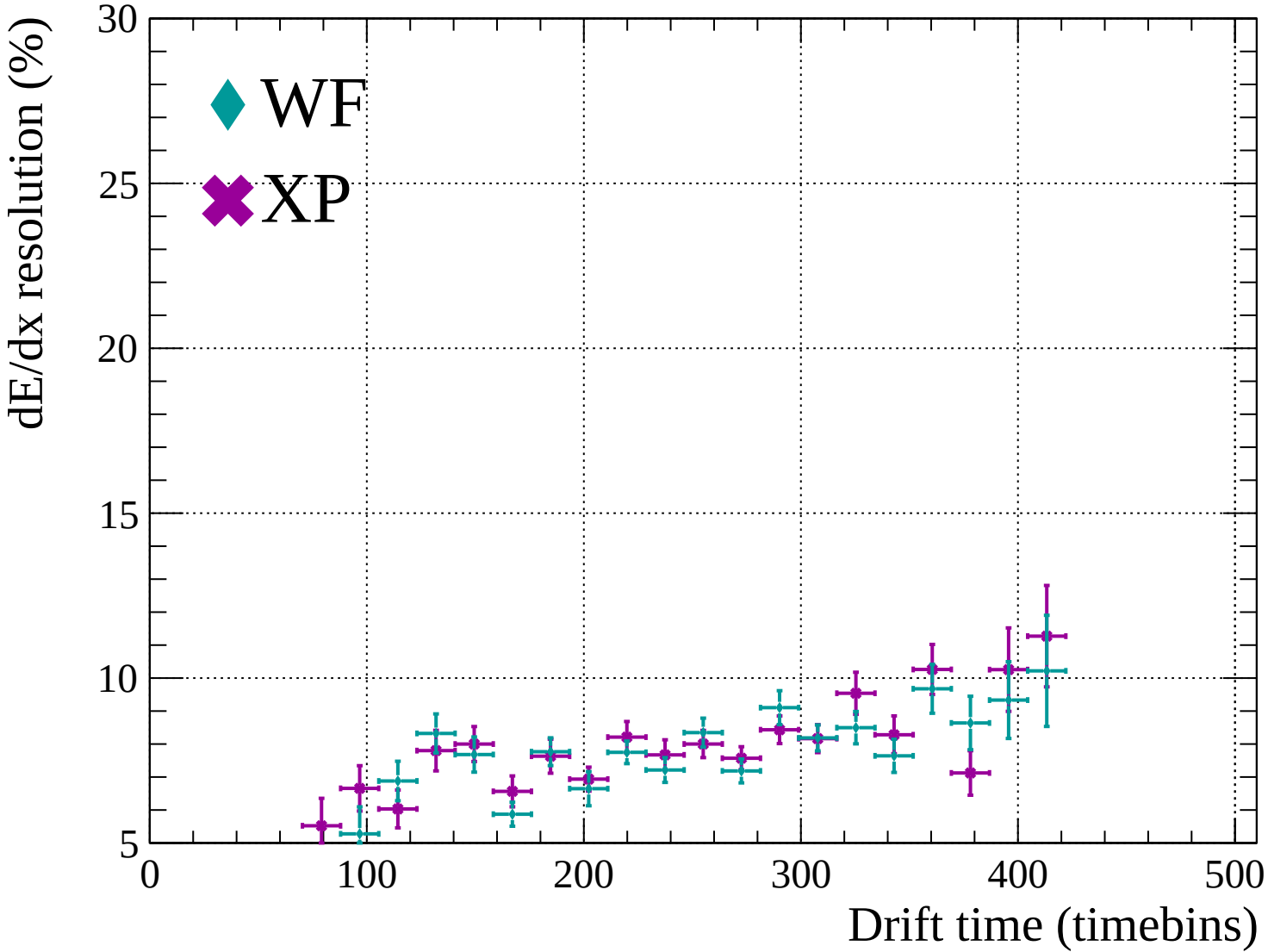


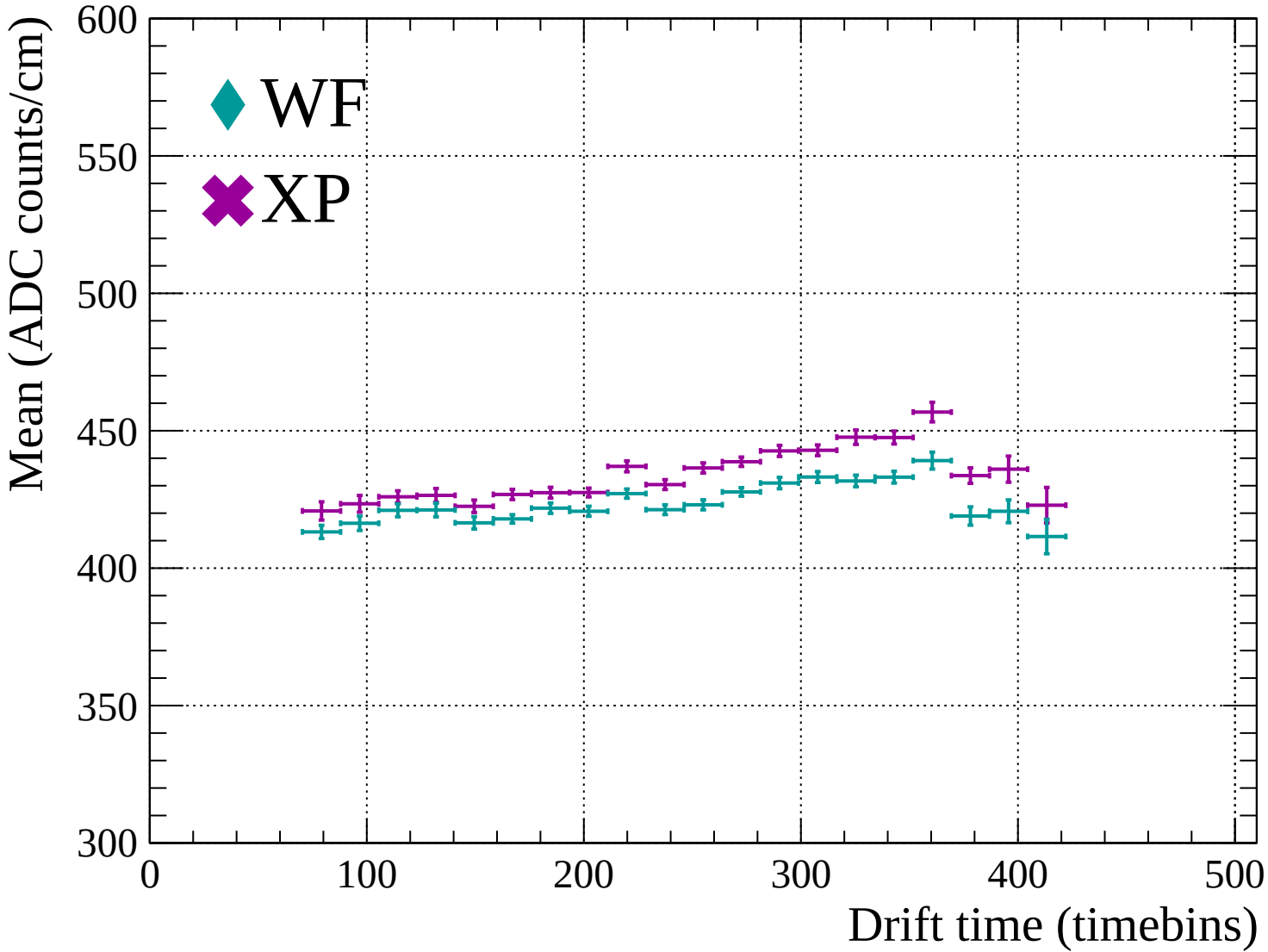


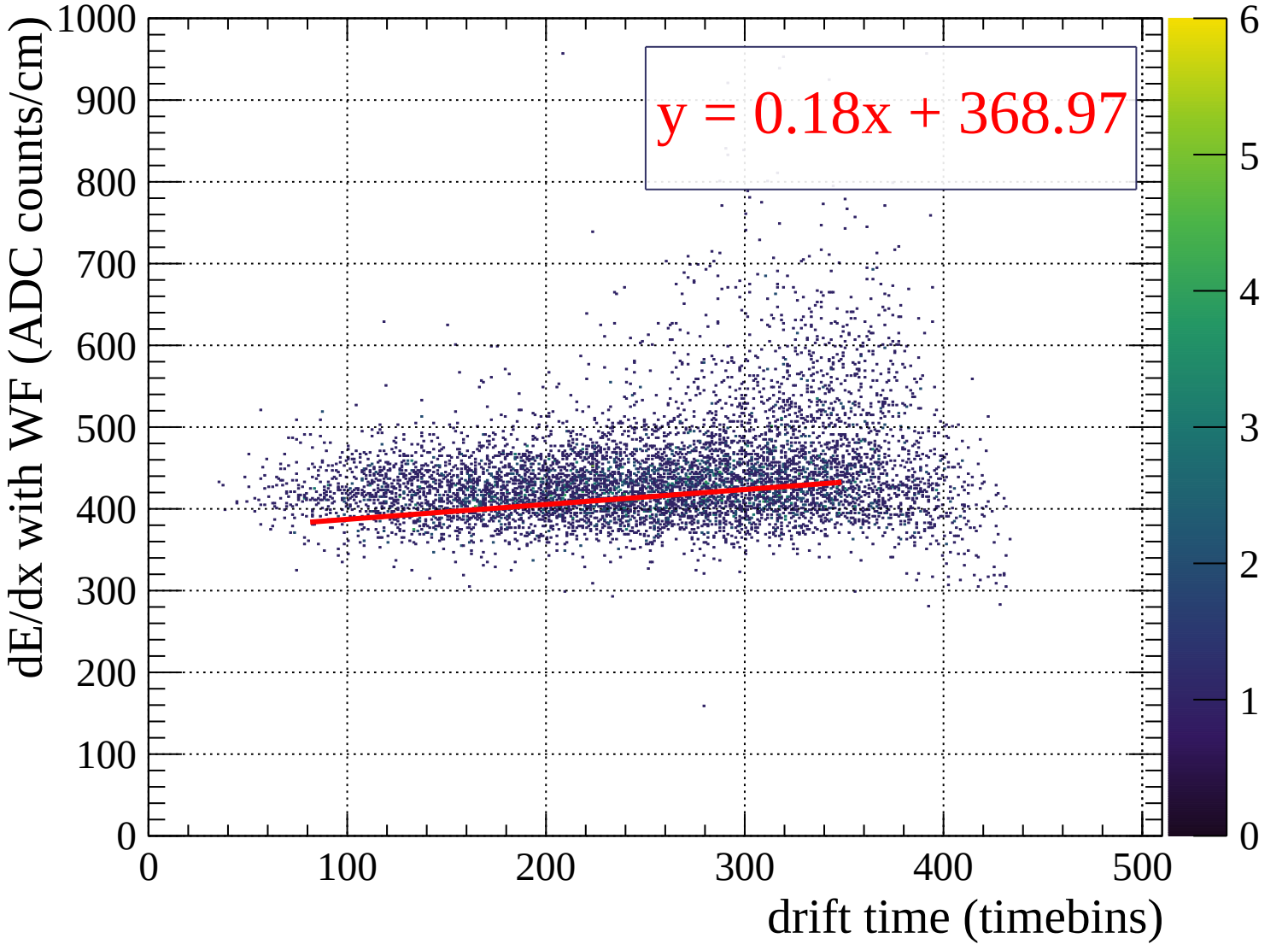


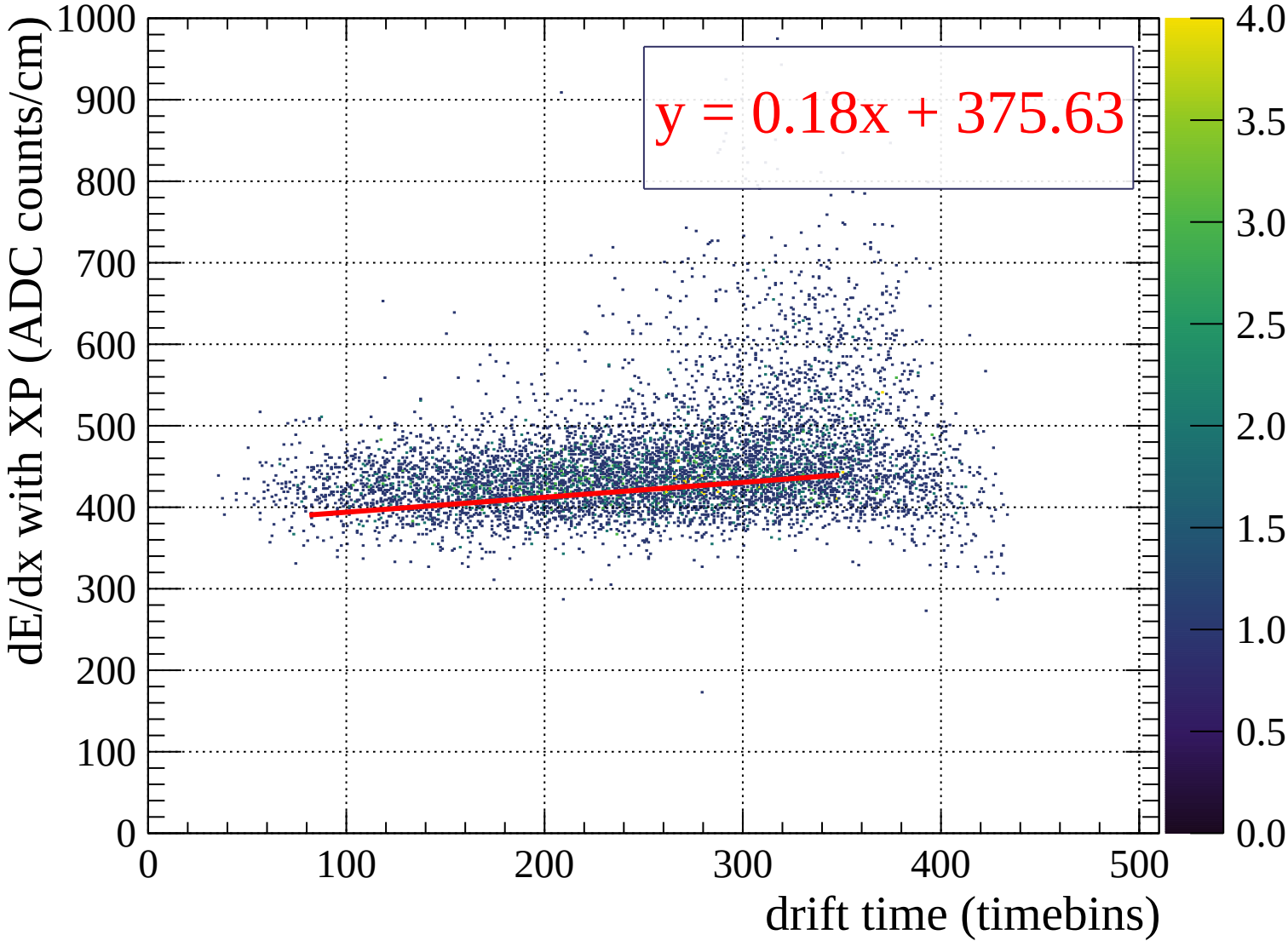


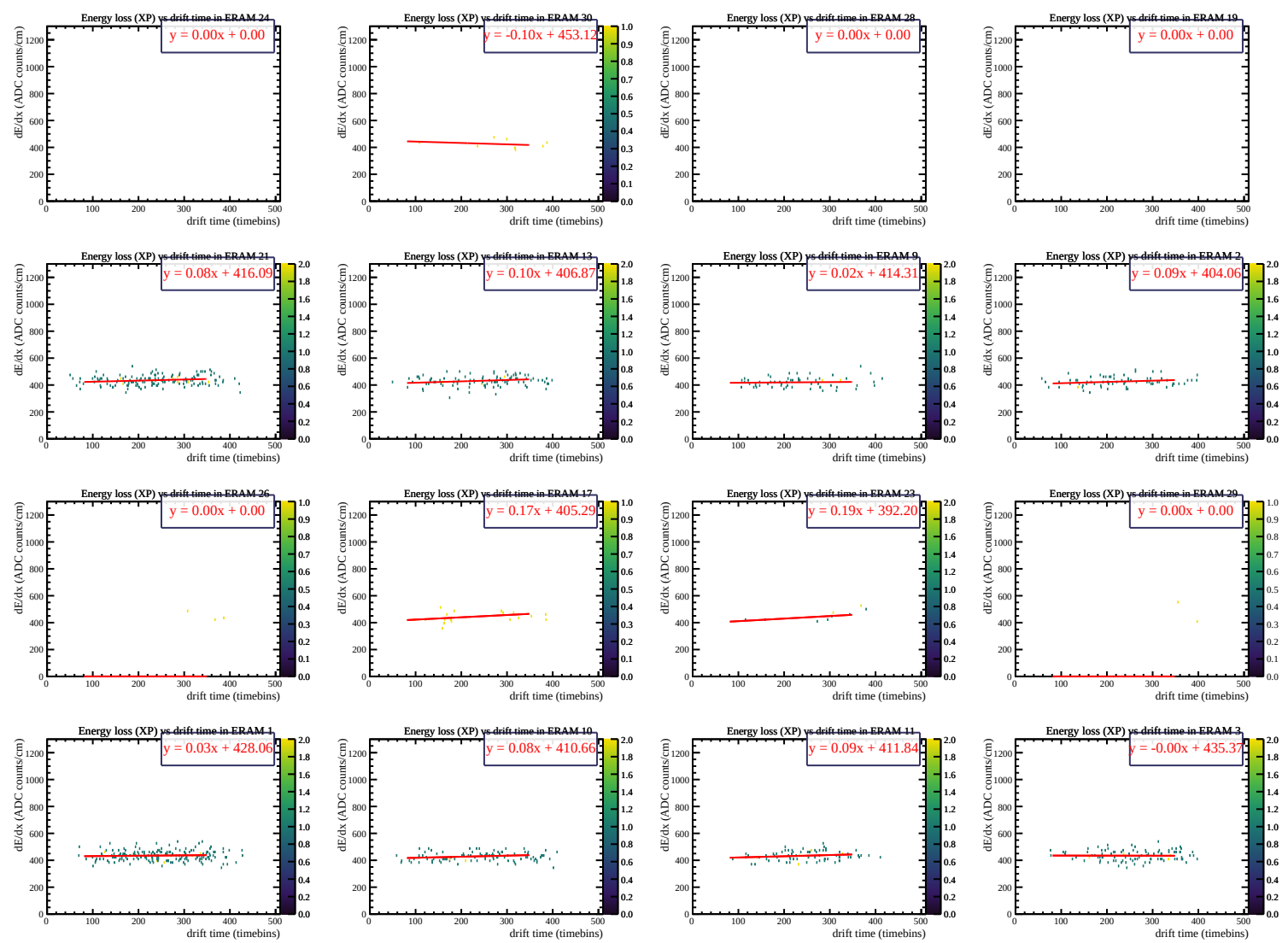


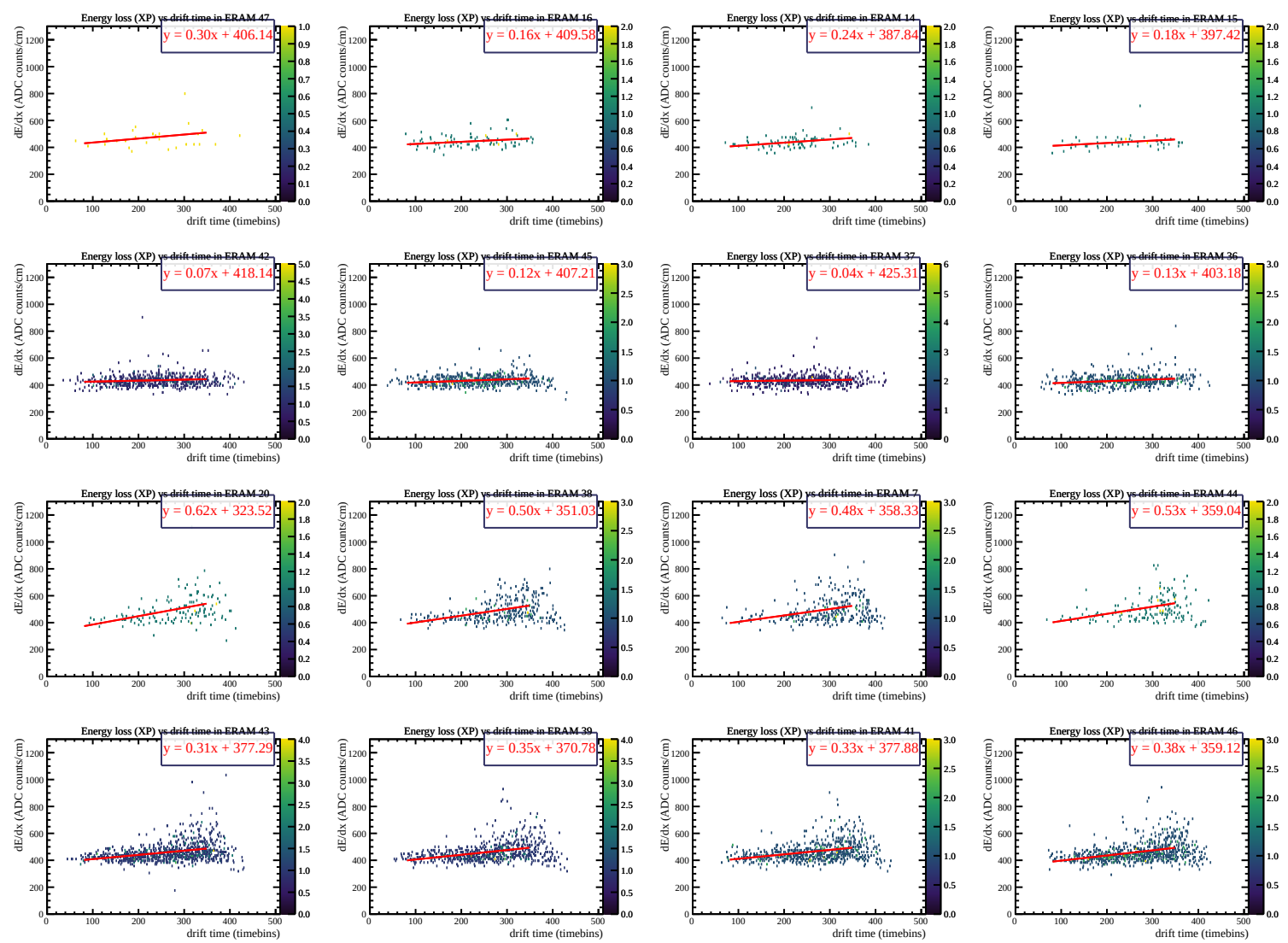


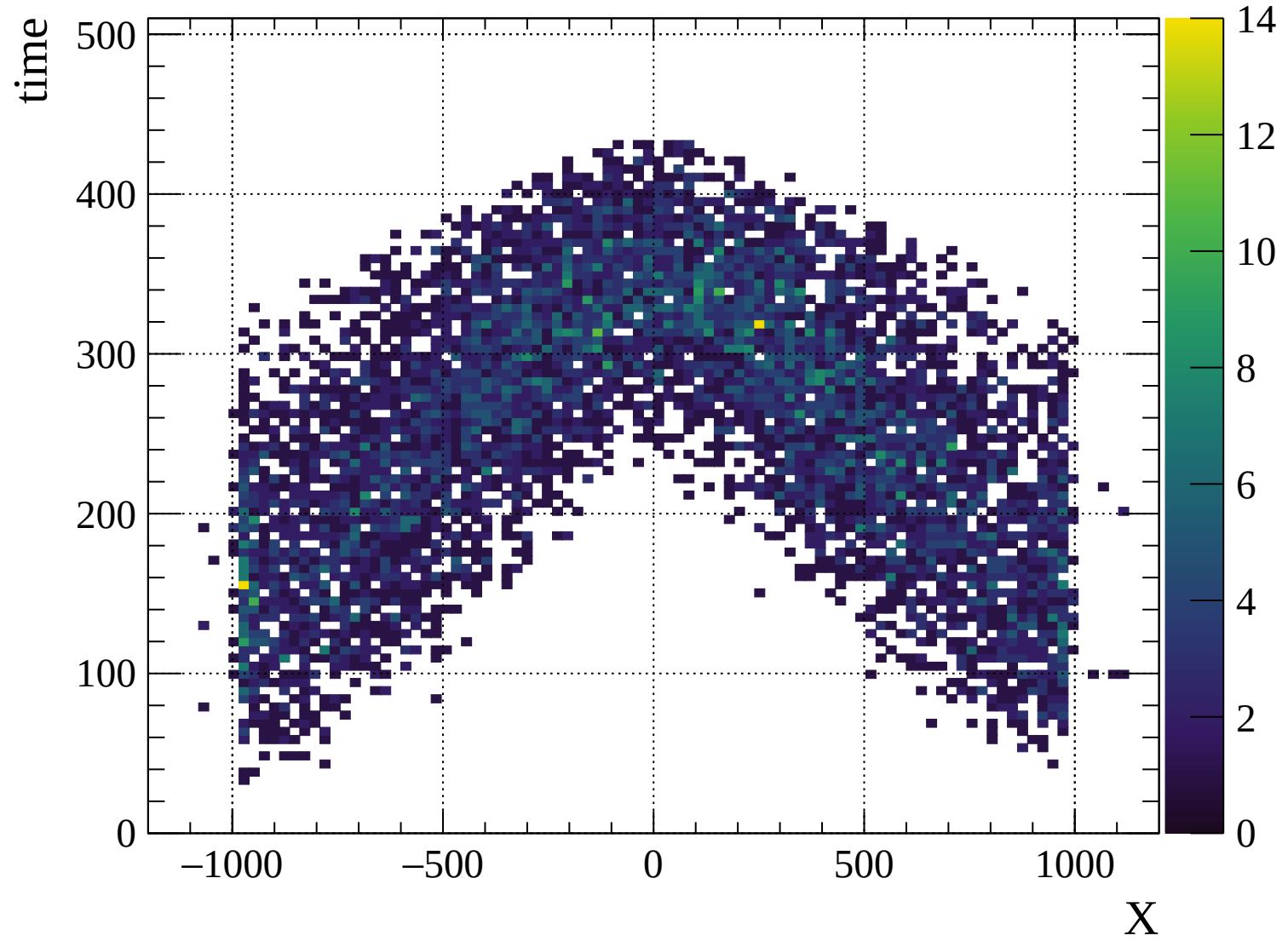


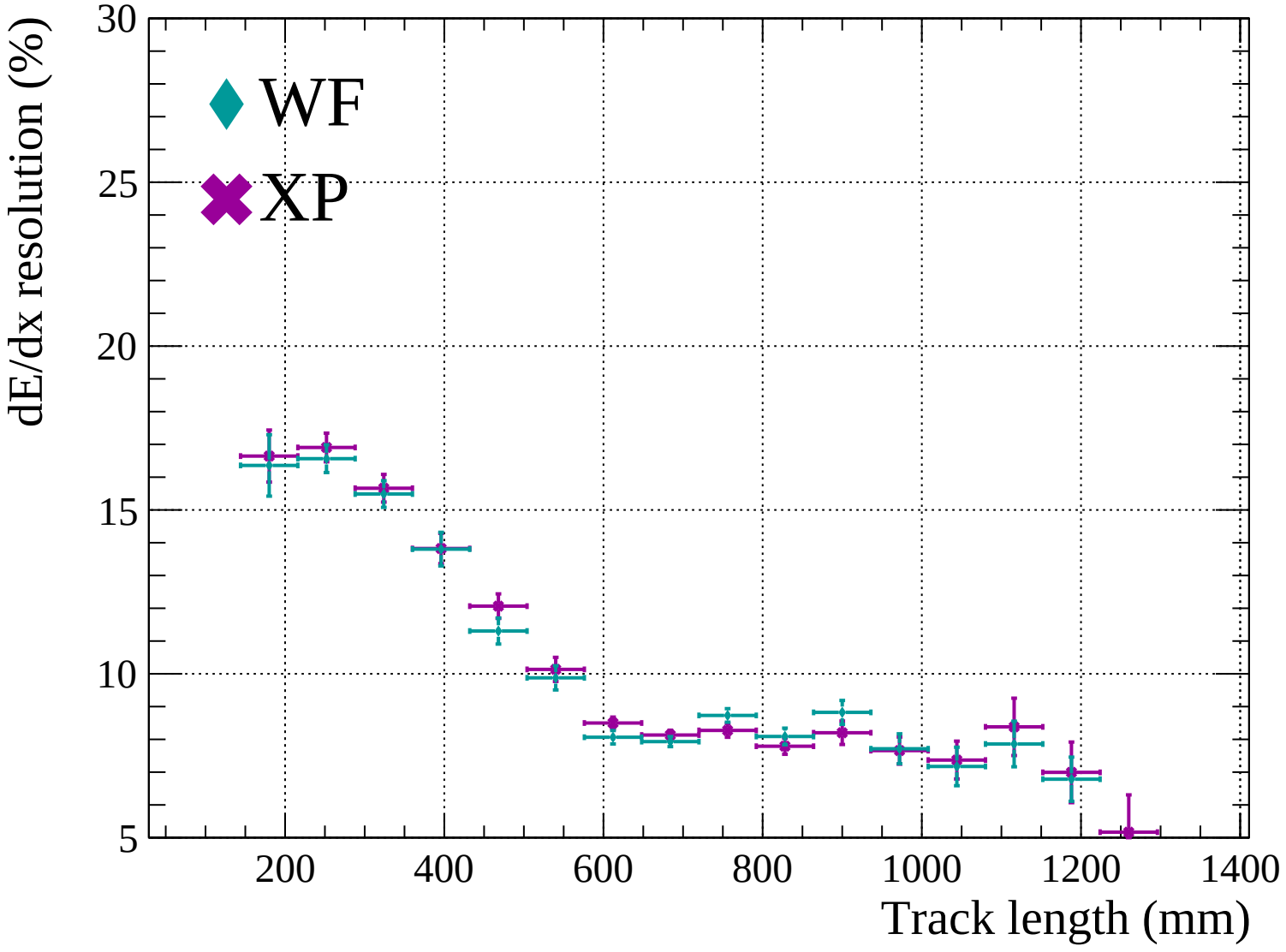


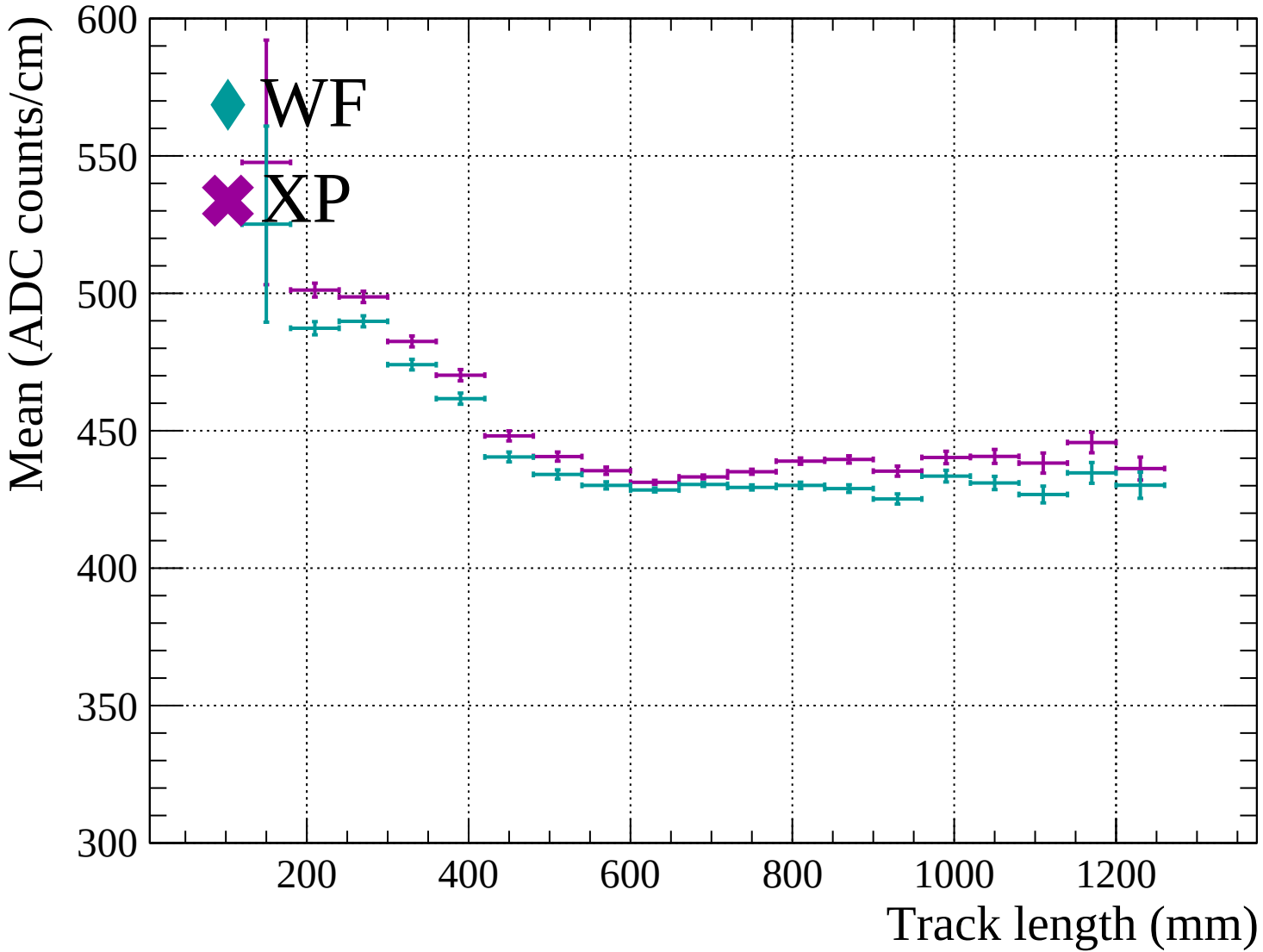


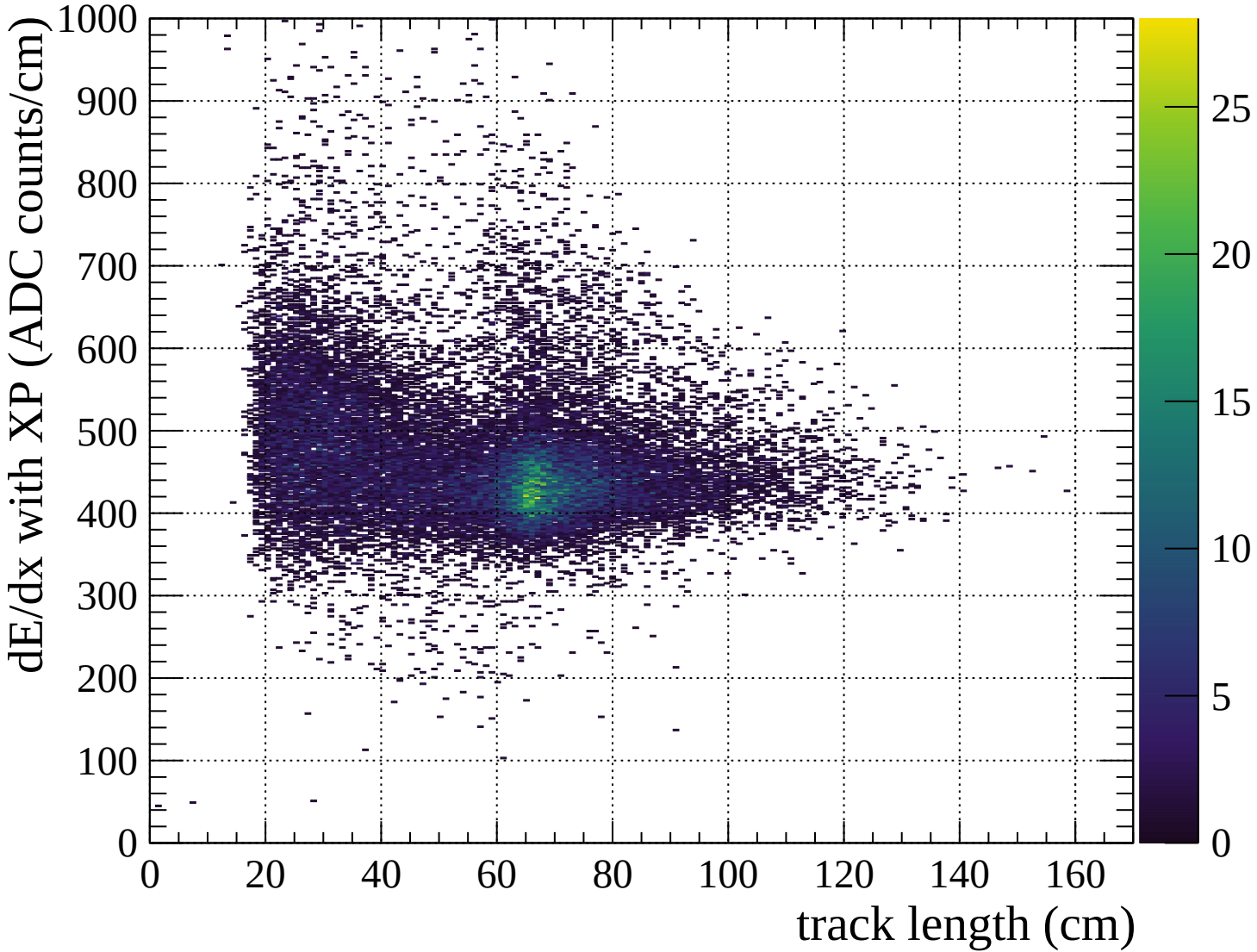


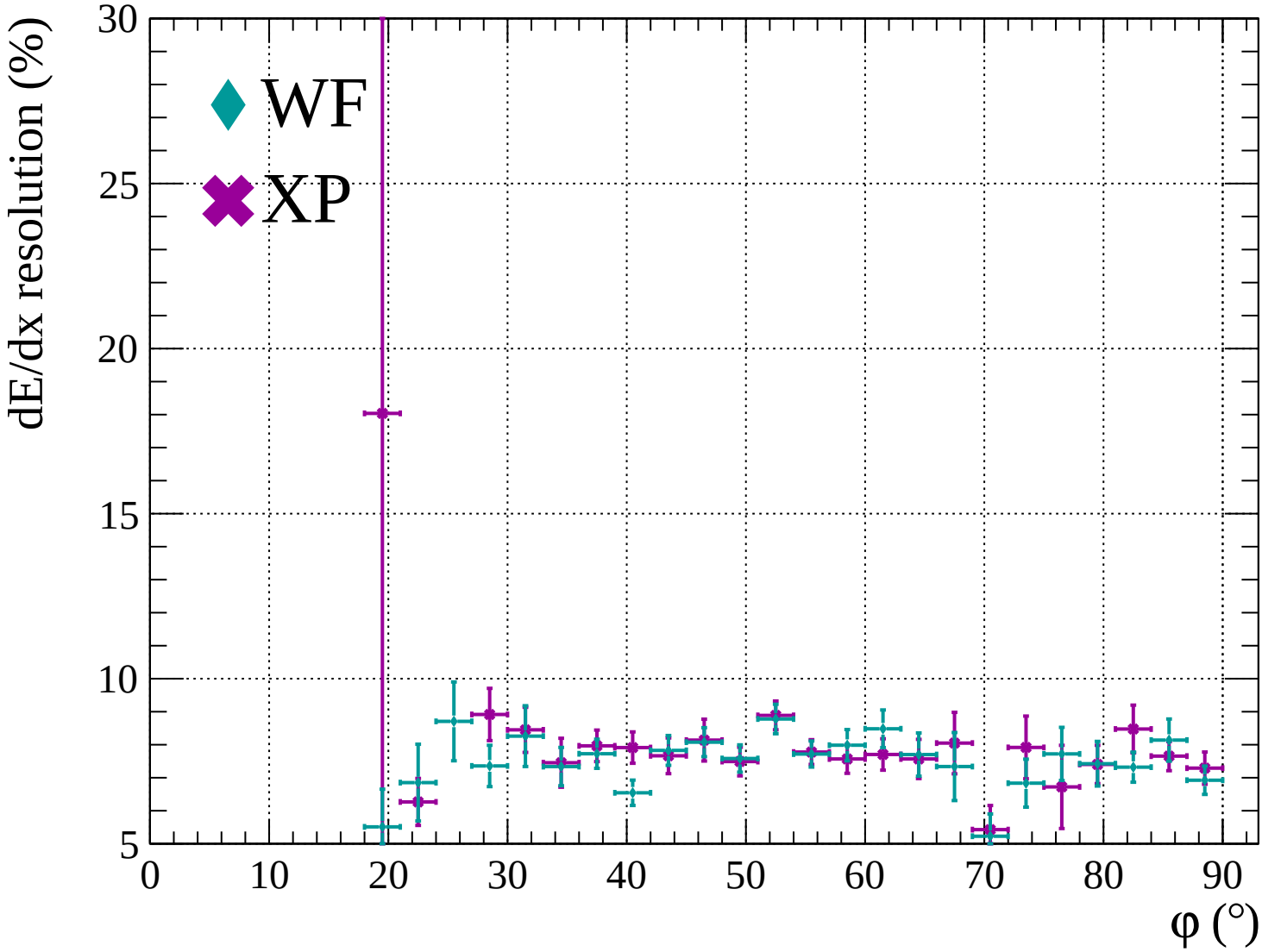


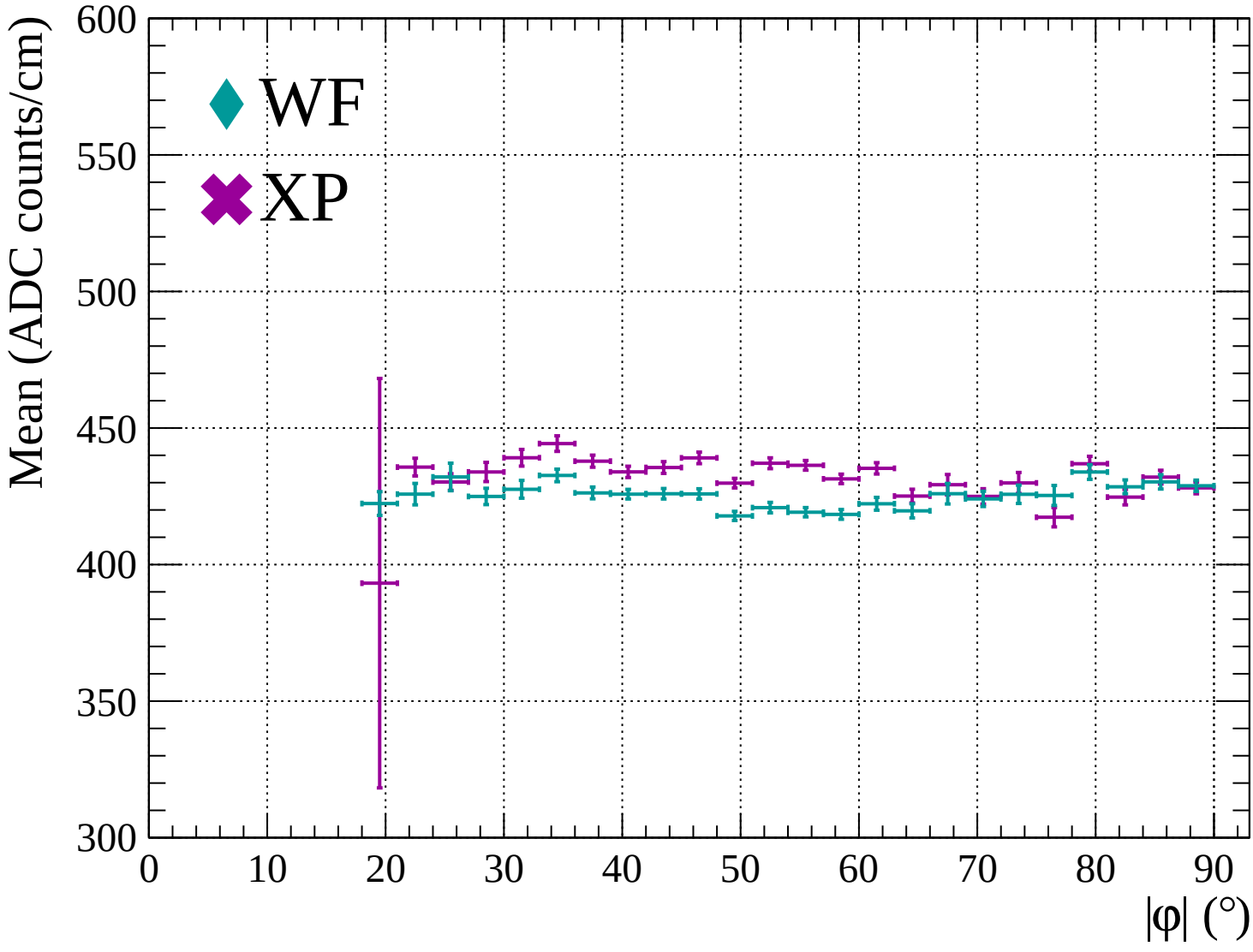


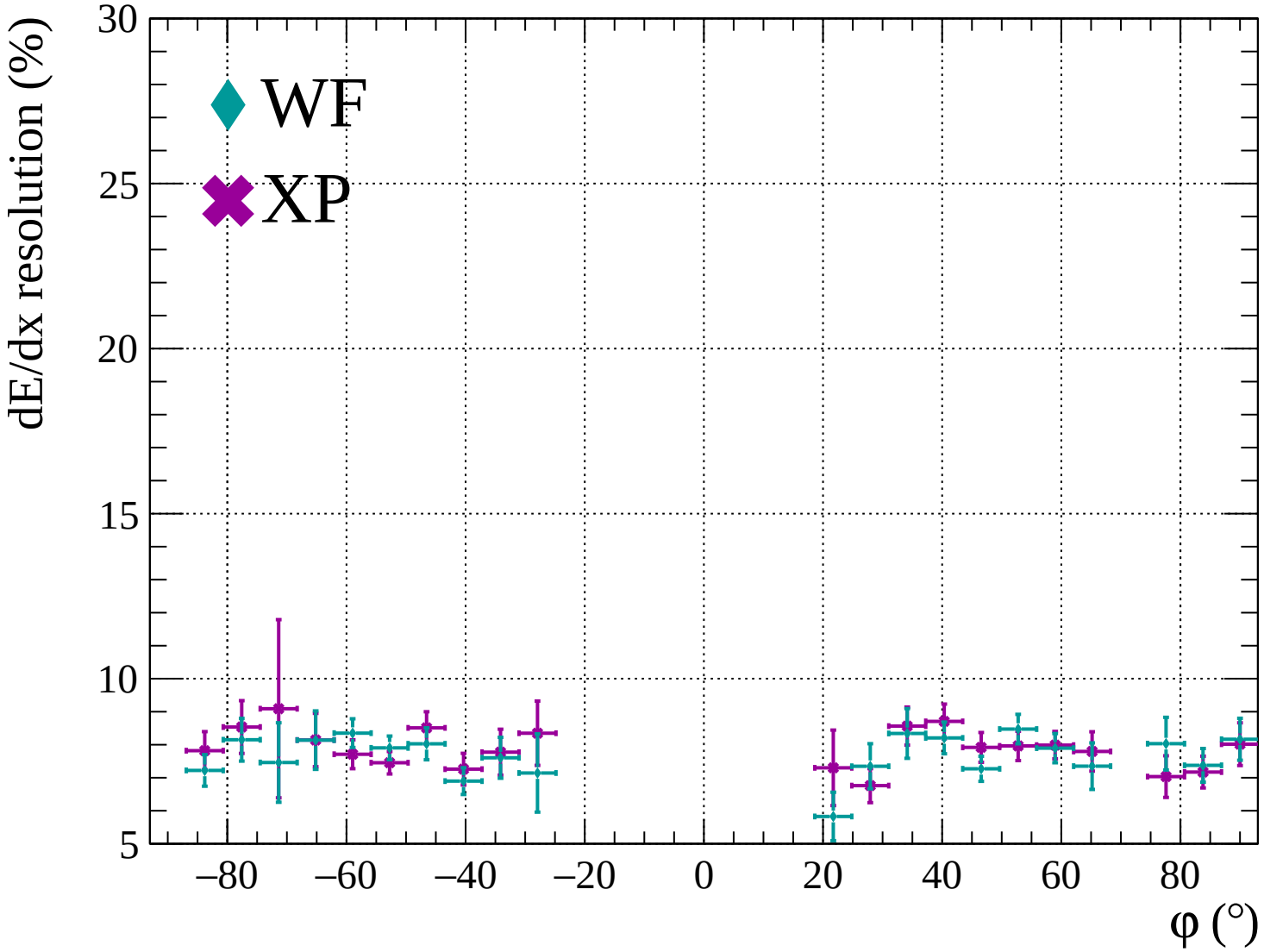


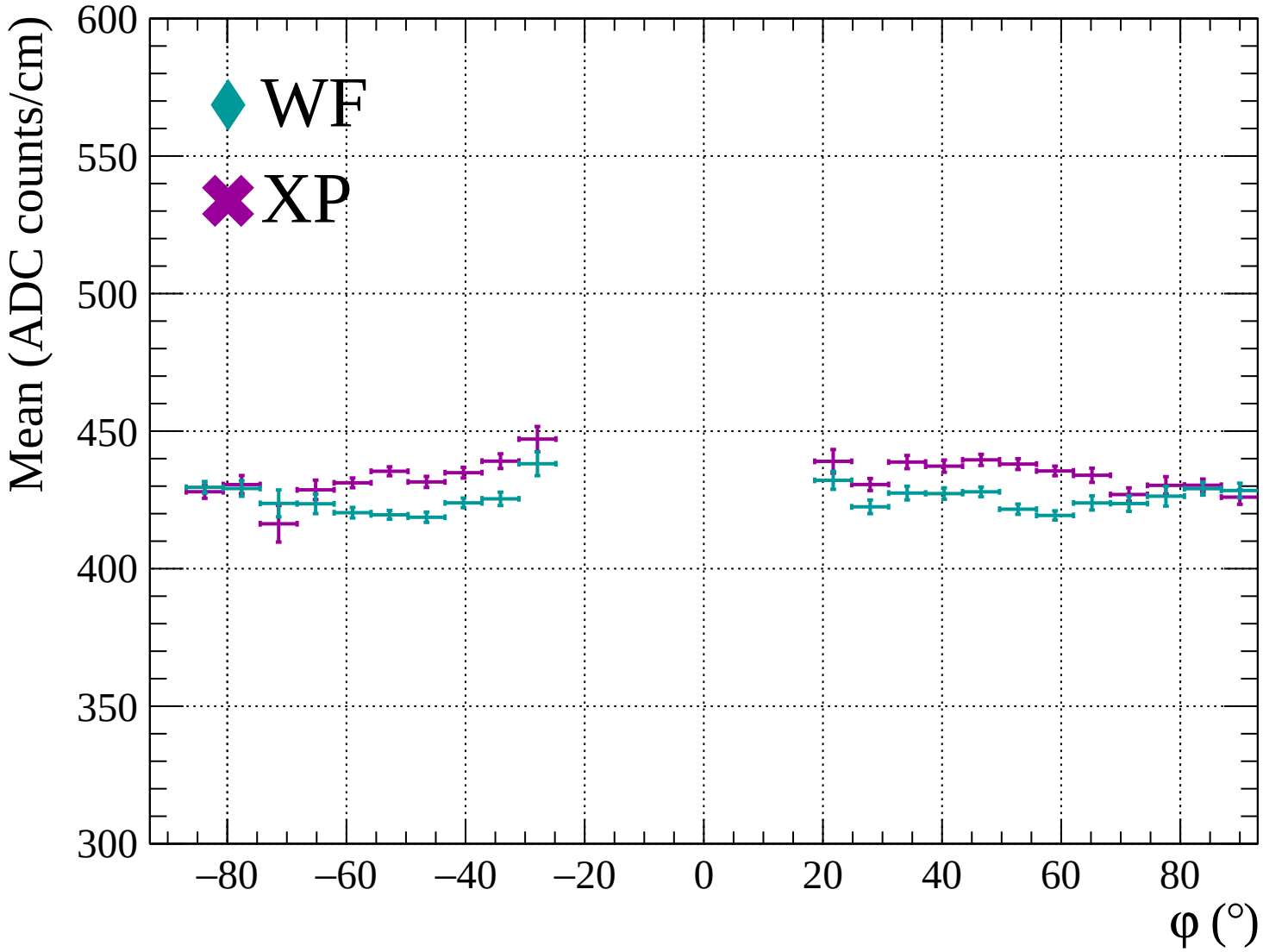


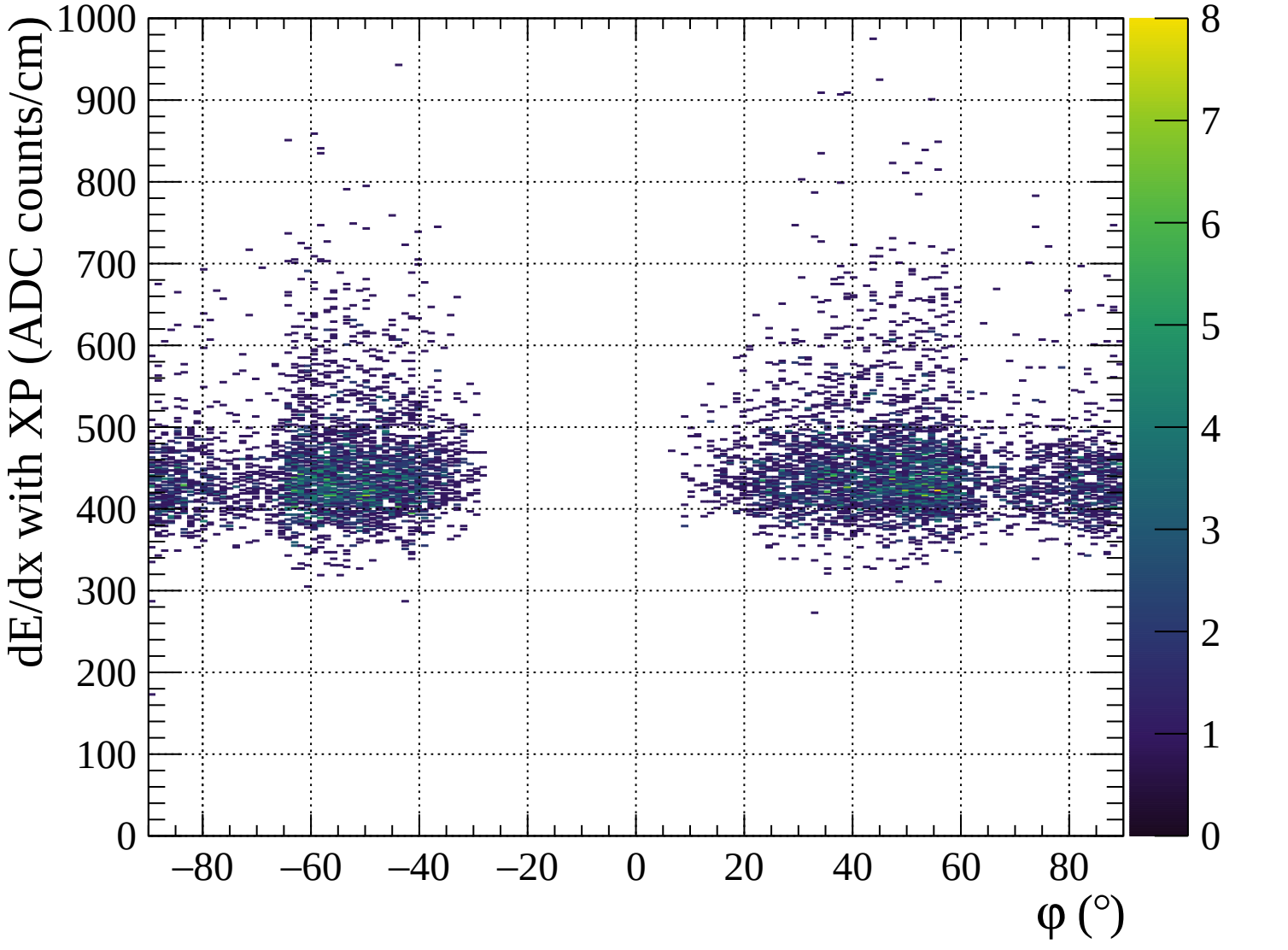


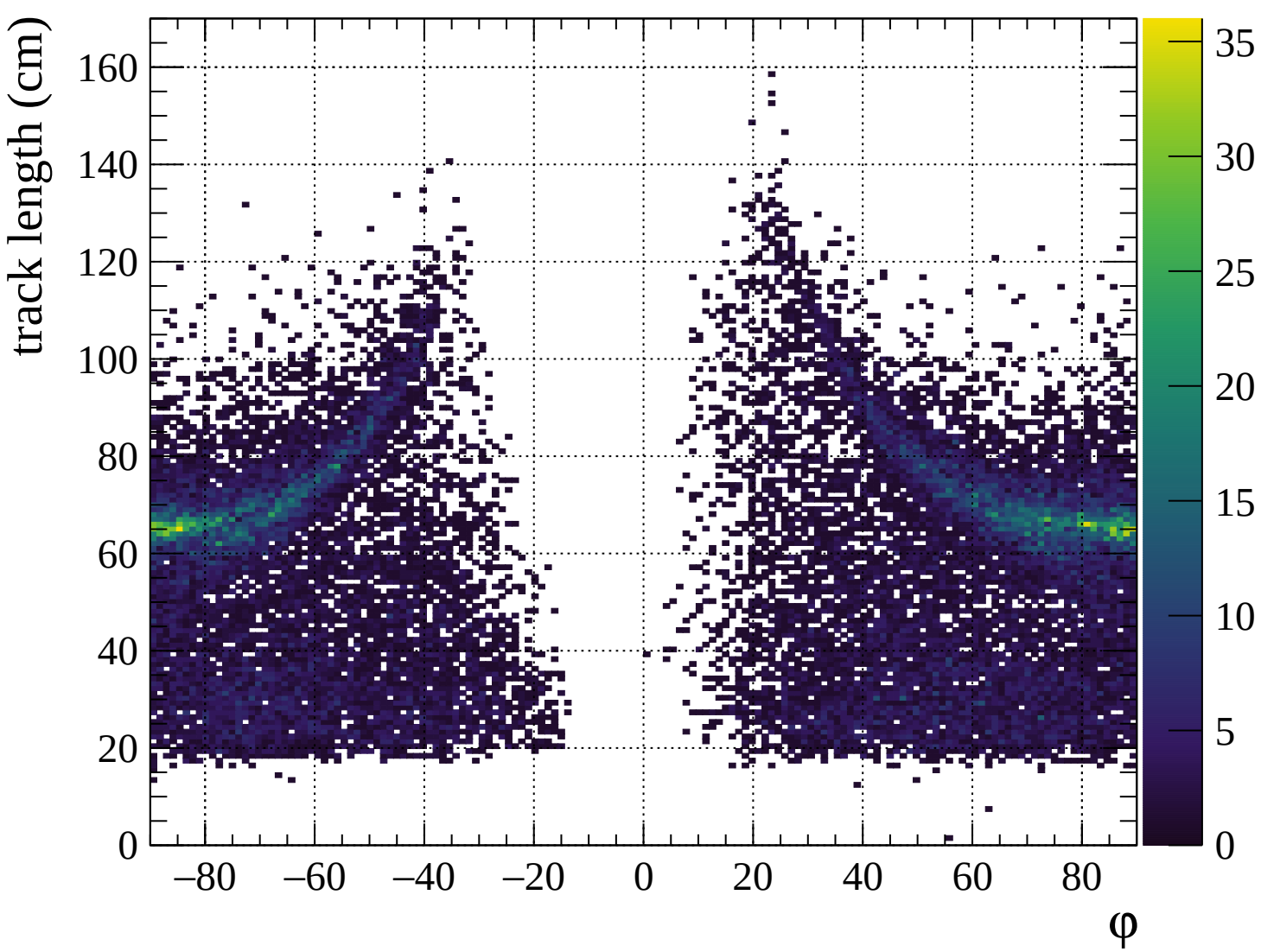


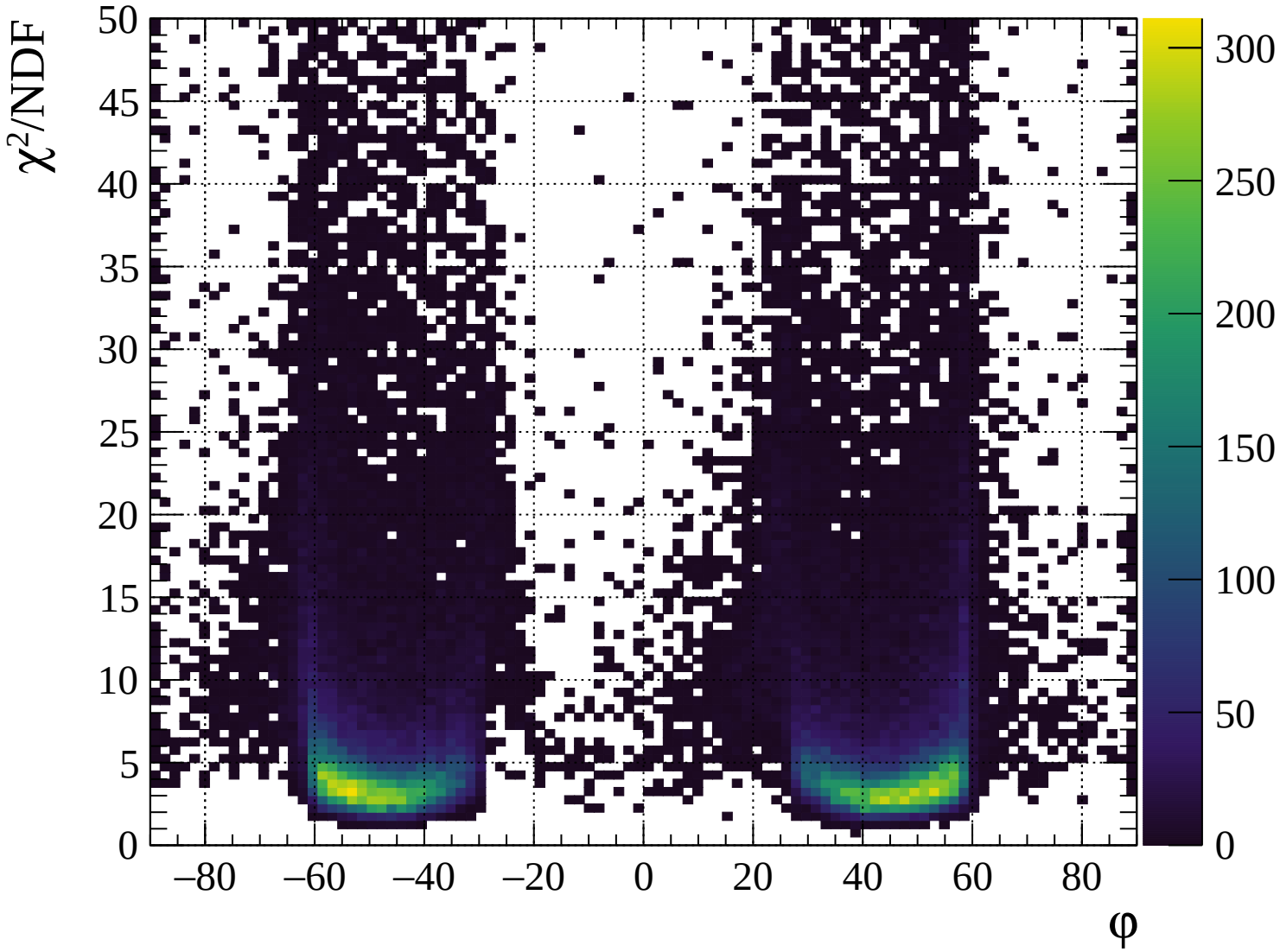


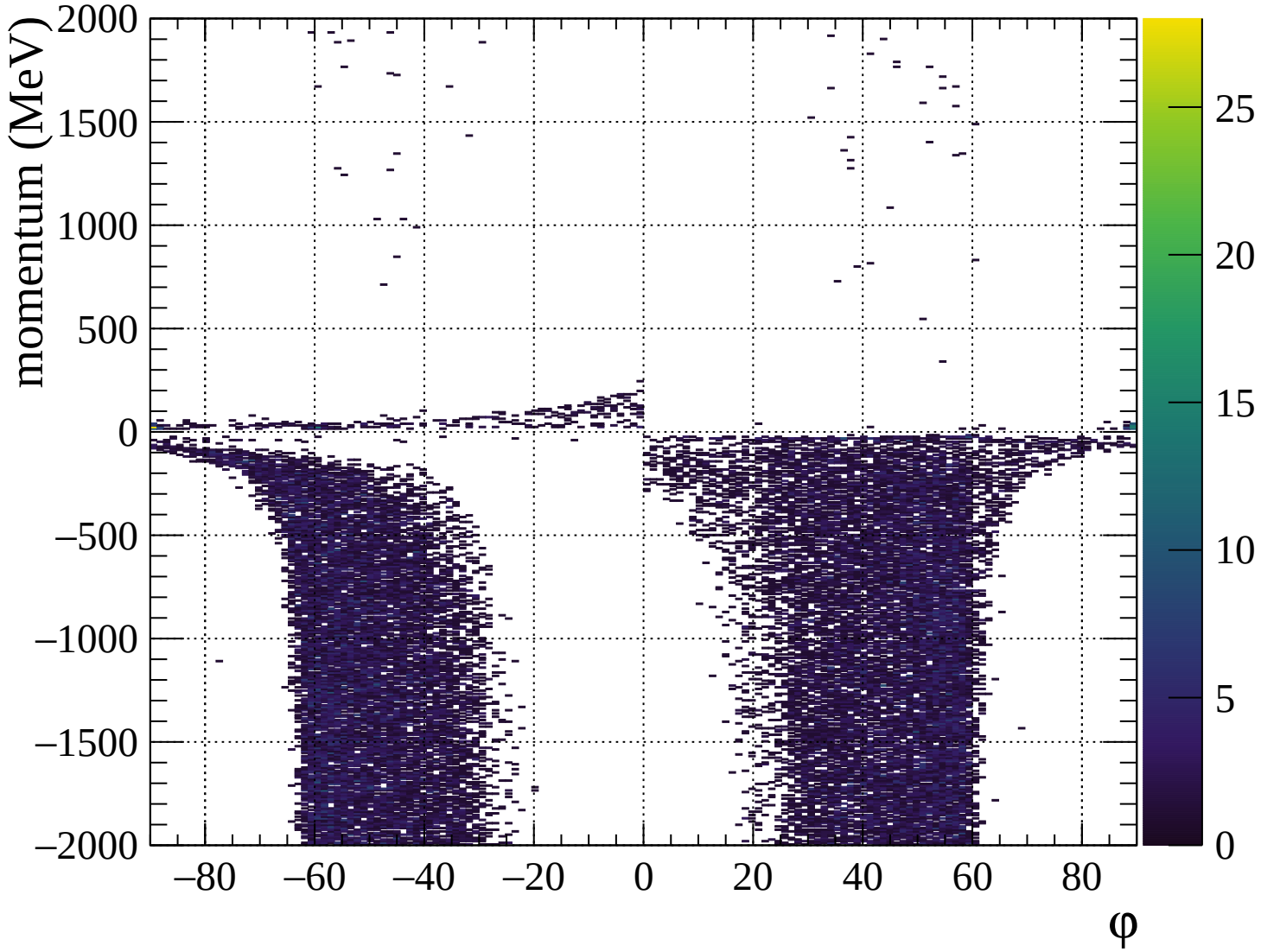


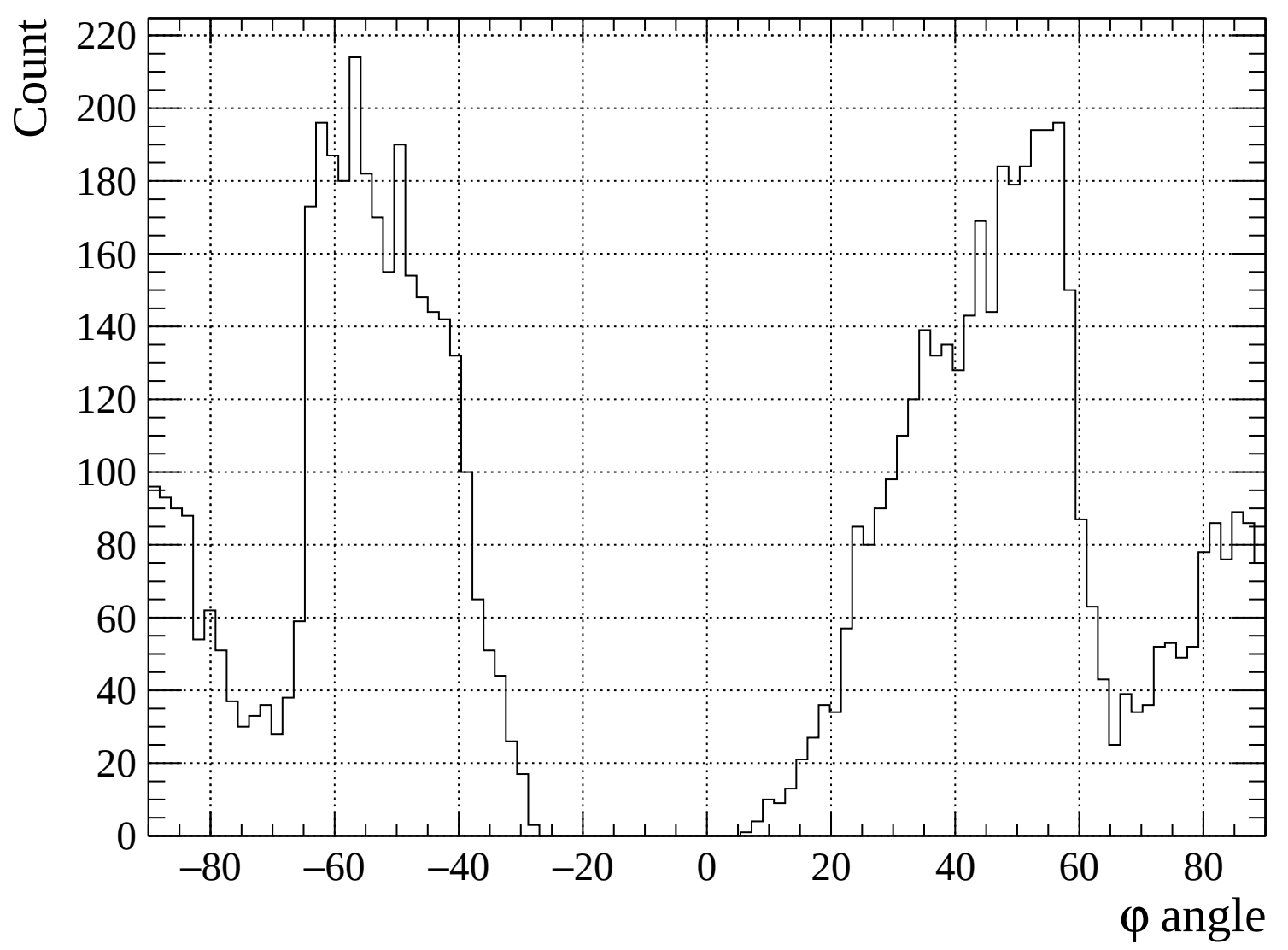


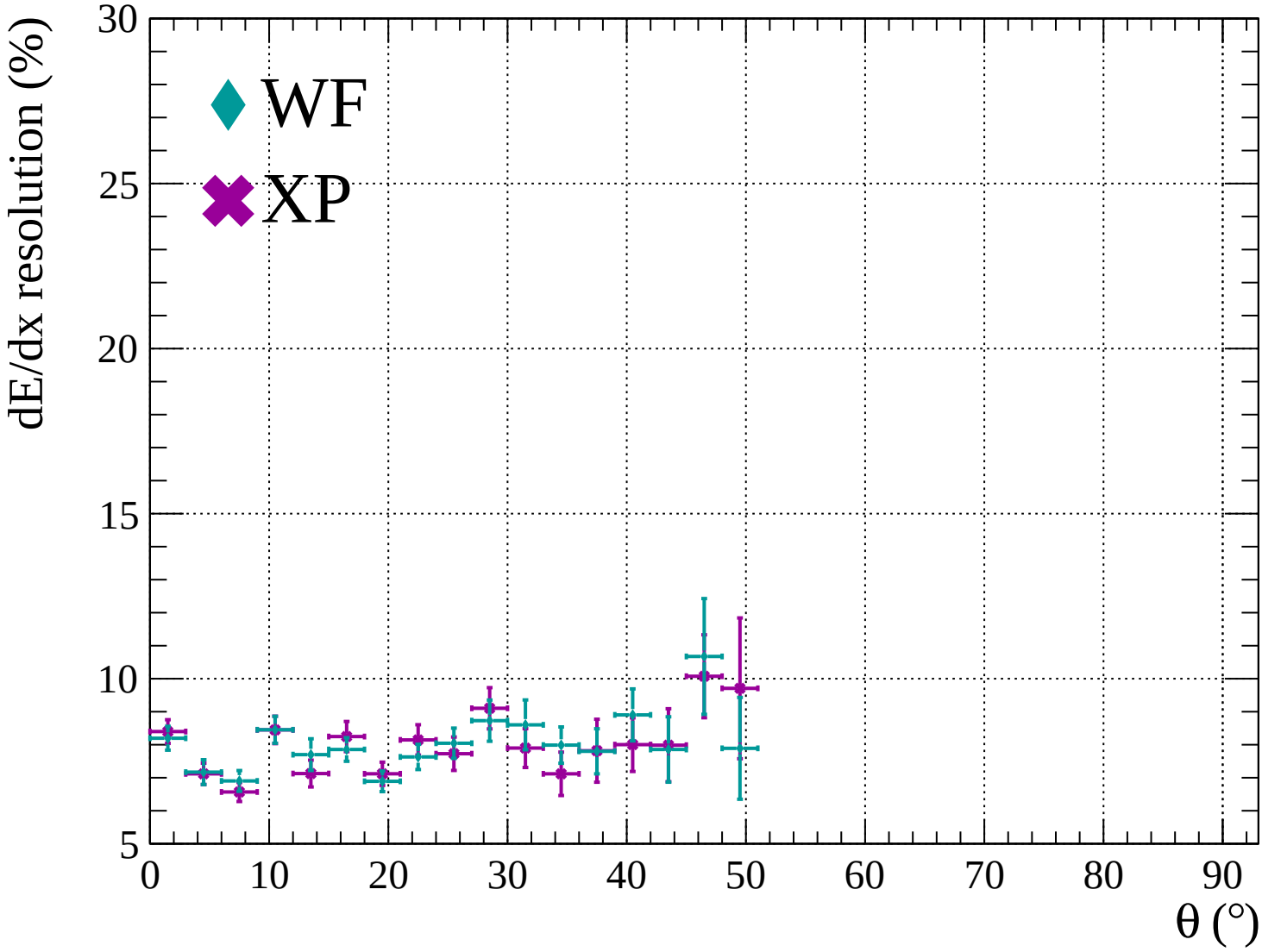


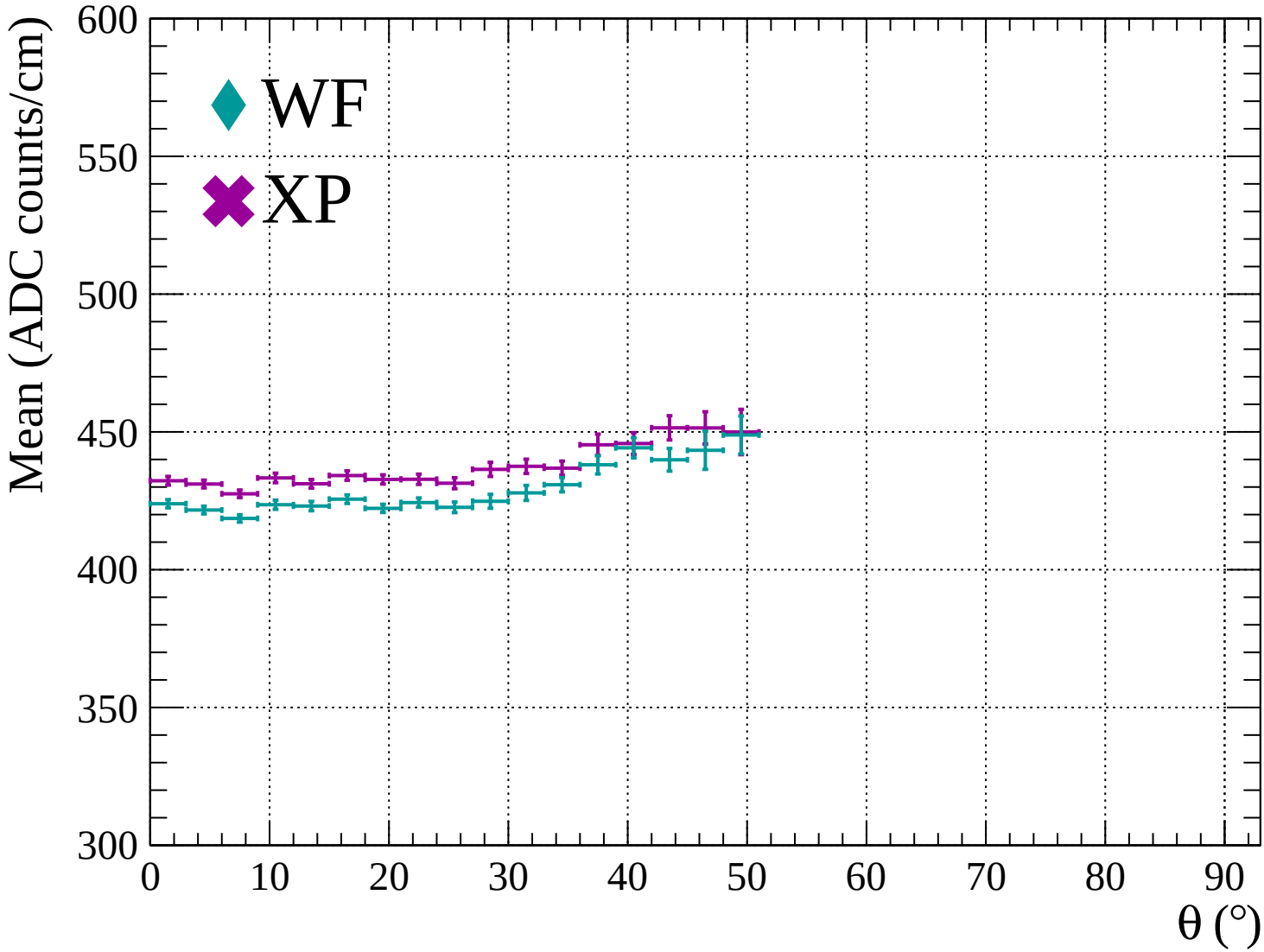


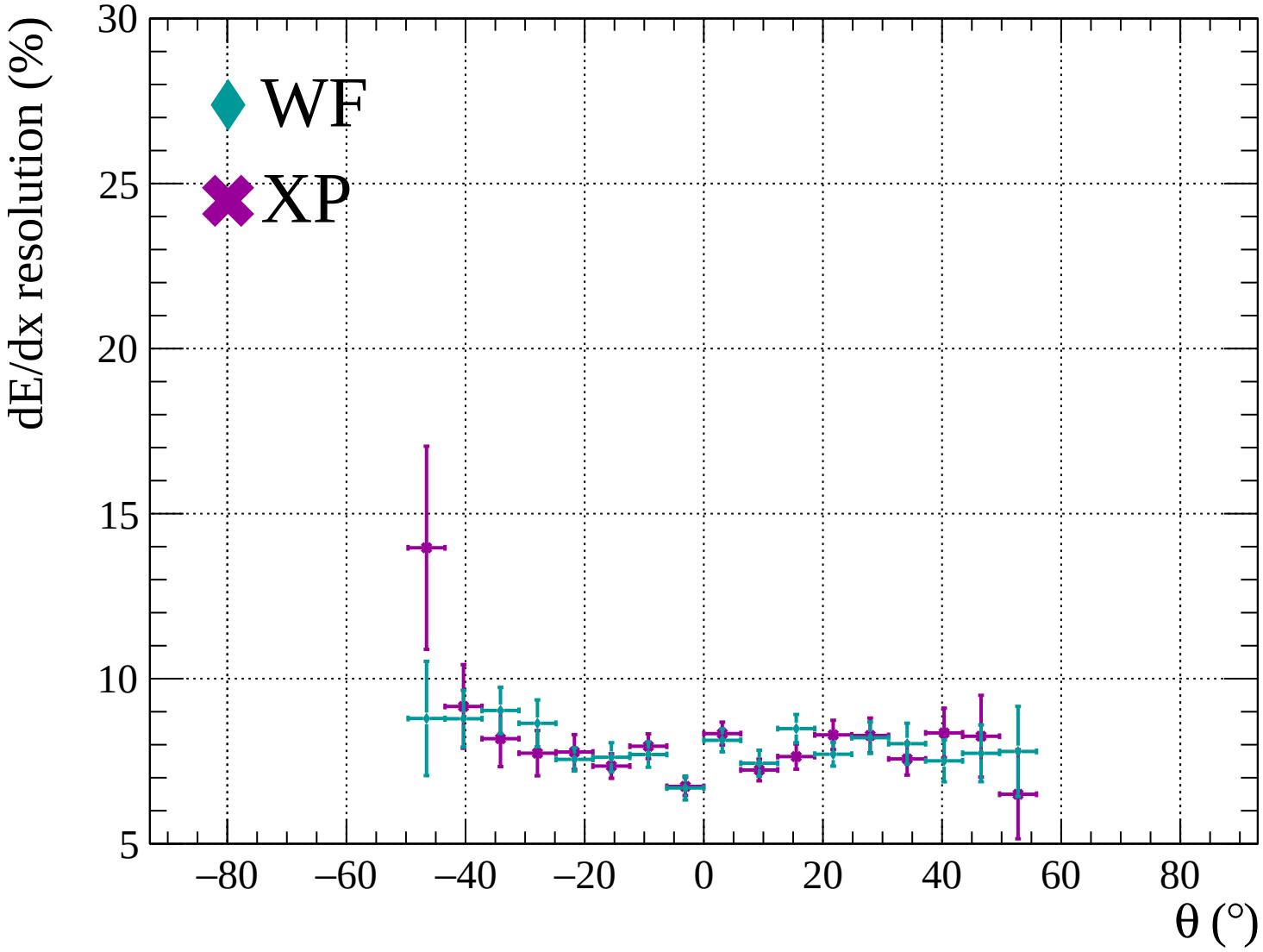


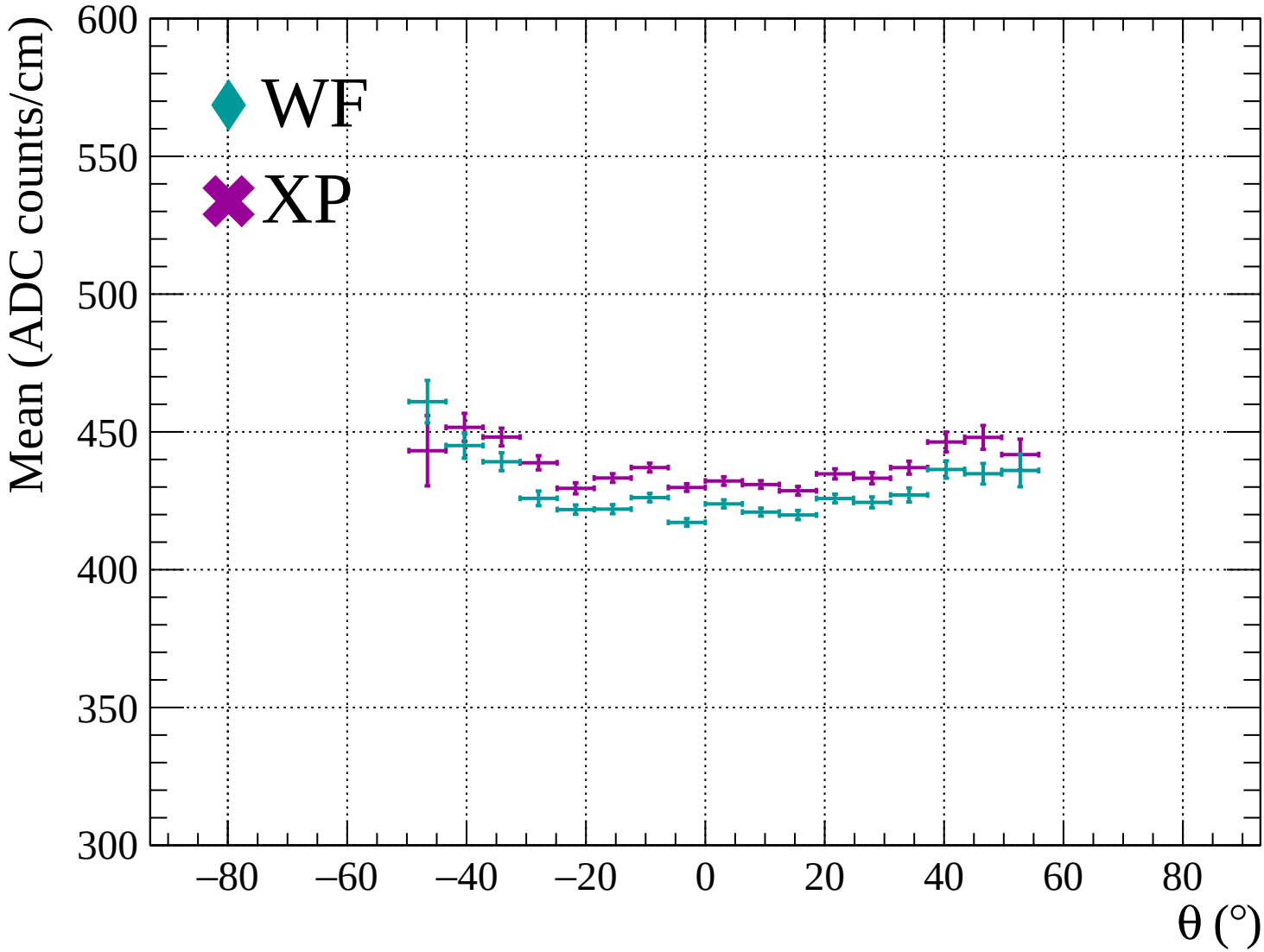


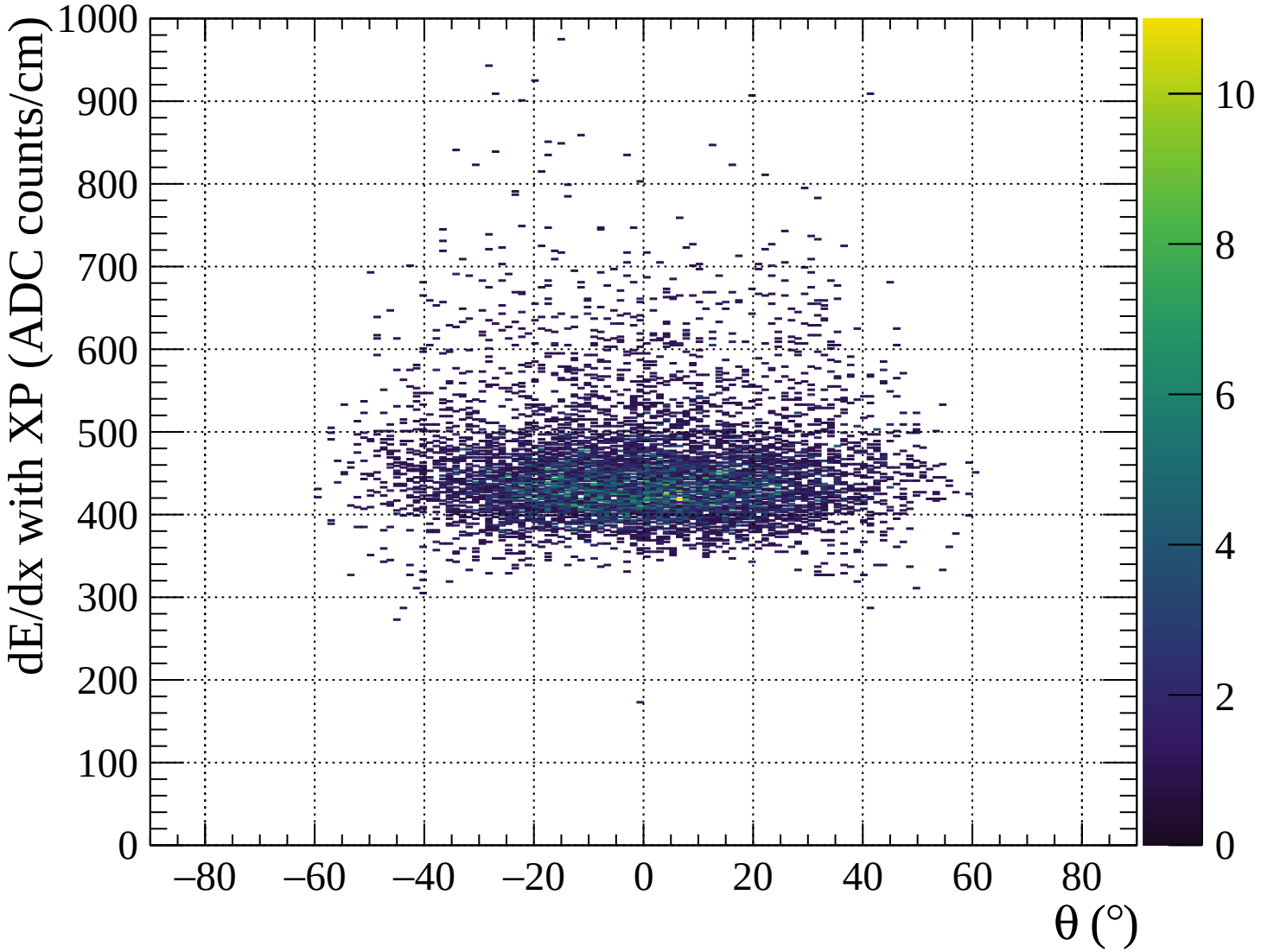


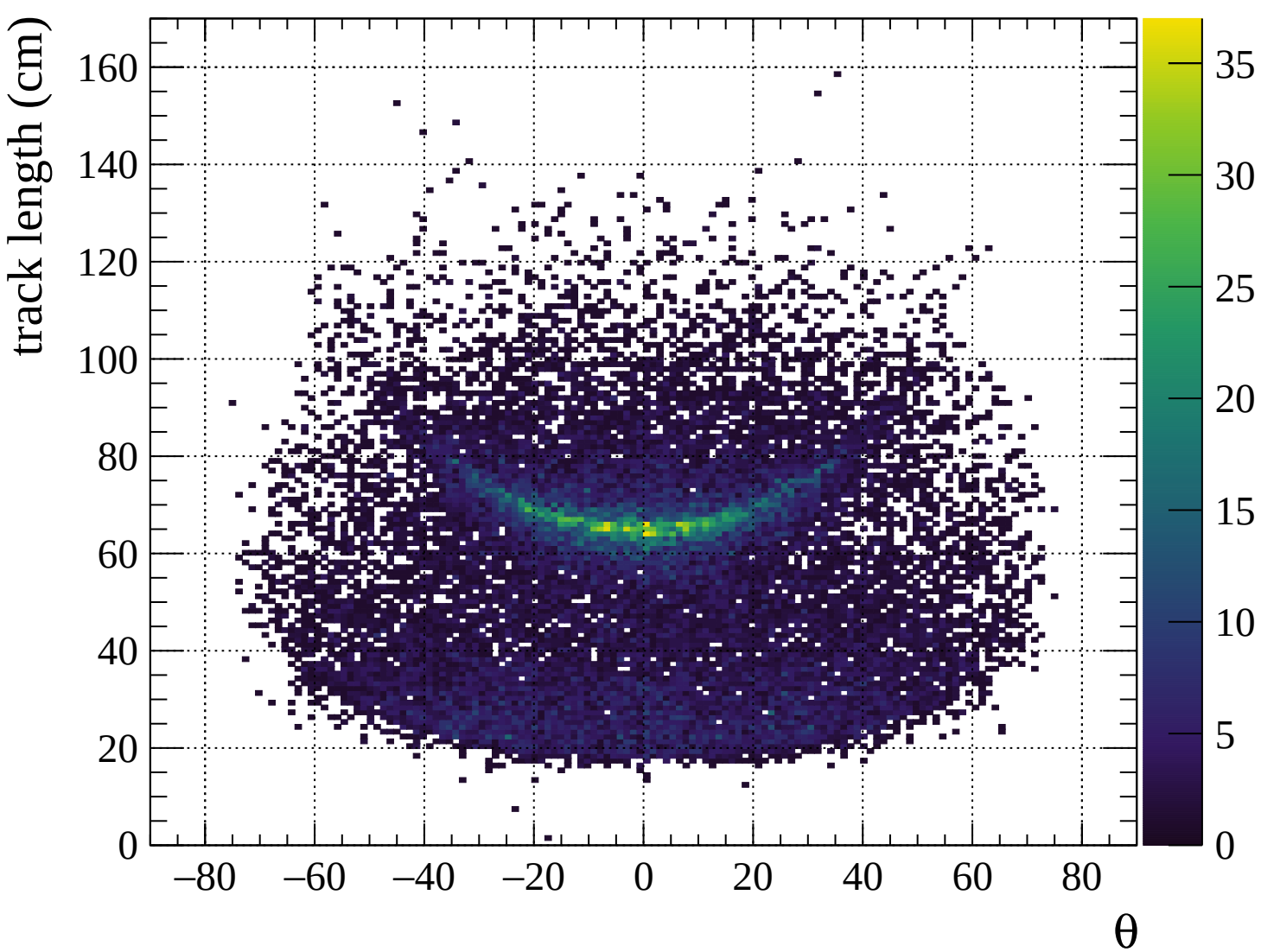


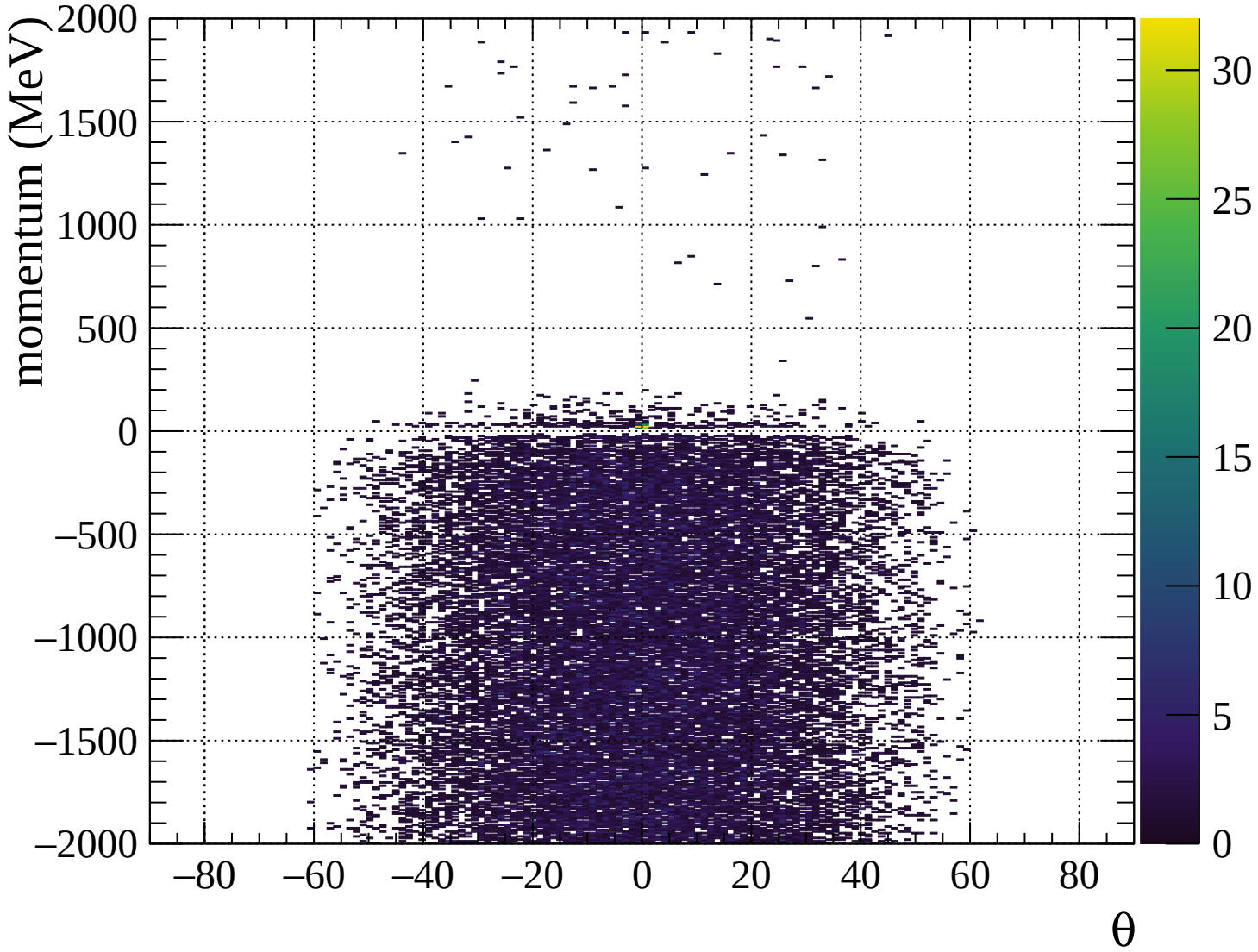


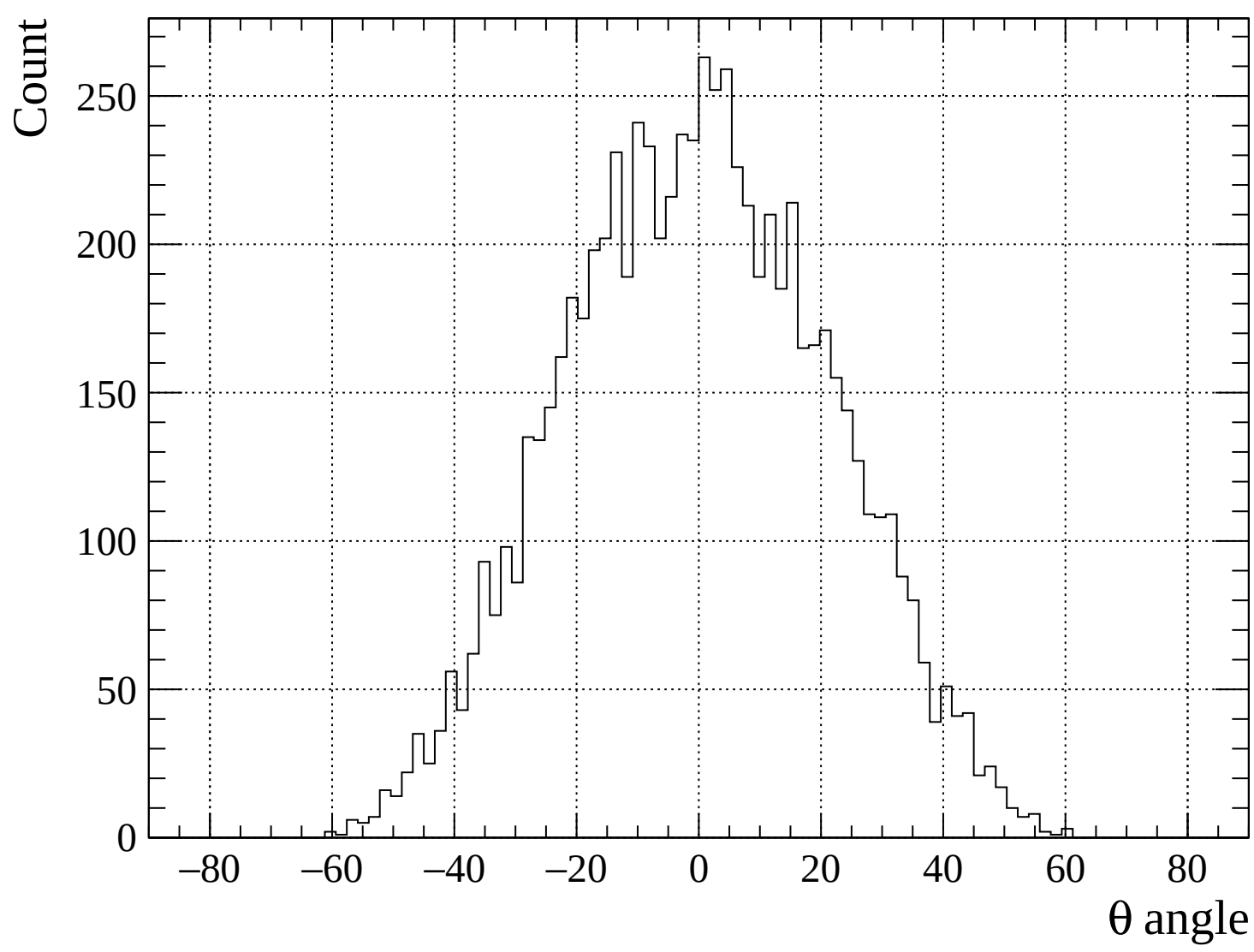


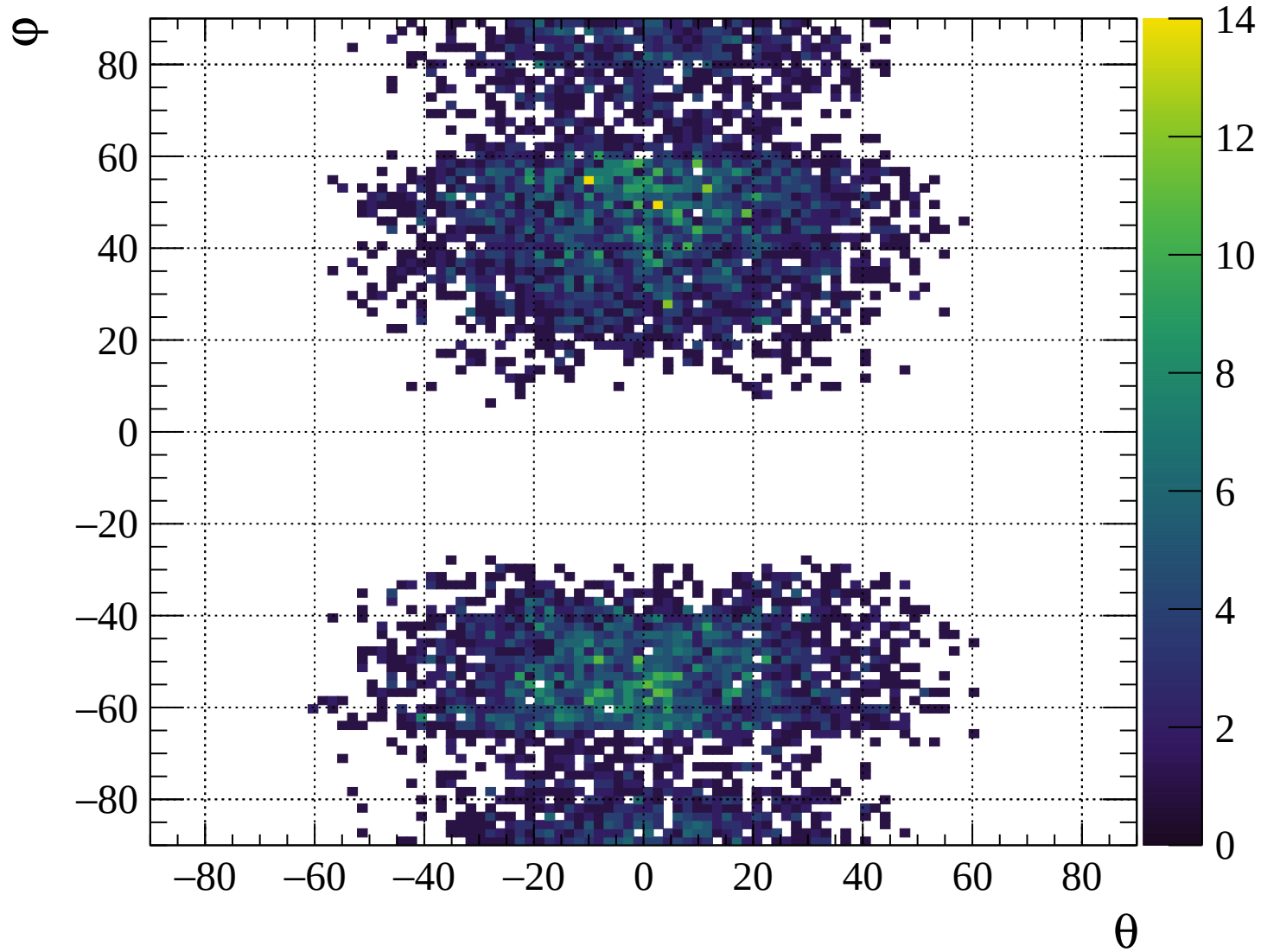


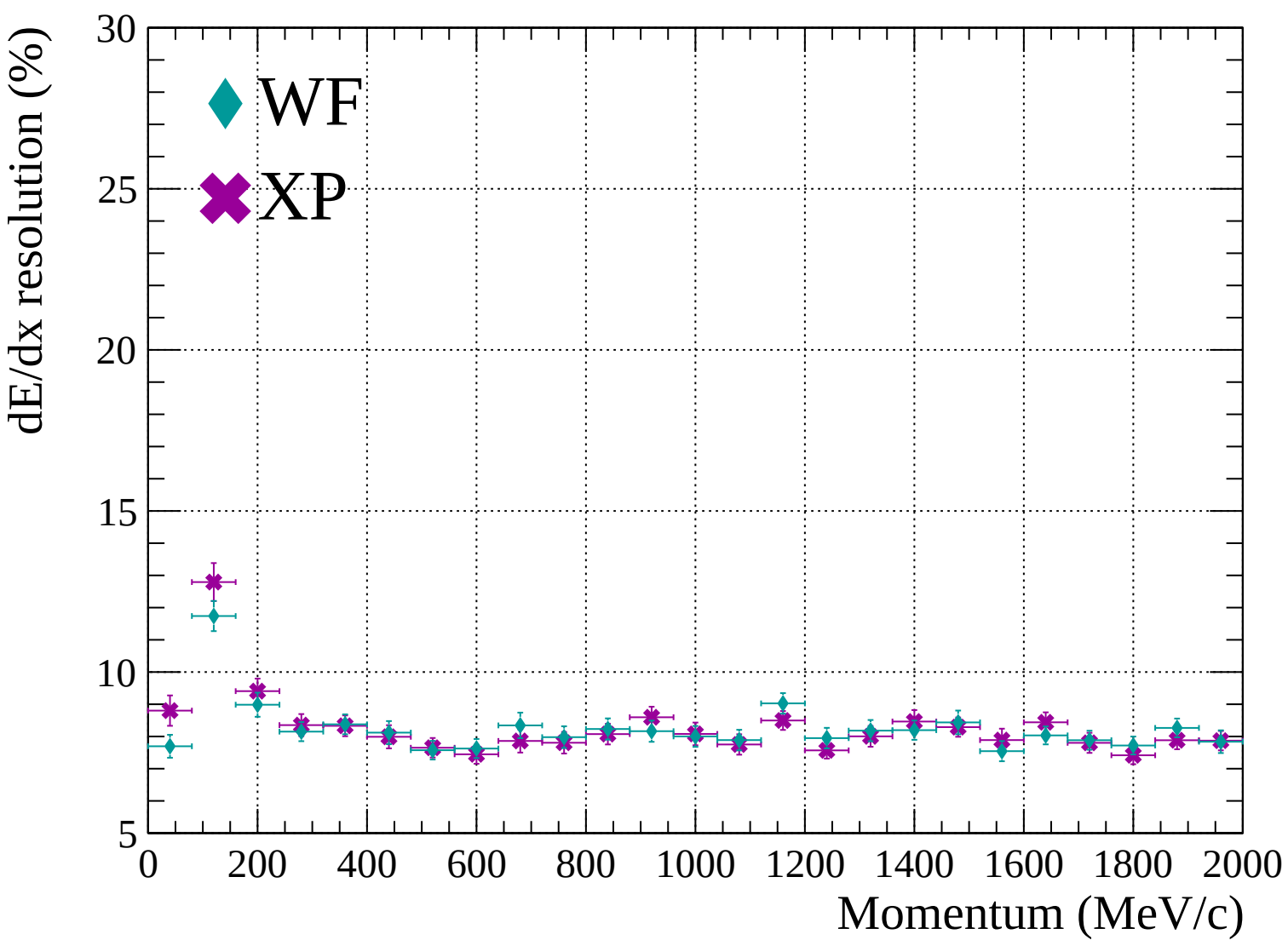


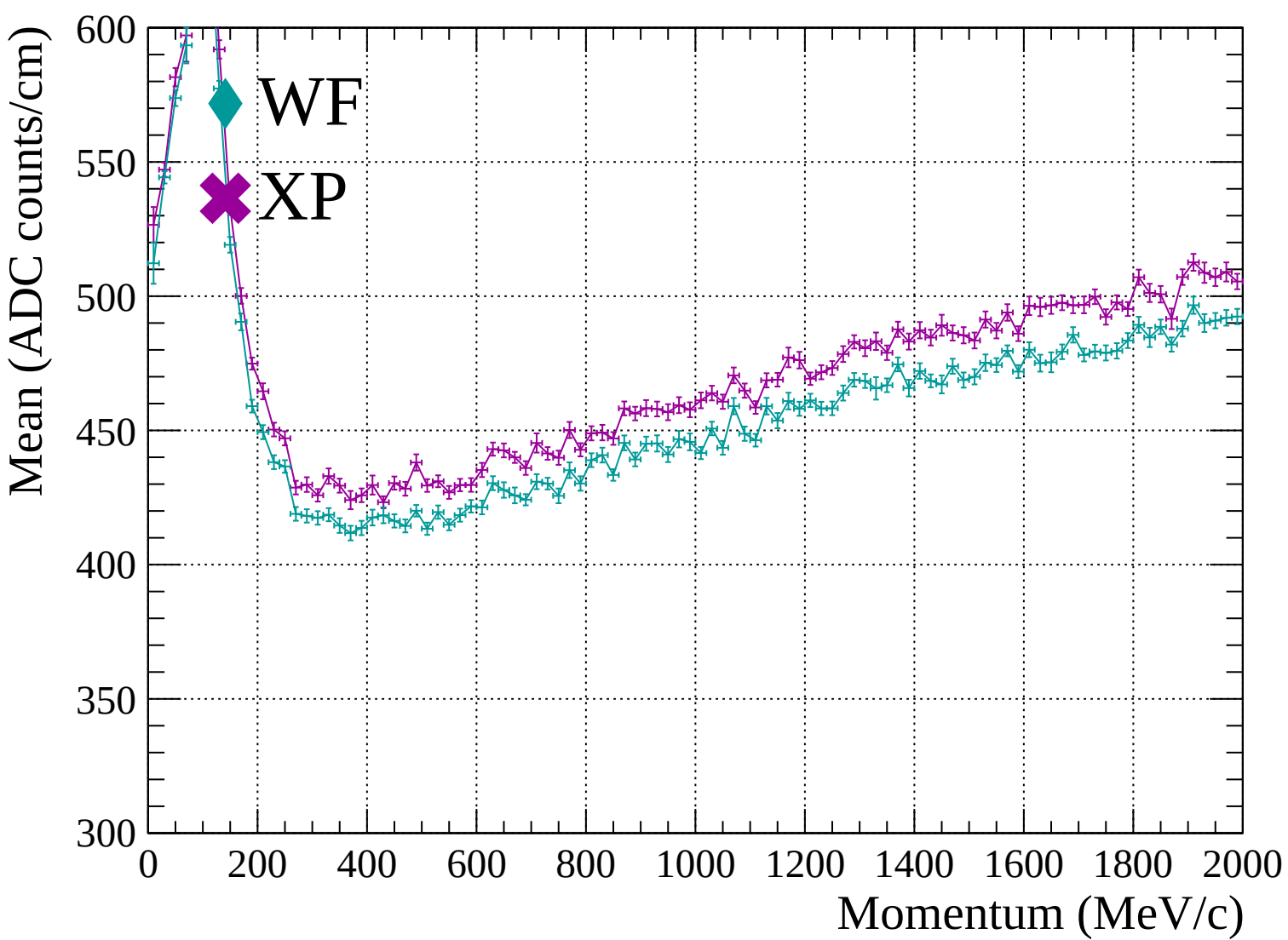


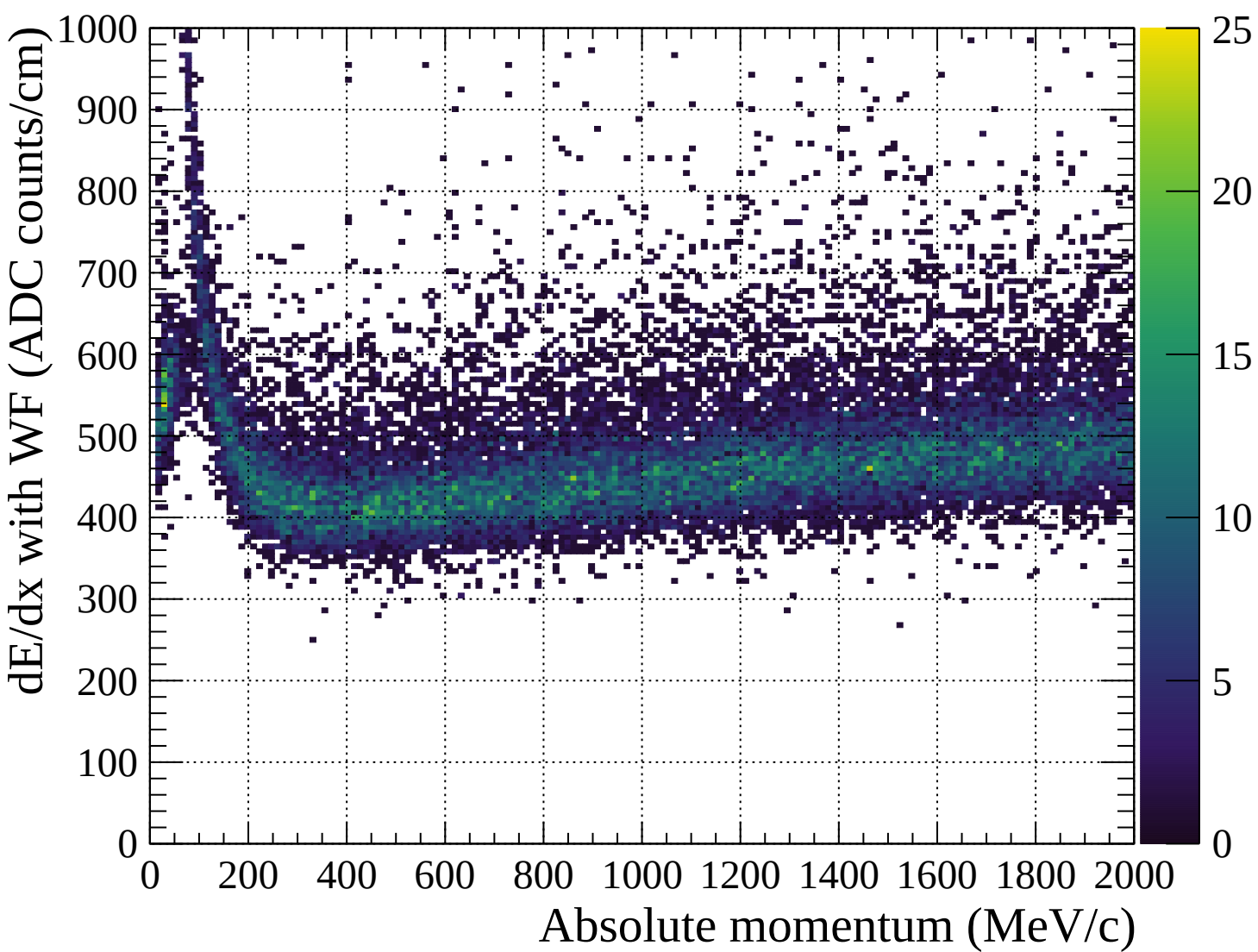


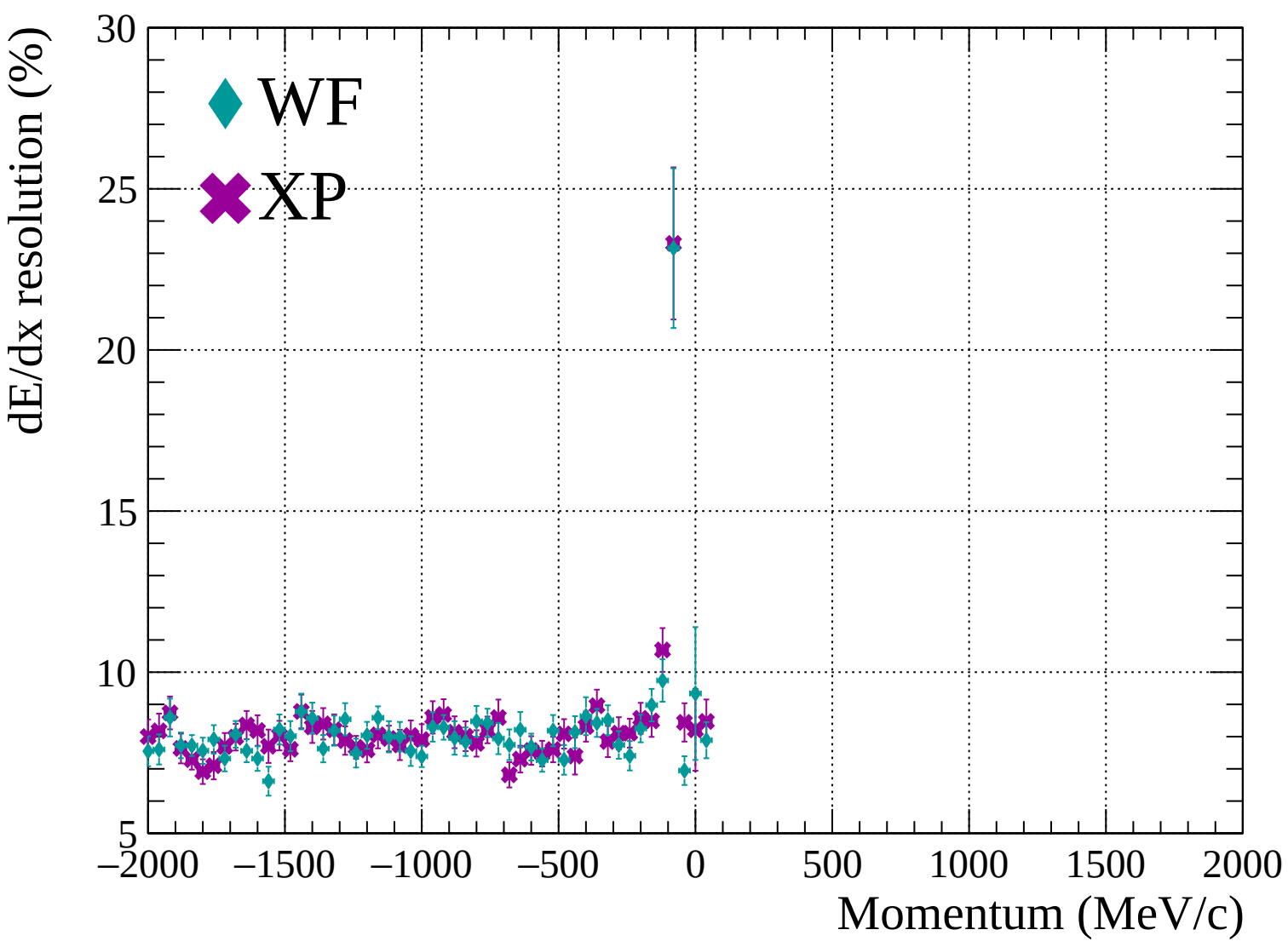


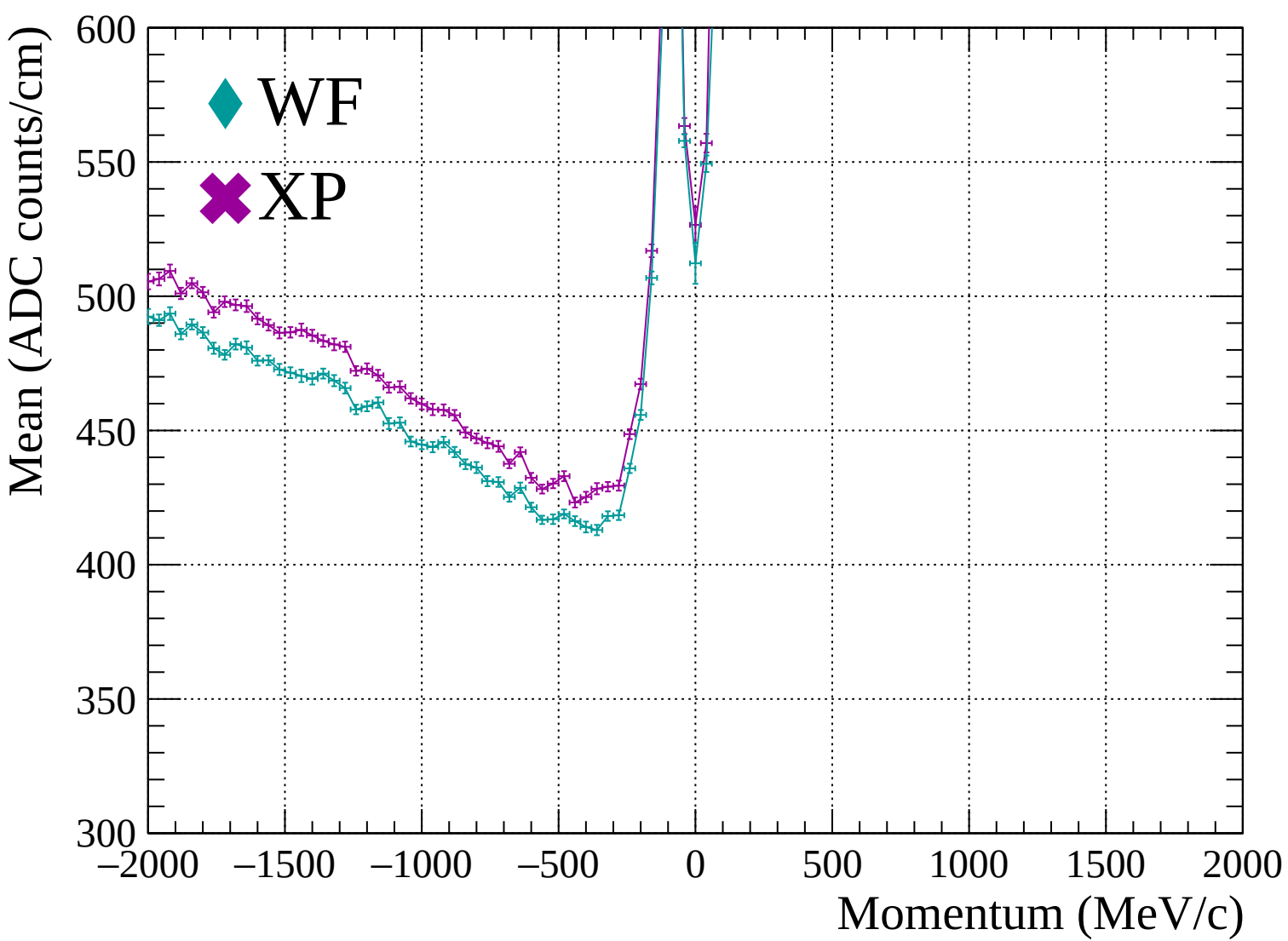


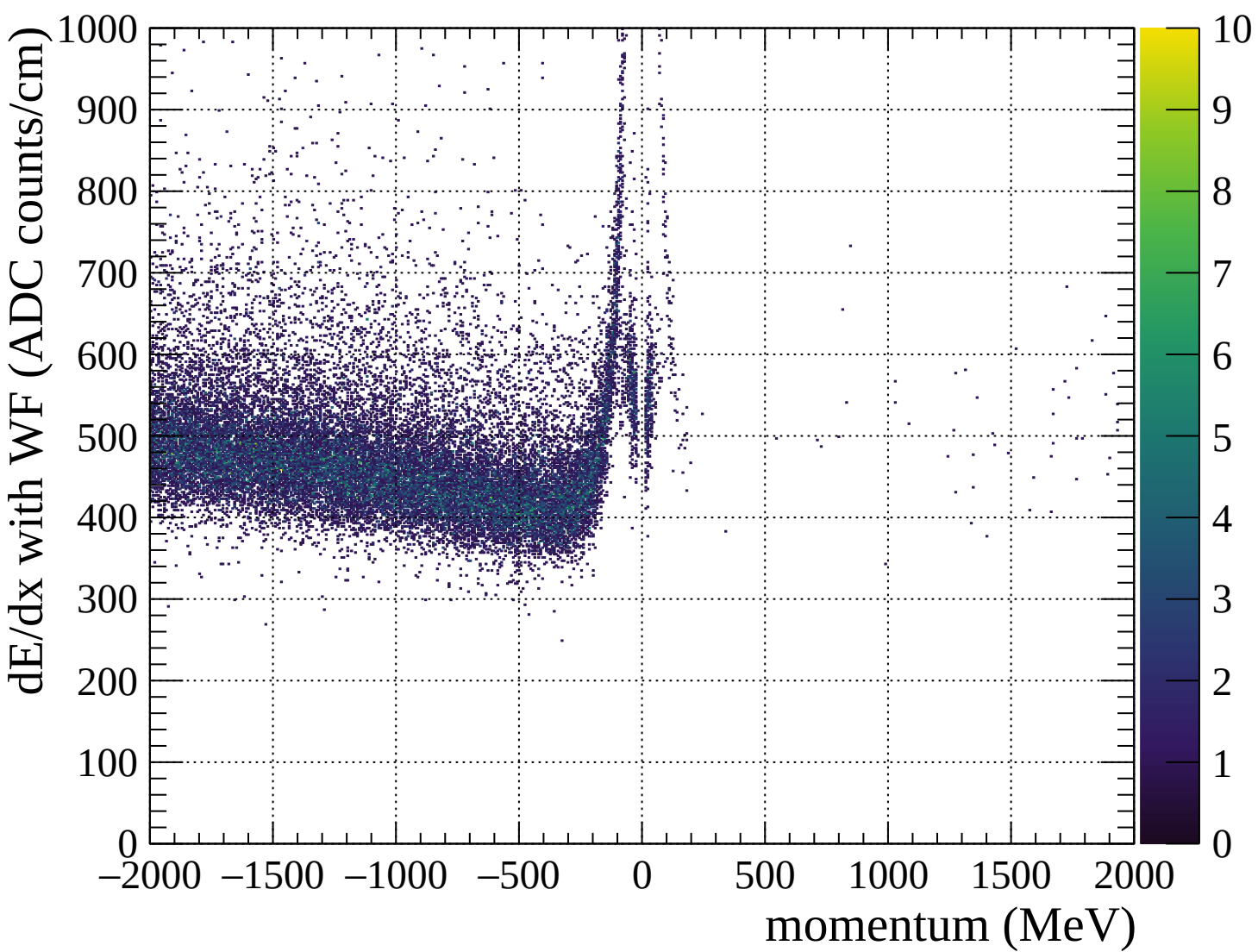


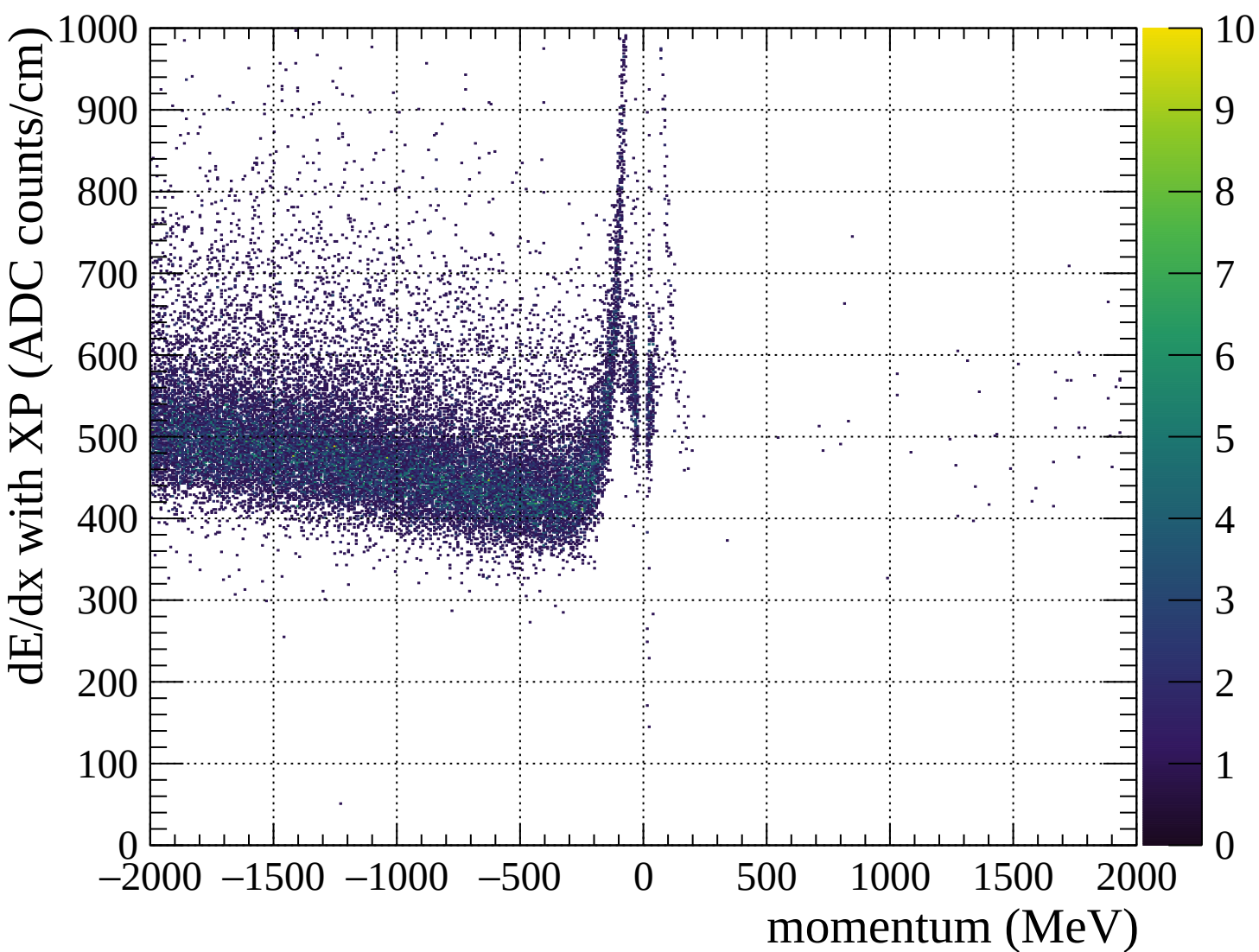


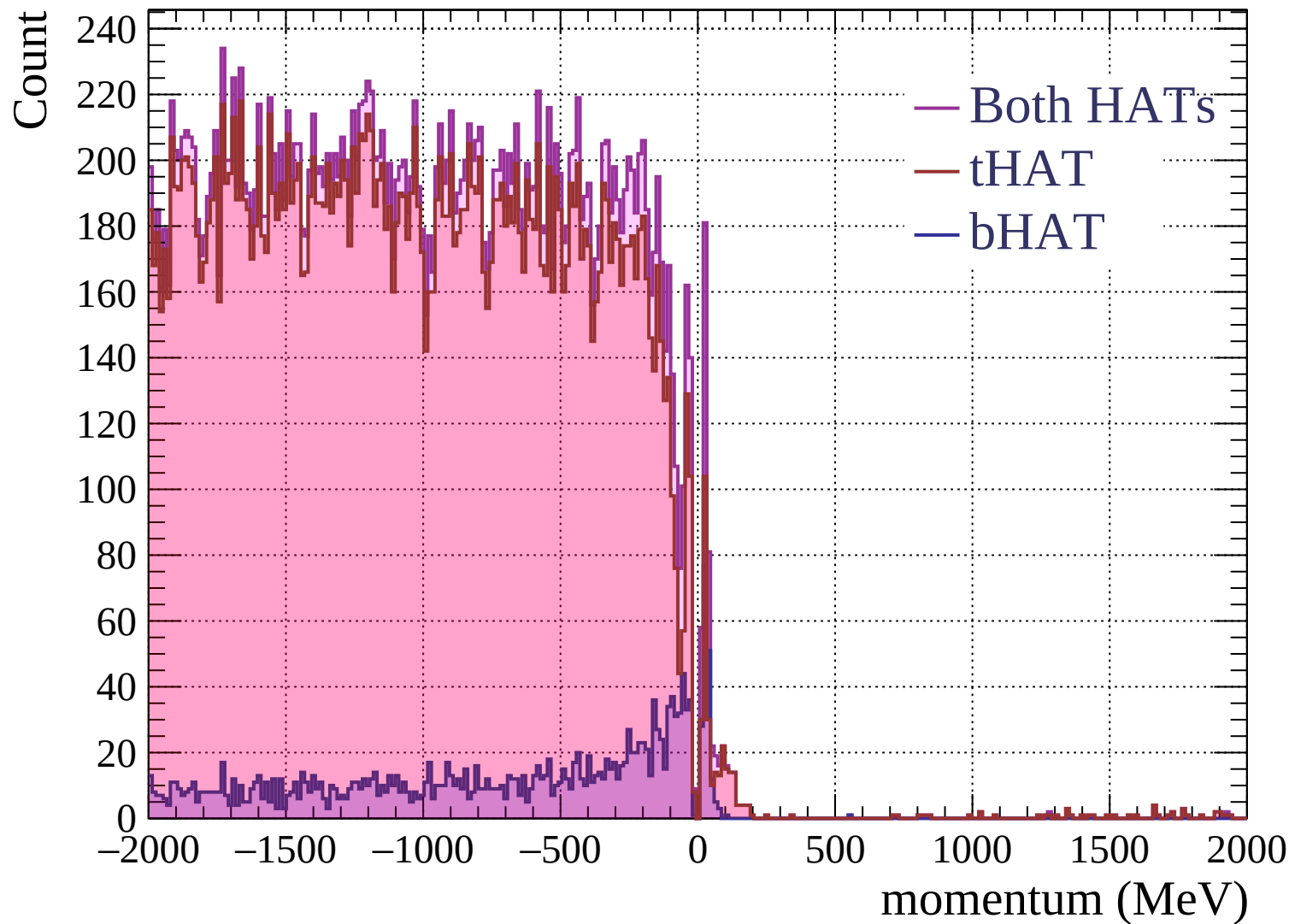


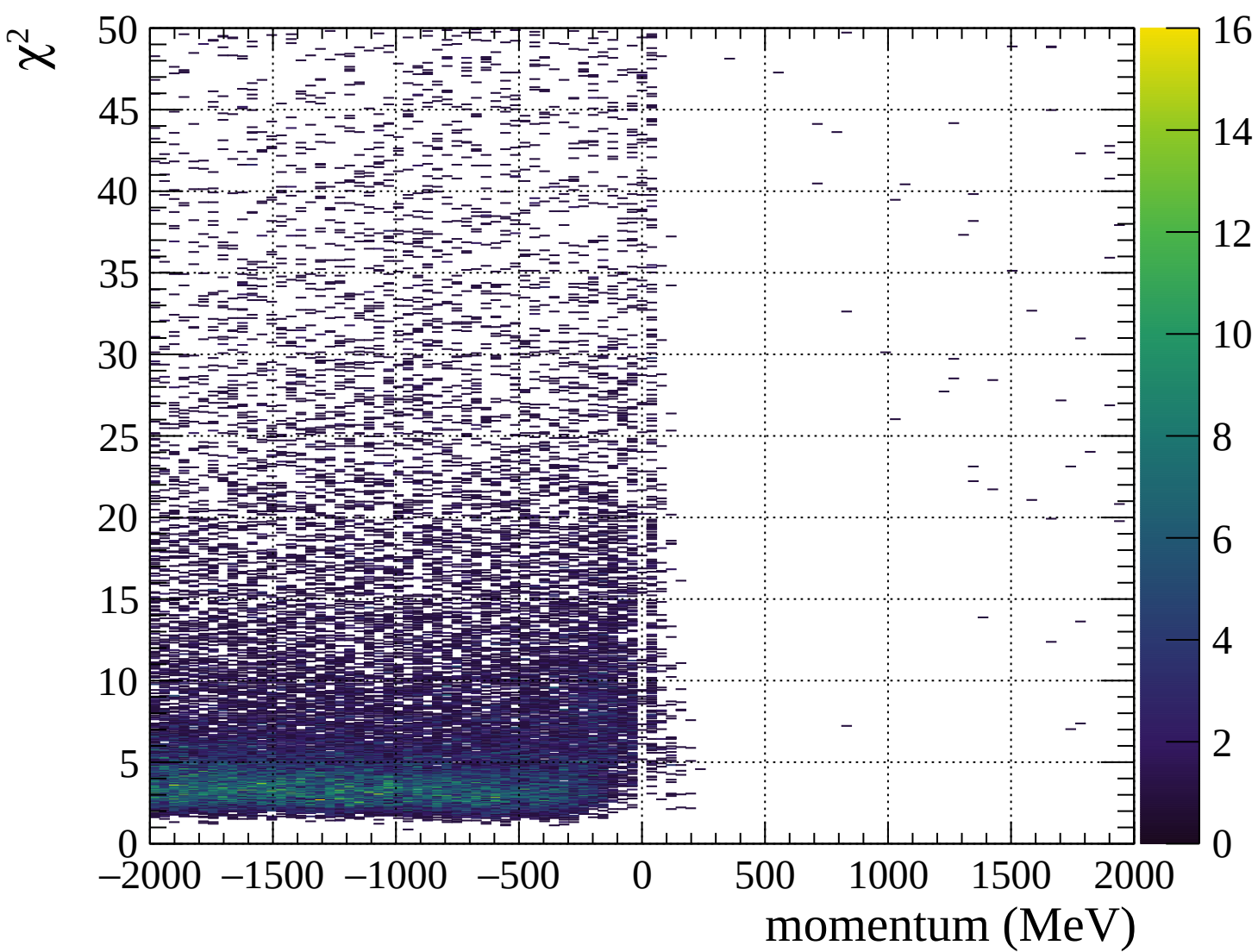




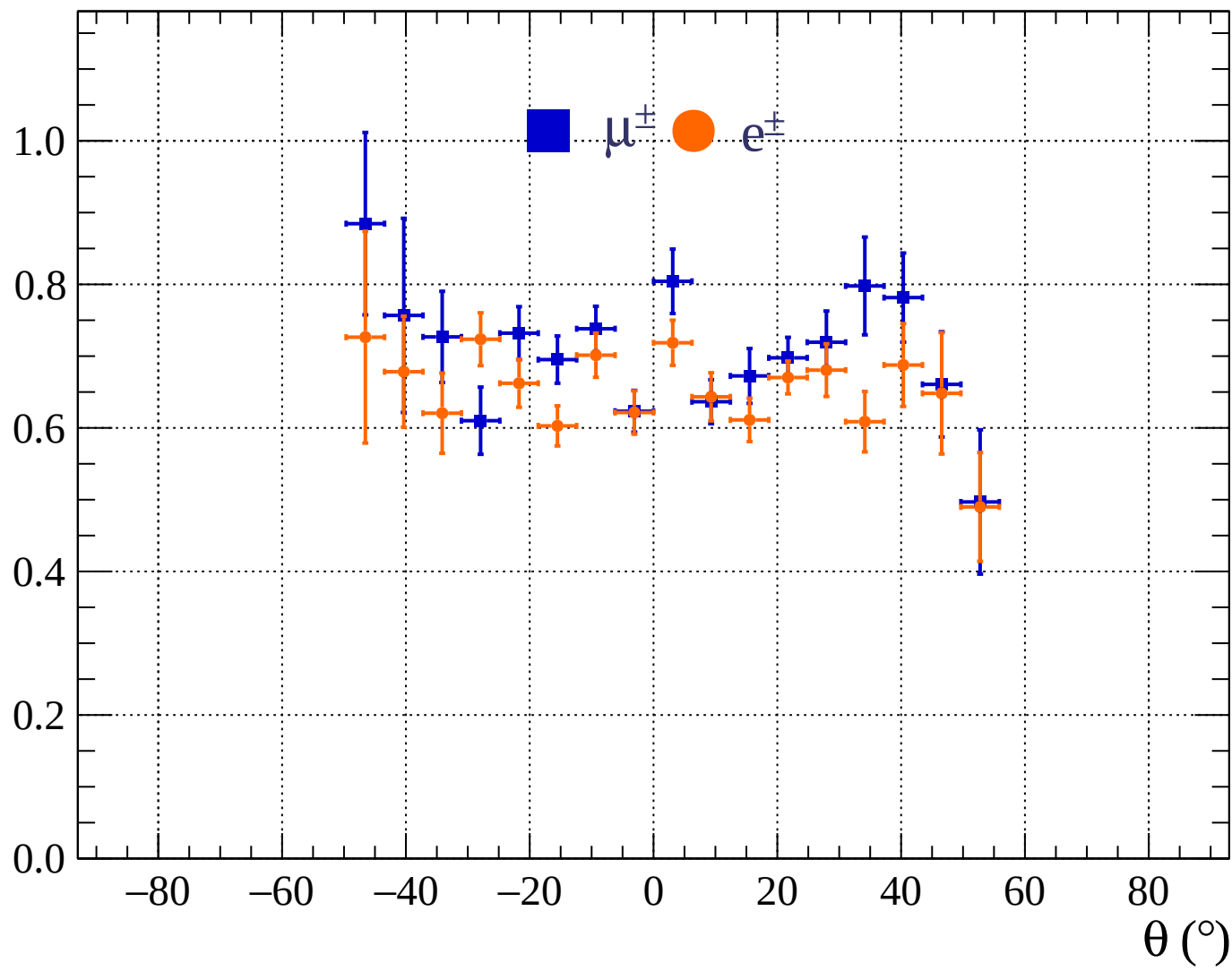




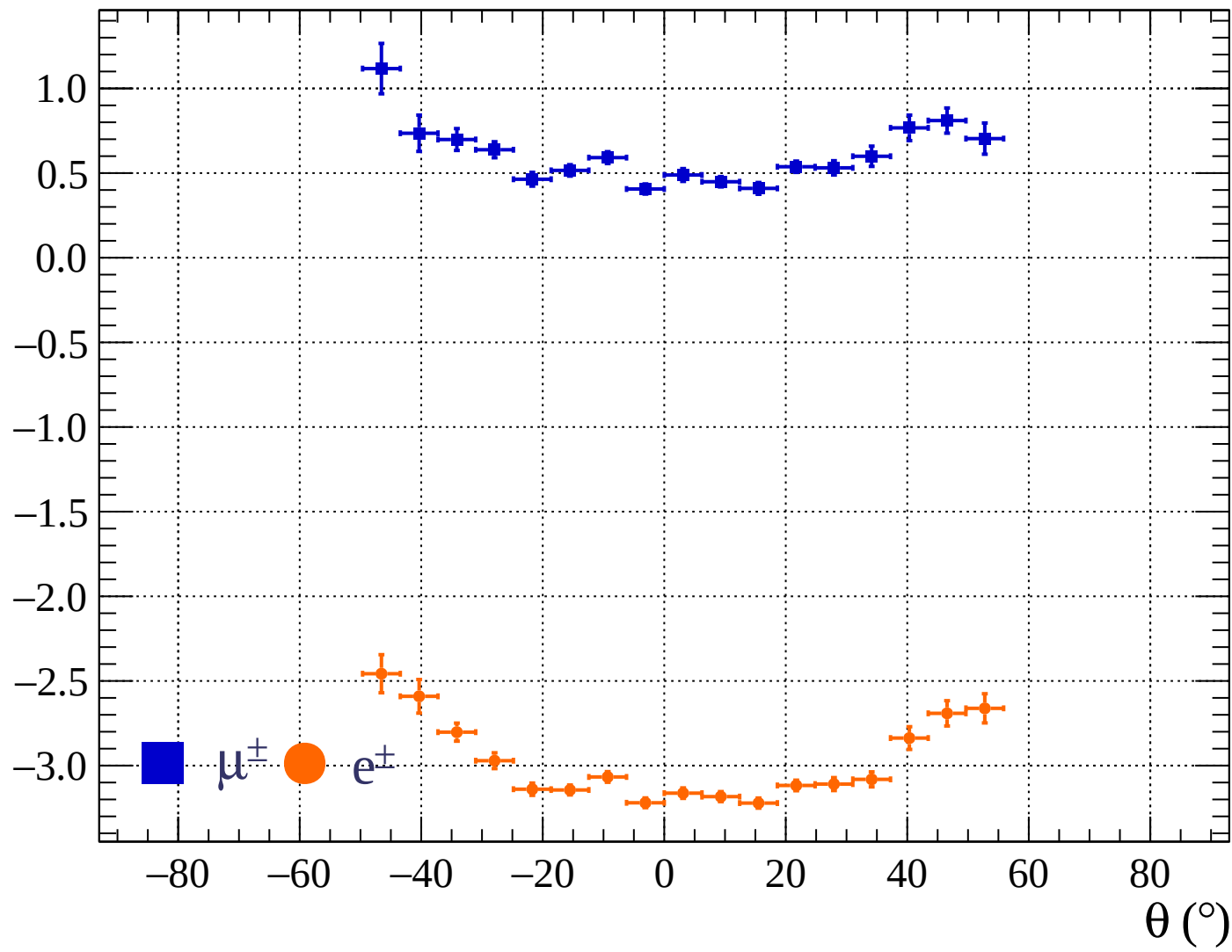




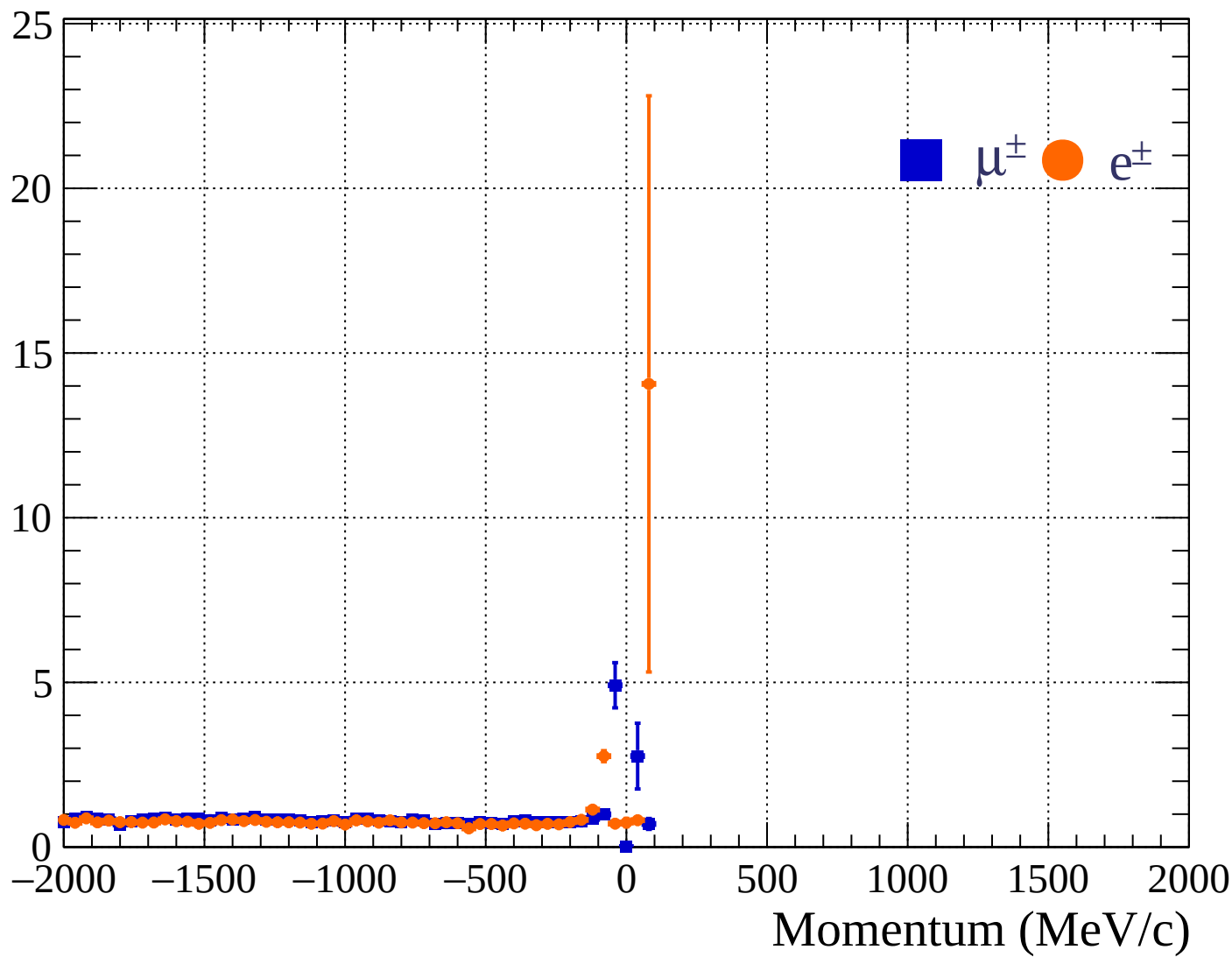
Pulls standard deviation



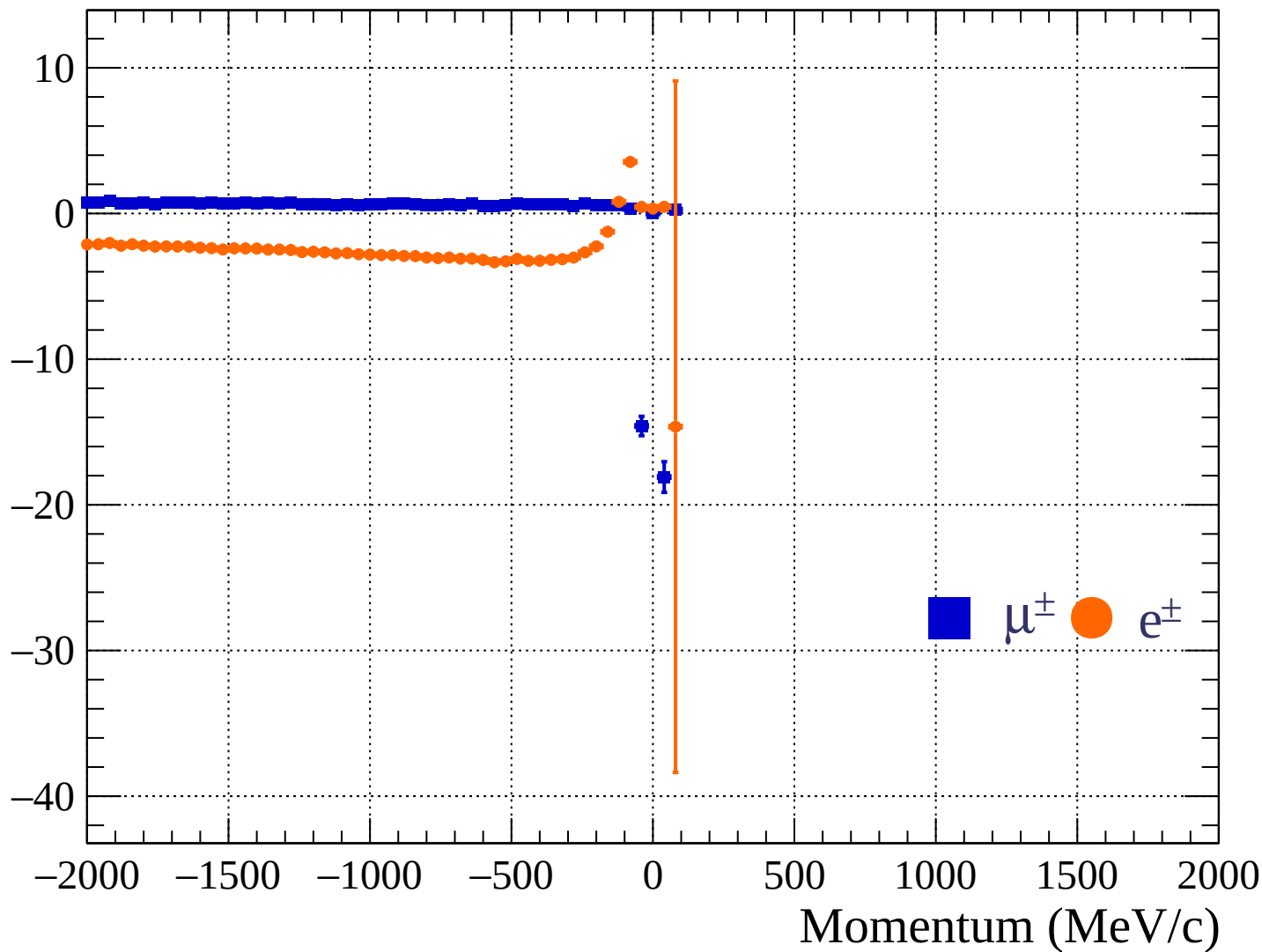
Pulls mean

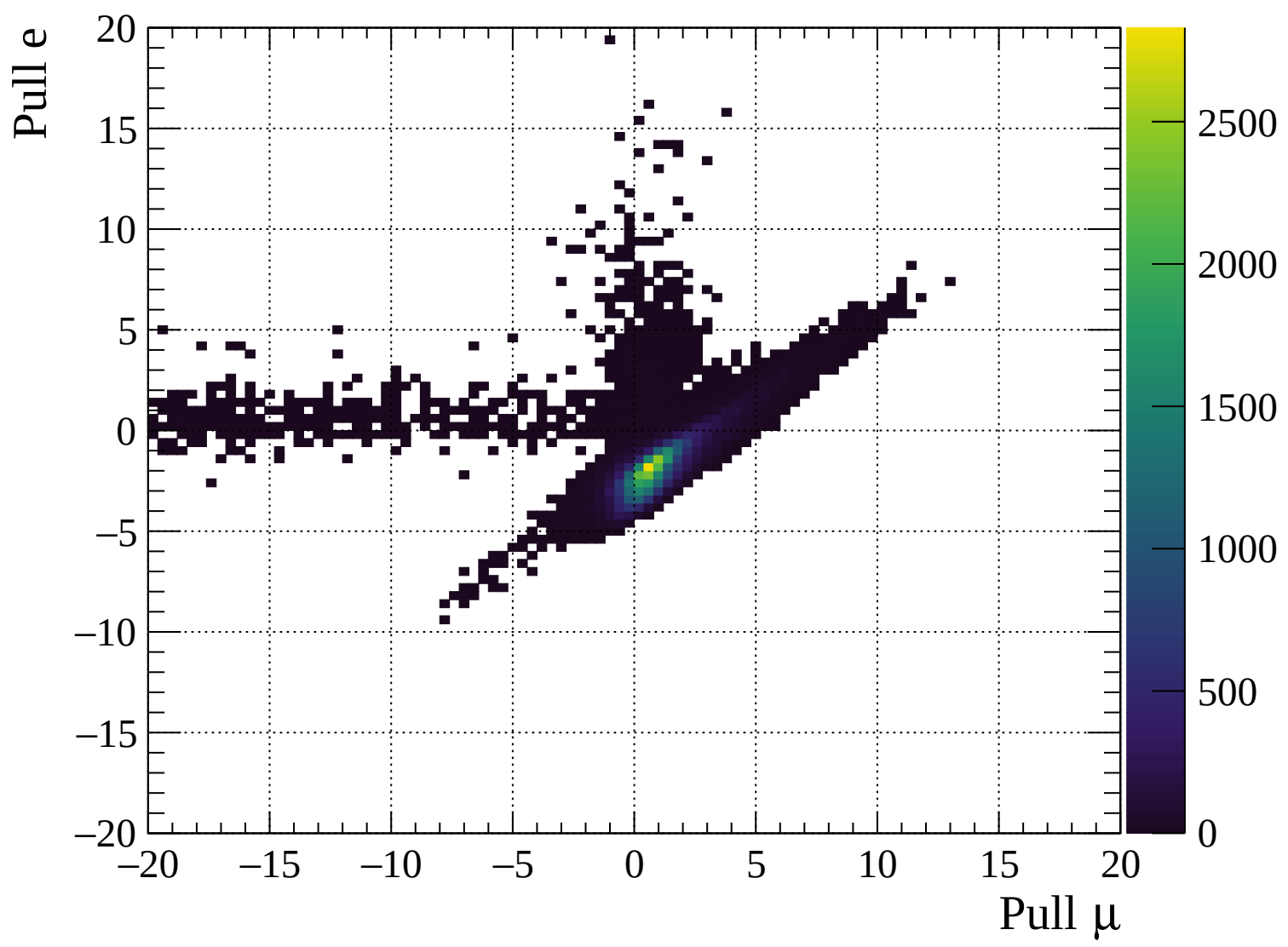


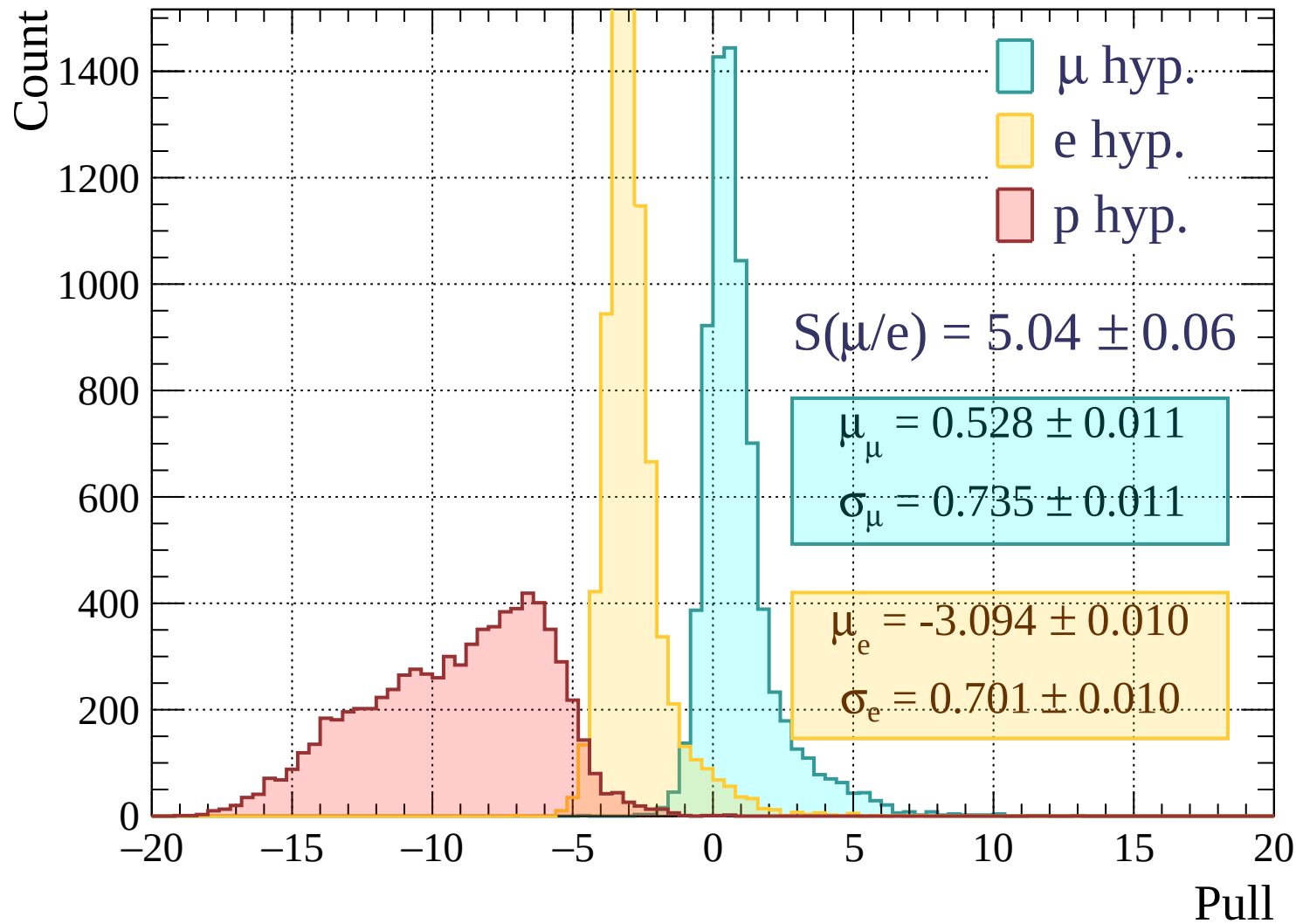
Pulls standard deviation

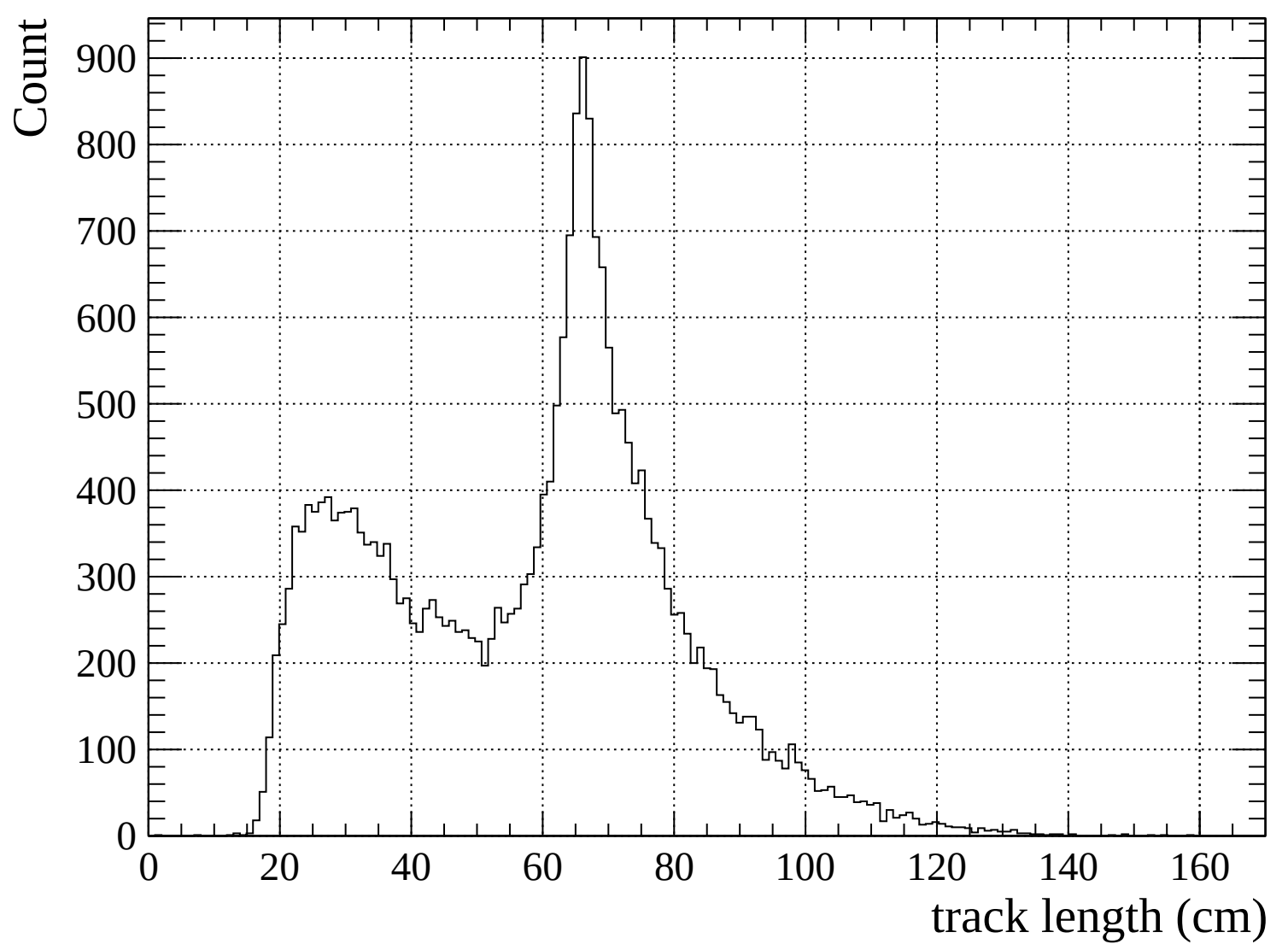


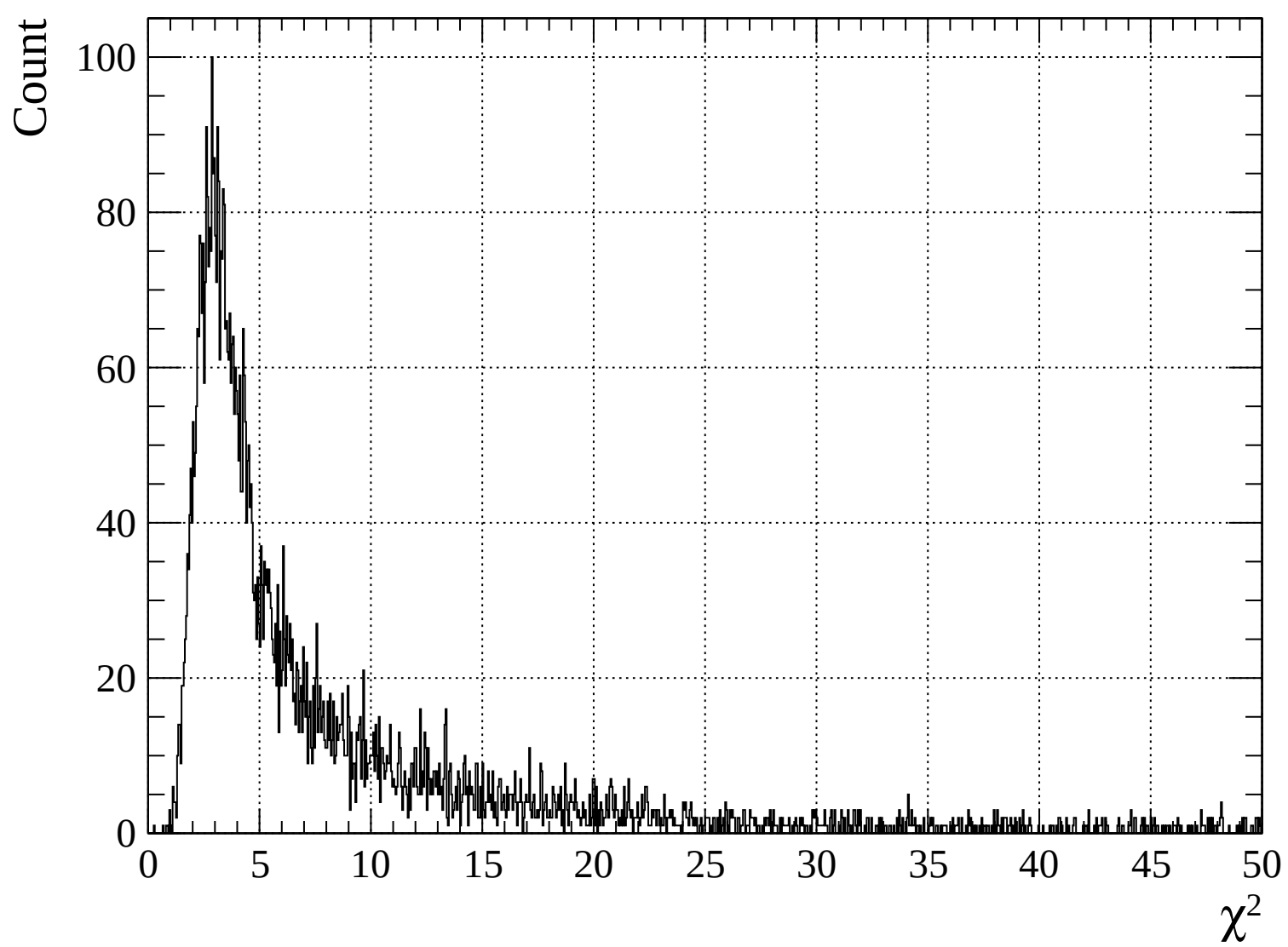
Pulls mean

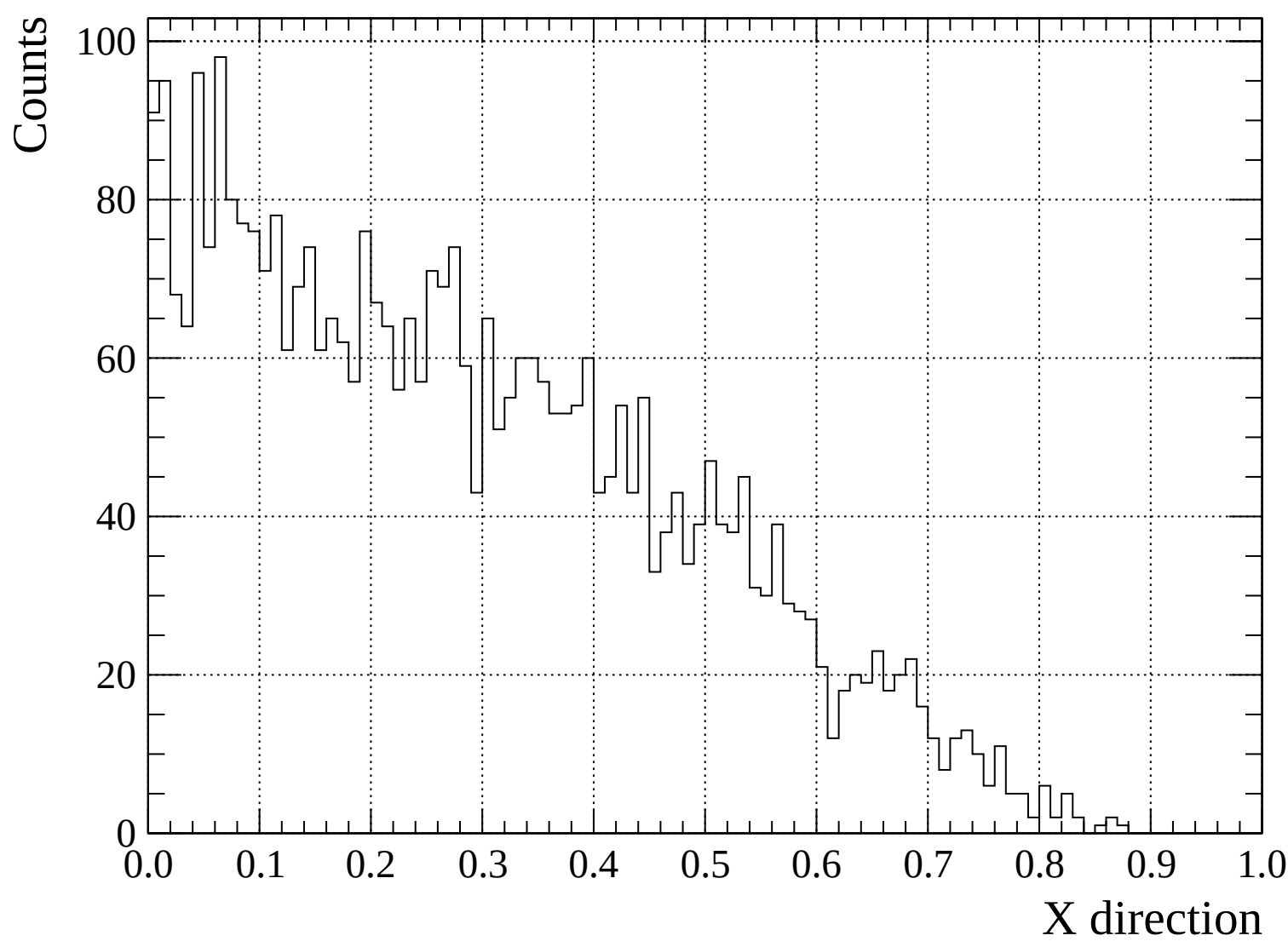


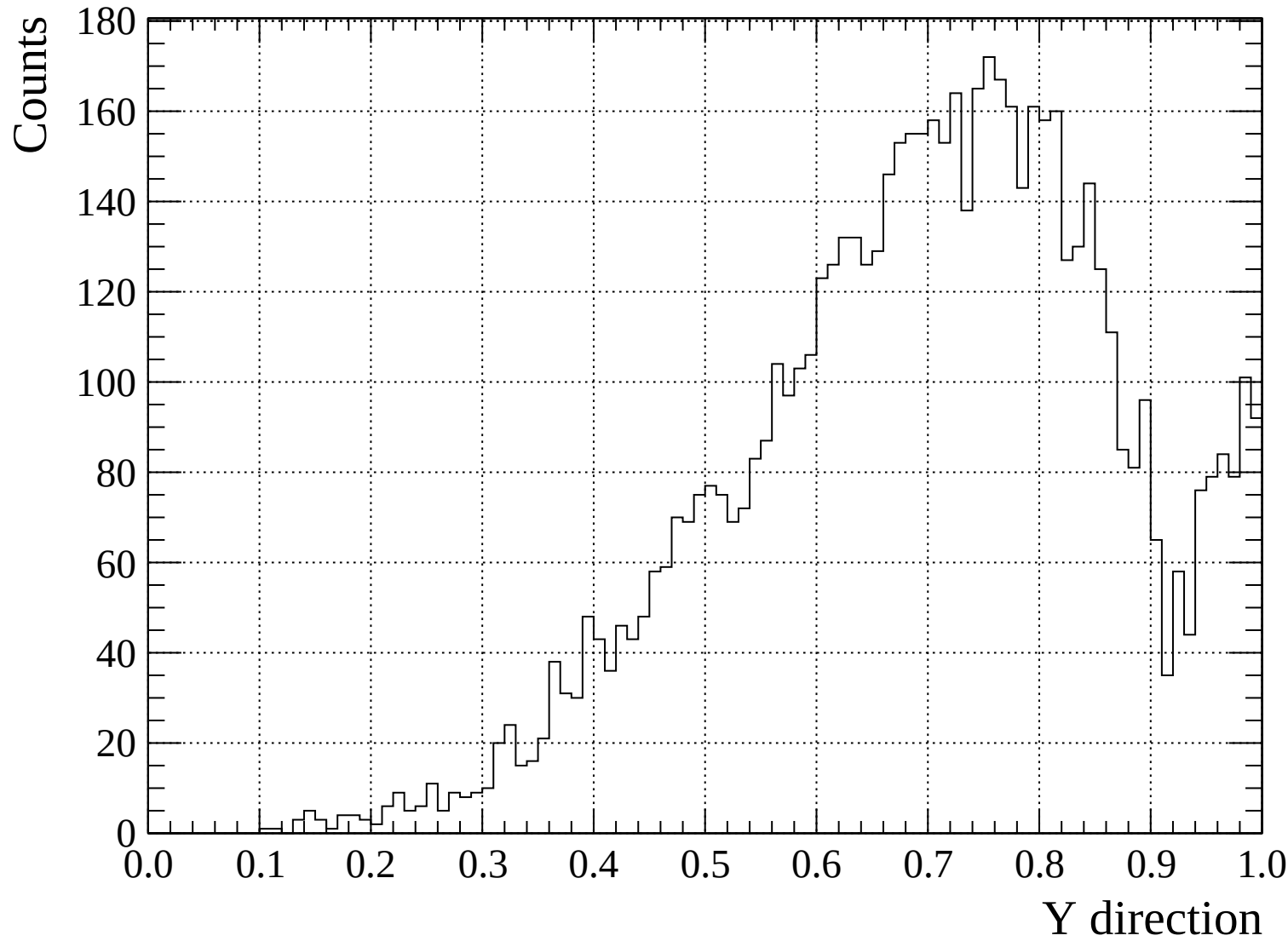


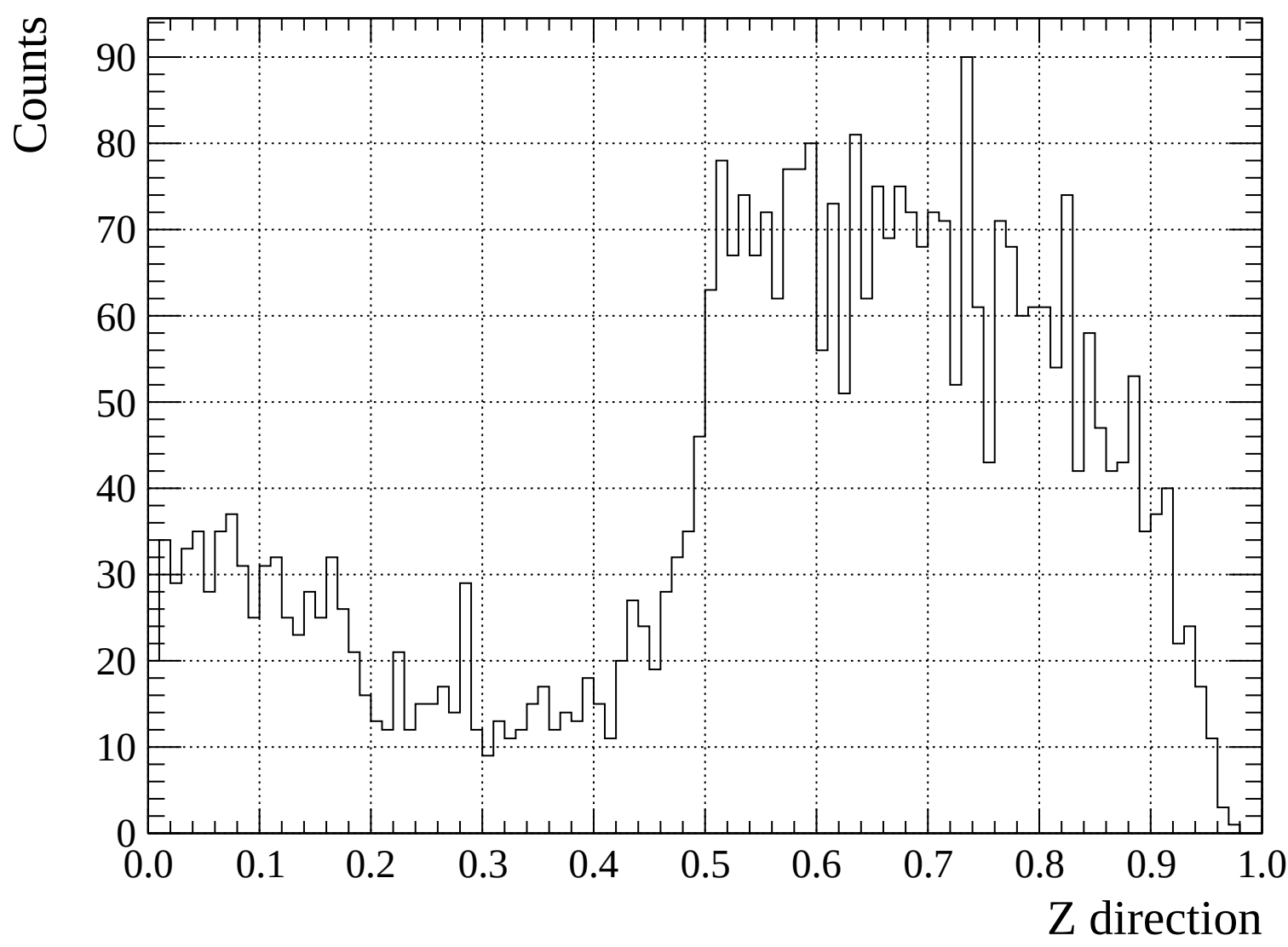


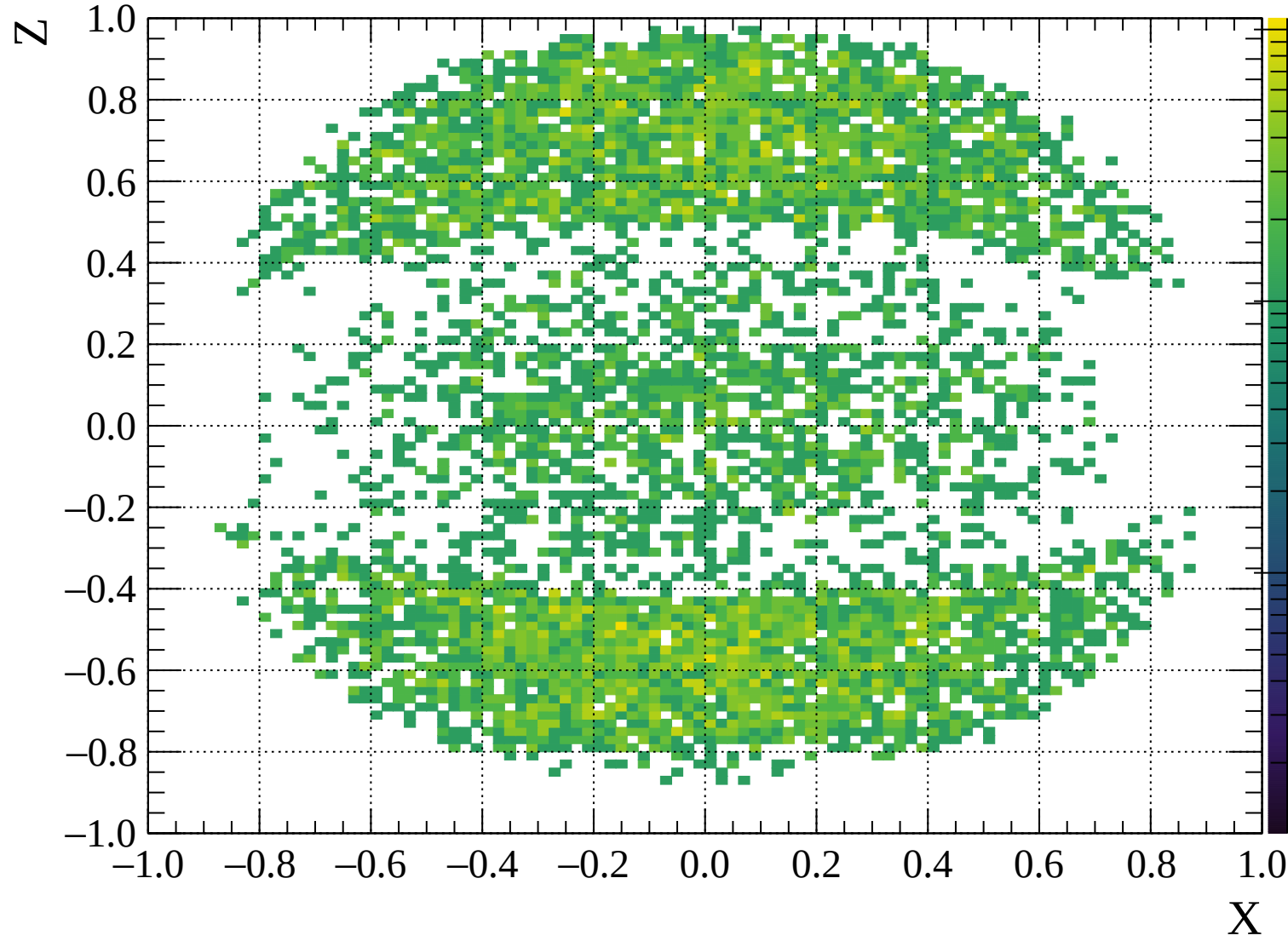


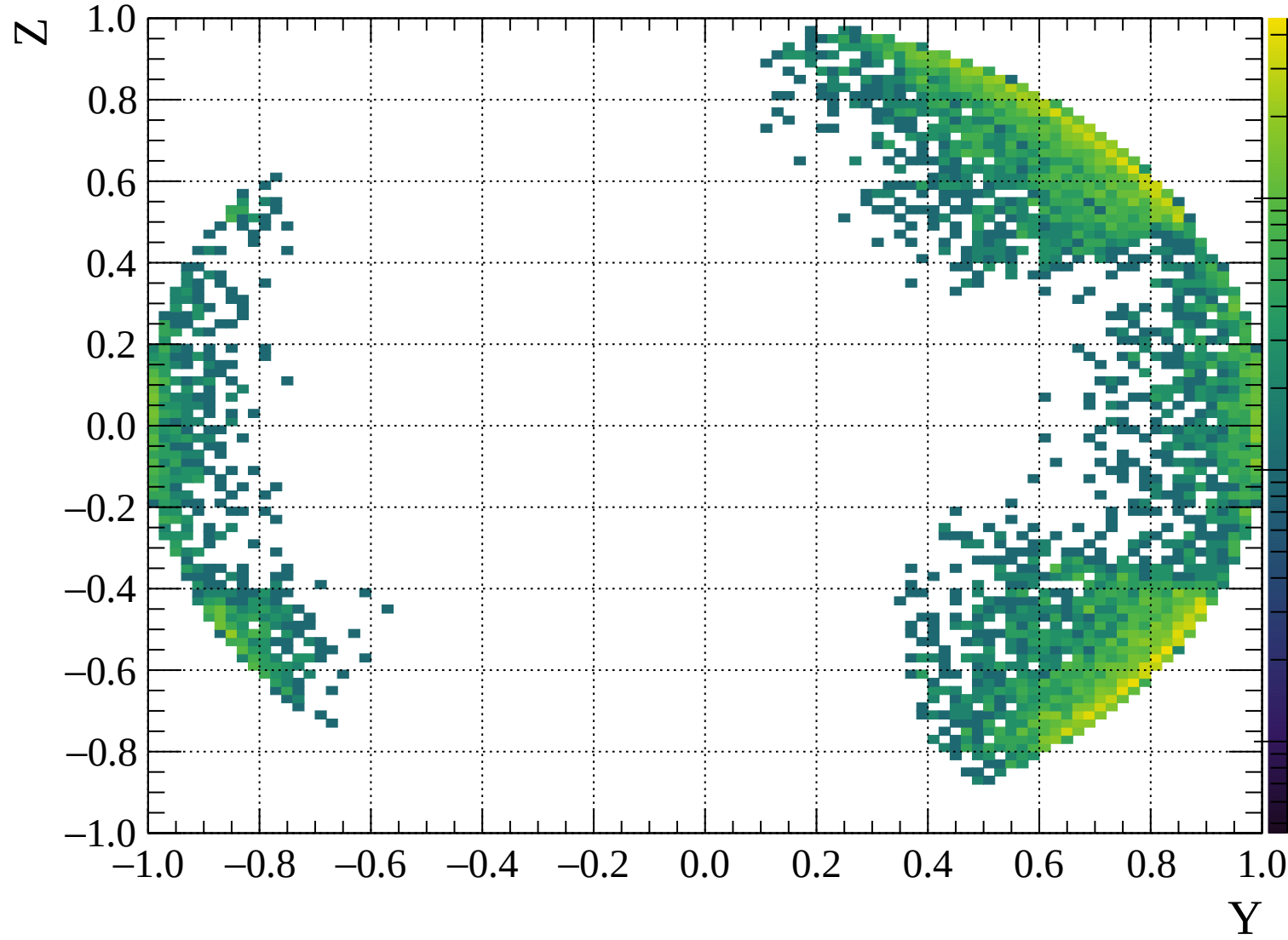


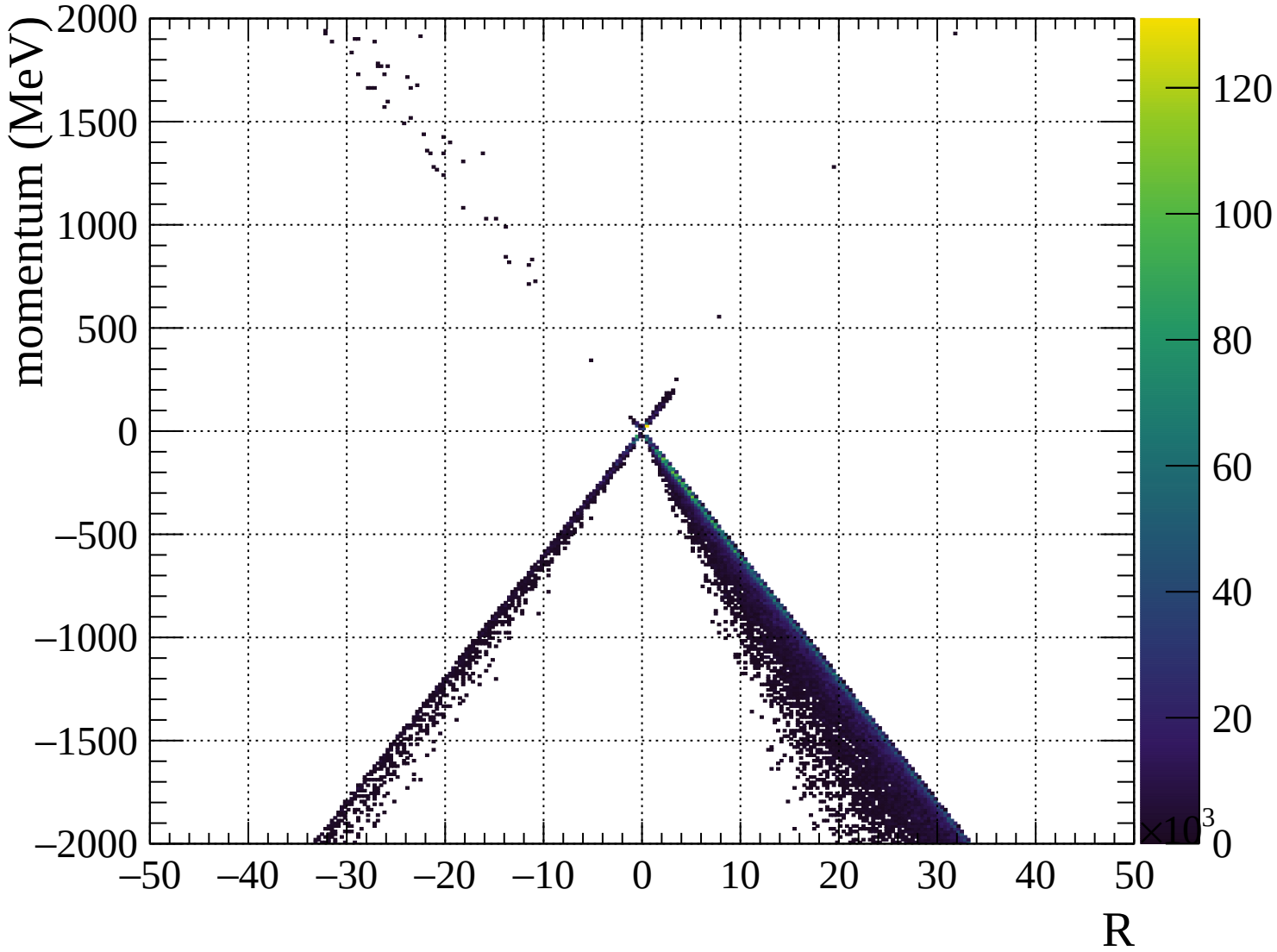


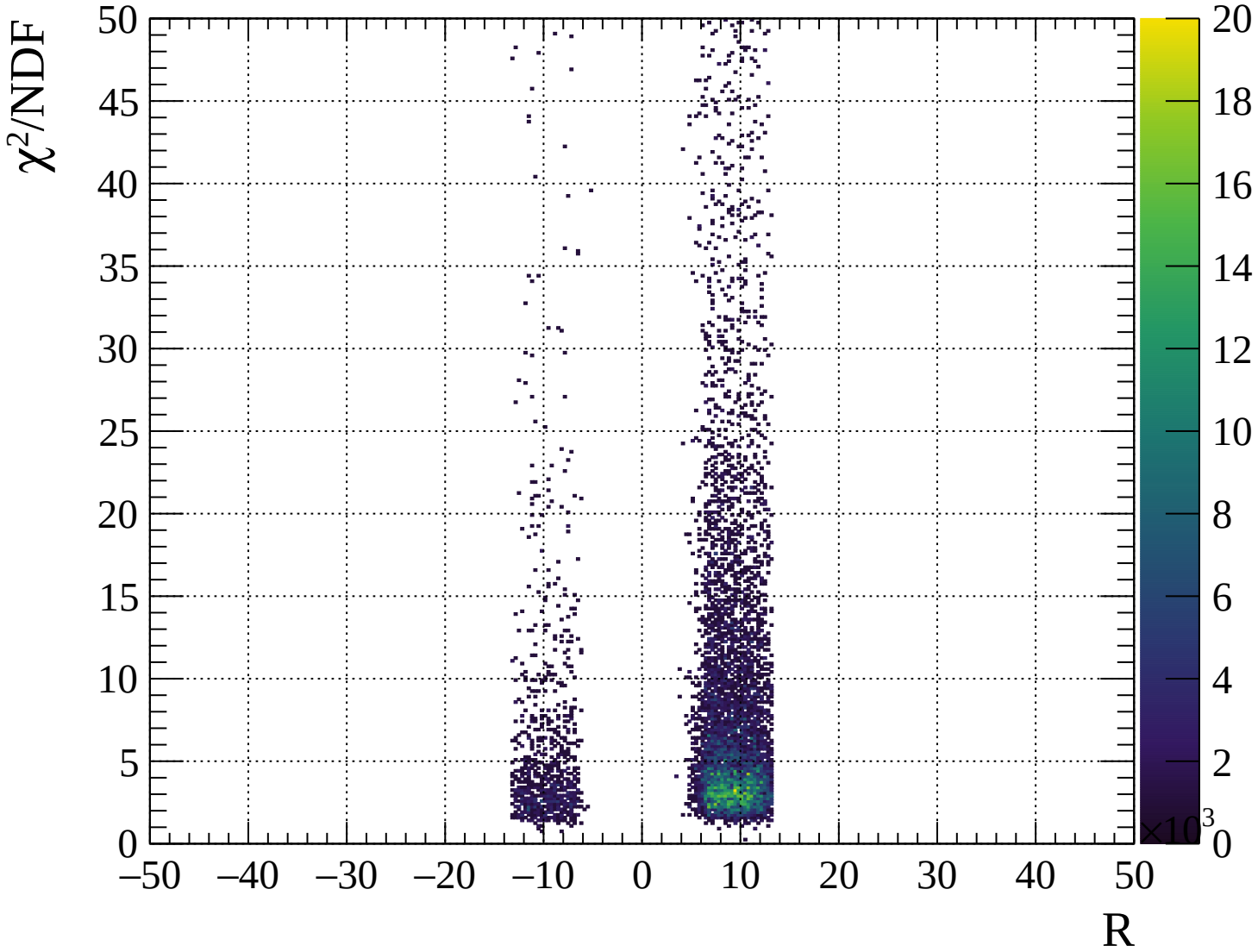






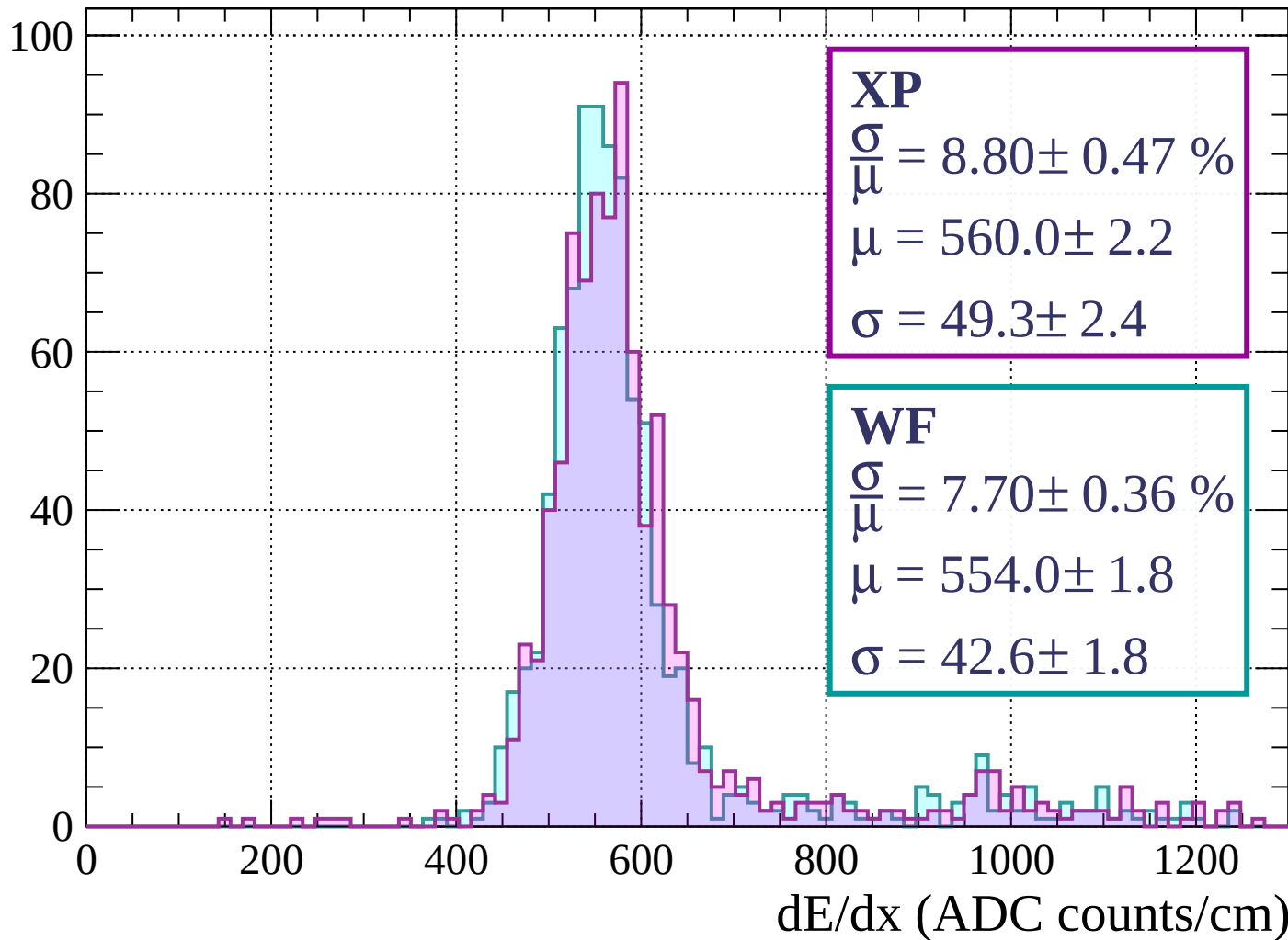






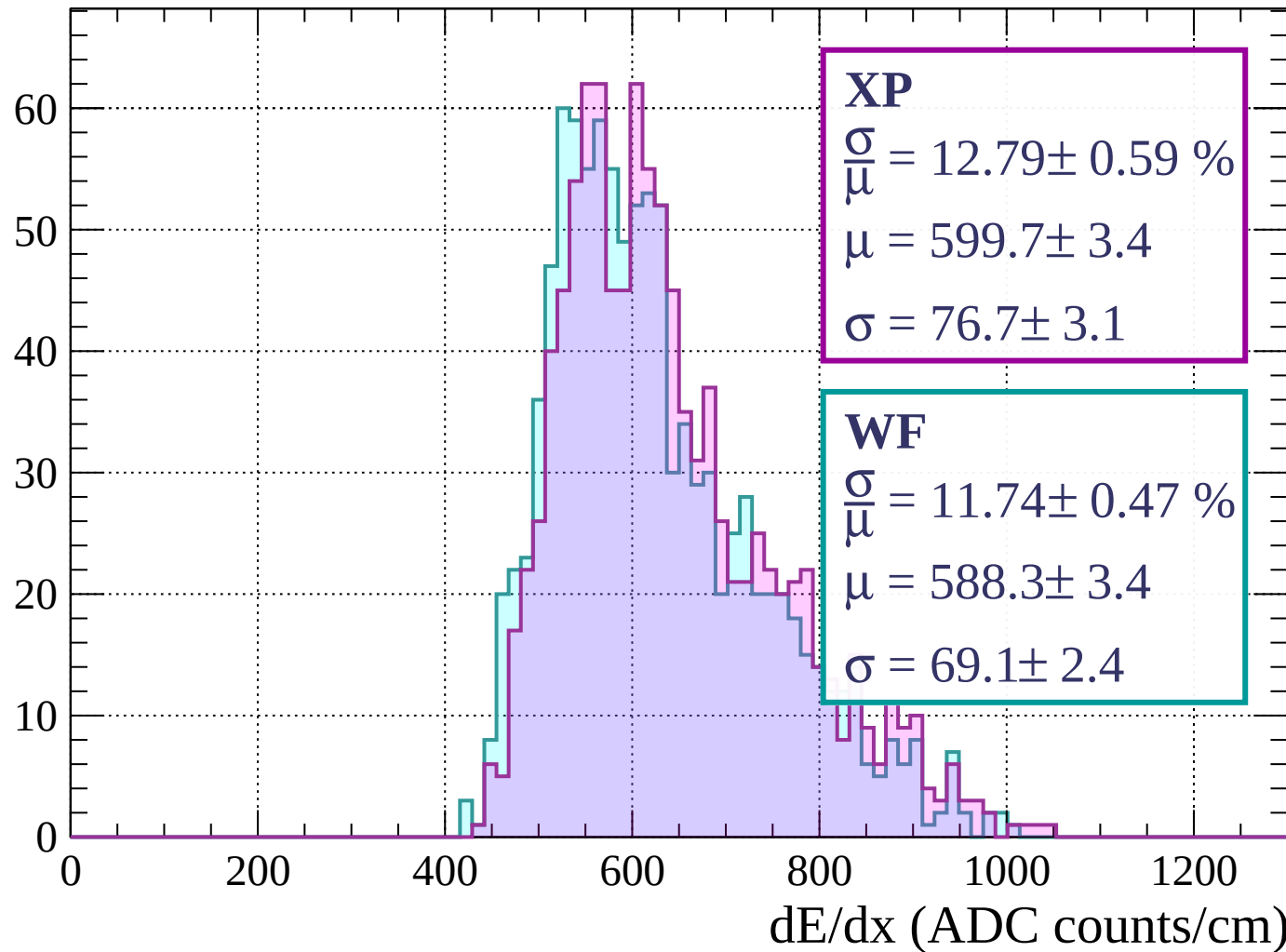
Energy loss | $0 < p < 80$

Count



Energy loss | $80 < p < 160$

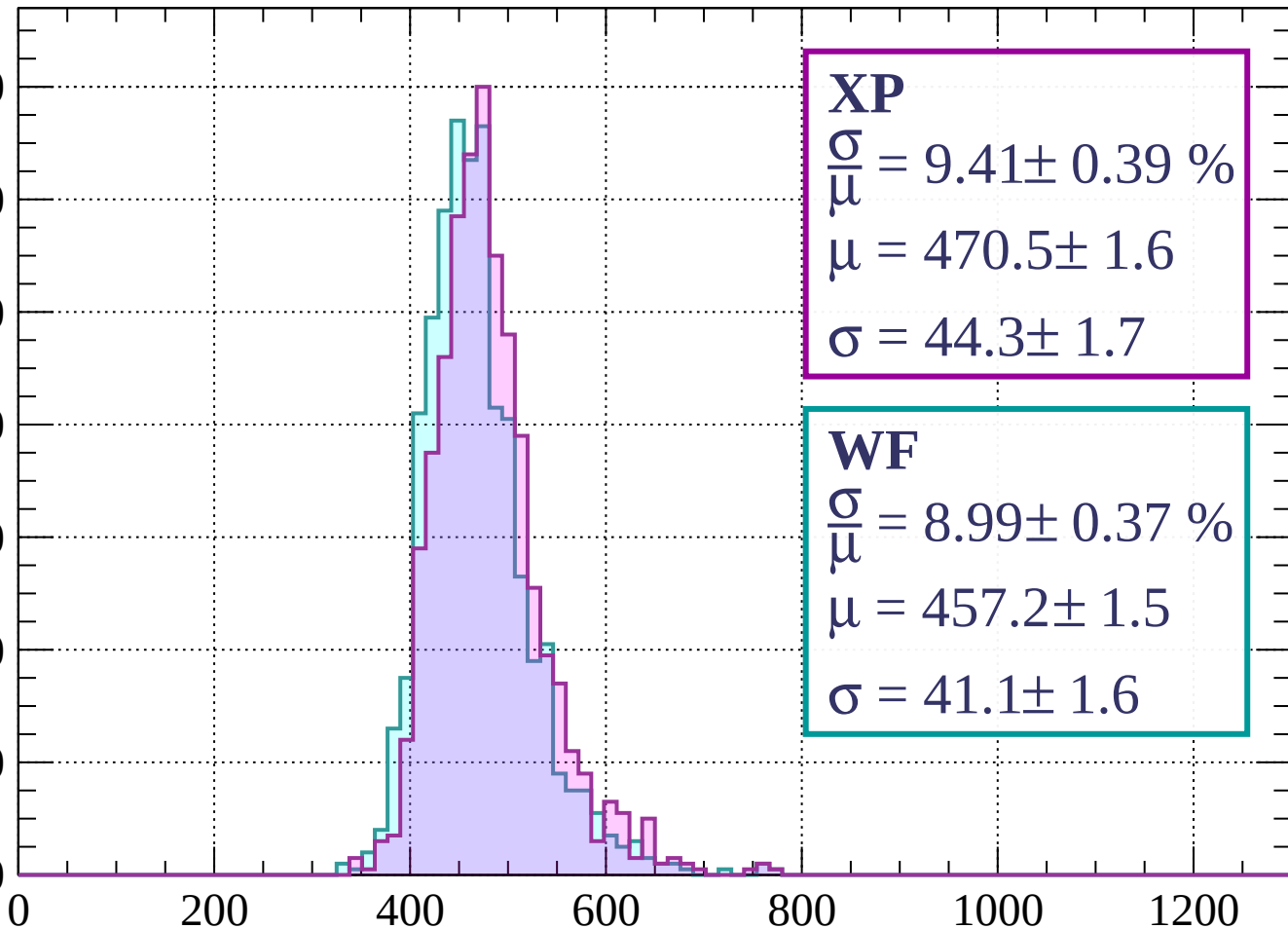
Count



Energy loss | $160 < p < 240$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.39 \%$$

$$\mu = 470.5 \pm 1.6$$

$$\sigma = 44.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.37 \%$$

$$\mu = 457.2 \pm 1.5$$

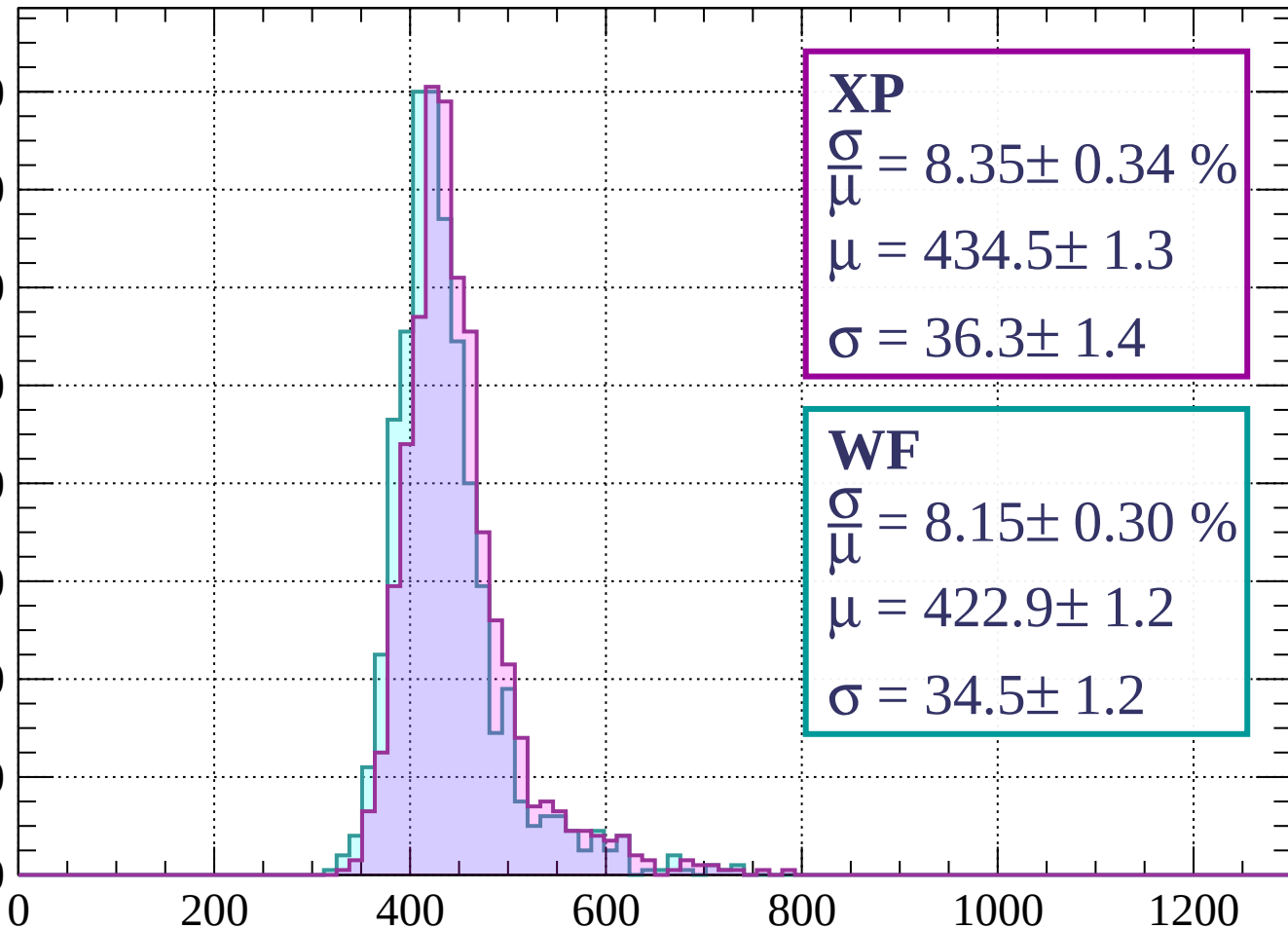
$$\sigma = 41.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.34 \%$$

$$\mu = 434.5 \pm 1.3$$

$$\sigma = 36.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.15 \pm 0.30 \%$$

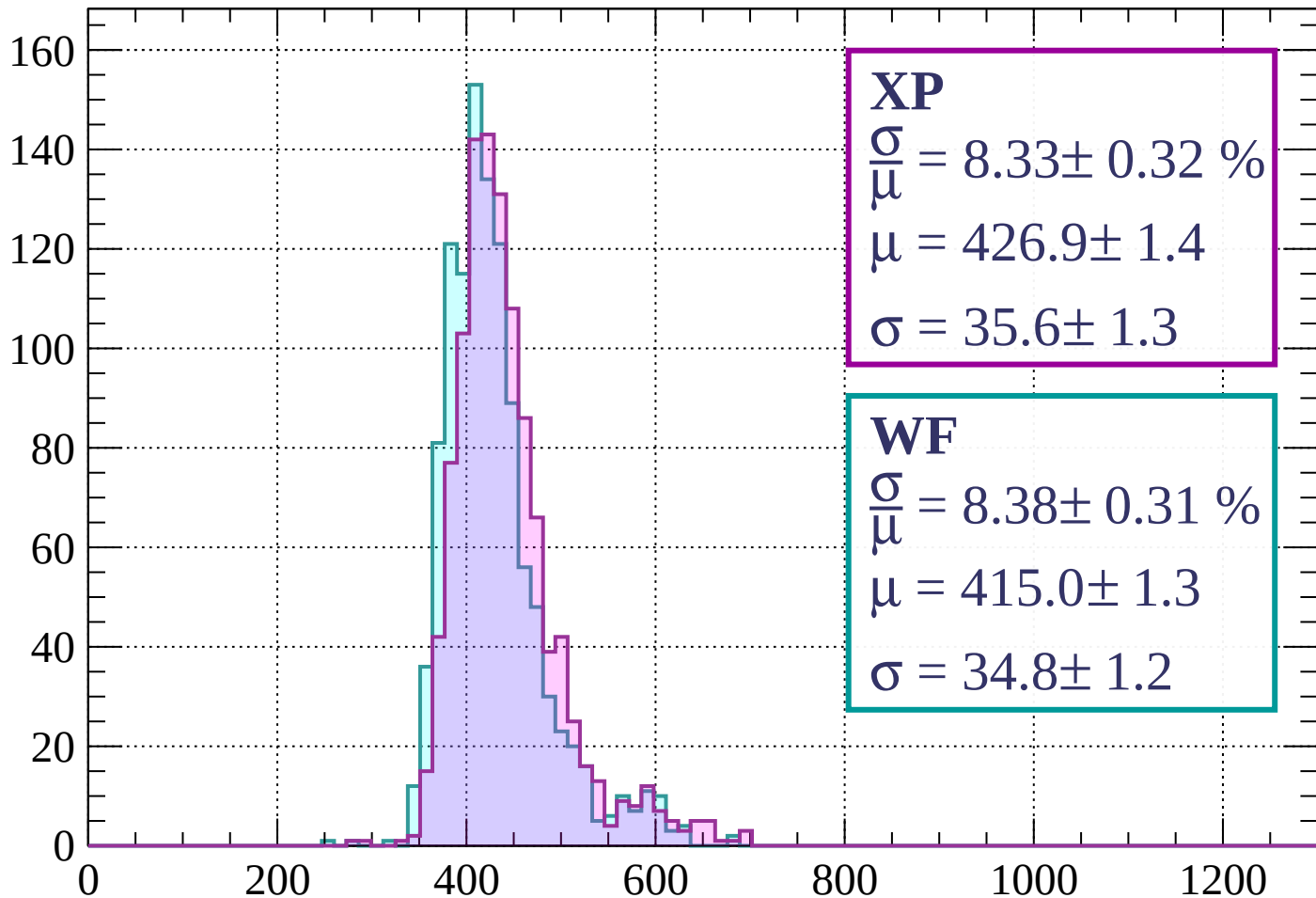
$$\mu = 422.9 \pm 1.2$$

$$\sigma = 34.5 \pm 1.2$$

dE/dx (ADC counts/cm)

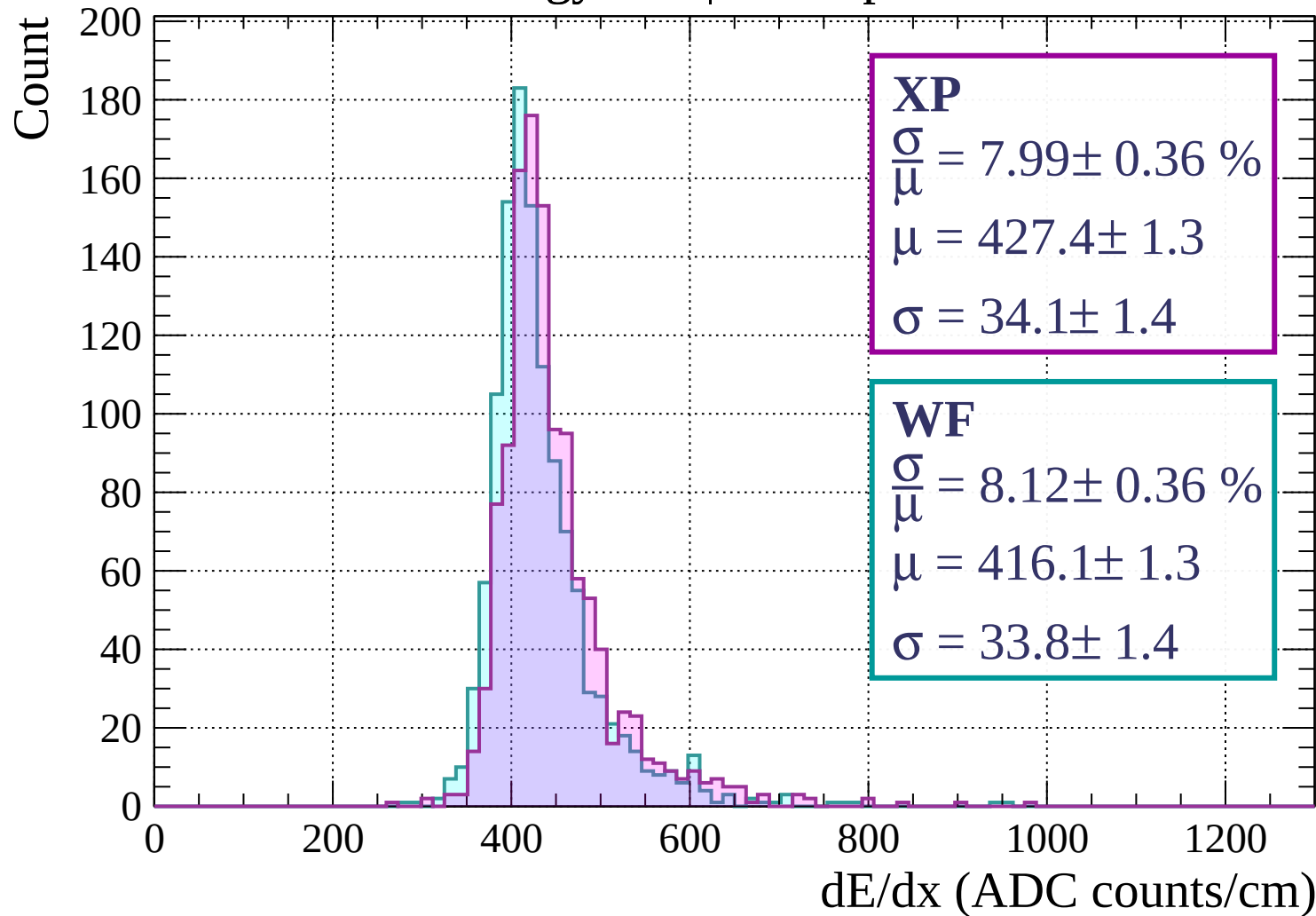
Energy loss | $320 < p < 400$

Count



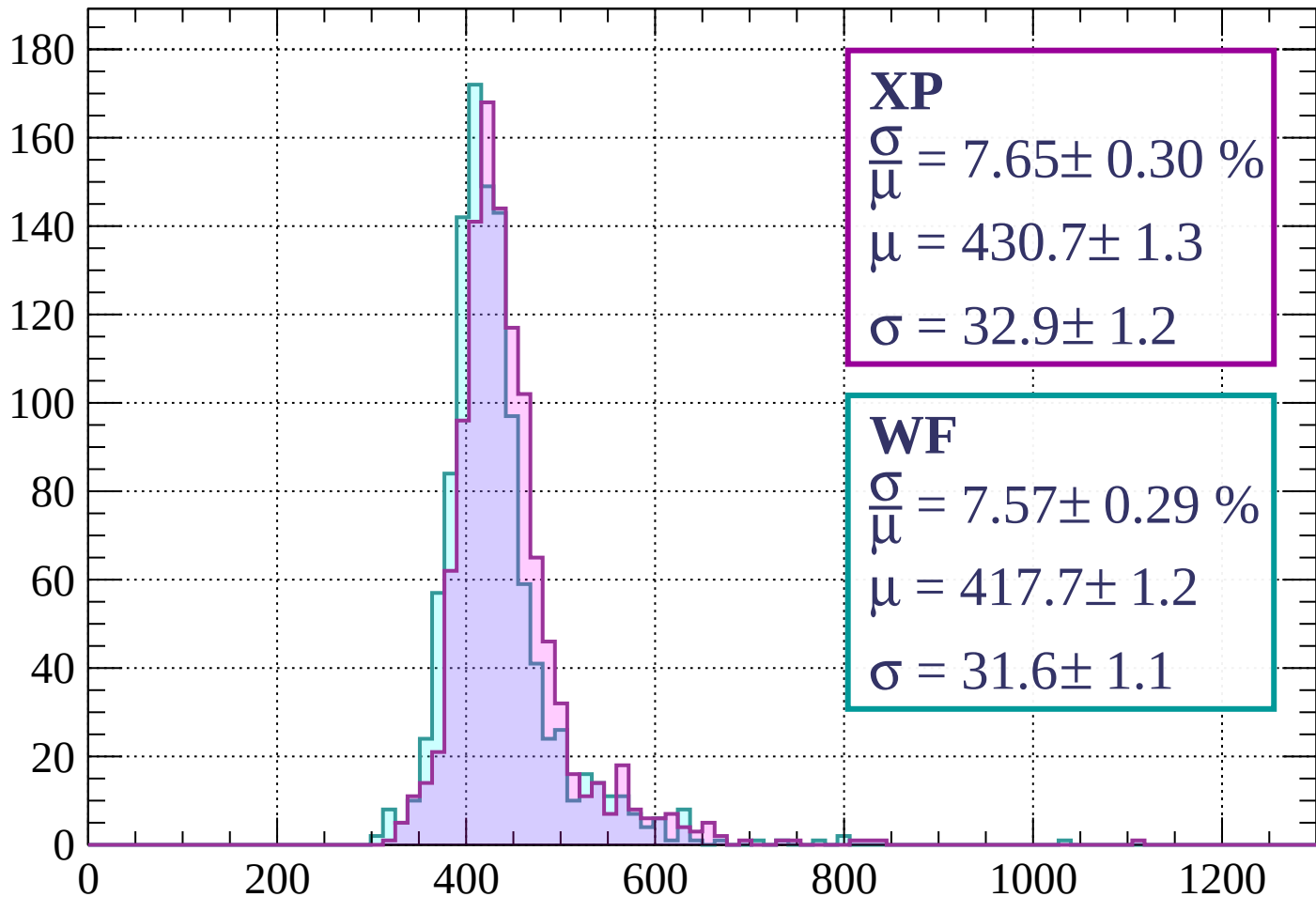
dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$



Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.65 \pm 0.30 \%$$

$$\mu = 430.7 \pm 1.3$$

$$\sigma = 32.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.57 \pm 0.29 \%$$

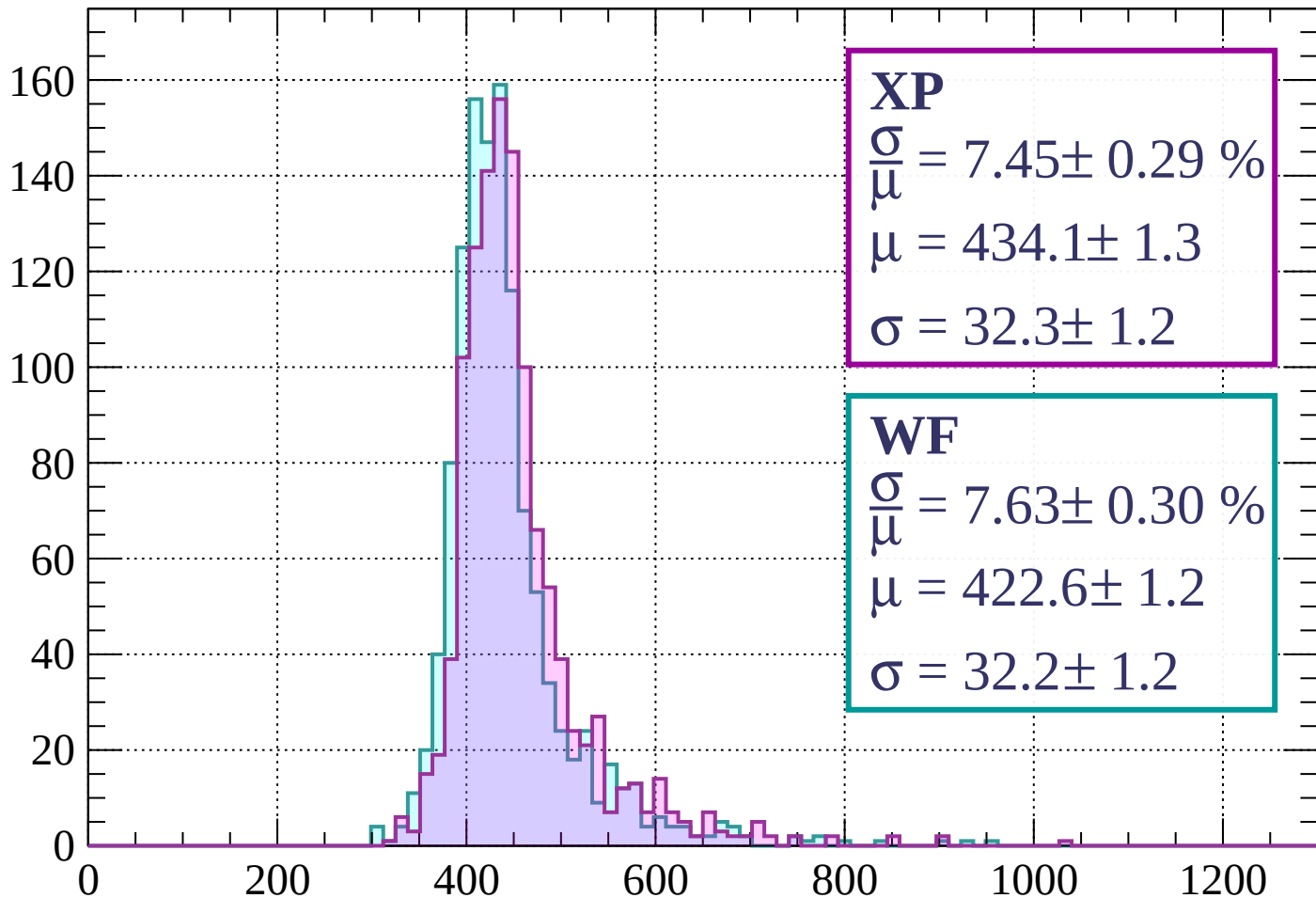
$$\mu = 417.7 \pm 1.2$$

$$\sigma = 31.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 7.45 \pm 0.29 \%$$

$$\mu = 434.1 \pm 1.3$$

$$\sigma = 32.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.63 \pm 0.30 \%$$

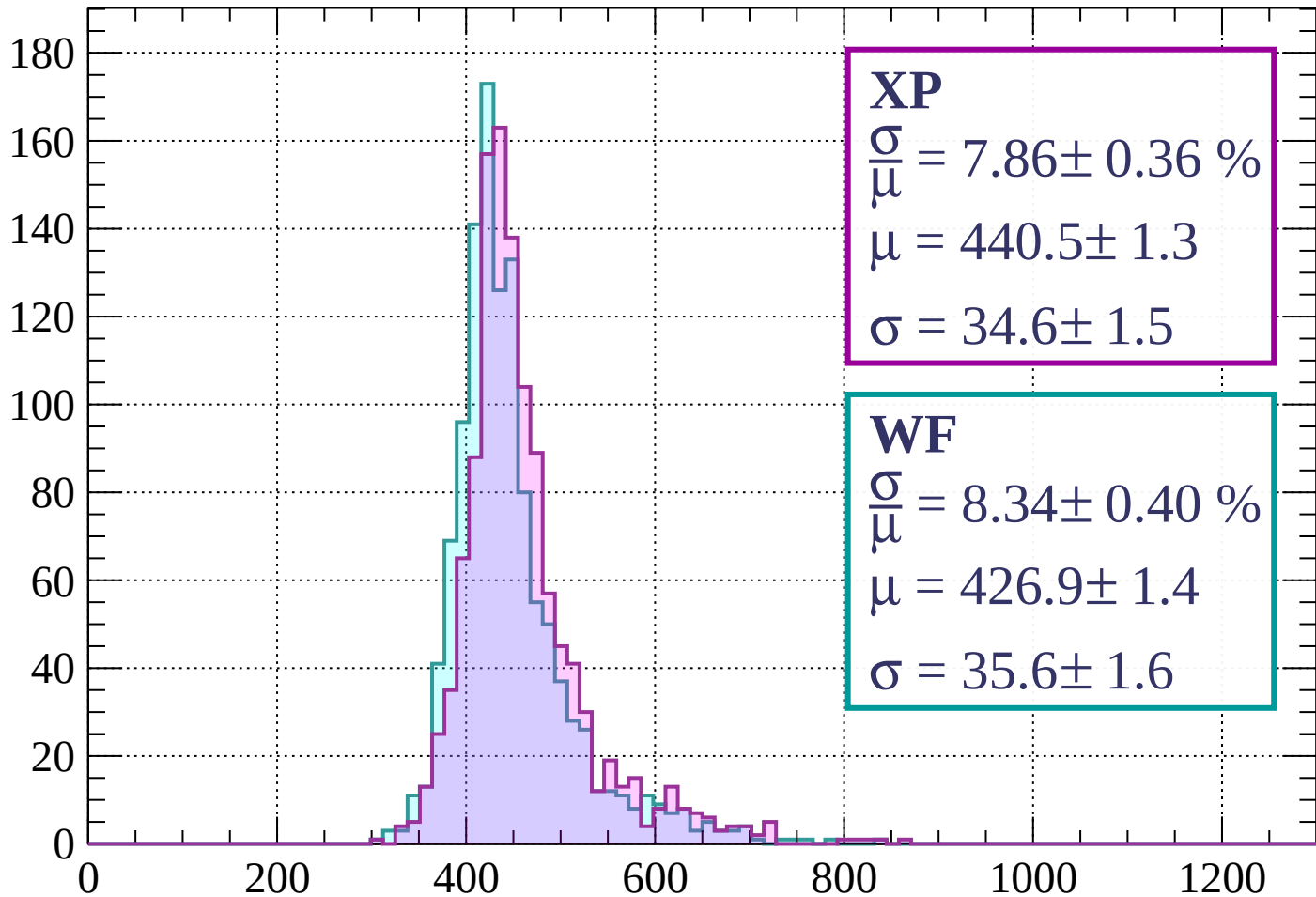
$$\mu = 422.6 \pm 1.2$$

$$\sigma = 32.2 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 7.86 \pm 0.36 \%$$

$$\mu = 440.5 \pm 1.3$$

$$\sigma = 34.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.34 \pm 0.40 \%$$

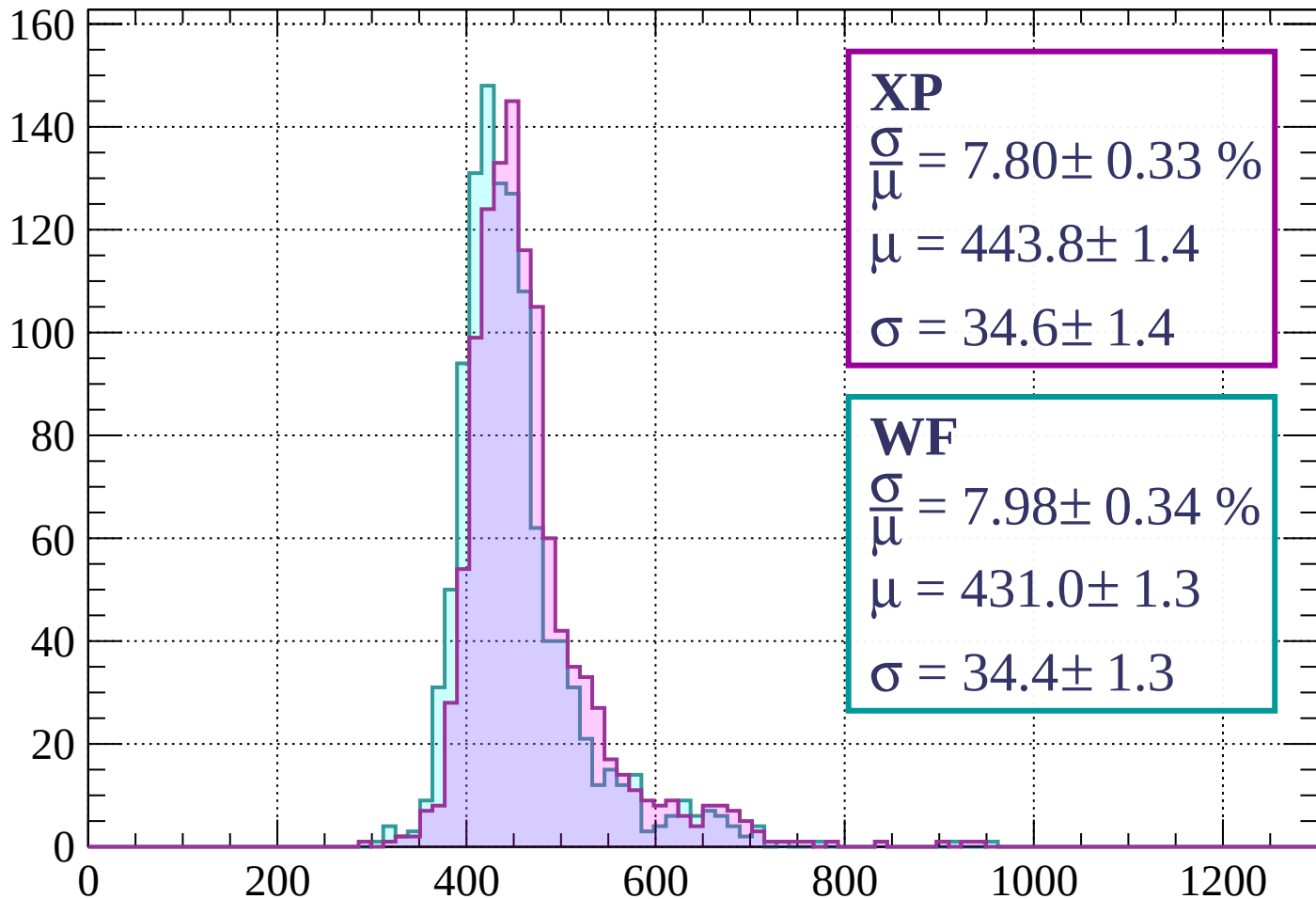
$$\mu = 426.9 \pm 1.4$$

$$\sigma = 35.6 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.80 \pm 0.33 \%$$

$$\mu = 443.8 \pm 1.4$$

$$\sigma = 34.6 \pm 1.4$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.98 \pm 0.34 \%$$

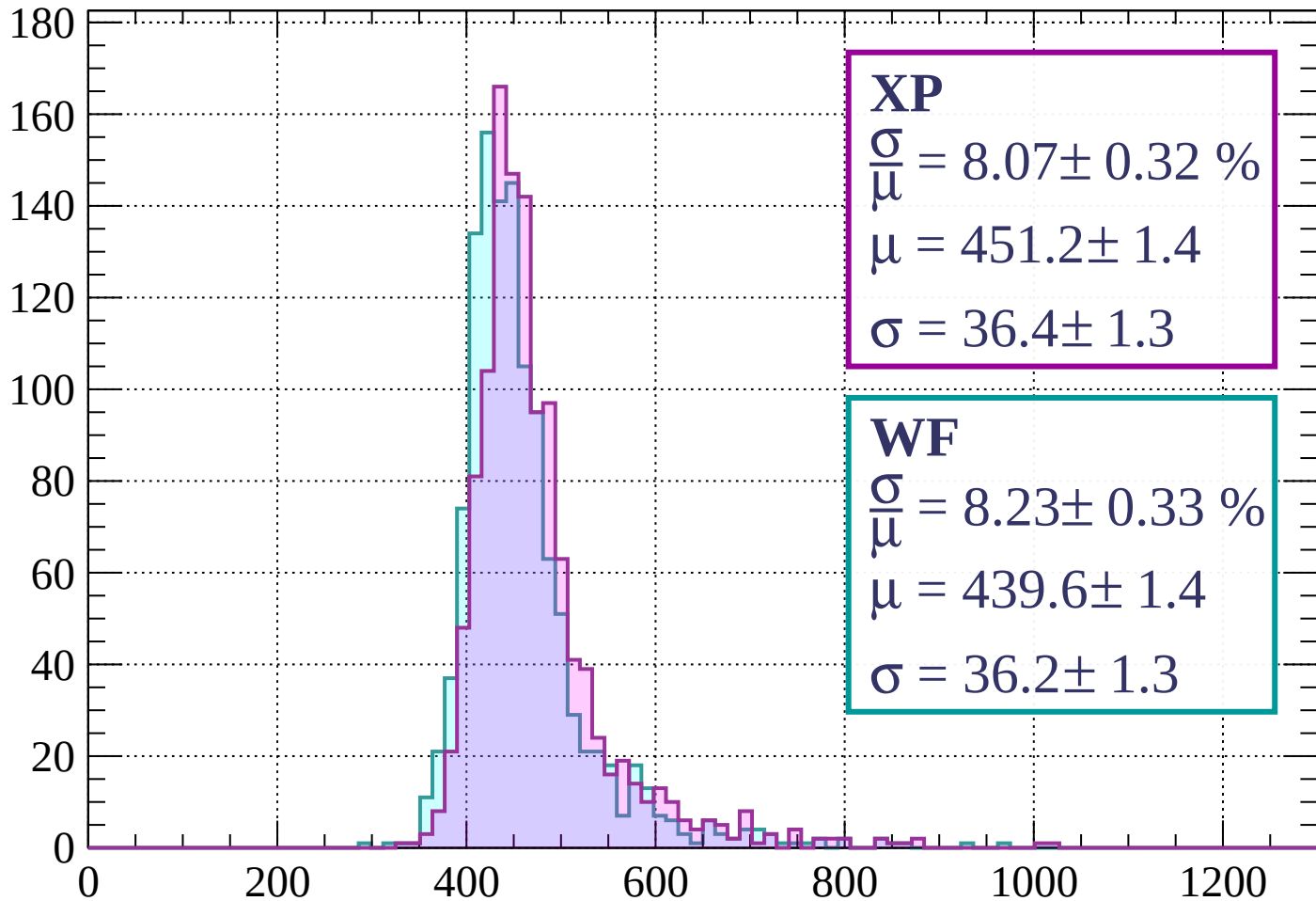
$$\mu = 431.0 \pm 1.3$$

$$\sigma = 34.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.32 \%$$

$$\mu = 451.2 \pm 1.4$$

$$\sigma = 36.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.23 \pm 0.33 \%$$

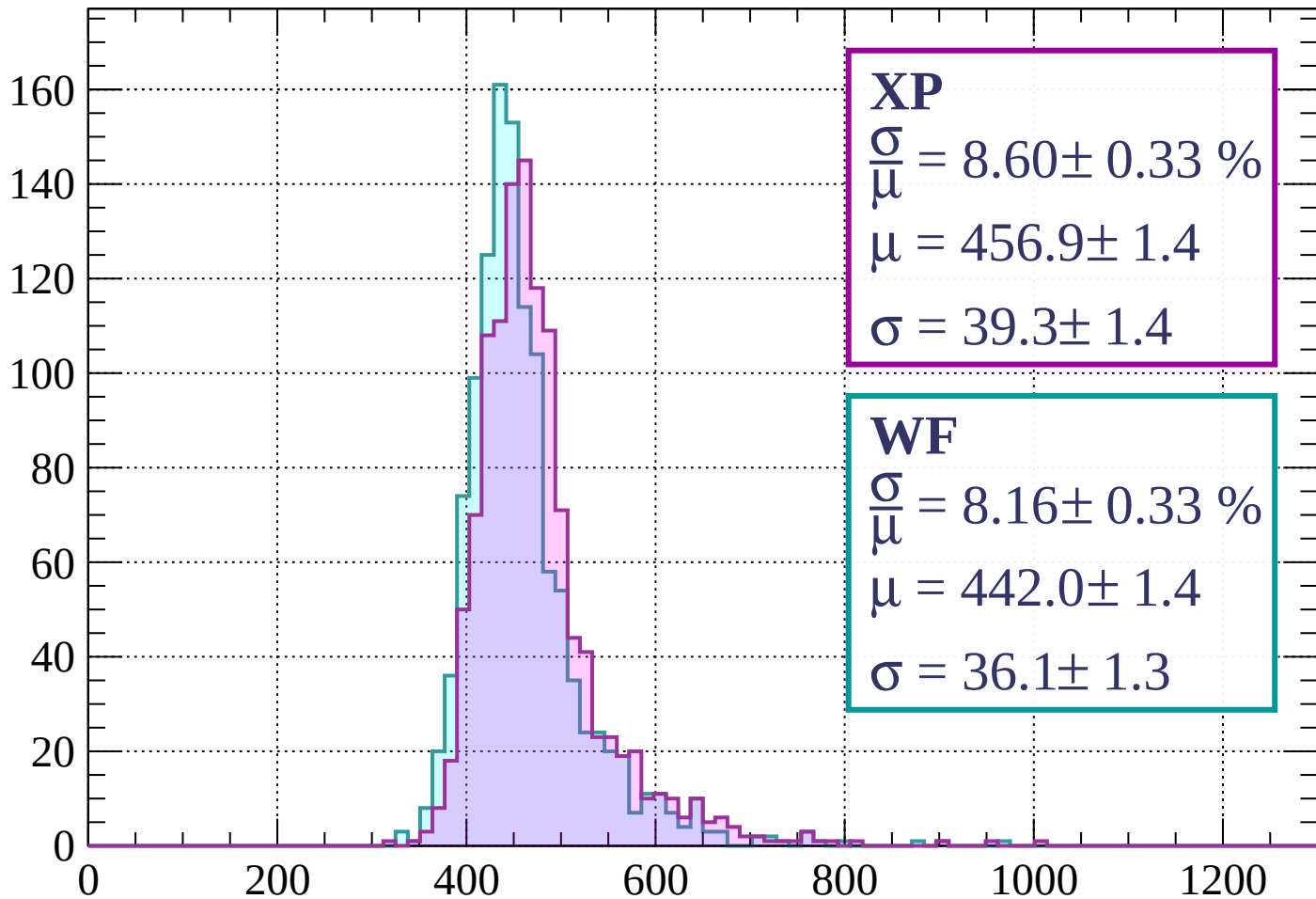
$$\mu = 439.6 \pm 1.4$$

$$\sigma = 36.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.60 \pm 0.33 \%$$

$$\mu = 456.9 \pm 1.4$$

$$\sigma = 39.3 \pm 1.4$$

WF

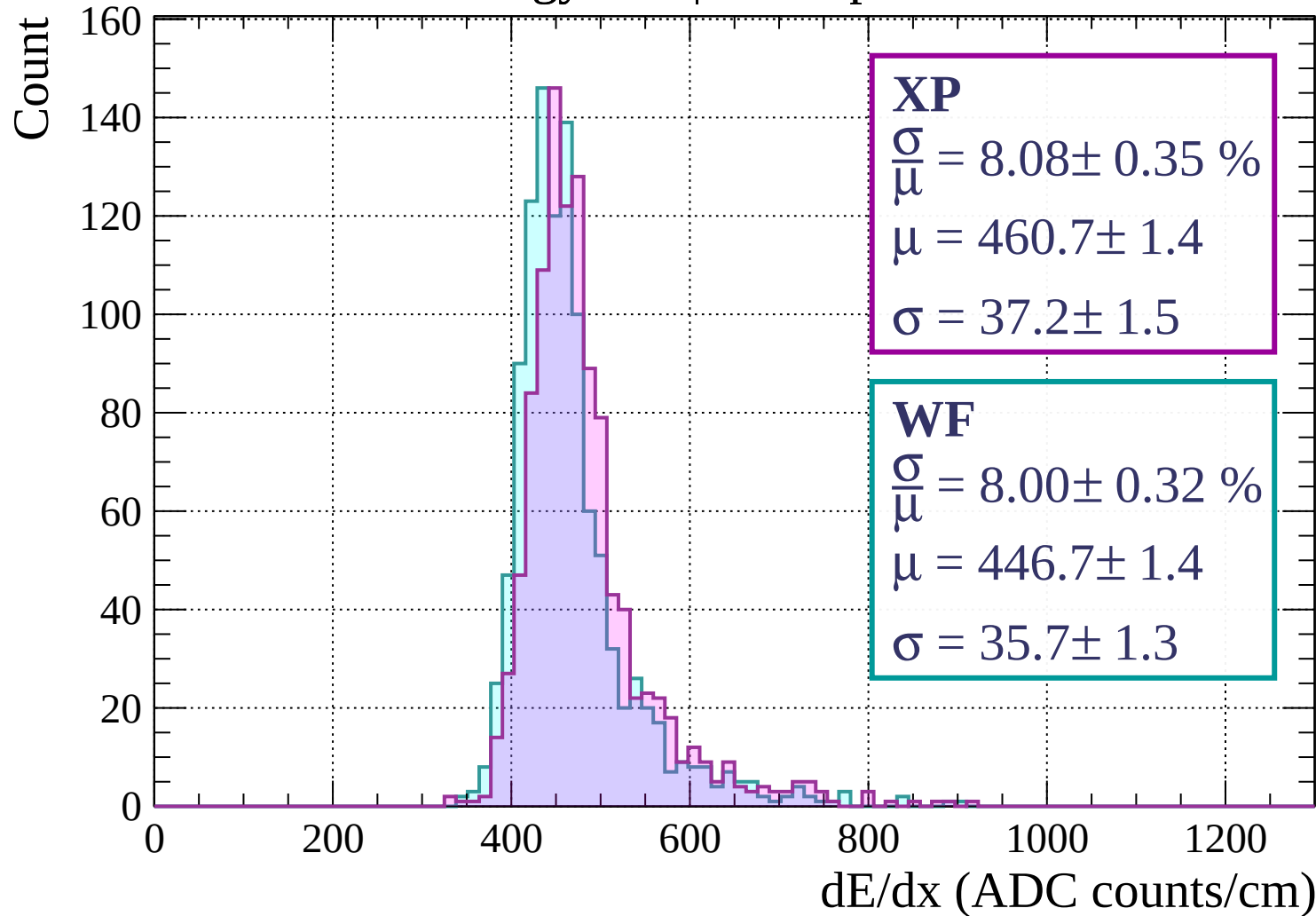
$$\frac{\sigma}{\mu} = 8.16 \pm 0.33 \%$$

$$\mu = 442.0 \pm 1.4$$

$$\sigma = 36.1 \pm 1.3$$

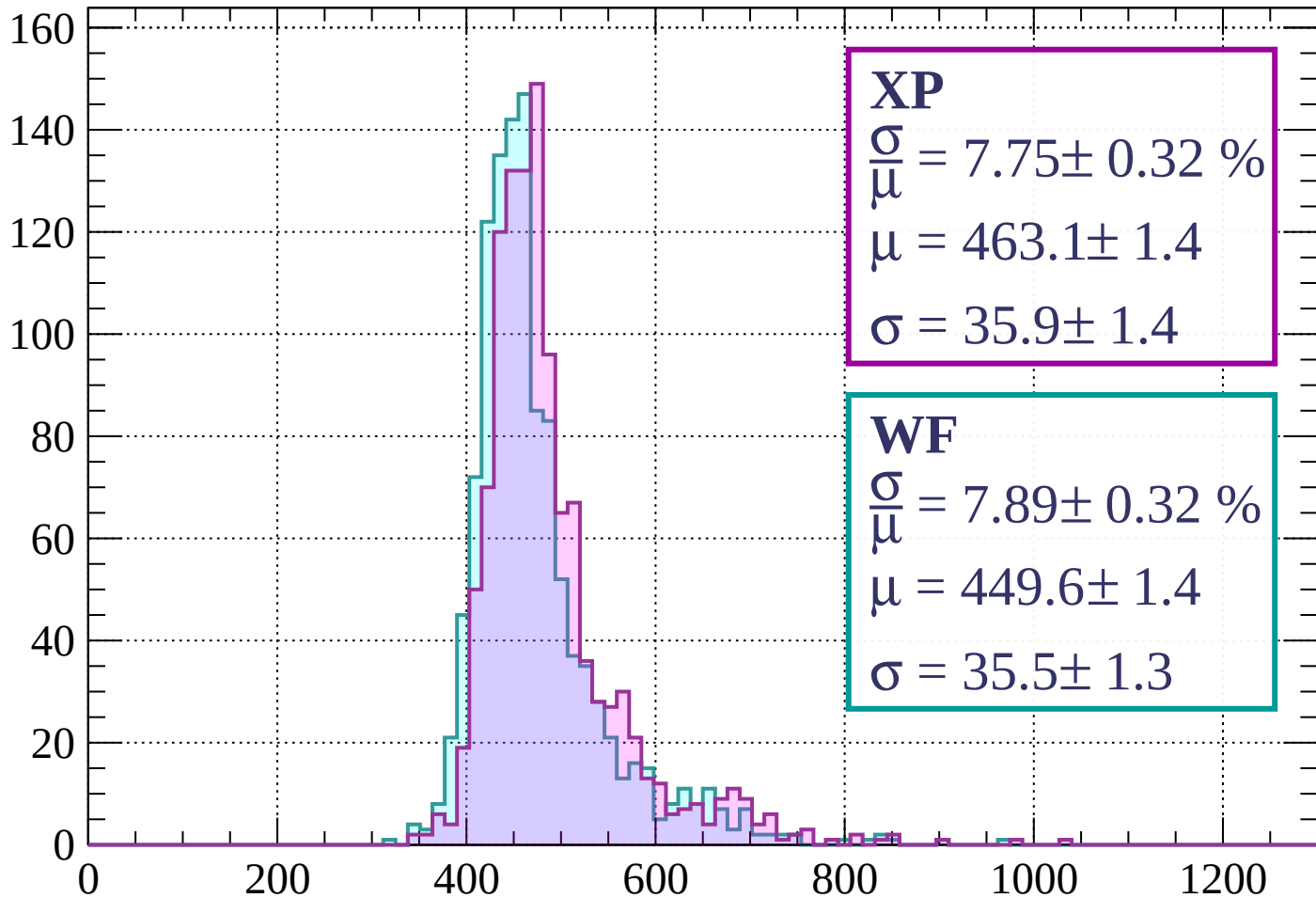
dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.75 \pm 0.32 \%$$

$$\mu = 463.1 \pm 1.4$$

$$\sigma = 35.9 \pm 1.4$$

WF

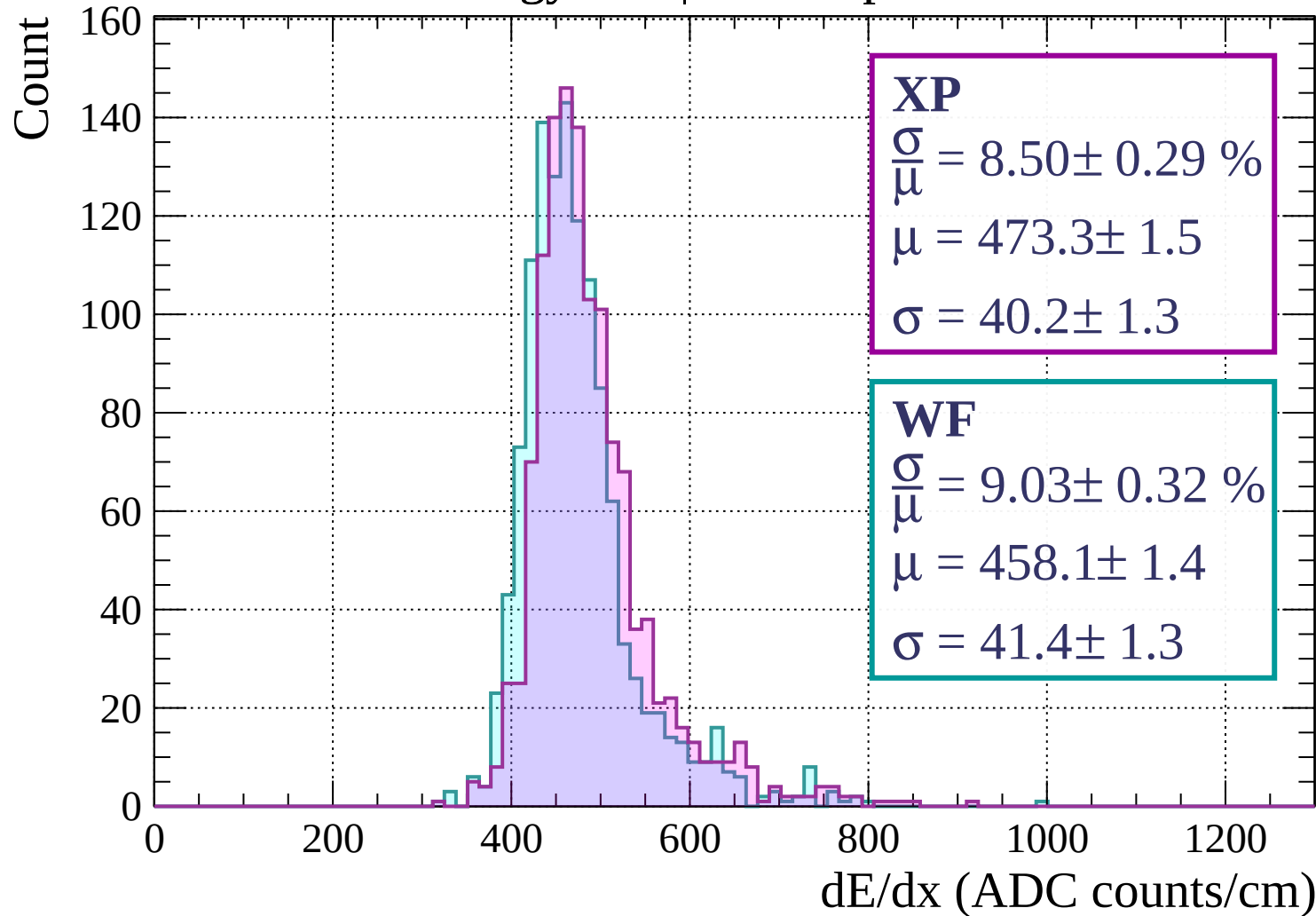
$$\frac{\sigma}{\mu} = 7.89 \pm 0.32 \%$$

$$\mu = 449.6 \pm 1.4$$

$$\sigma = 35.5 \pm 1.3$$

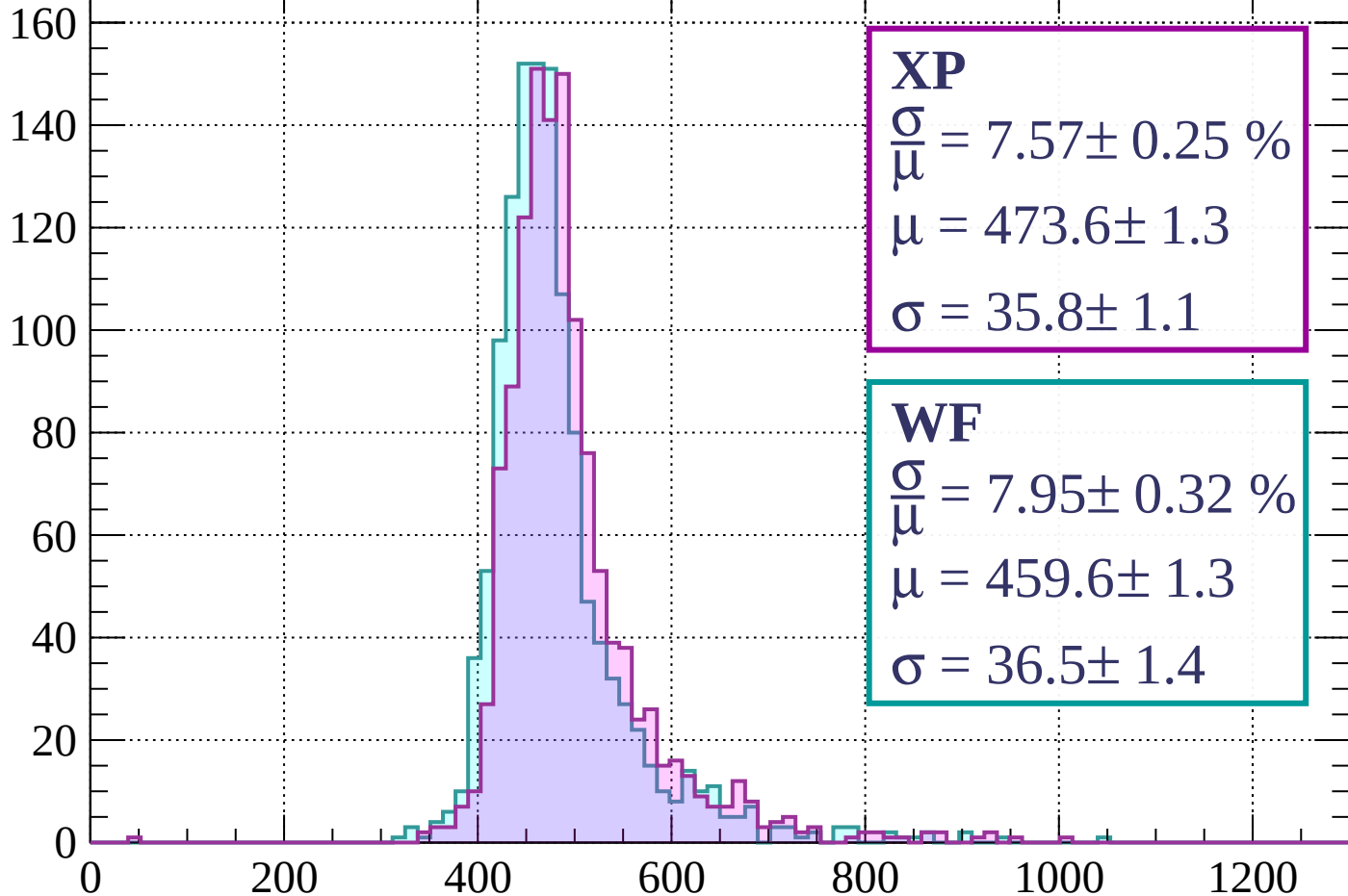
dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$



Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.25 \%$$

$$\mu = 473.6 \pm 1.3$$

$$\sigma = 35.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.32 \%$$

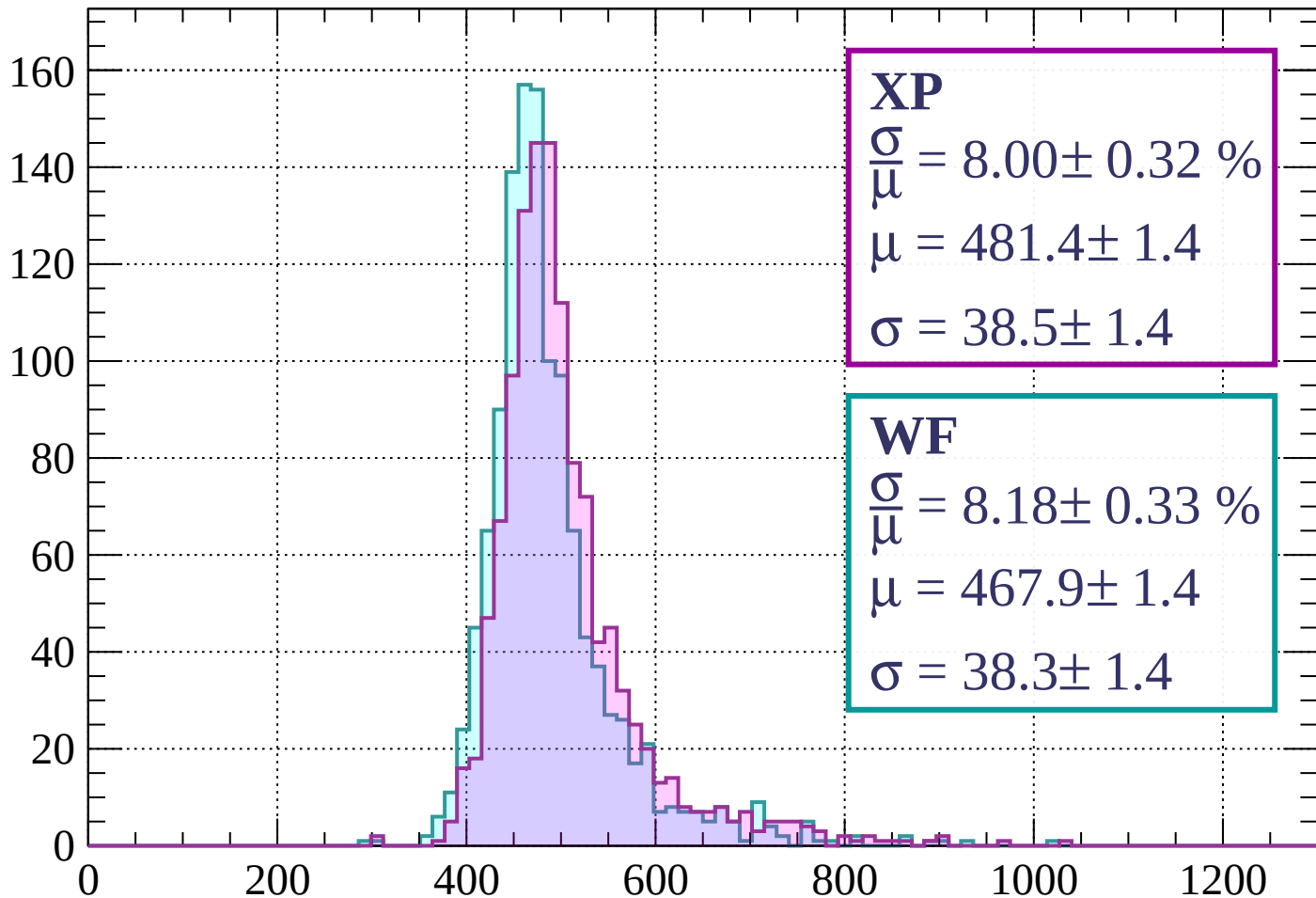
$$\mu = 459.6 \pm 1.3$$

$$\sigma = 36.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.32 \%$$

$$\mu = 481.4 \pm 1.4$$

$$\sigma = 38.5 \pm 1.4$$

WF

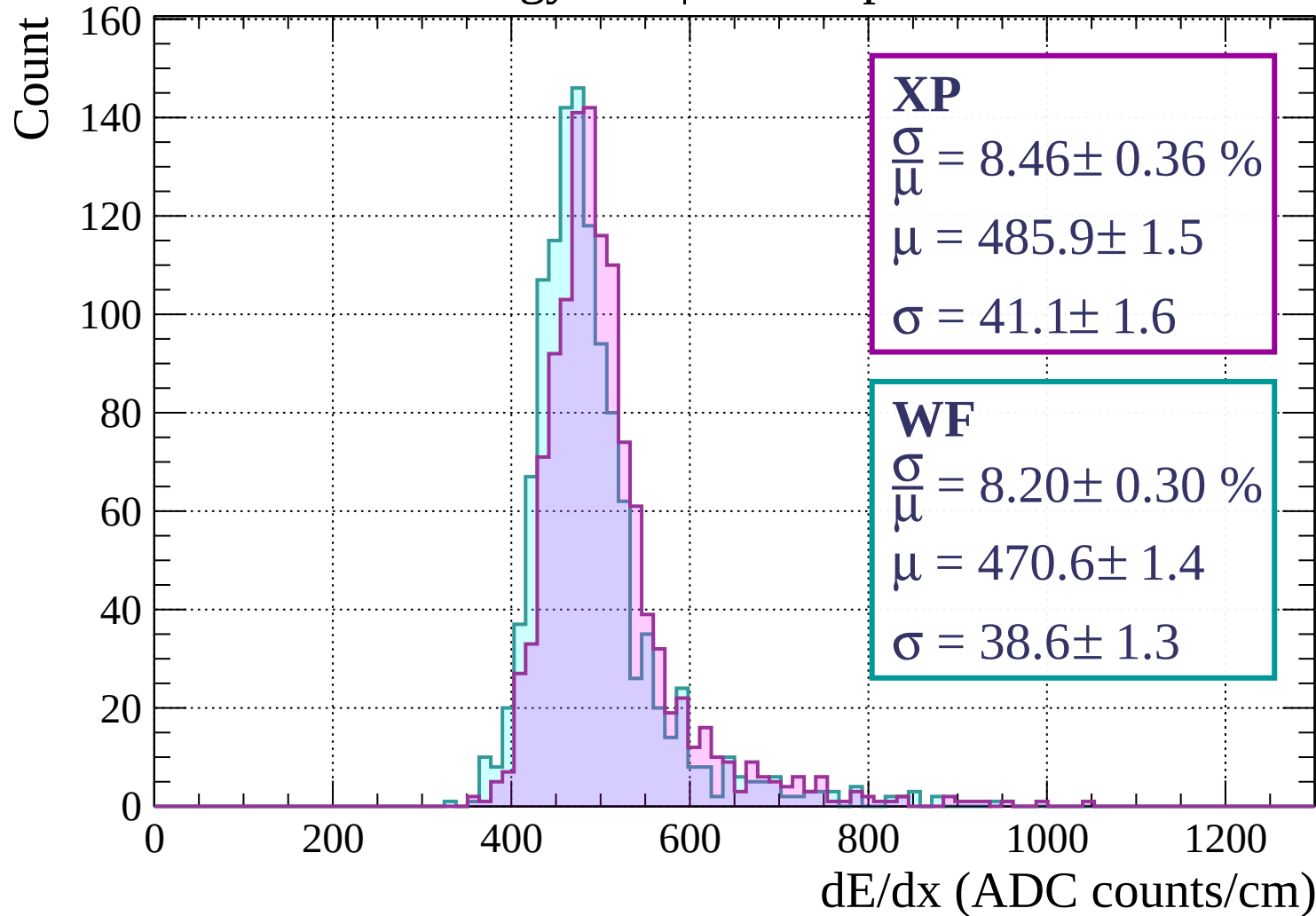
$$\frac{\sigma}{\mu} = 8.18 \pm 0.33 \%$$

$$\mu = 467.9 \pm 1.4$$

$$\sigma = 38.3 \pm 1.4$$

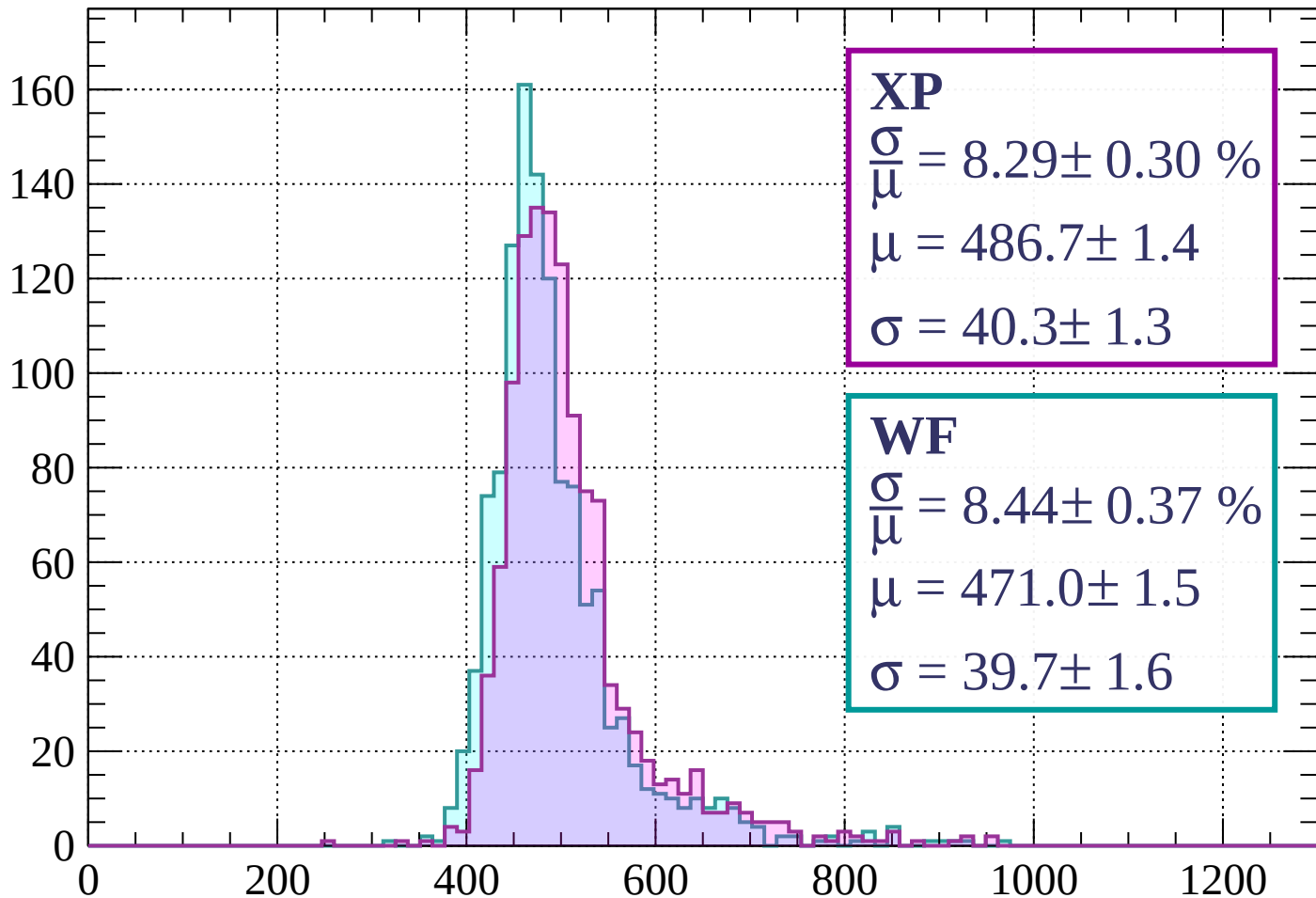
dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.30 \%$$

$$\mu = 486.7 \pm 1.4$$

$$\sigma = 40.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.44 \pm 0.37 \%$$

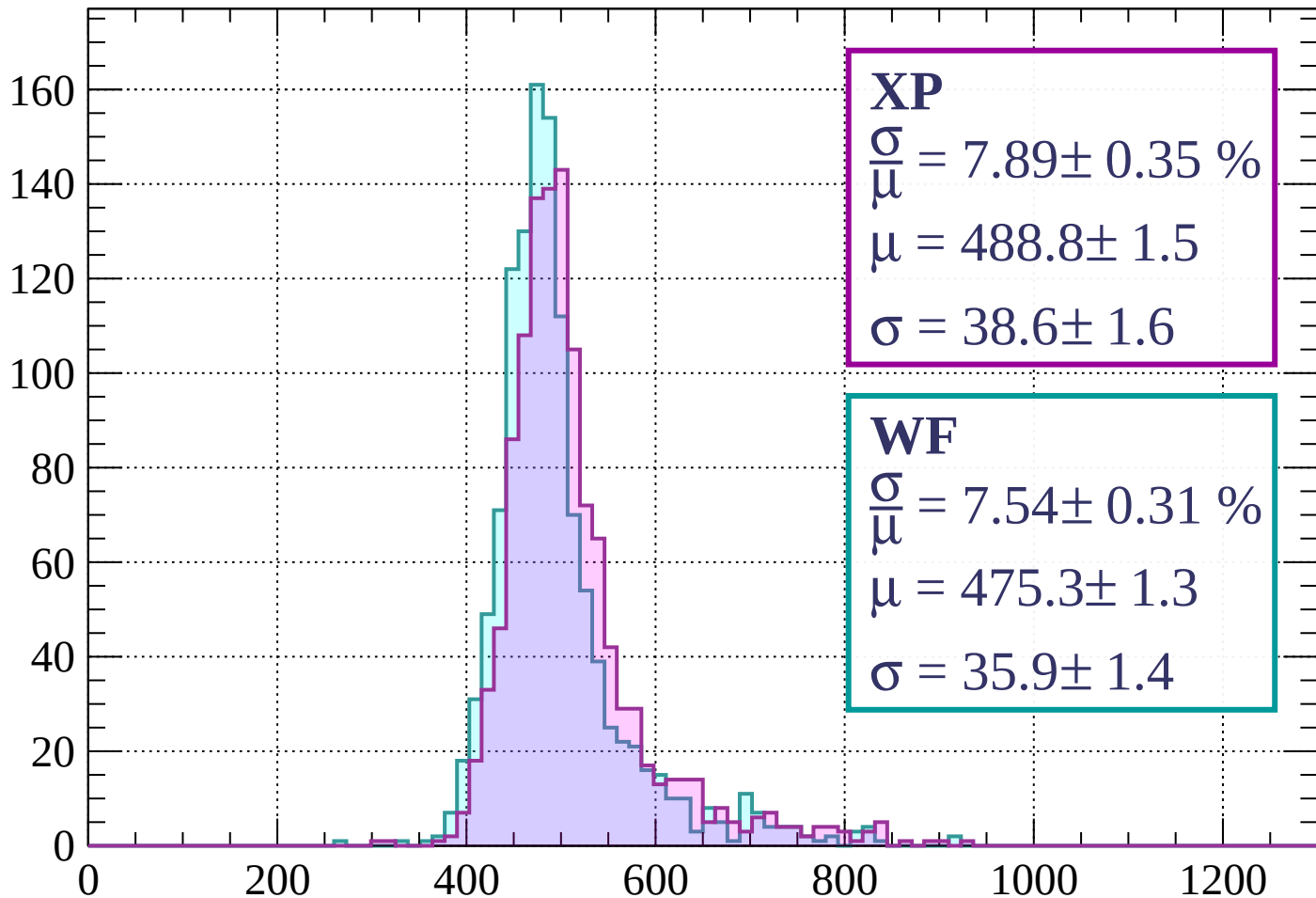
$$\mu = 471.0 \pm 1.5$$

$$\sigma = 39.7 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.35 \%$$

$$\mu = 488.8 \pm 1.5$$

$$\sigma = 38.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.31 \%$$

$$\mu = 475.3 \pm 1.3$$

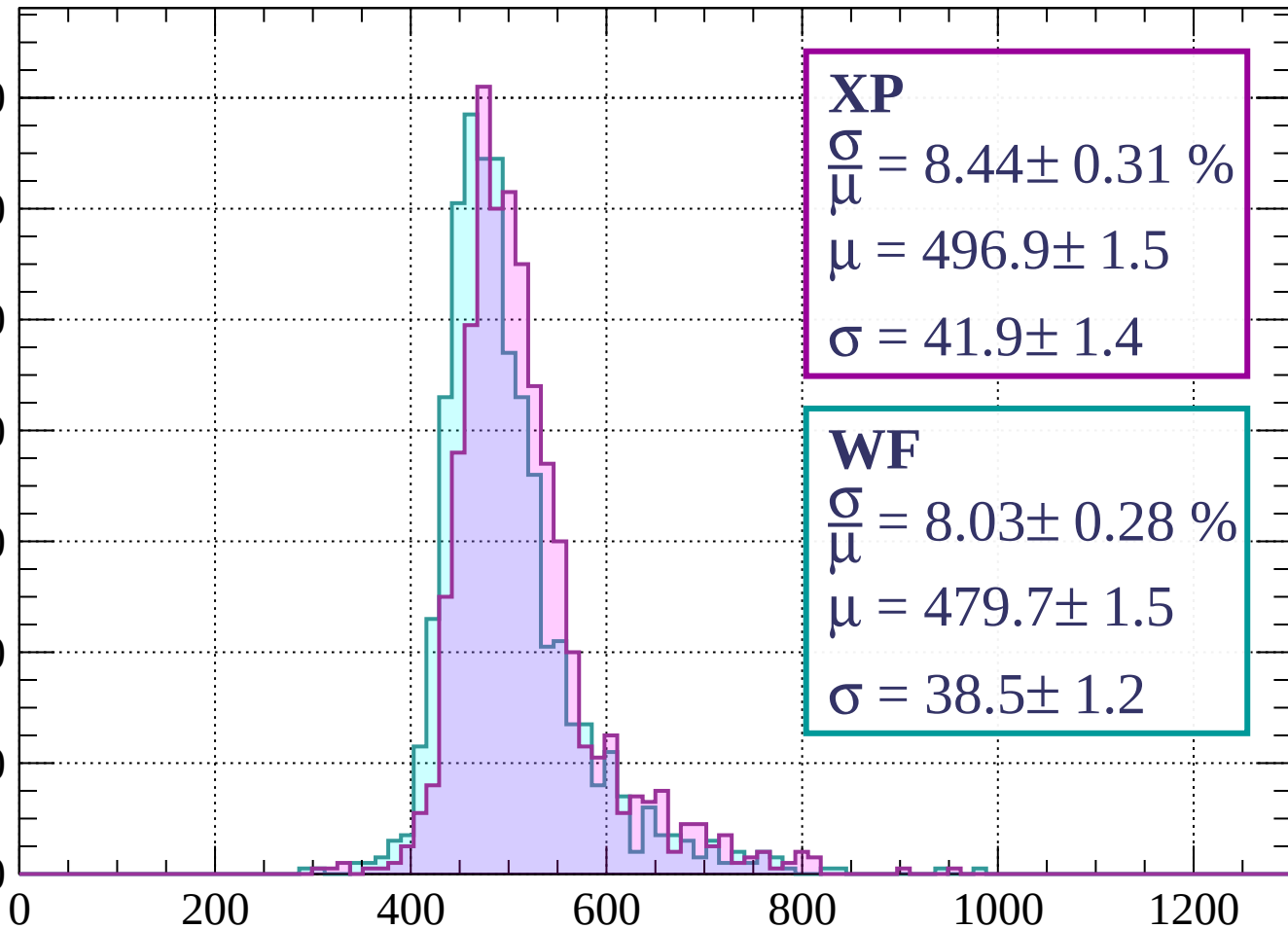
$$\sigma = 35.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.31 \%$$

$$\mu = 496.9 \pm 1.5$$

$$\sigma = 41.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.03 \pm 0.28 \%$$

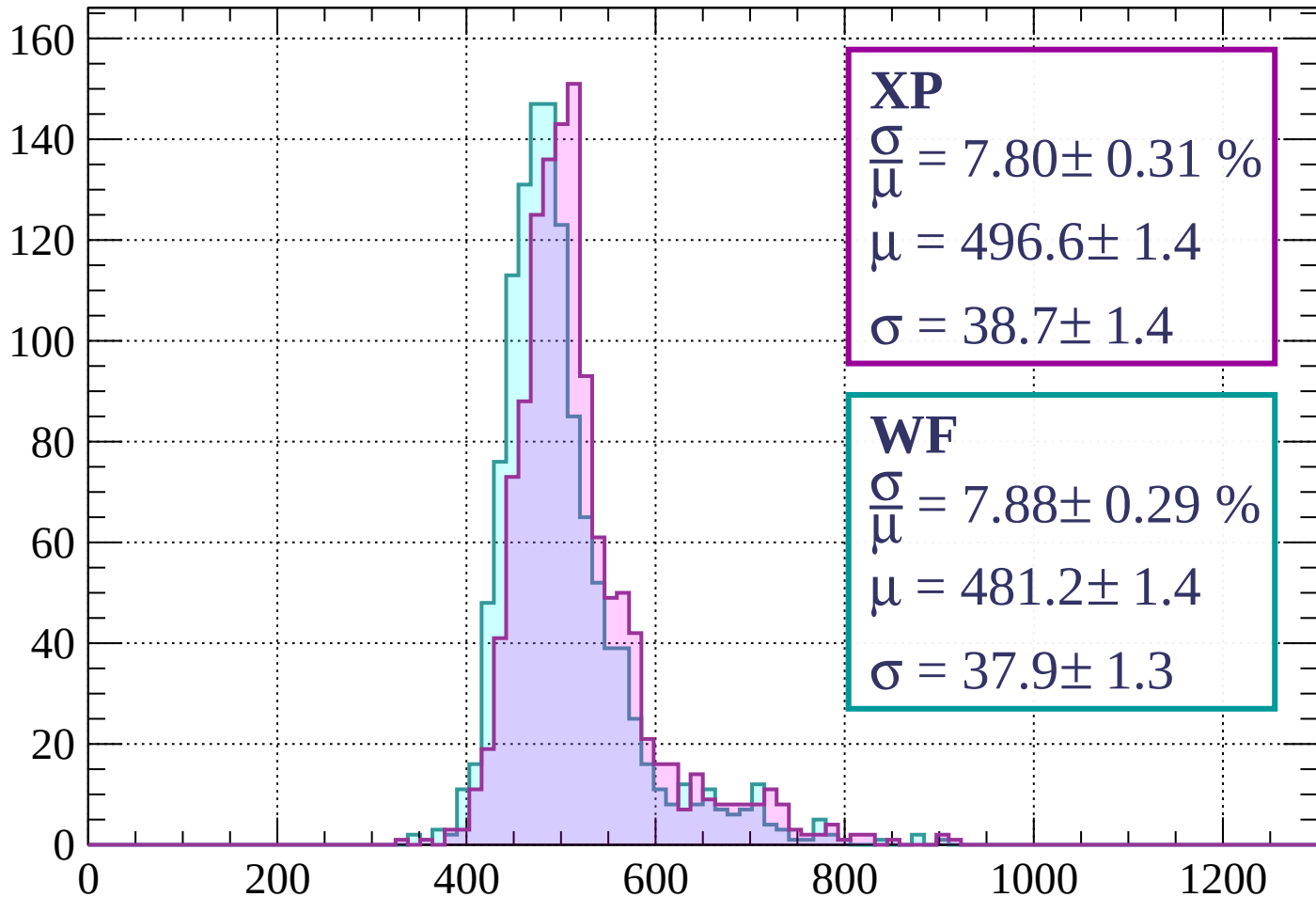
$$\mu = 479.7 \pm 1.5$$

$$\sigma = 38.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 7.80 \pm 0.31 \%$$

$$\mu = 496.6 \pm 1.4$$

$$\sigma = 38.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.29 \%$$

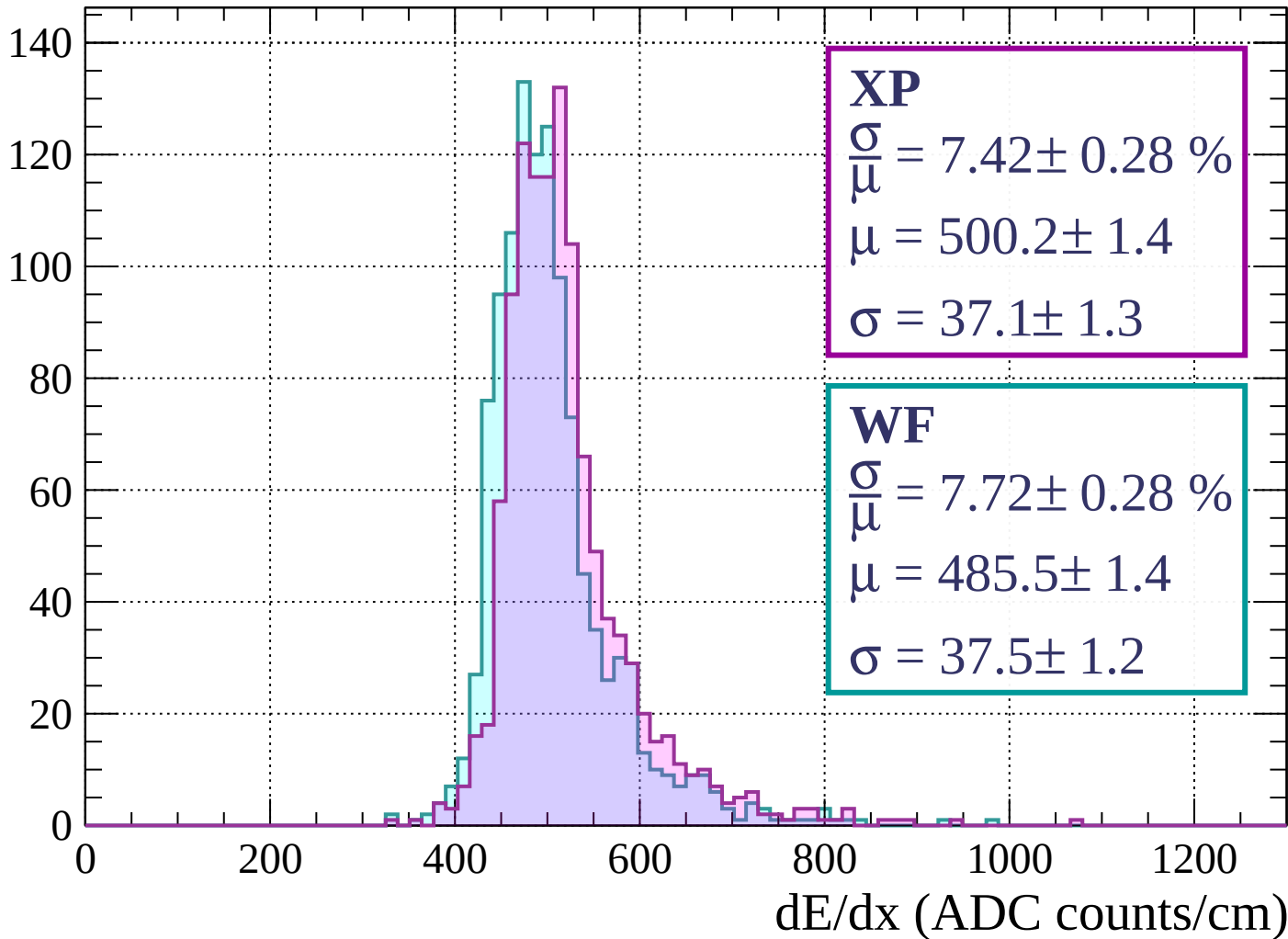
$$\mu = 481.2 \pm 1.4$$

$$\sigma = 37.9 \pm 1.3$$

dE/dx (ADC counts/cm)

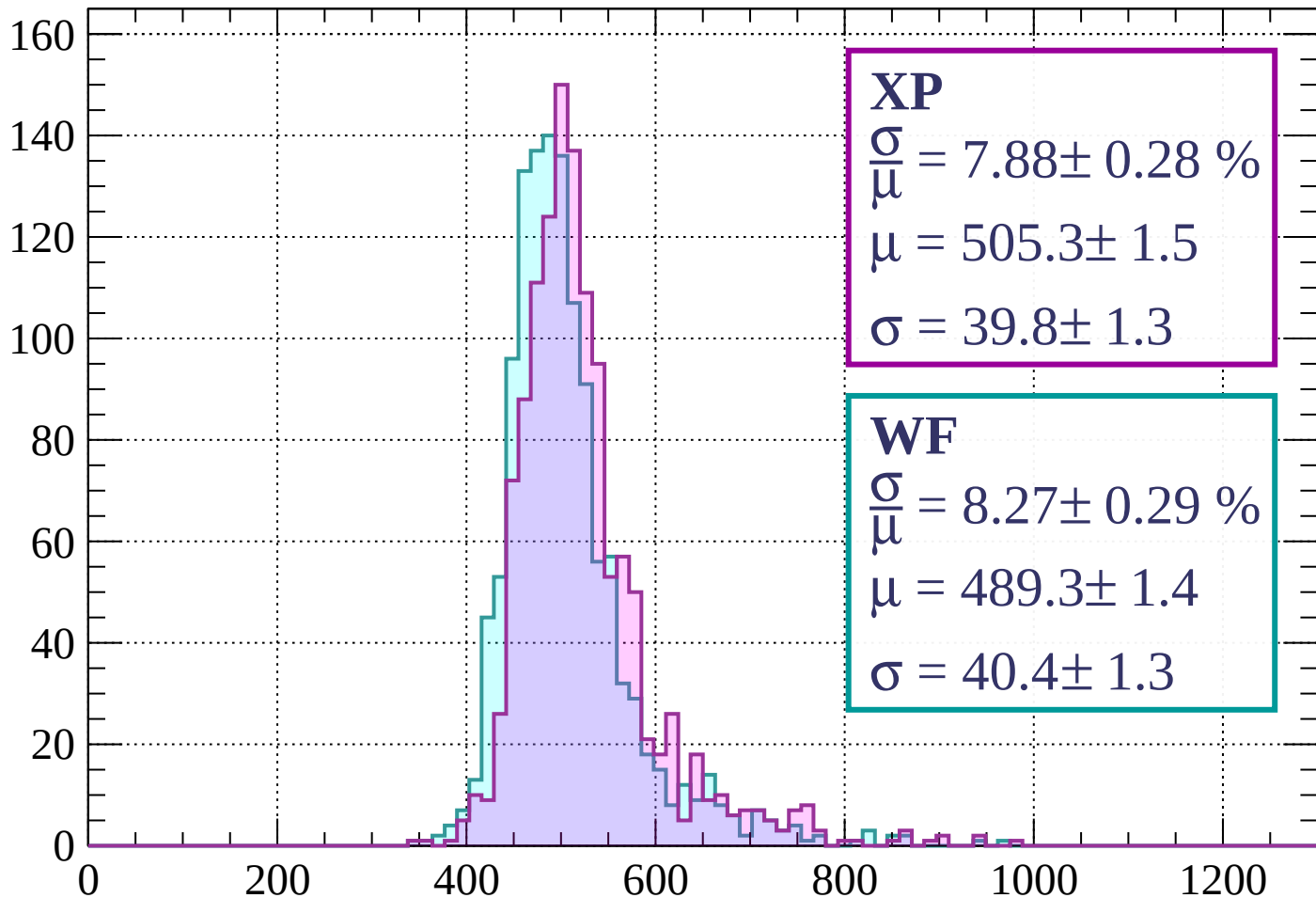
Energy loss | $1760 < p < 1840$

Count



Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 7.88 \pm 0.28 \%$$

$$\mu = 505.3 \pm 1.5$$

$$\sigma = 39.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.29 \%$$

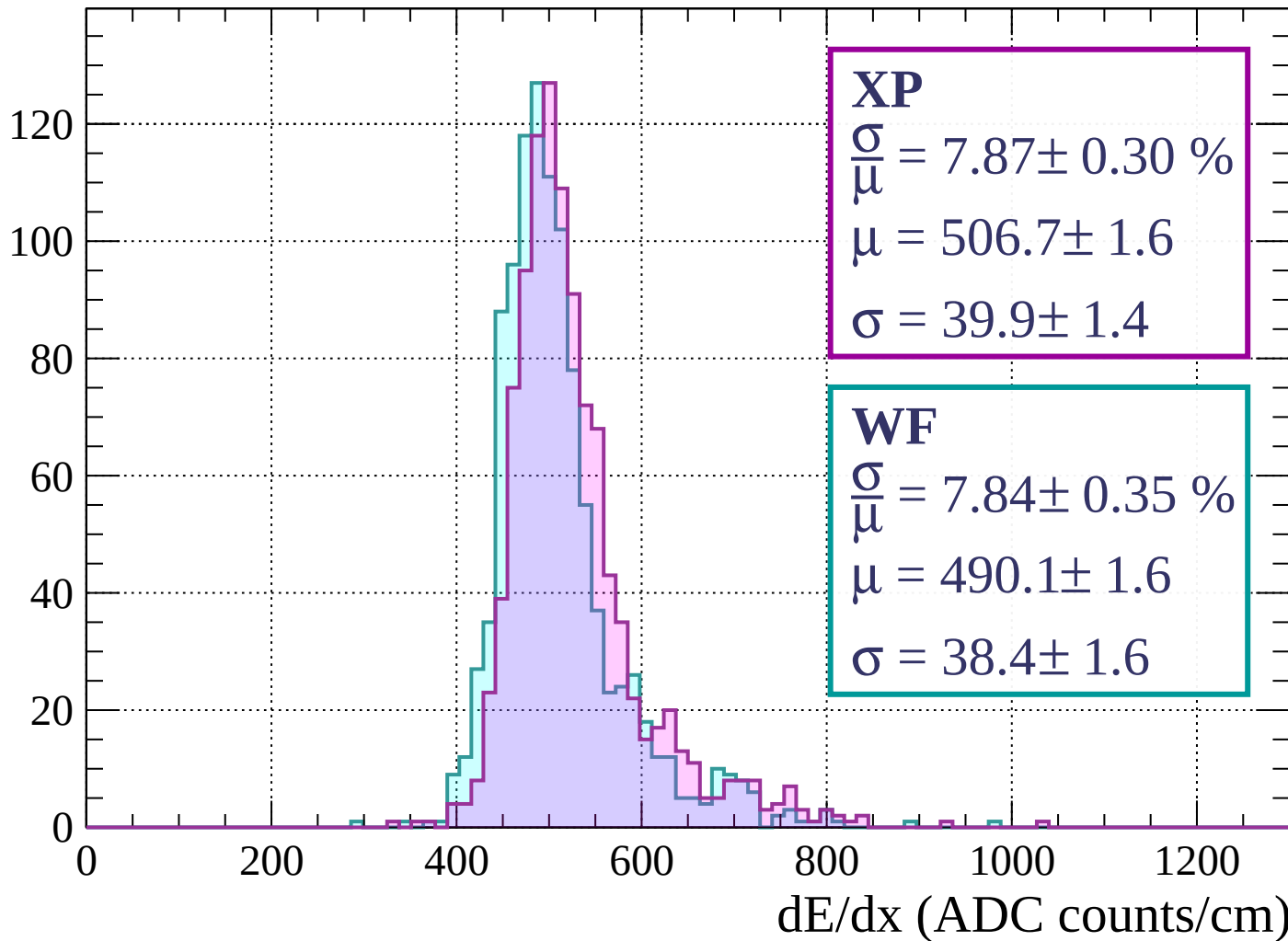
$$\mu = 489.3 \pm 1.4$$

$$\sigma = 40.4 \pm 1.3$$

dE/dx (ADC counts/cm)

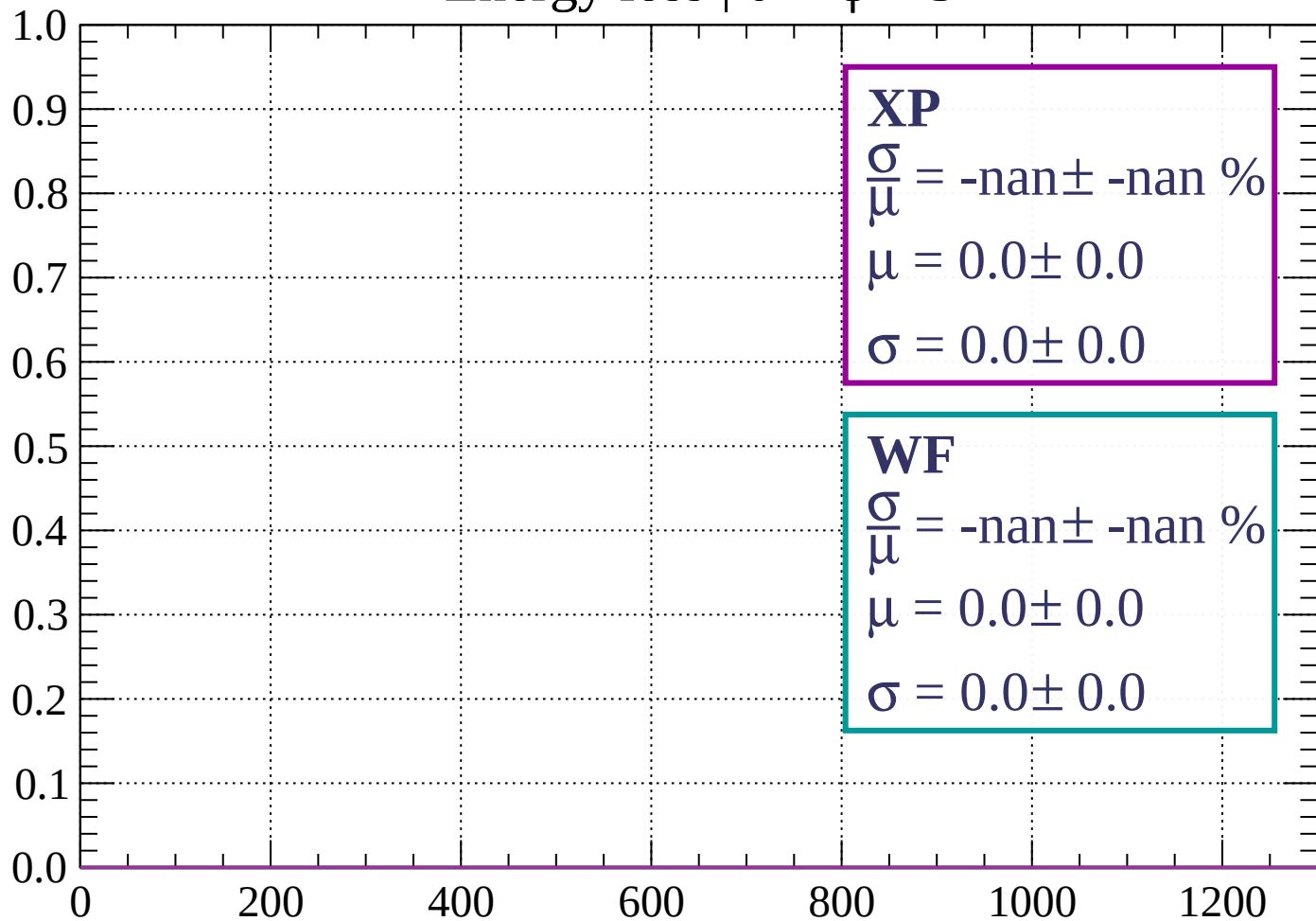
Energy loss | $1920 < p < 2000$

Count



Energy loss | $0 < \phi < 3$

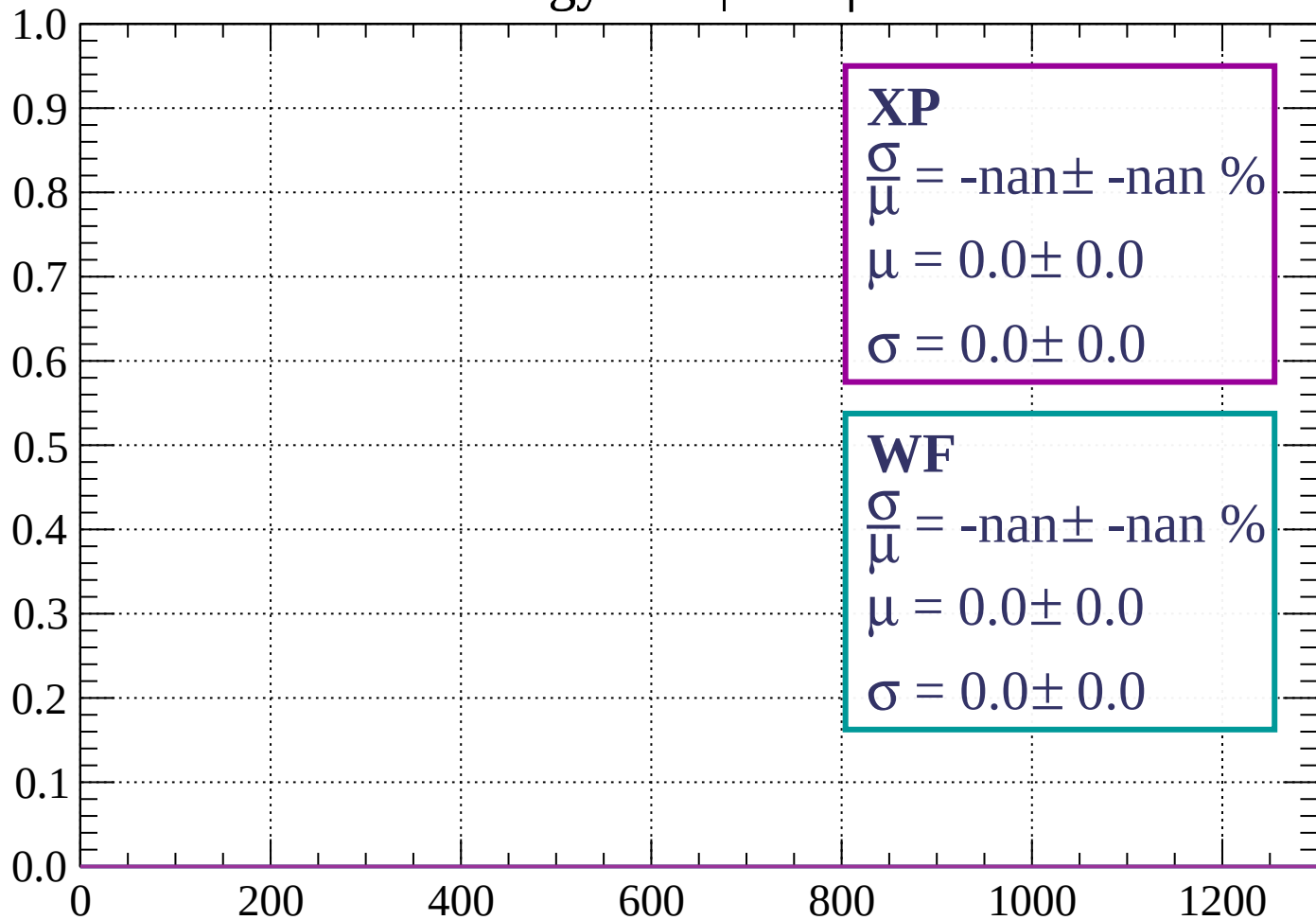
Count



dE/dx (ADC counts/cm)

Energy loss | $3 < \phi < 6$

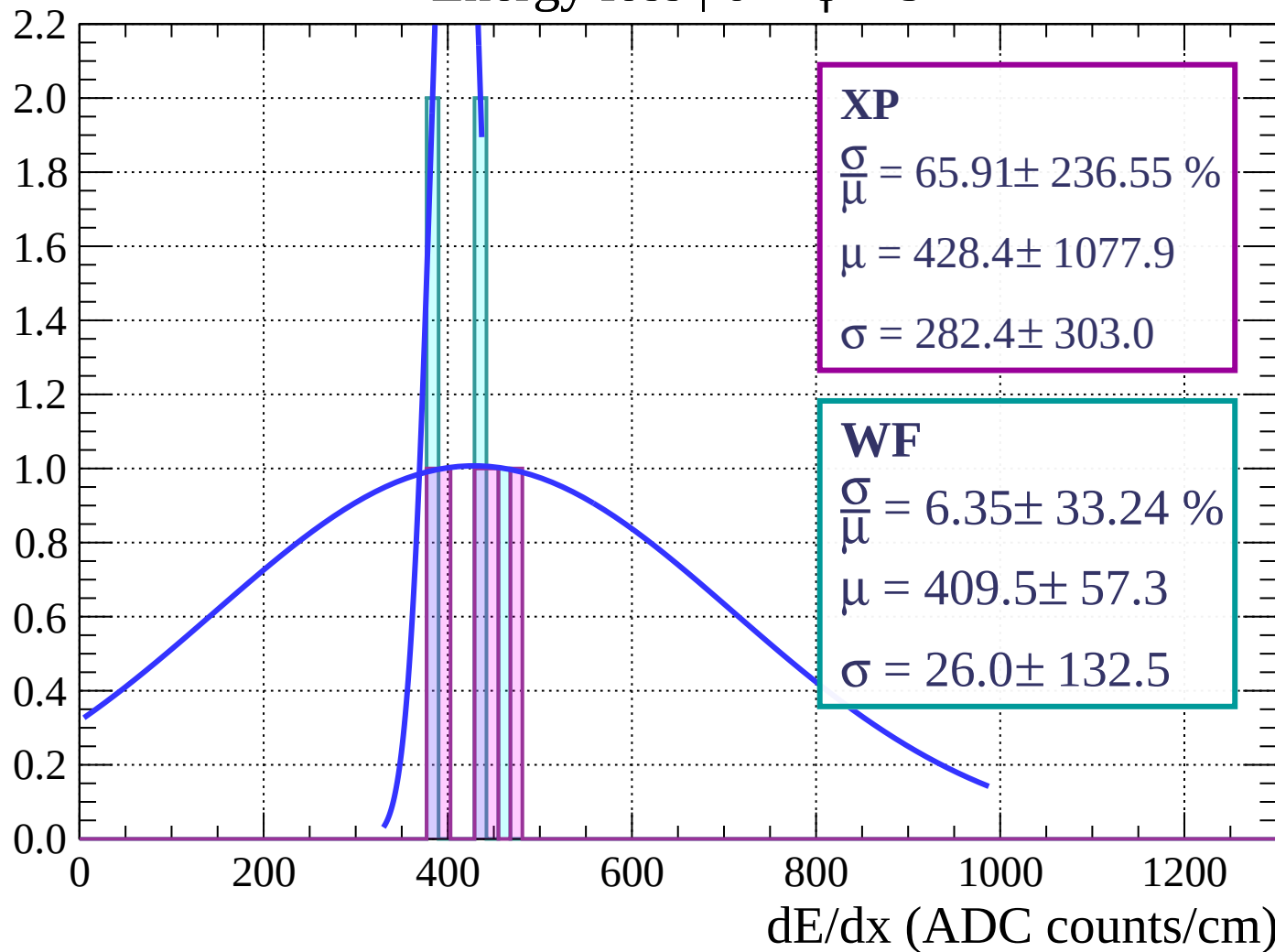
Count



dE/dx (ADC counts/cm)

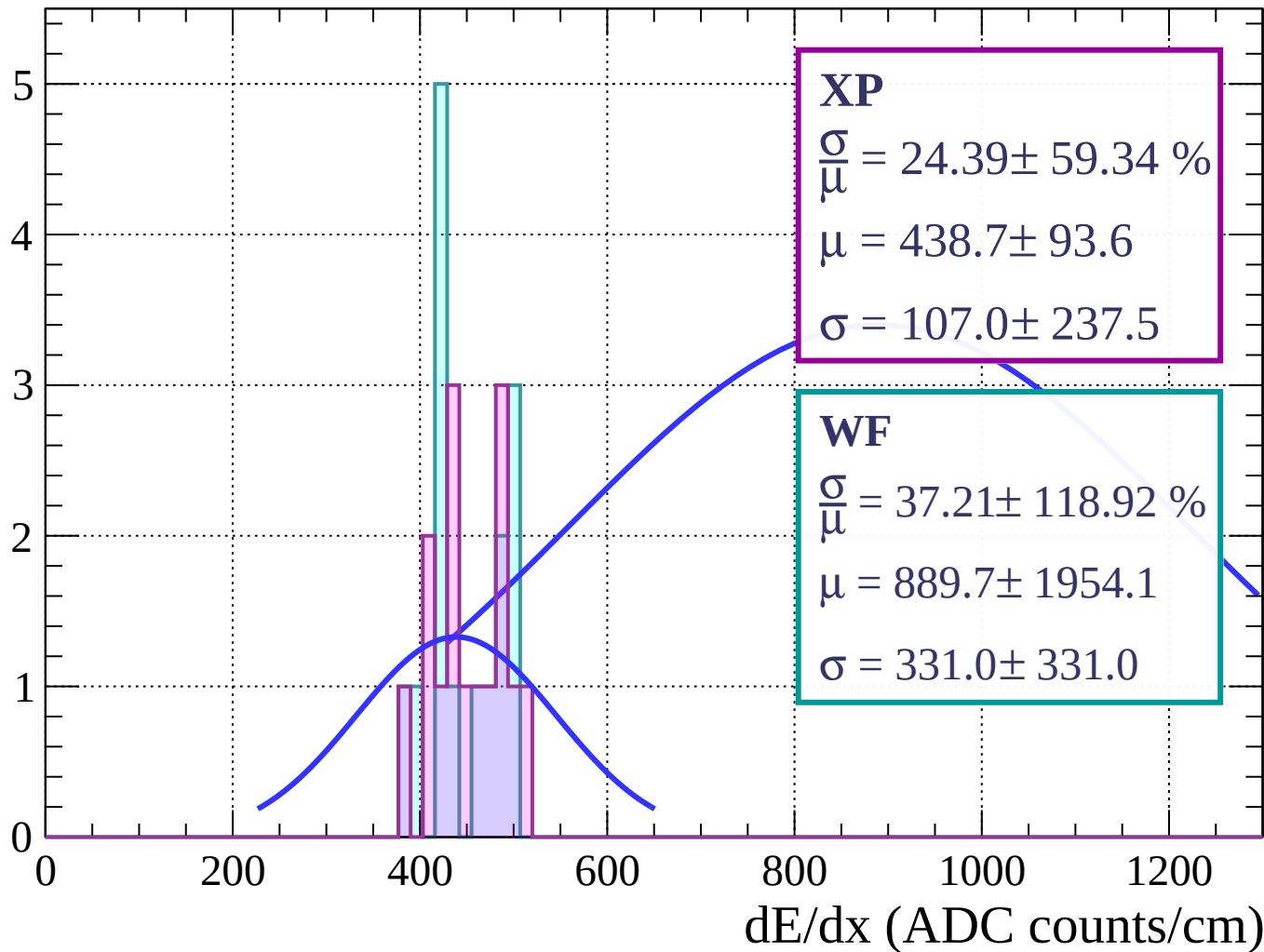
Energy loss | $6 < \phi < 9$

Count



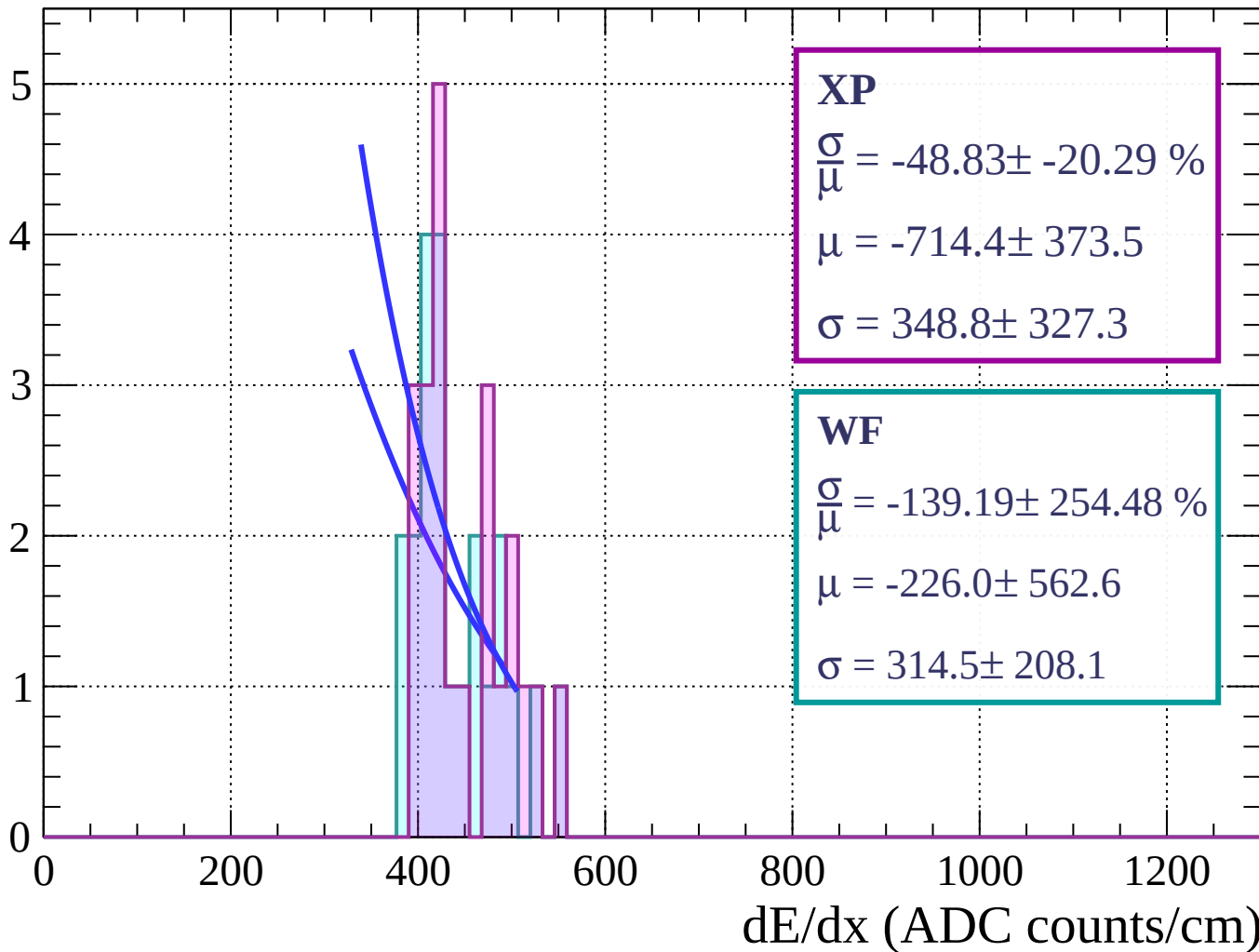
Energy loss | $9 < \phi < 12$

Count



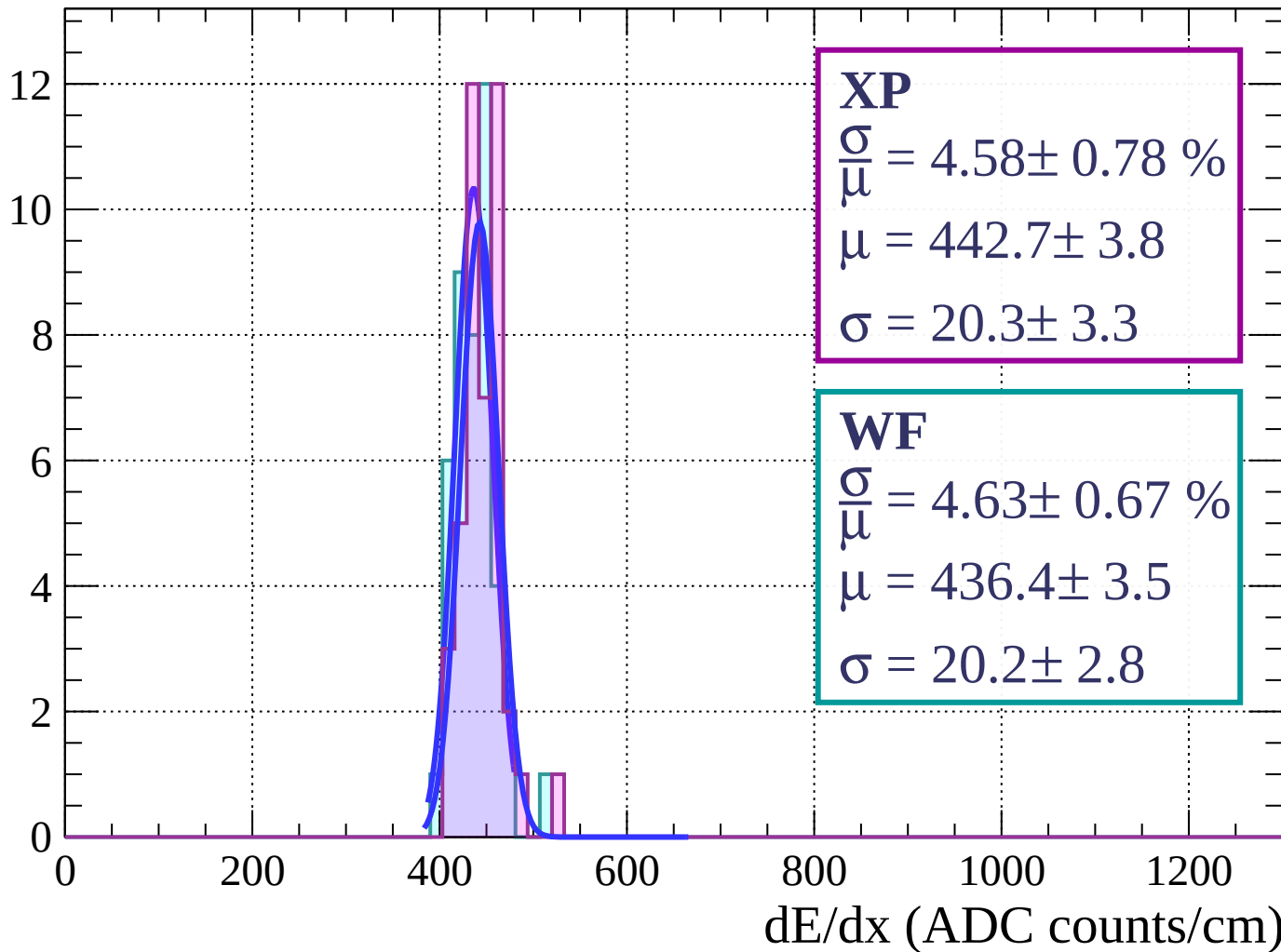
Energy loss | $12 < \phi < 15$

Count



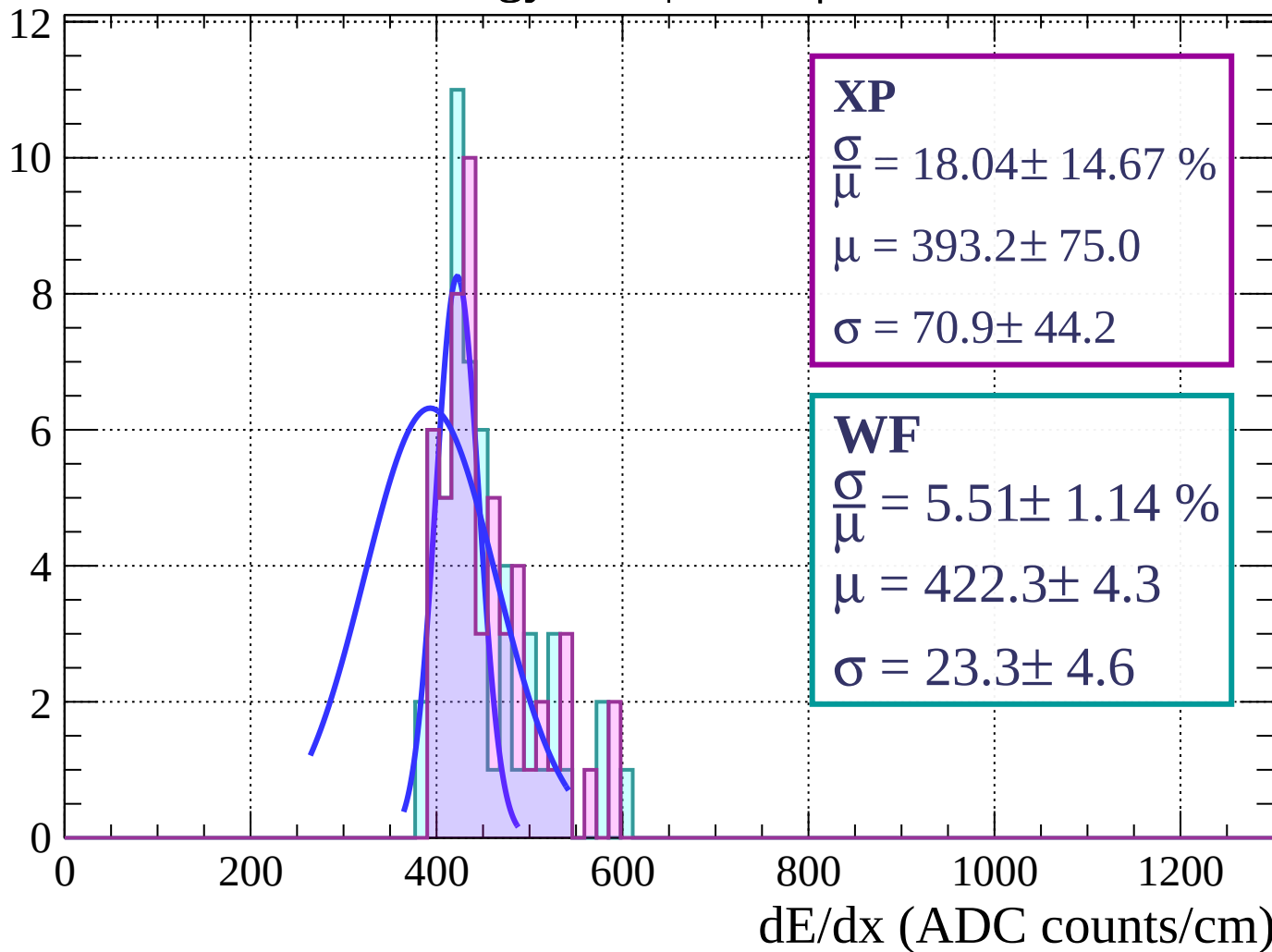
Energy loss | $15 < \phi < 18$

Count



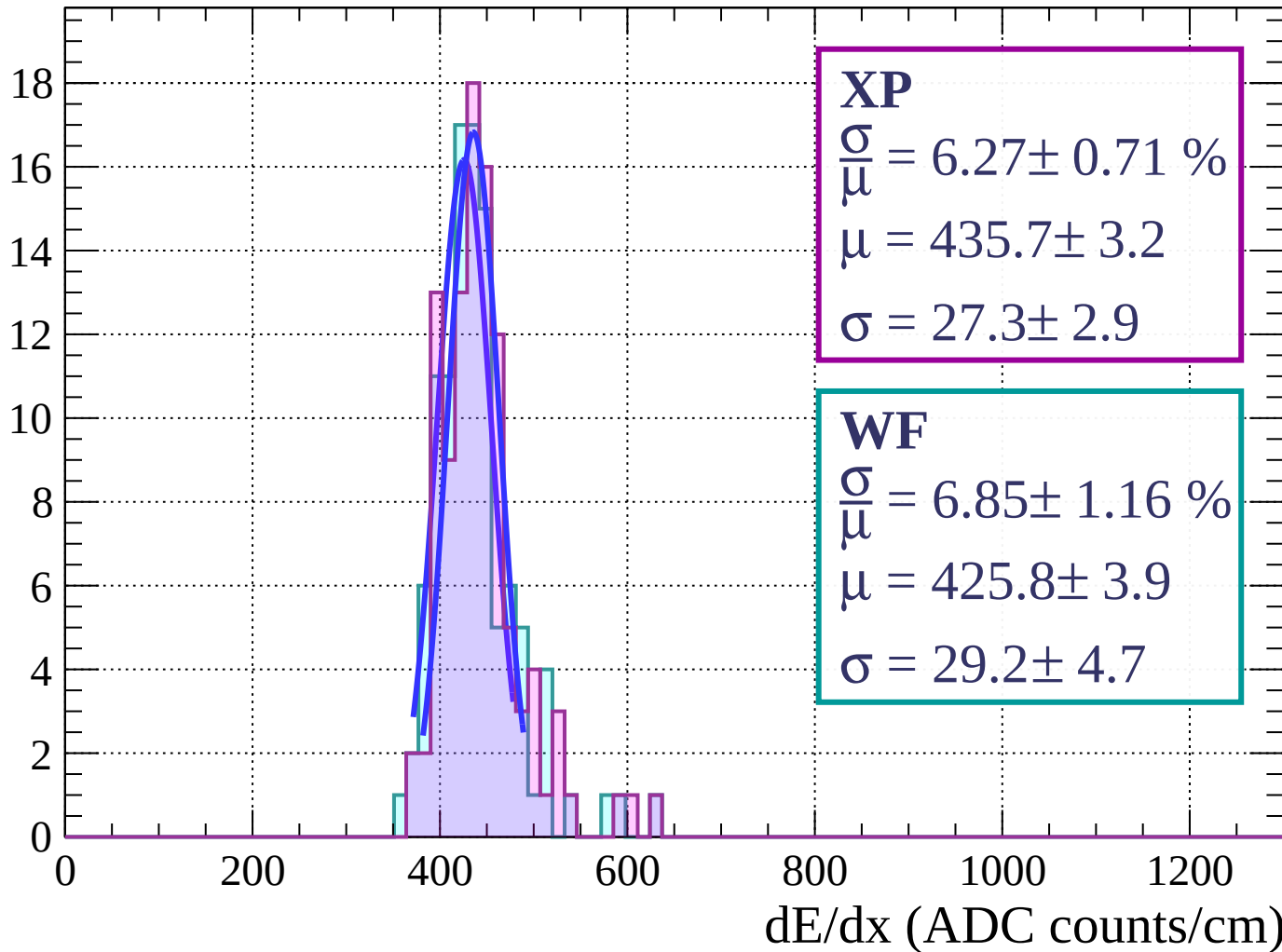
Energy loss | $18 < \phi < 21$

Count



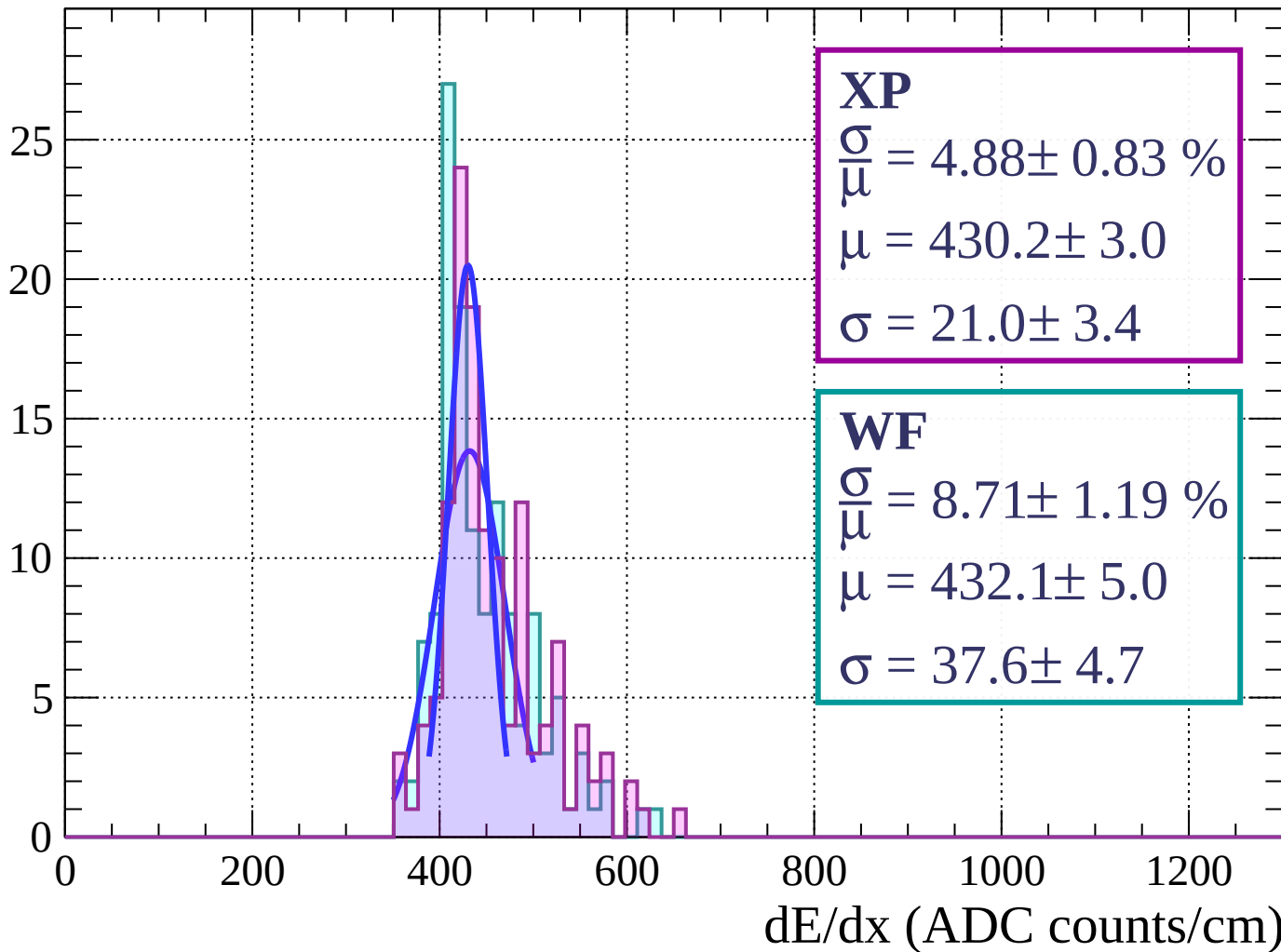
Energy loss | $21 < \phi < 24$

Count



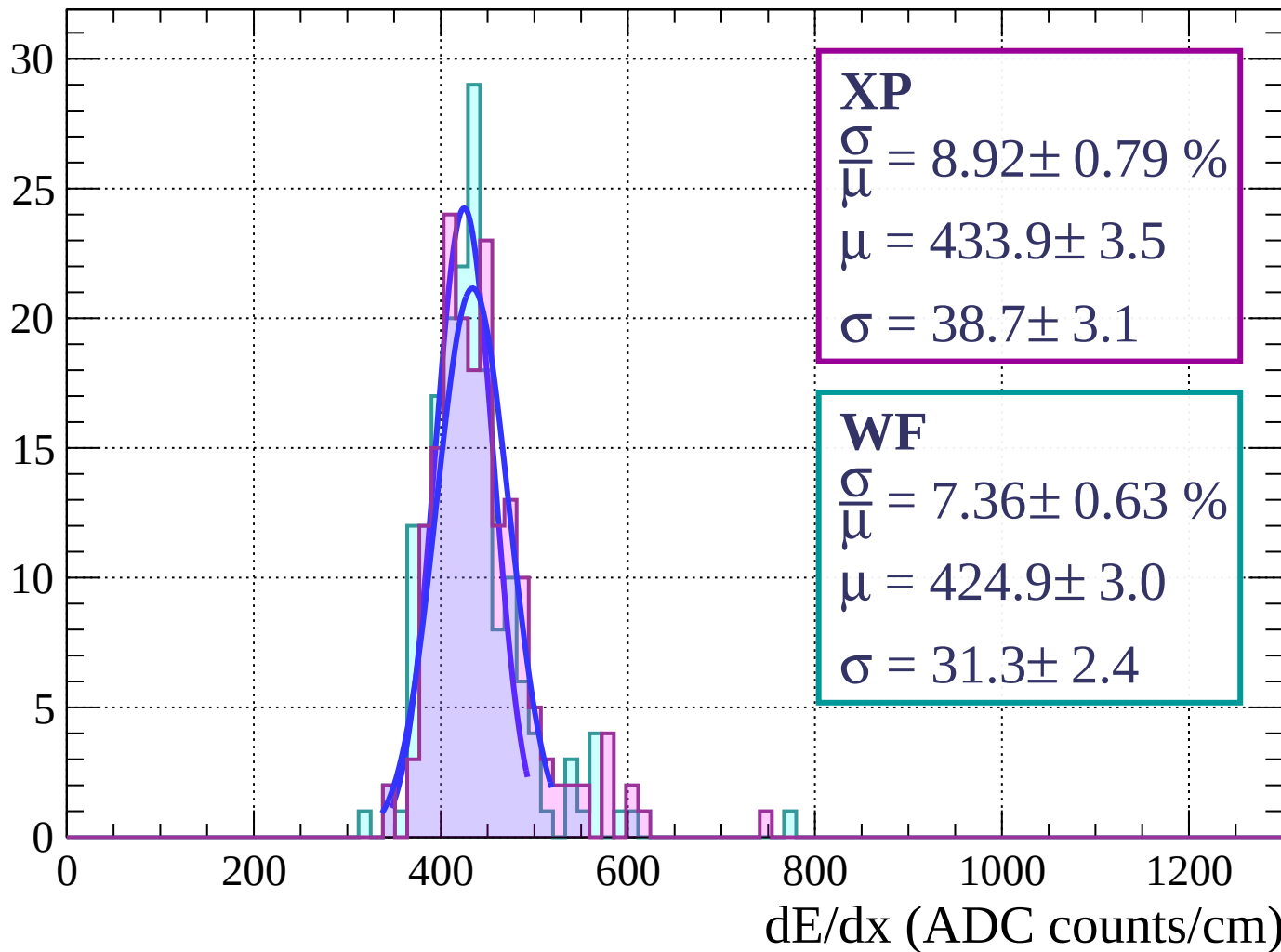
Energy loss | $24 < \phi < 27$

Count



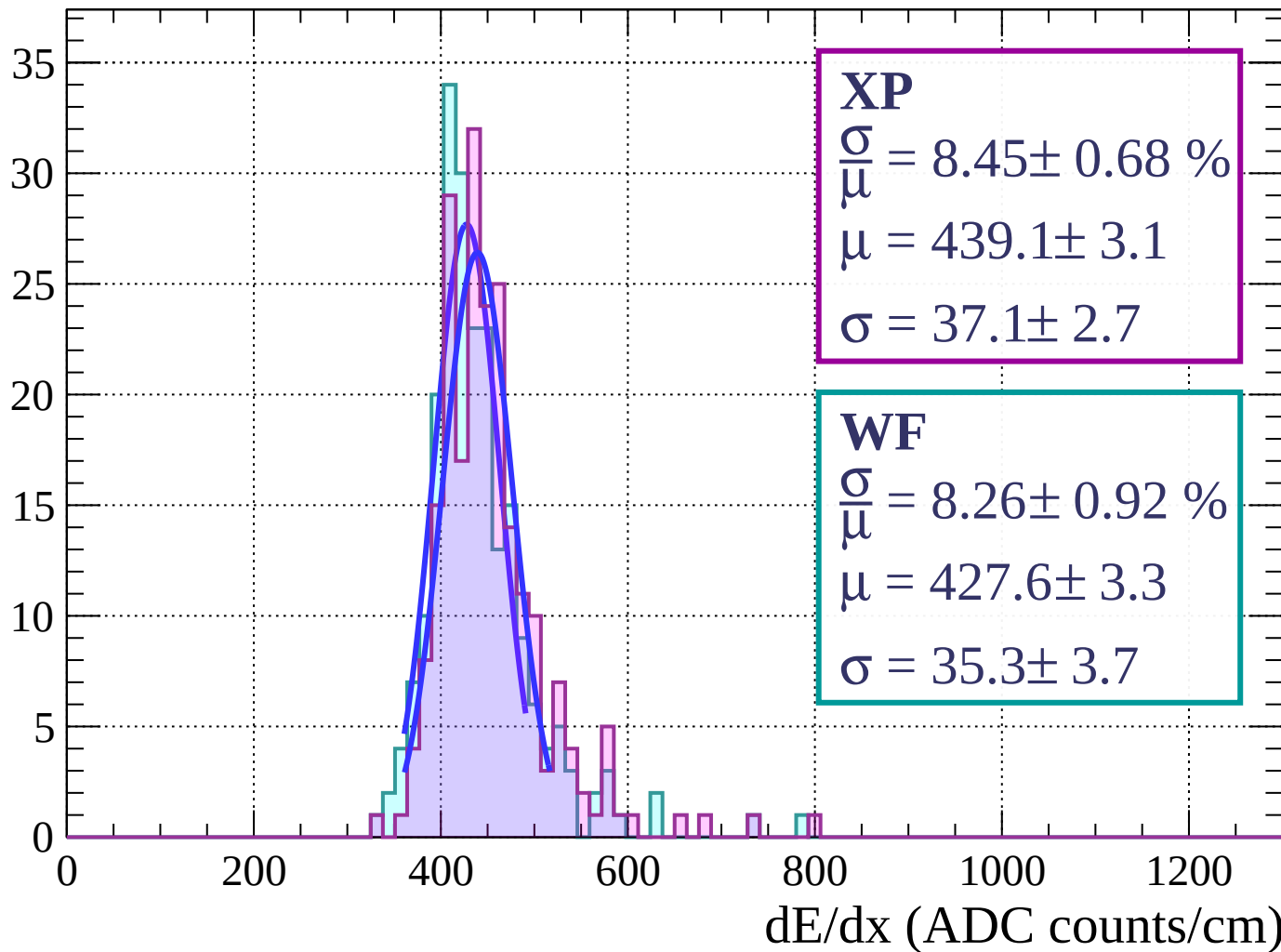
Energy loss | $27 < \phi < 30$

Count



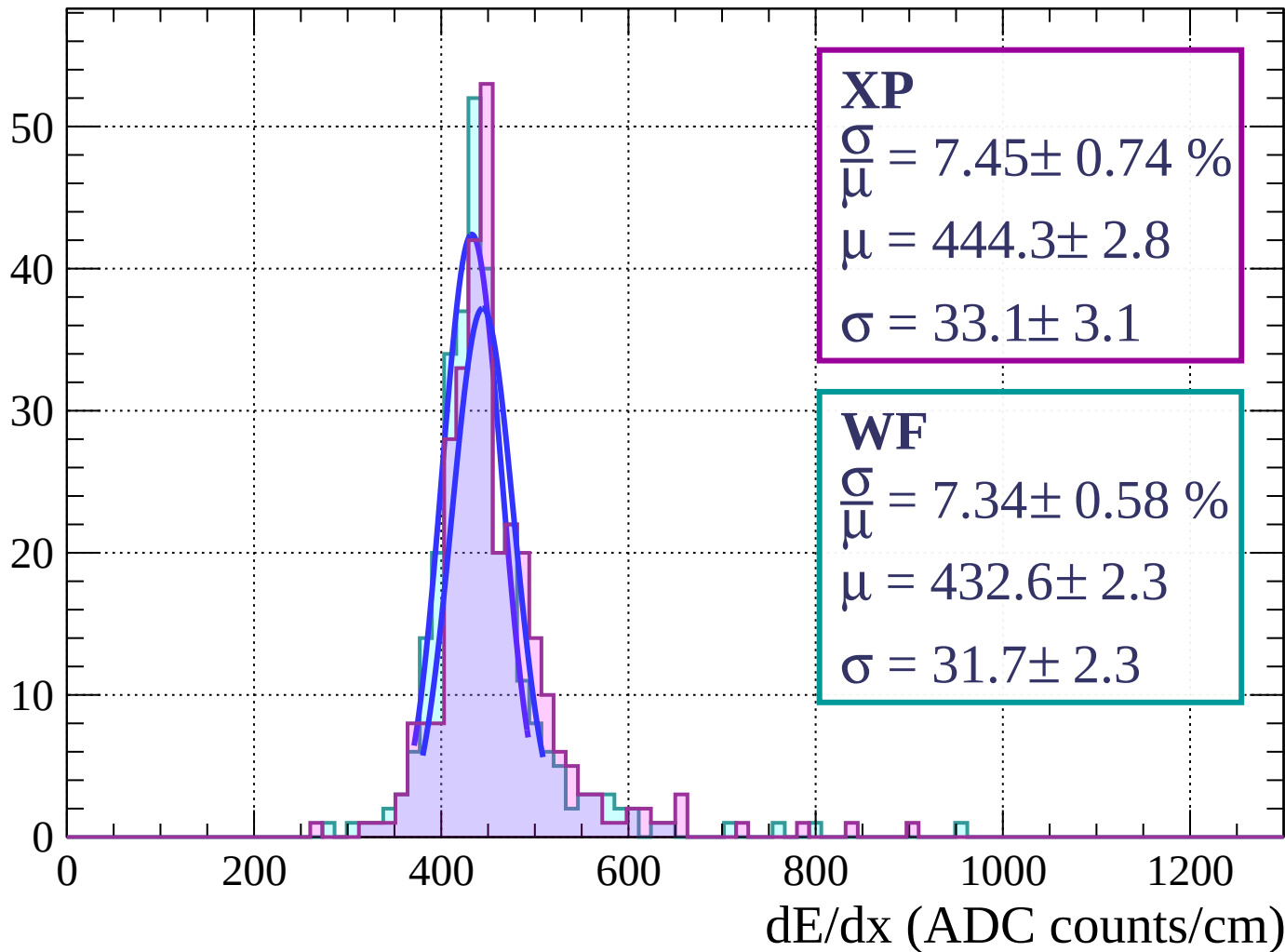
Energy loss | $30 < \phi < 33$

Count



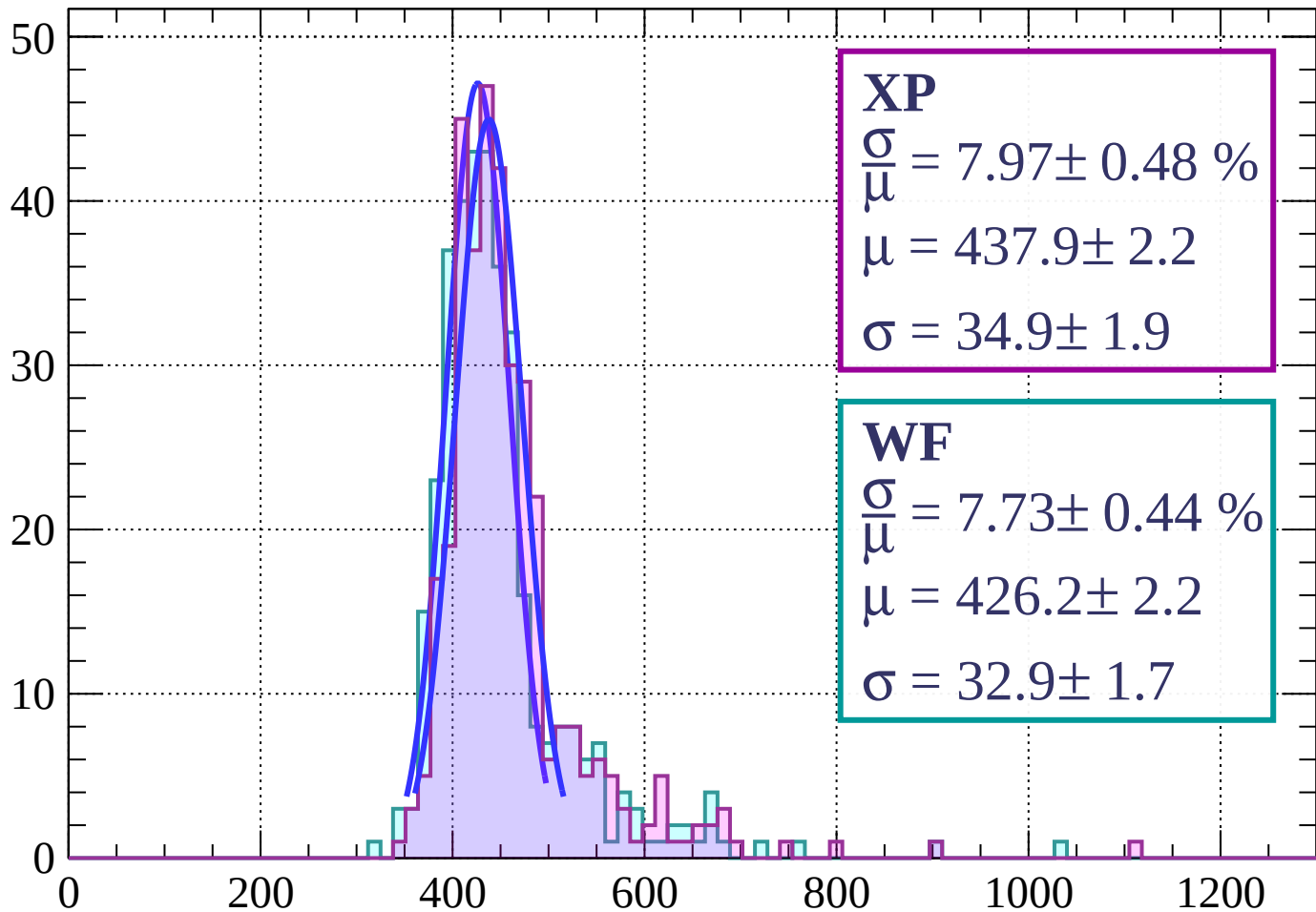
Energy loss | $33 < \phi < 36$

Count



Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 7.97 \pm 0.48 \%$$

$$\mu = 437.9 \pm 2.2$$

$$\sigma = 34.9 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 7.73 \pm 0.44 \%$$

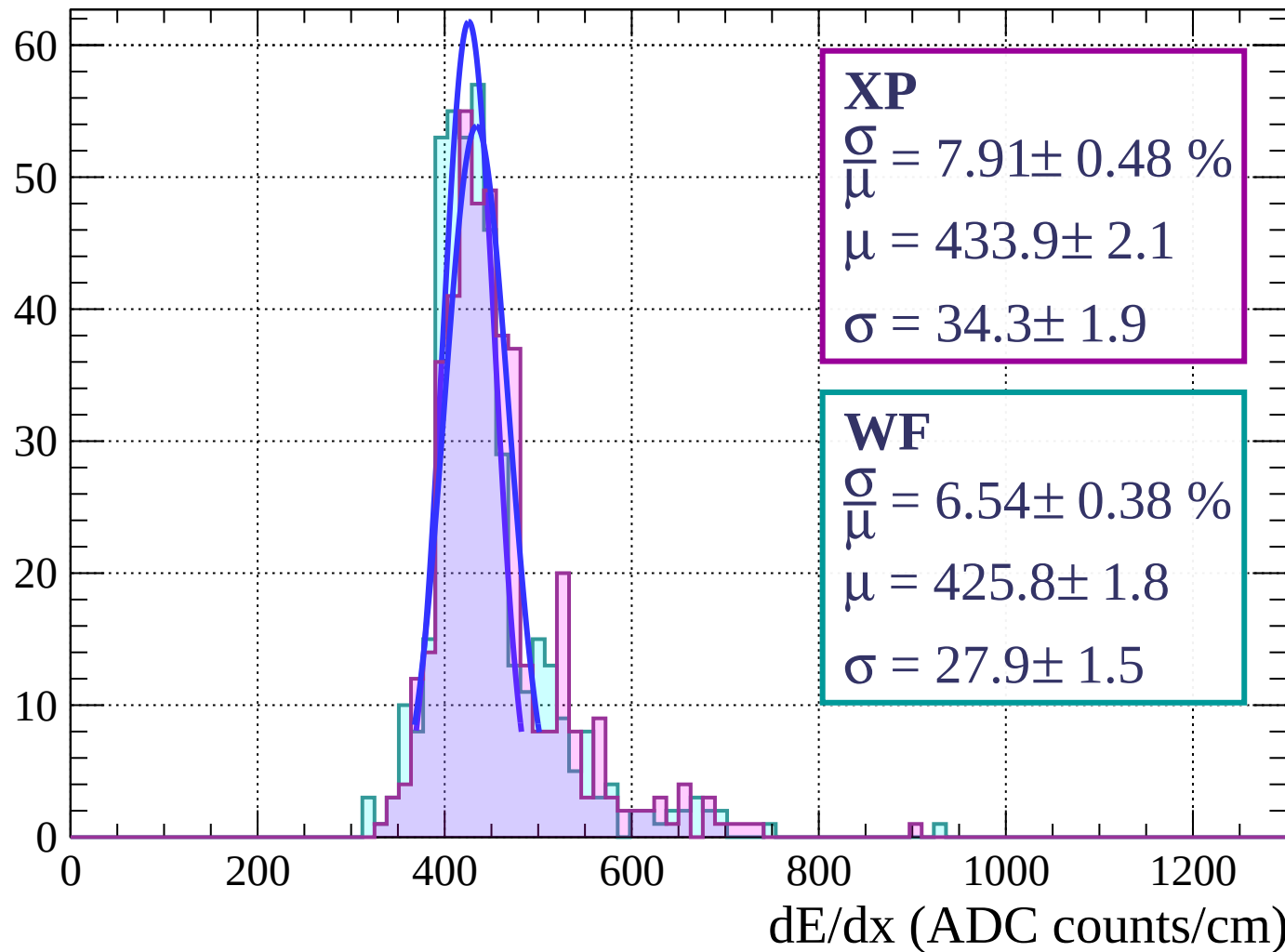
$$\mu = 426.2 \pm 2.2$$

$$\sigma = 32.9 \pm 1.7$$

dE/dx (ADC counts/cm)

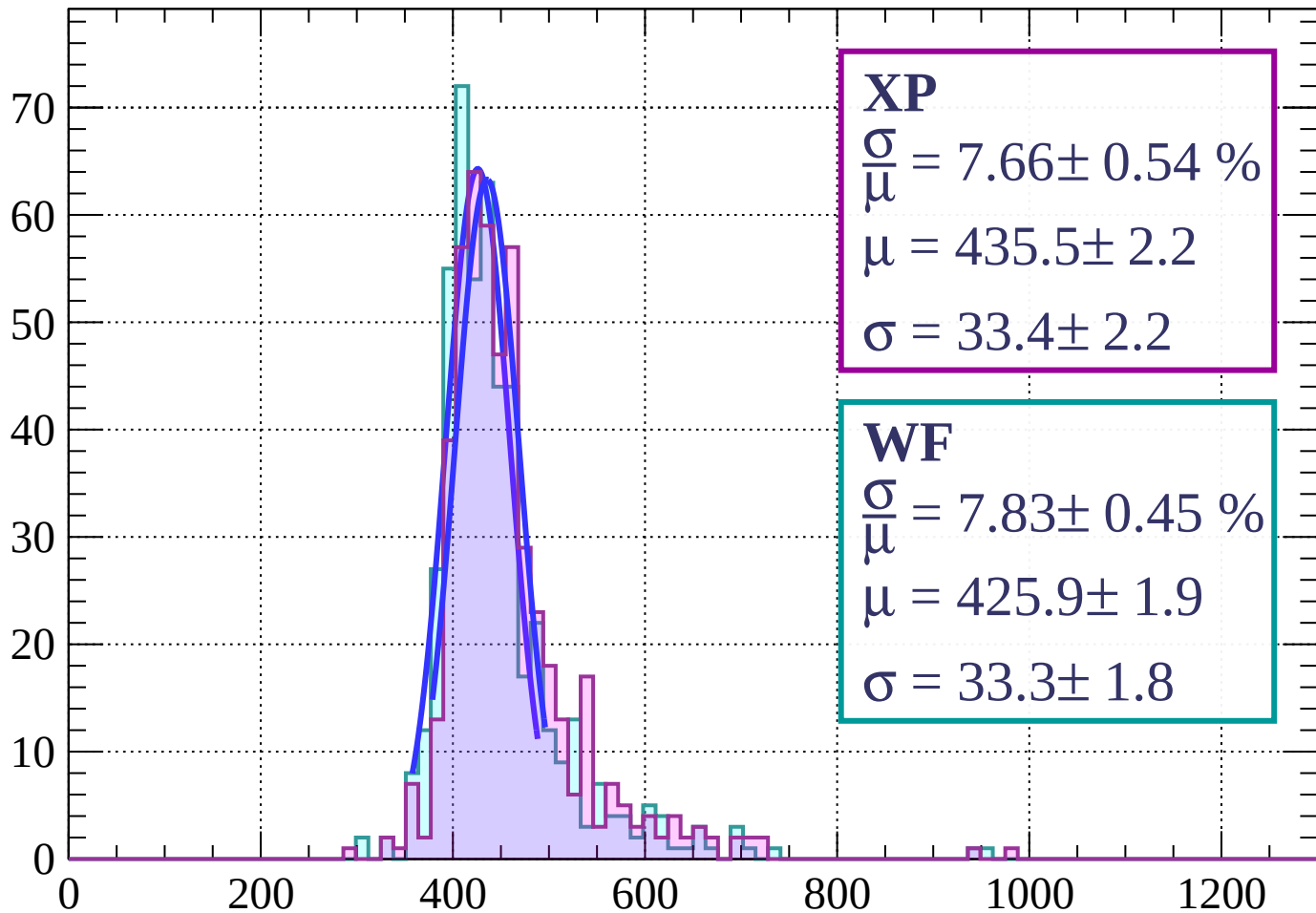
Energy loss | $39 < \phi < 42$

Count



Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.66 \pm 0.54 \%$$

$$\mu = 435.5 \pm 2.2$$

$$\sigma = 33.4 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.83 \pm 0.45 \%$$

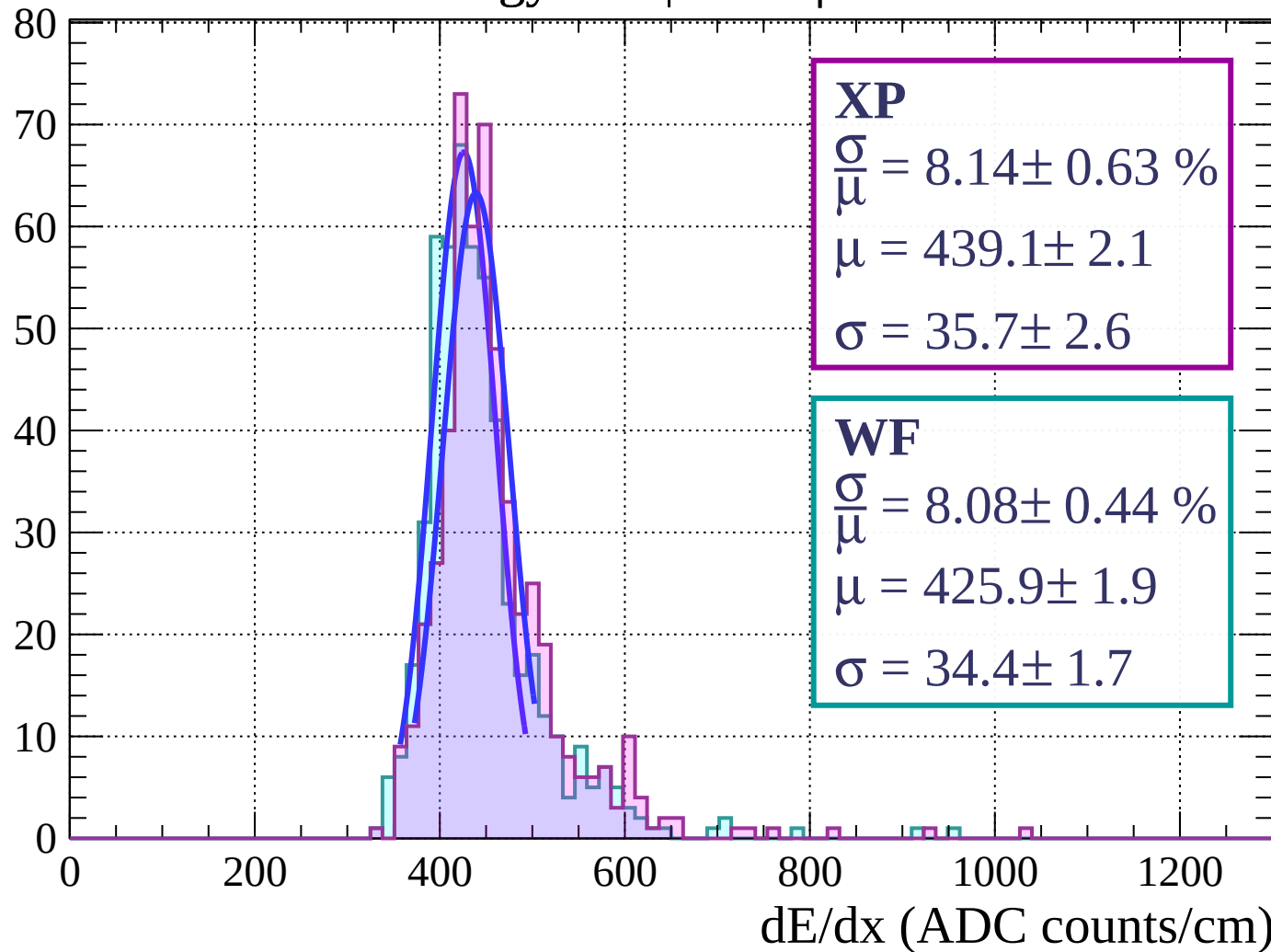
$$\mu = 425.9 \pm 1.9$$

$$\sigma = 33.3 \pm 1.8$$

dE/dx (ADC counts/cm)

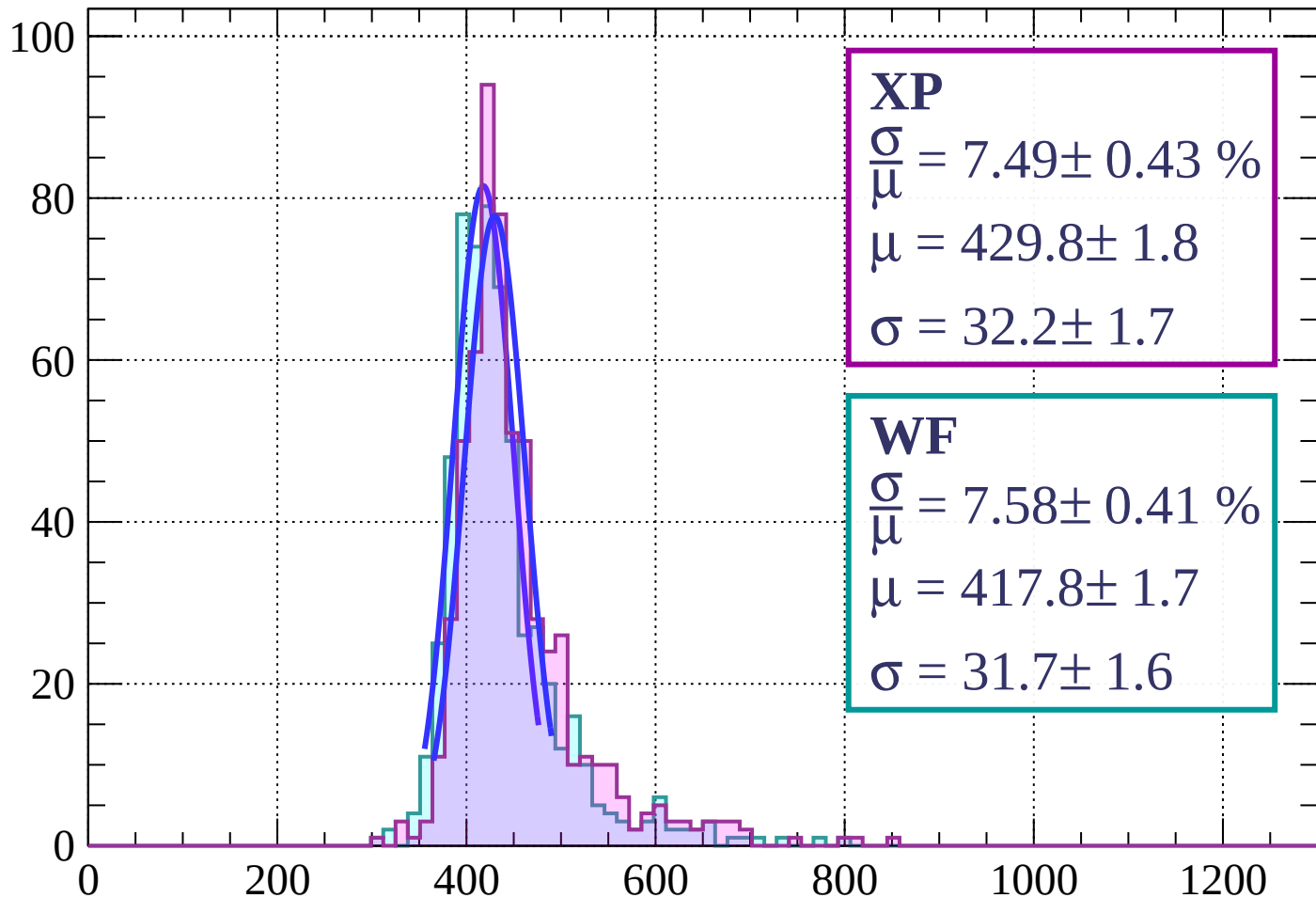
Energy loss | $45 < \phi < 48$

Count



Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.43 \%$$

$$\mu = 429.8 \pm 1.8$$

$$\sigma = 32.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.58 \pm 0.41 \%$$

$$\mu = 417.8 \pm 1.7$$

$$\sigma = 31.7 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

Count

XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.43 \%$$

$$\mu = 437.1 \pm 2.0$$

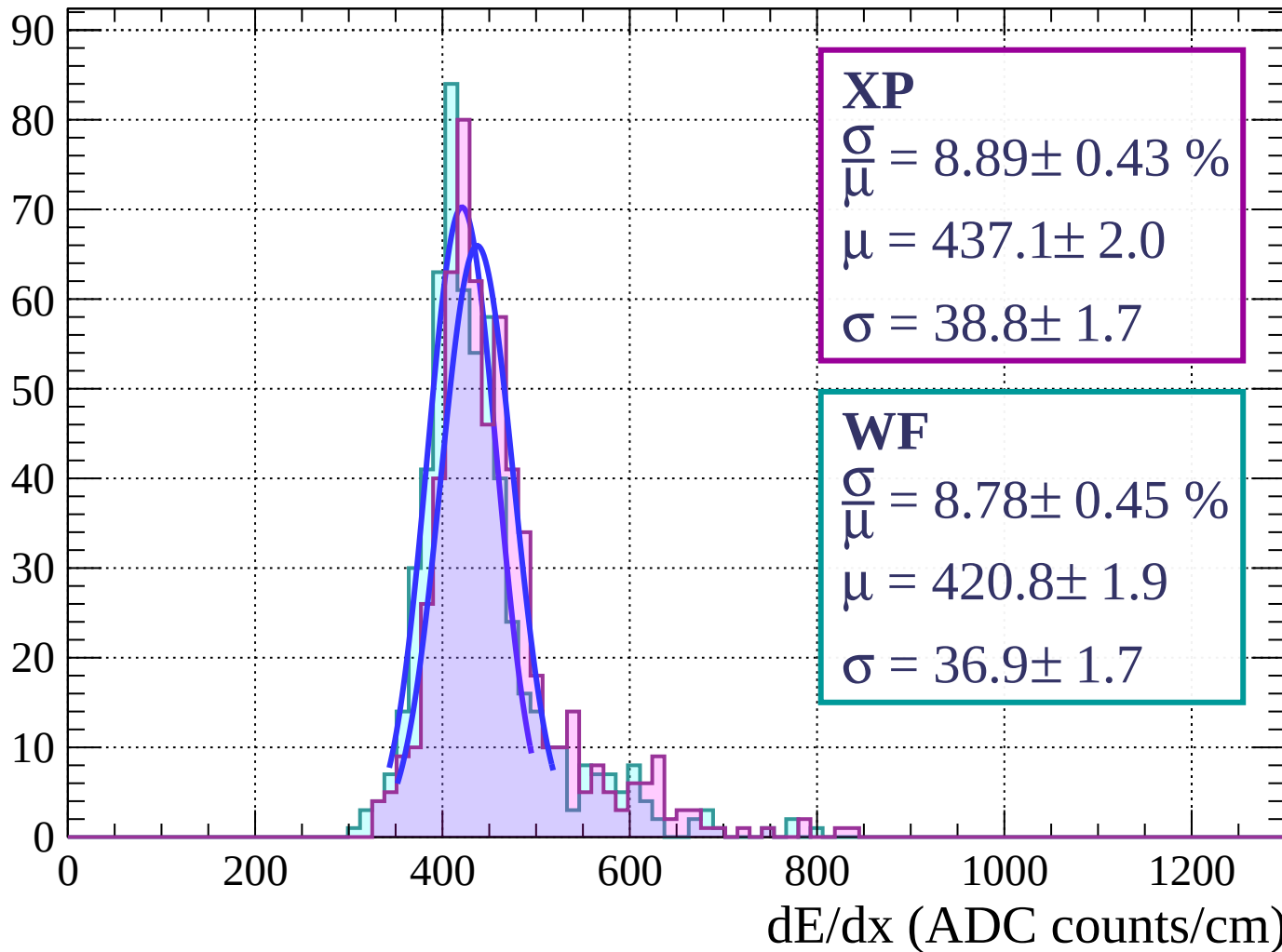
$$\sigma = 38.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.78 \pm 0.45 \%$$

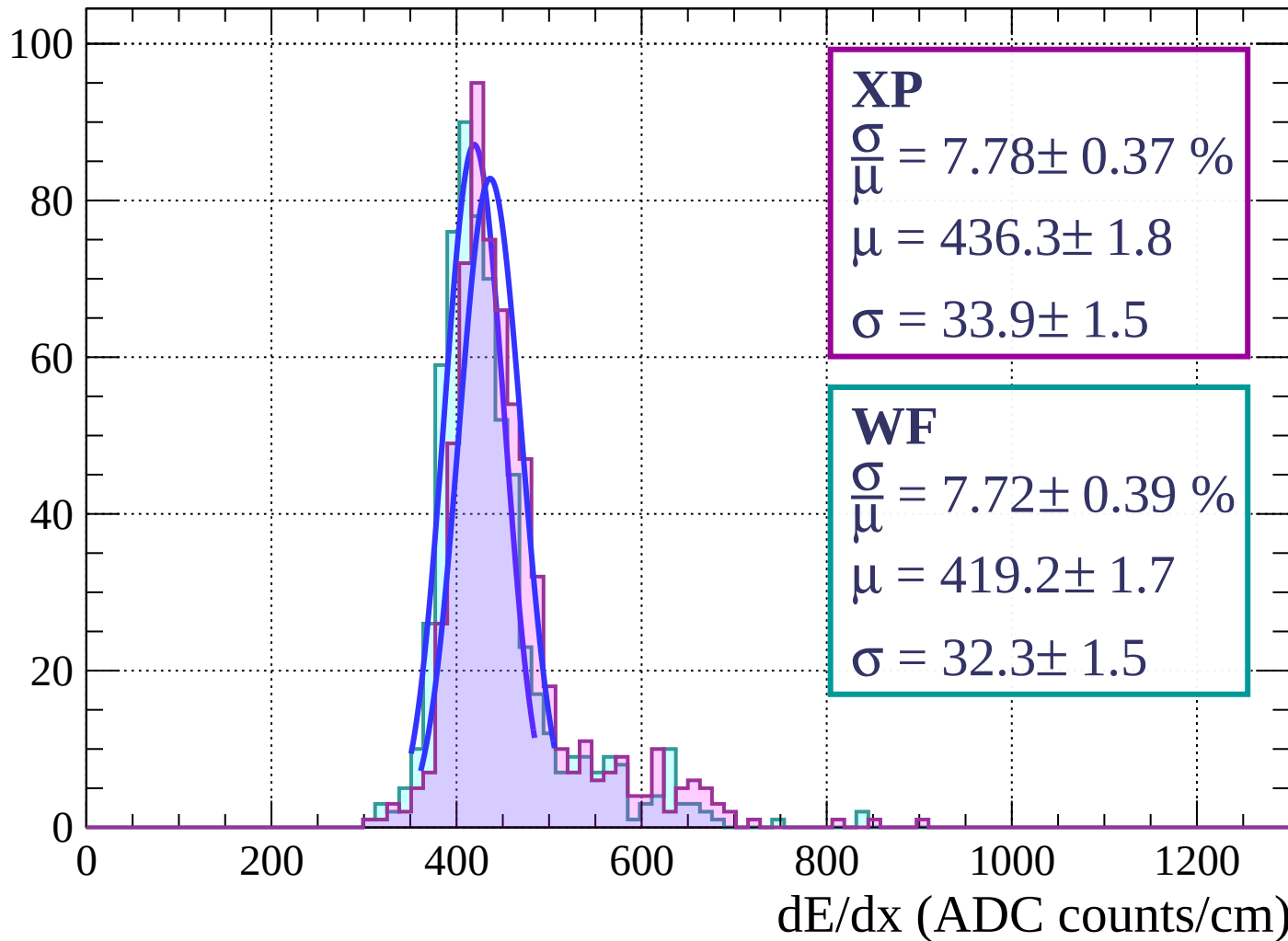
$$\mu = 420.8 \pm 1.9$$

$$\sigma = 36.9 \pm 1.7$$



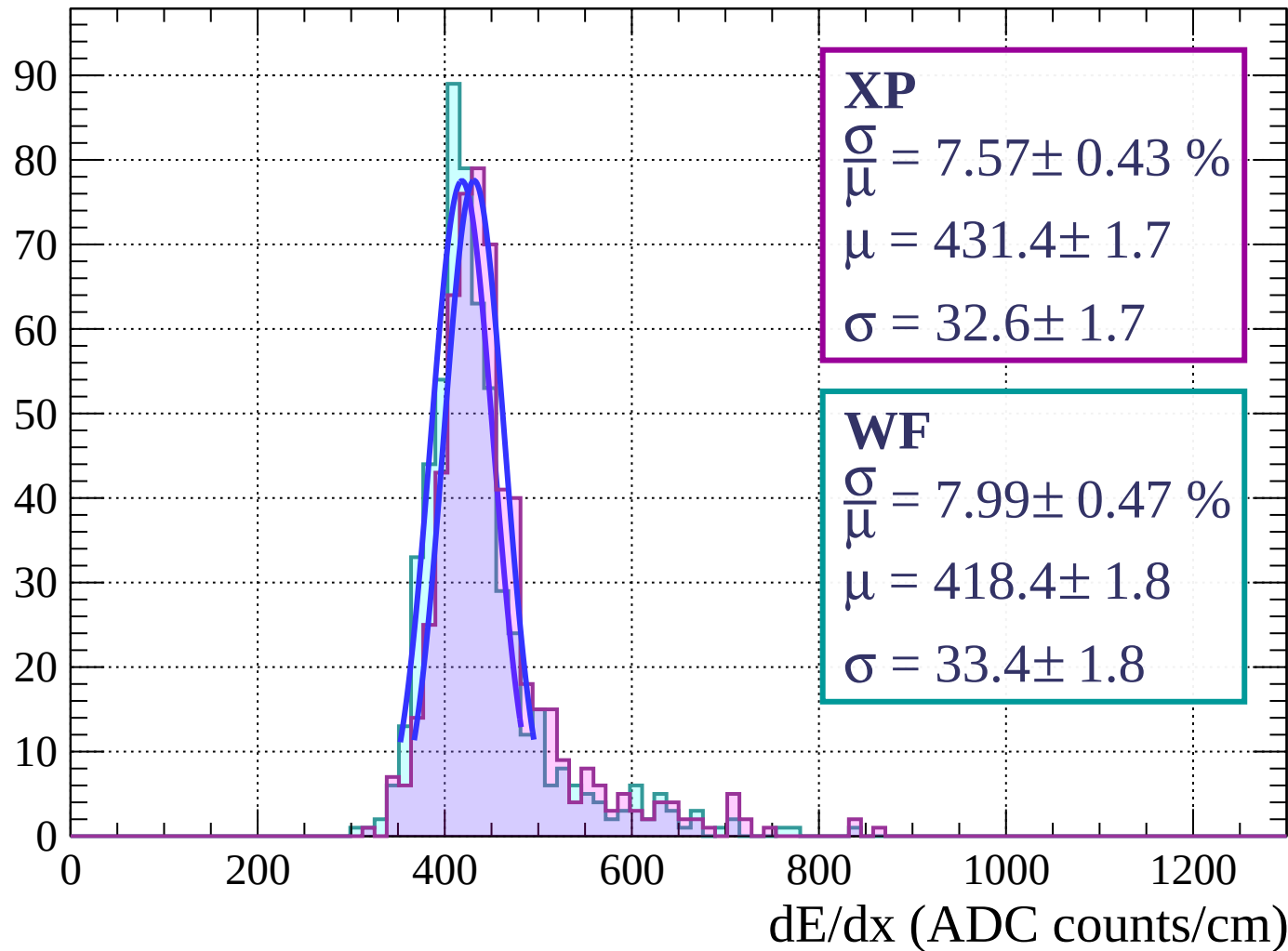
Energy loss | $54 < \phi < 57$

Count



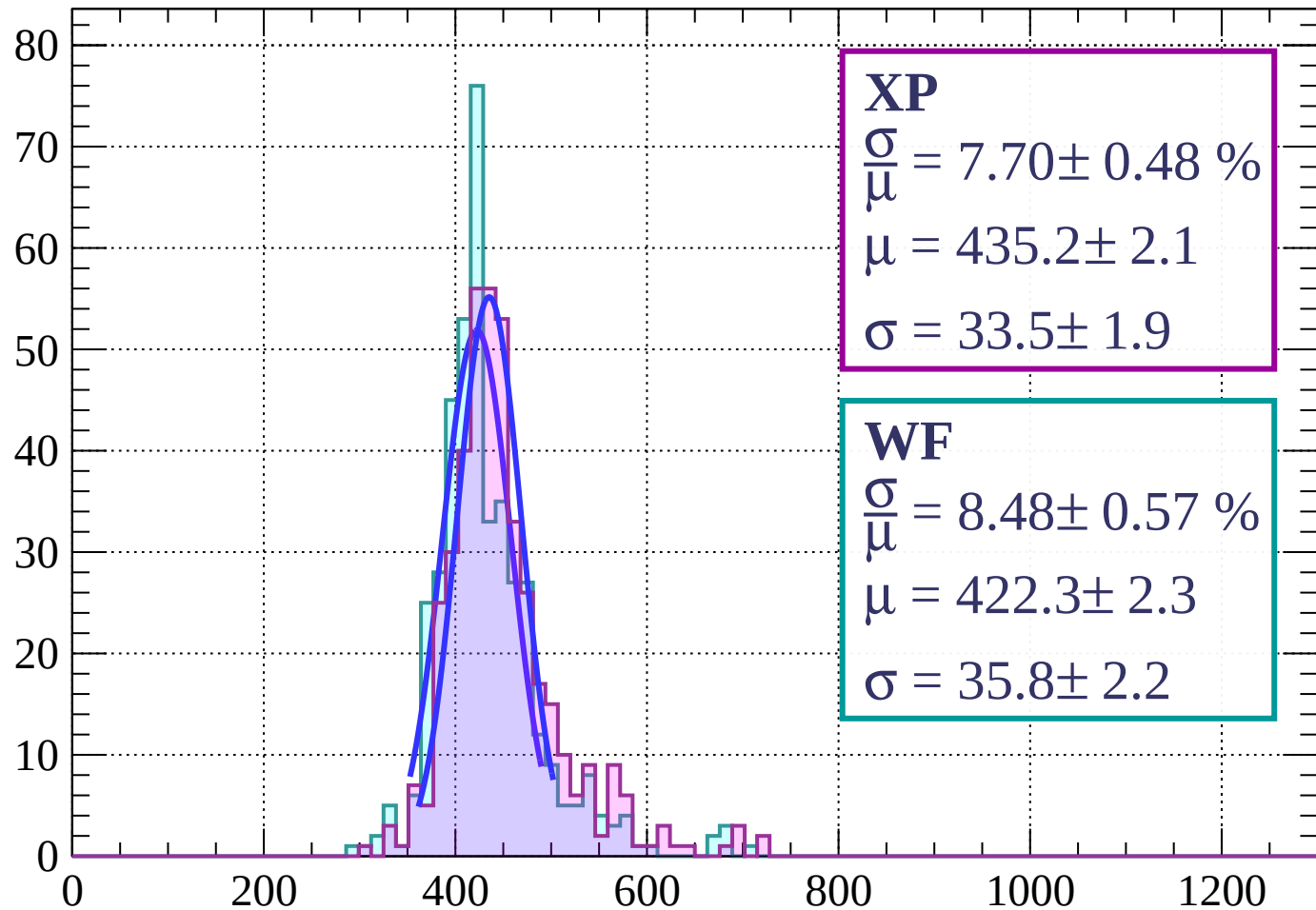
Energy loss | $57 < \phi < 60$

Count



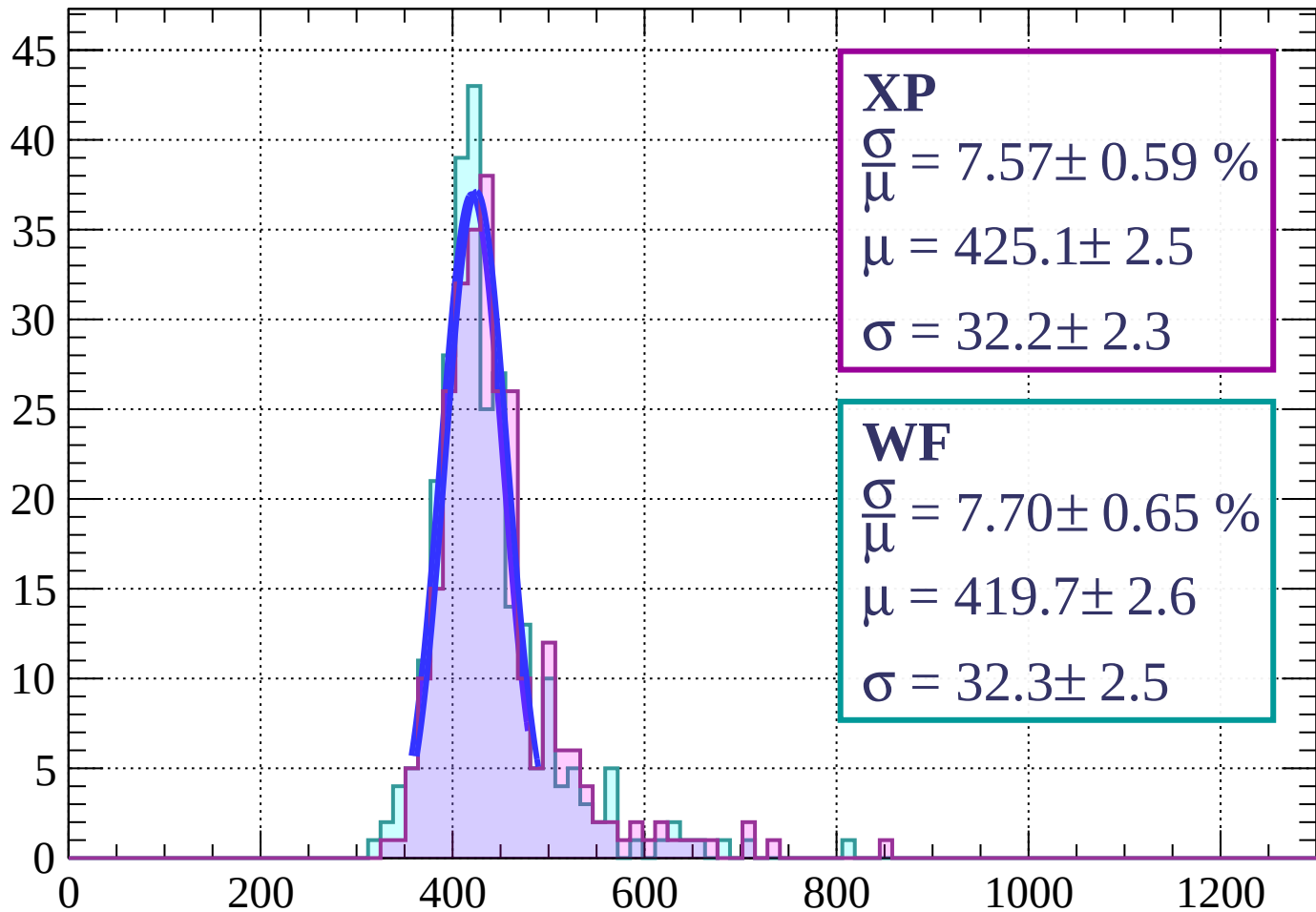
Energy loss | $60 < \phi < 63$

Count



Energy loss | $63 < \phi < 66$

Count



XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.59 \%$$

$$\mu = 425.1 \pm 2.5$$

$$\sigma = 32.2 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.70 \pm 0.65 \%$$

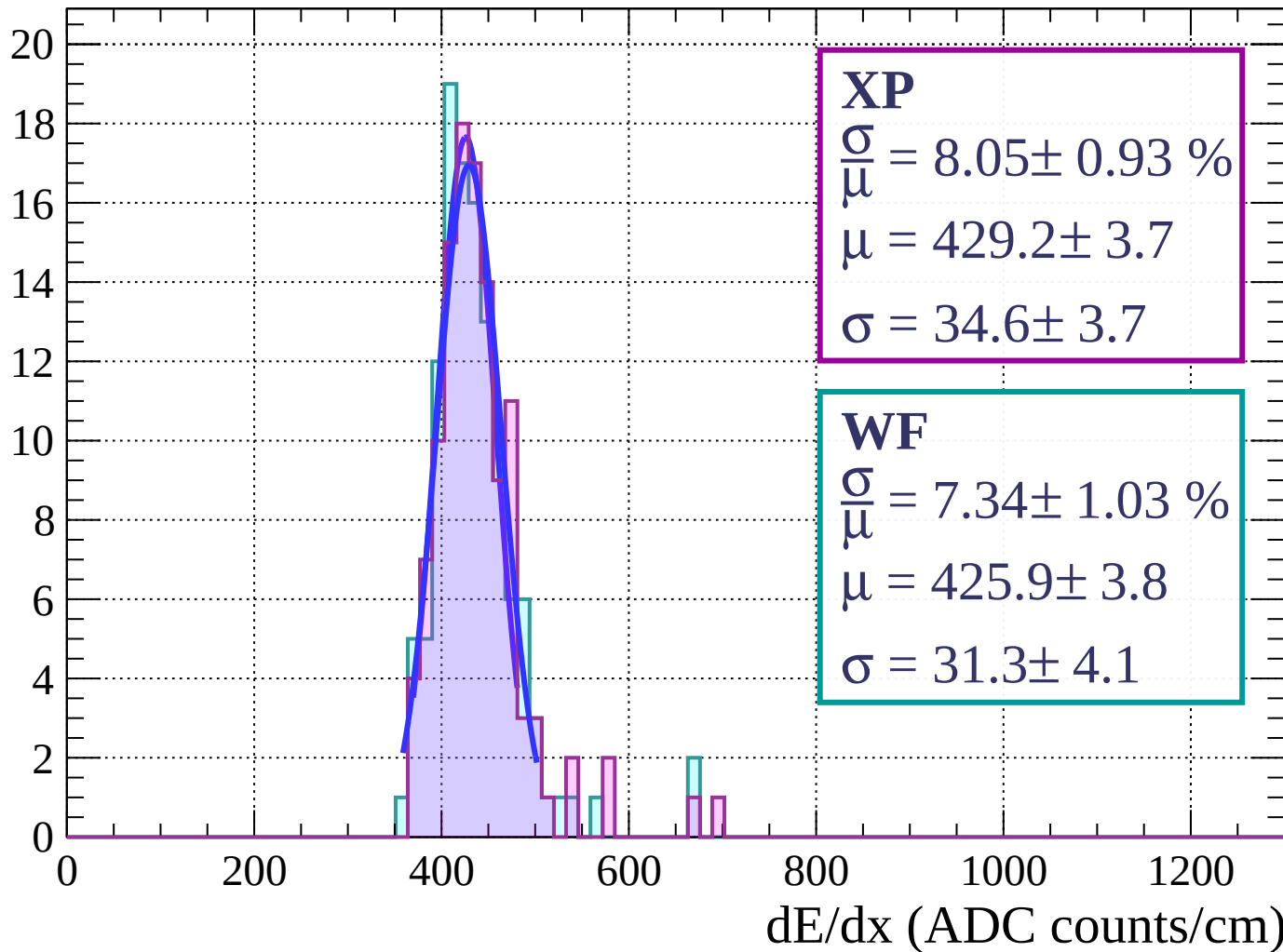
$$\mu = 419.7 \pm 2.6$$

$$\sigma = 32.3 \pm 2.5$$

dE/dx (ADC counts/cm)

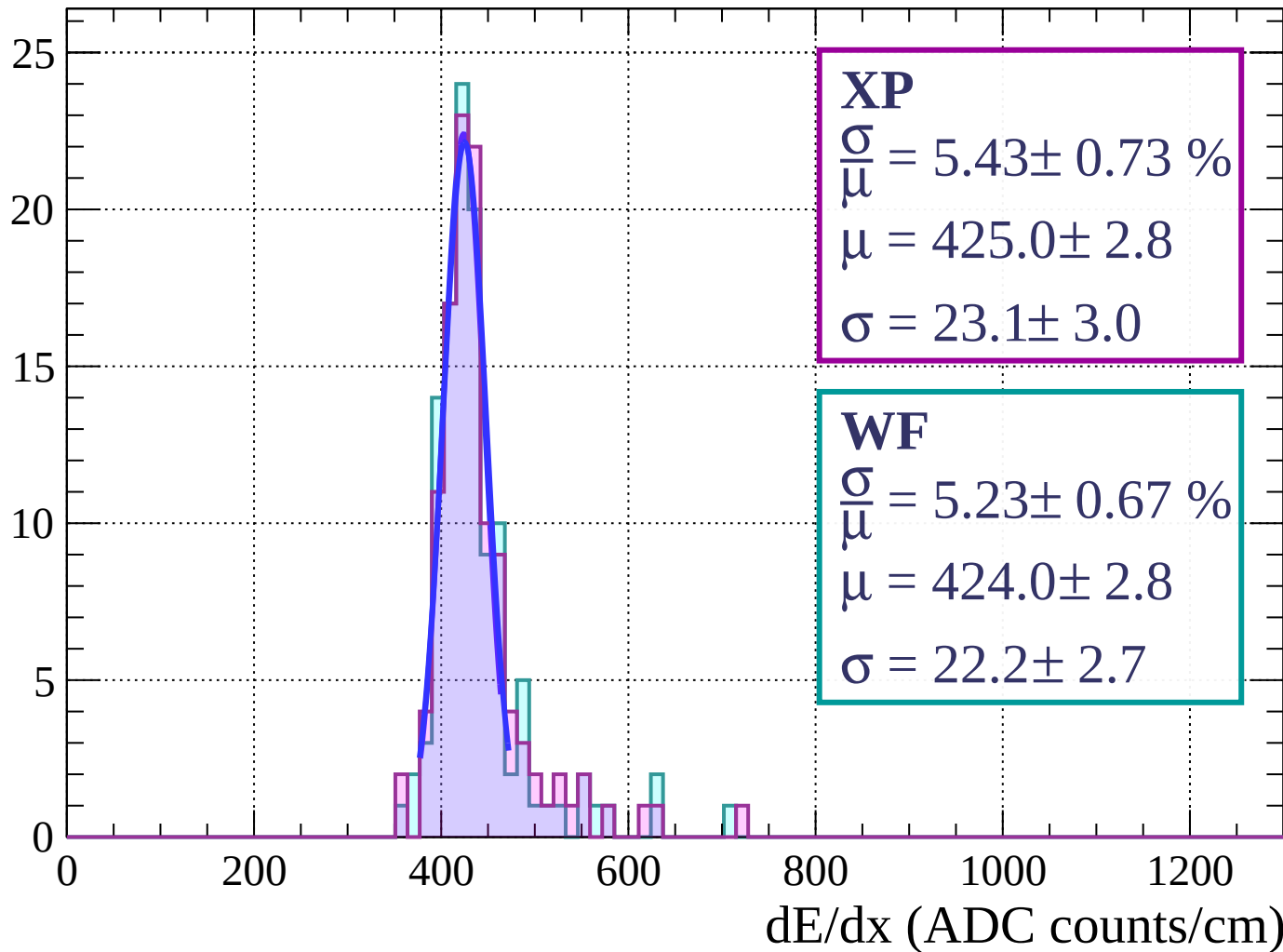
Energy loss | $66 < \phi < 69$

Count



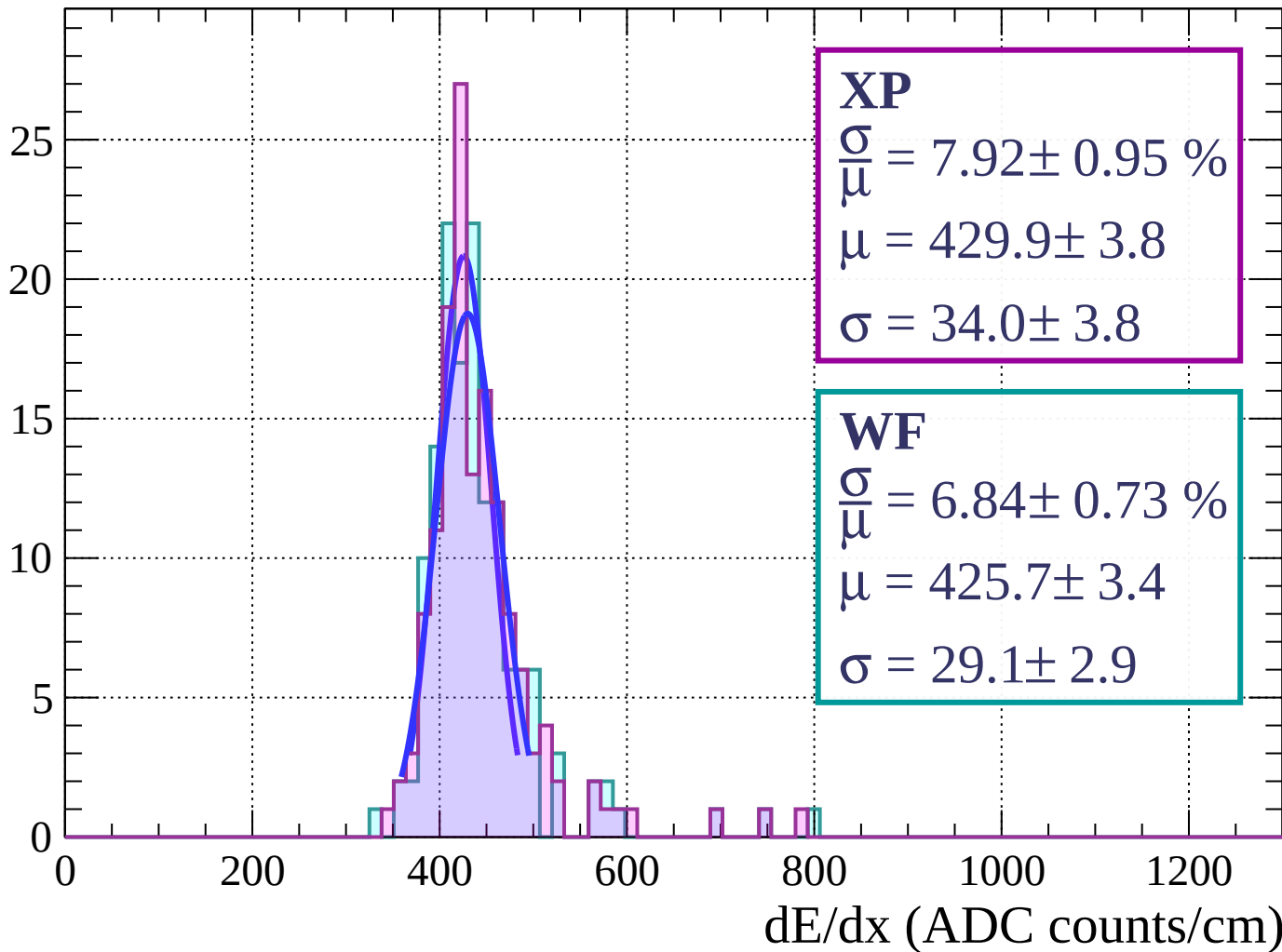
Energy loss | $69 < \phi < 72$

Count



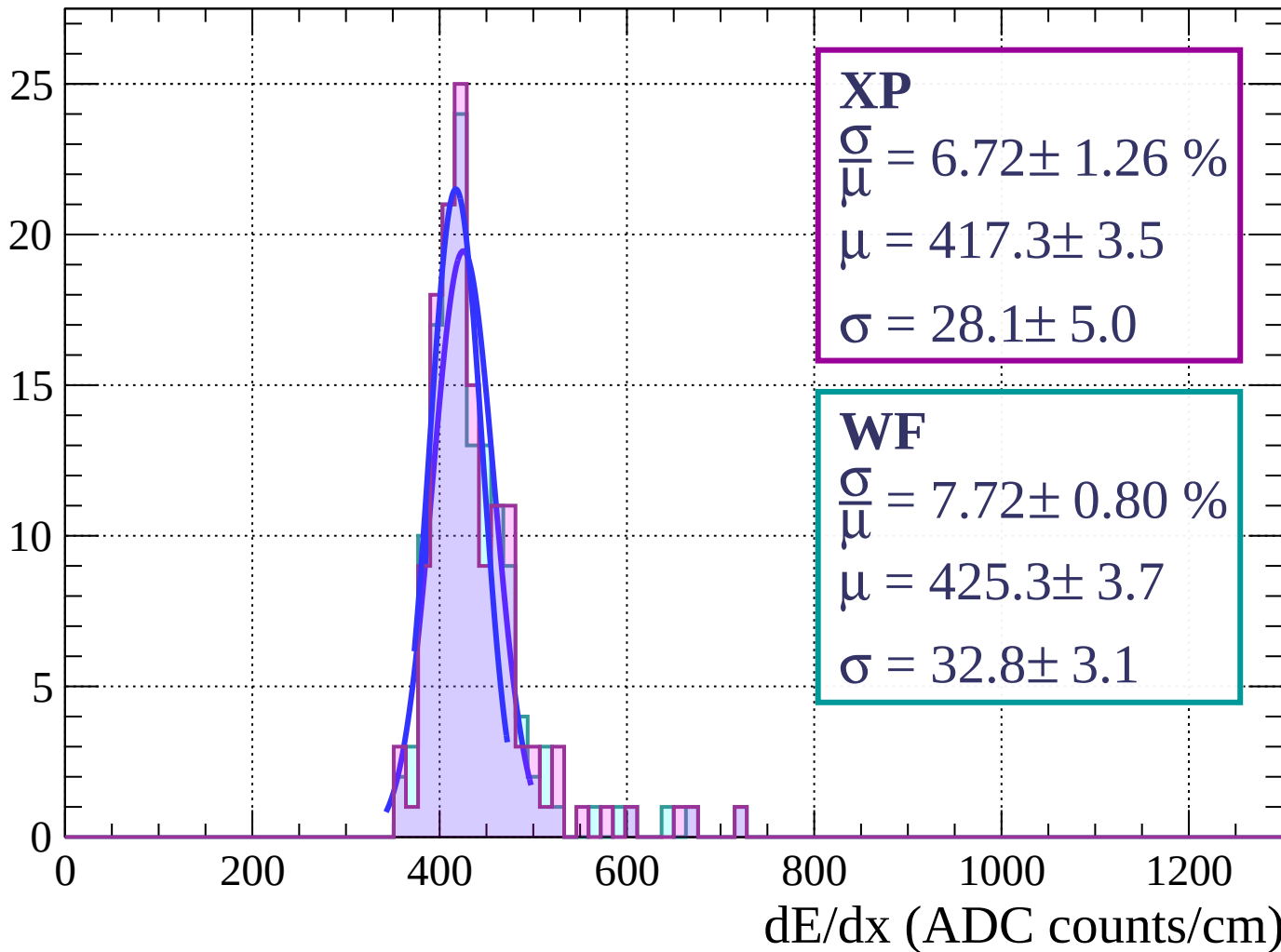
Energy loss | $72 < \phi < 75$

Count



Energy loss | $75 < \phi < 78$

Count



Energy loss | $78 < \phi < 81$

Count

XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.58 \%$$

$$\mu = 436.9 \pm 2.7$$

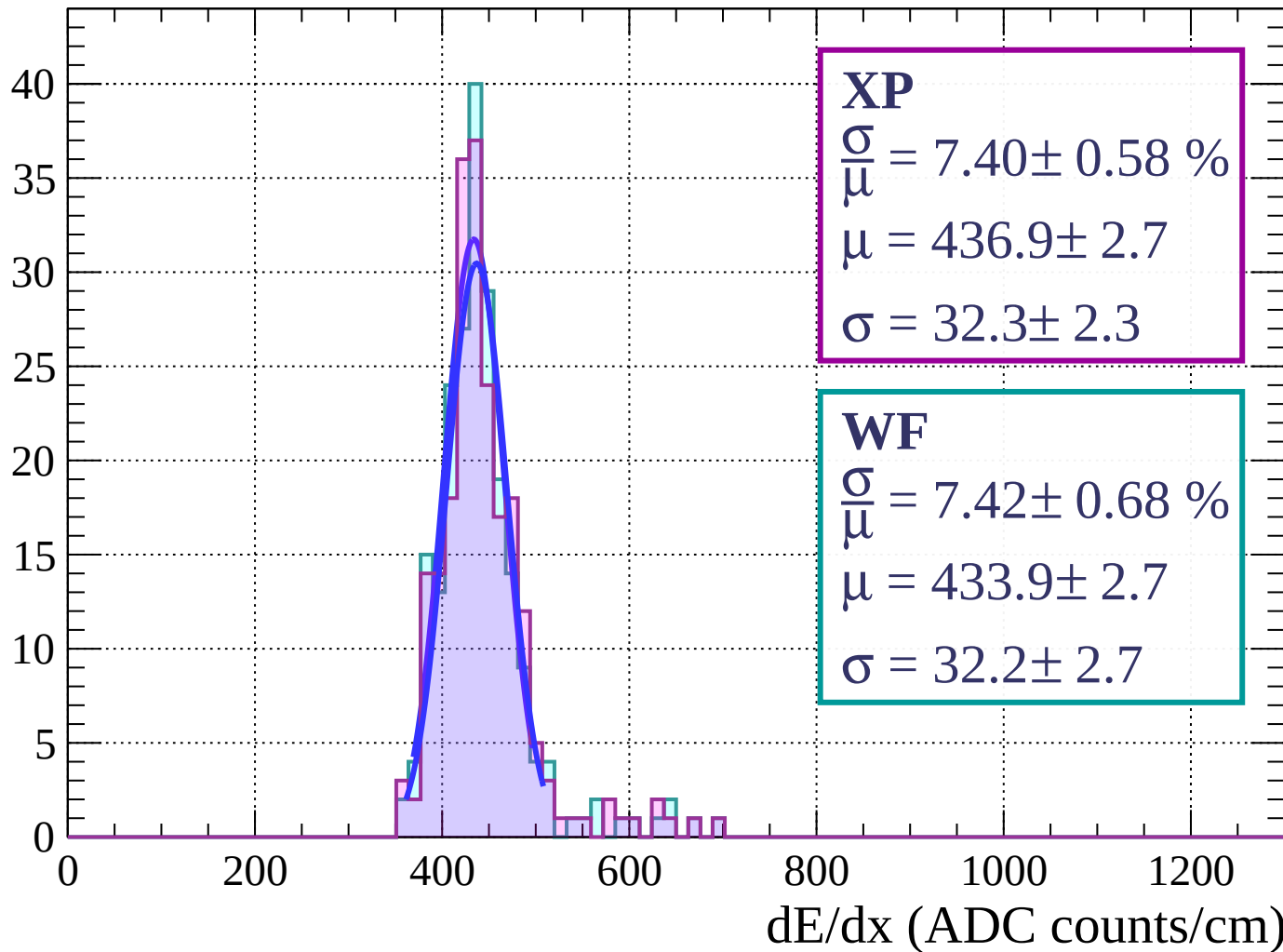
$$\sigma = 32.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.42 \pm 0.68 \%$$

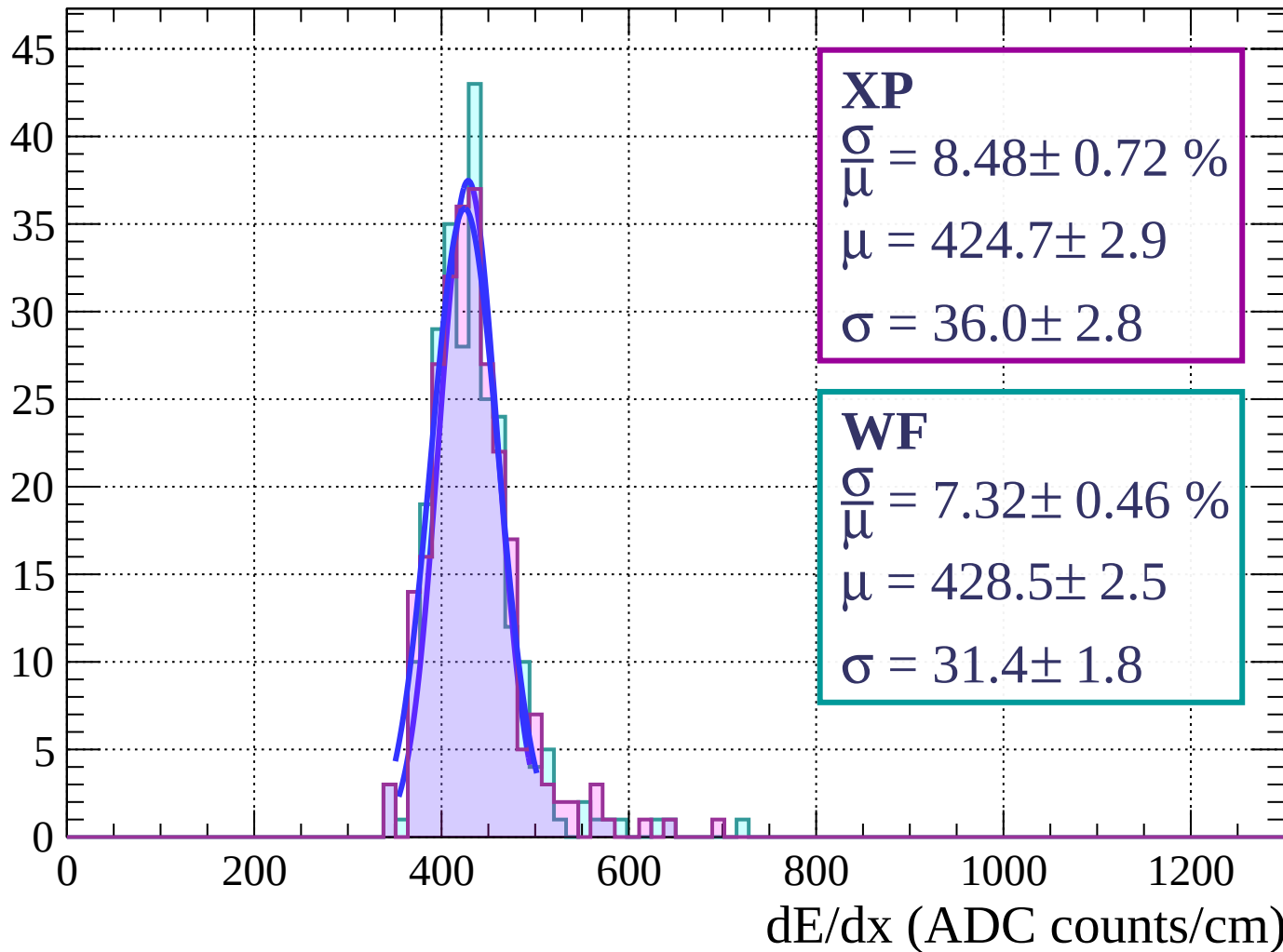
$$\mu = 433.9 \pm 2.7$$

$$\sigma = 32.2 \pm 2.7$$



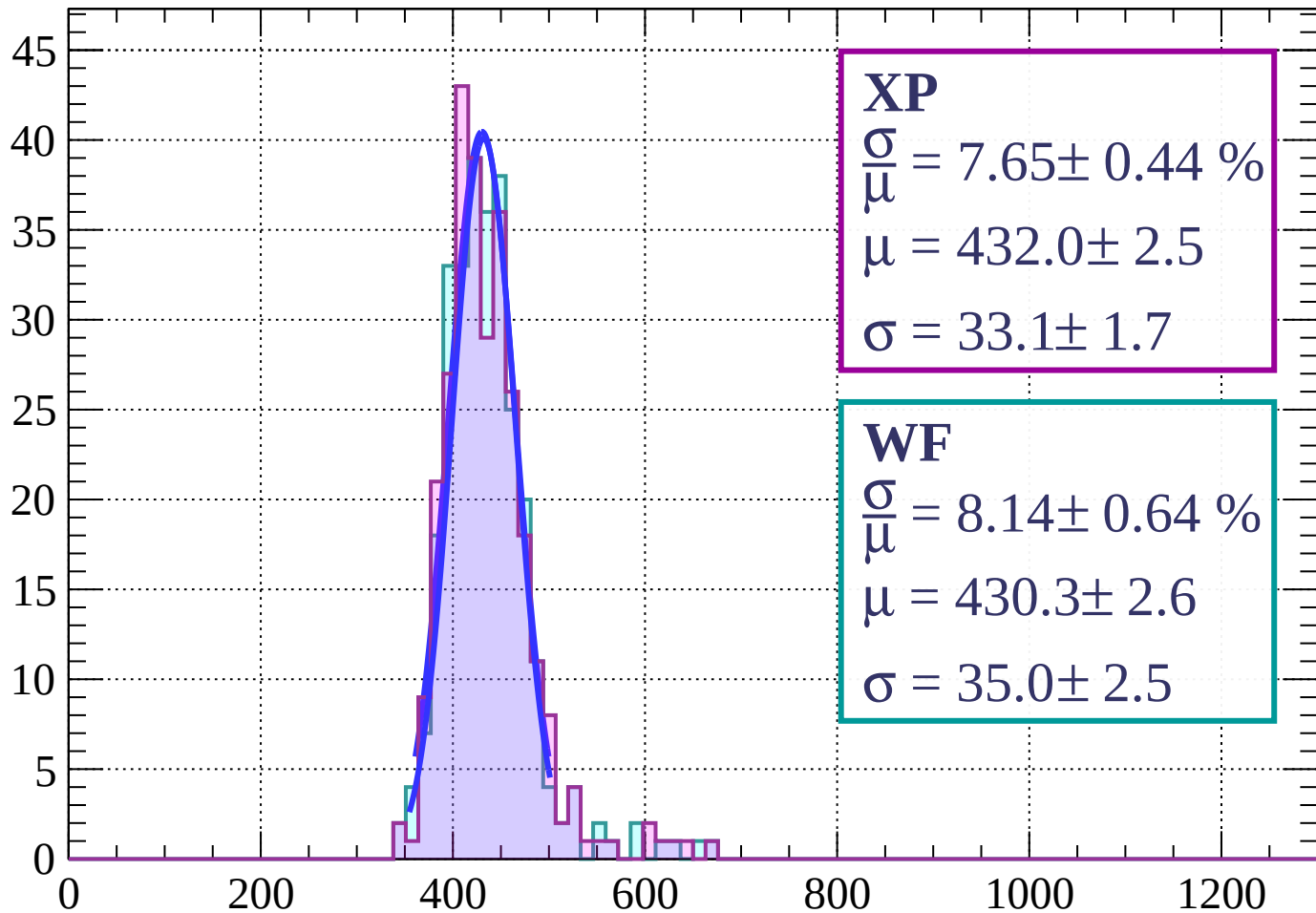
Energy loss | $81 < \phi < 84$

Count



Energy loss | $84 < \phi < 87$

Count



XP

$$\frac{\sigma}{\mu} = 7.65 \pm 0.44 \%$$

$$\mu = 432.0 \pm 2.5$$

$$\sigma = 33.1 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.14 \pm 0.64 \%$$

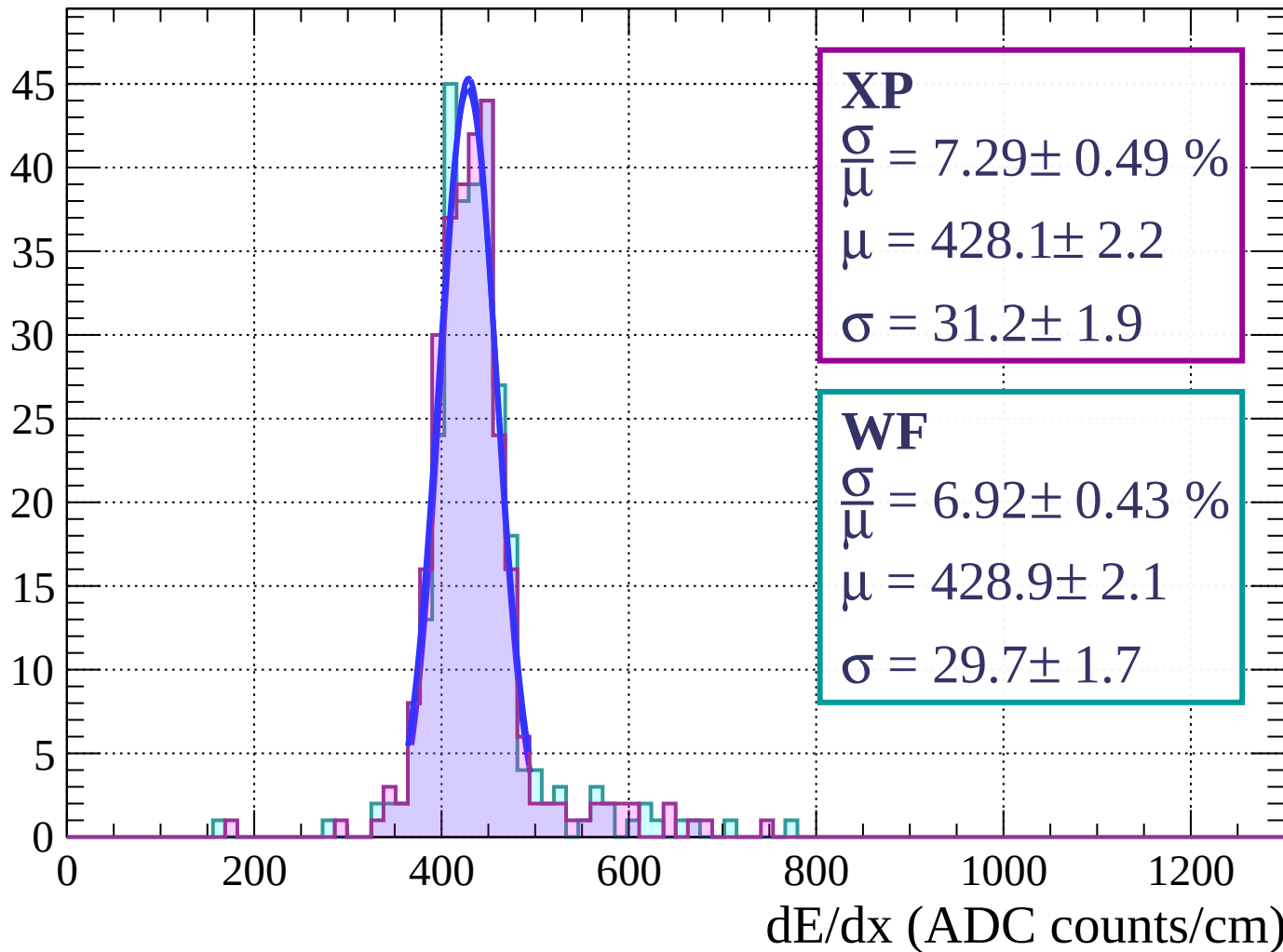
$$\mu = 430.3 \pm 2.6$$

$$\sigma = 35.0 \pm 2.5$$

dE/dx (ADC counts/cm)

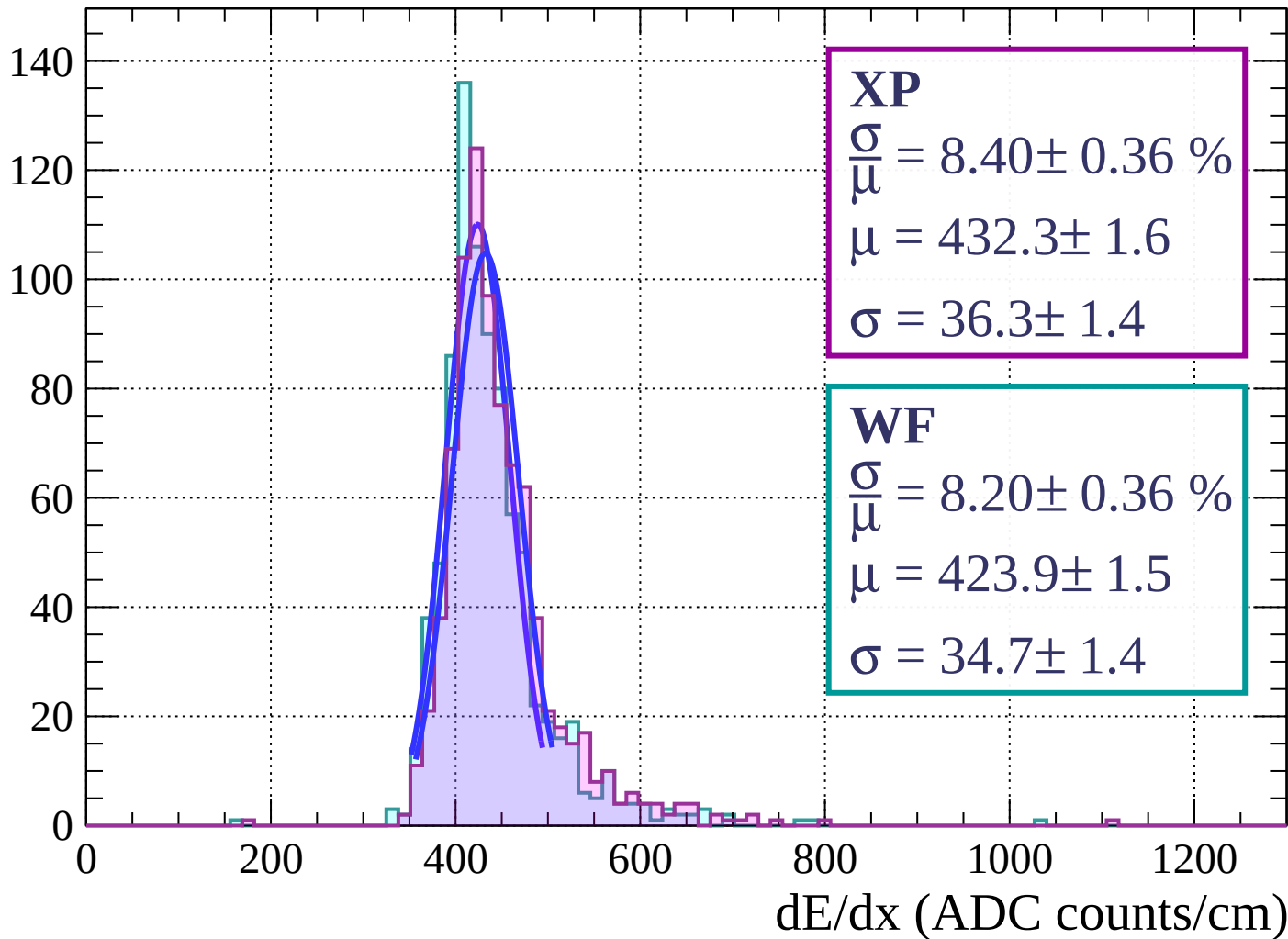
Energy loss | $87 < \phi < 90$

Count



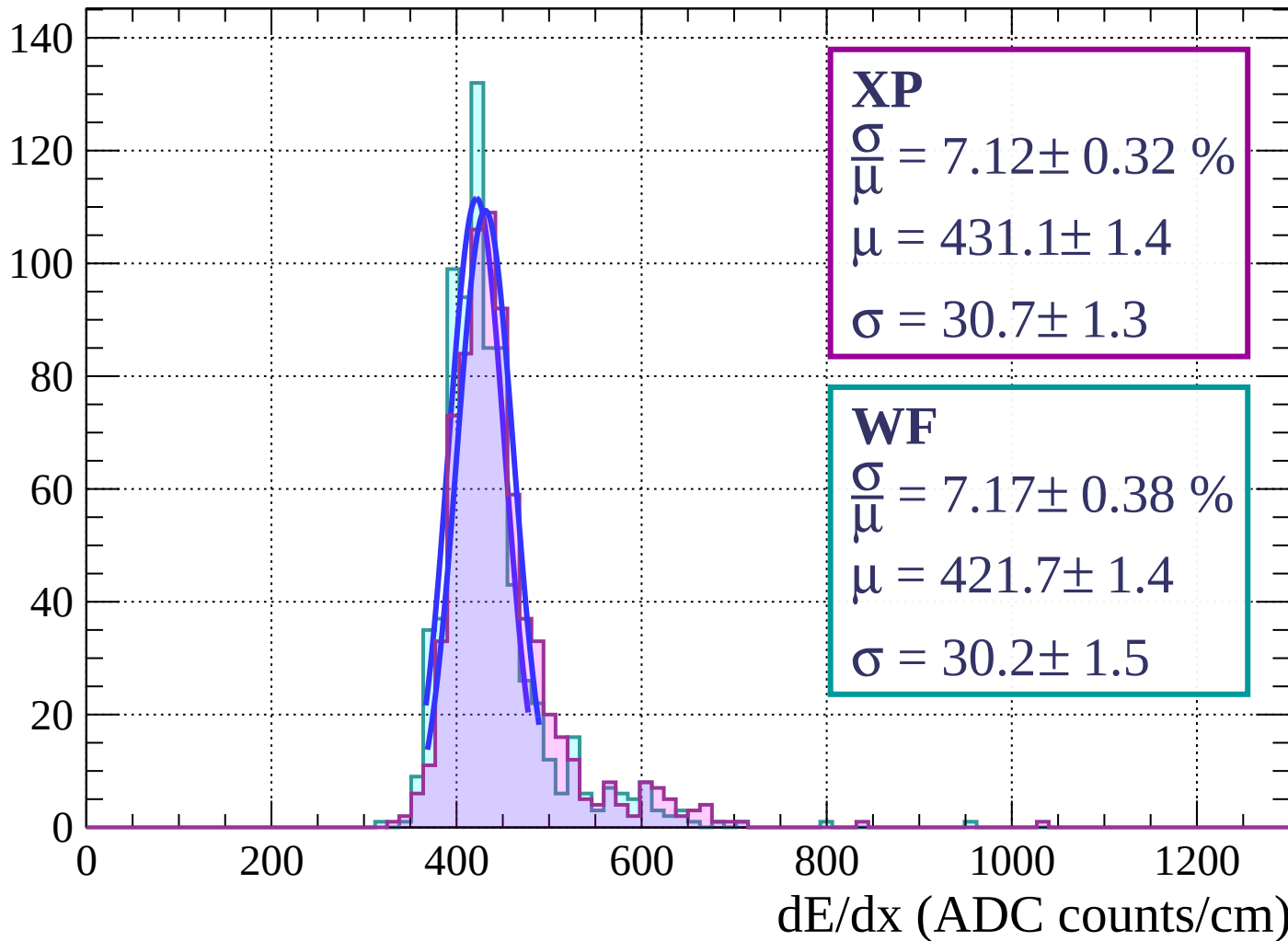
Energy loss | $0 < \theta < 3$

Count



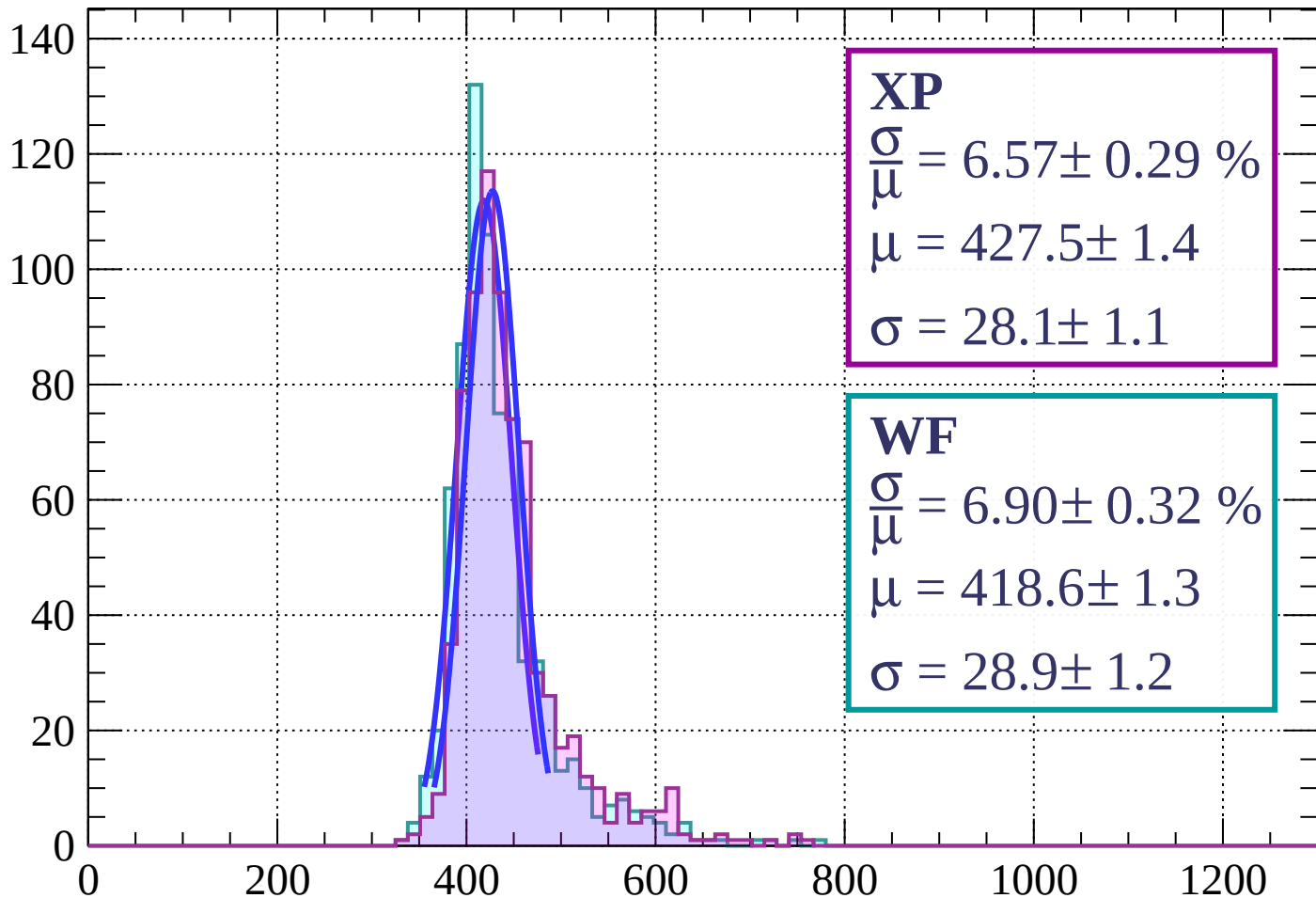
Energy loss | $3 < \theta < 6$

Count



Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 6.57 \pm 0.29 \%$$

$$\mu = 427.5 \pm 1.4$$

$$\sigma = 28.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.90 \pm 0.32 \%$$

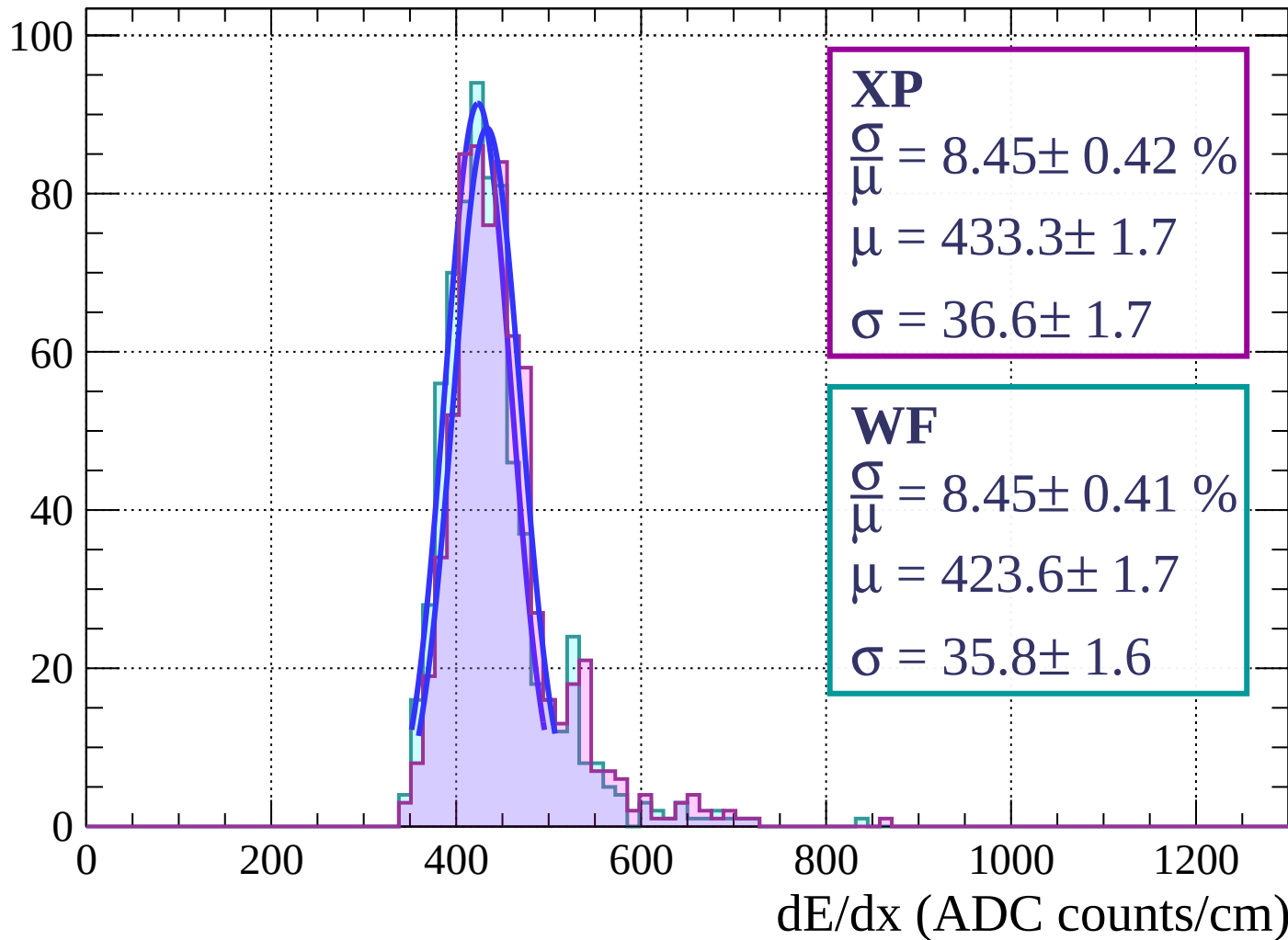
$$\mu = 418.6 \pm 1.3$$

$$\sigma = 28.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $9 < \theta < 12$

Count



Energy loss | $12 < \theta < 15$

Count

100

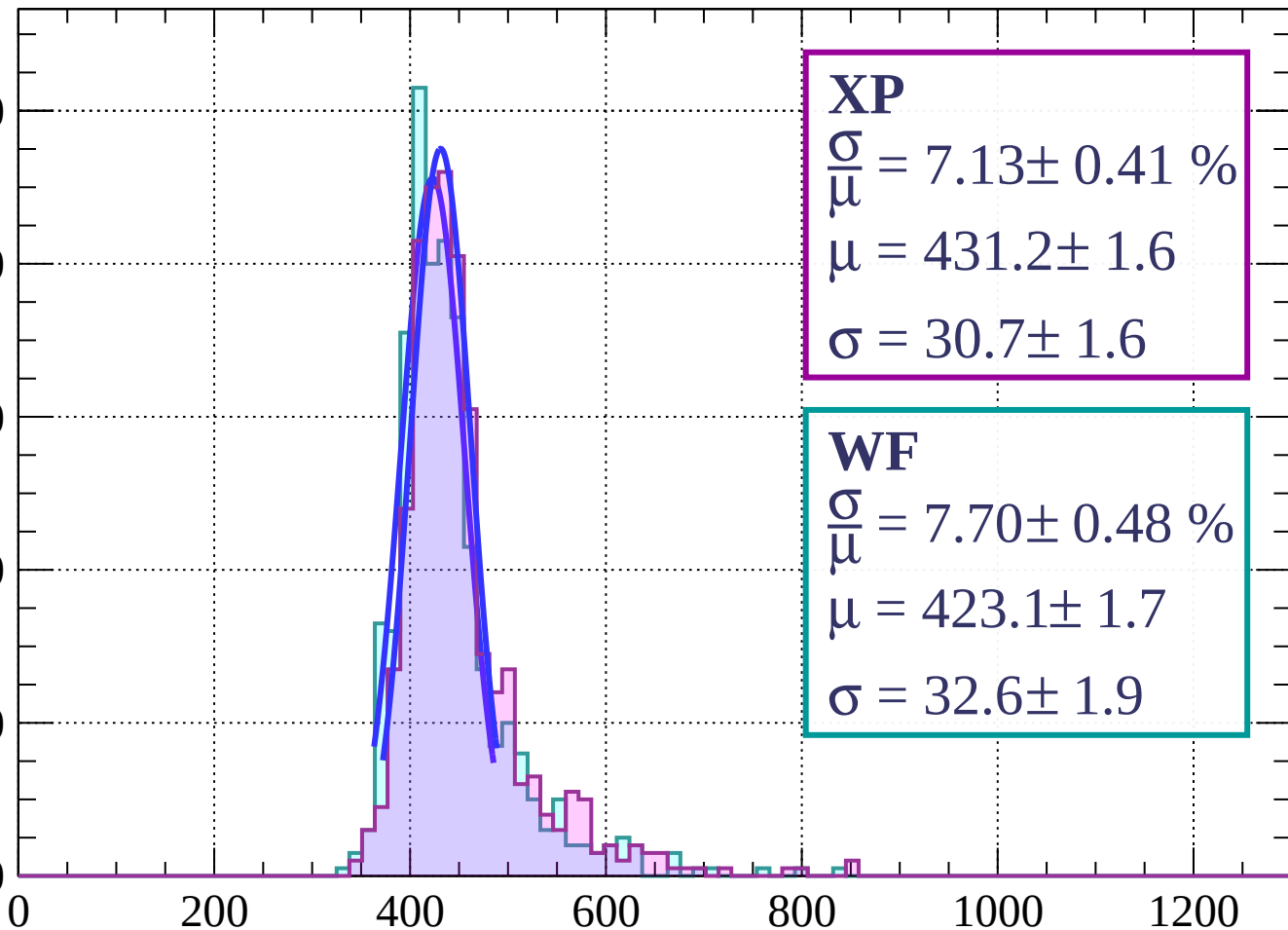
80

60

40

20

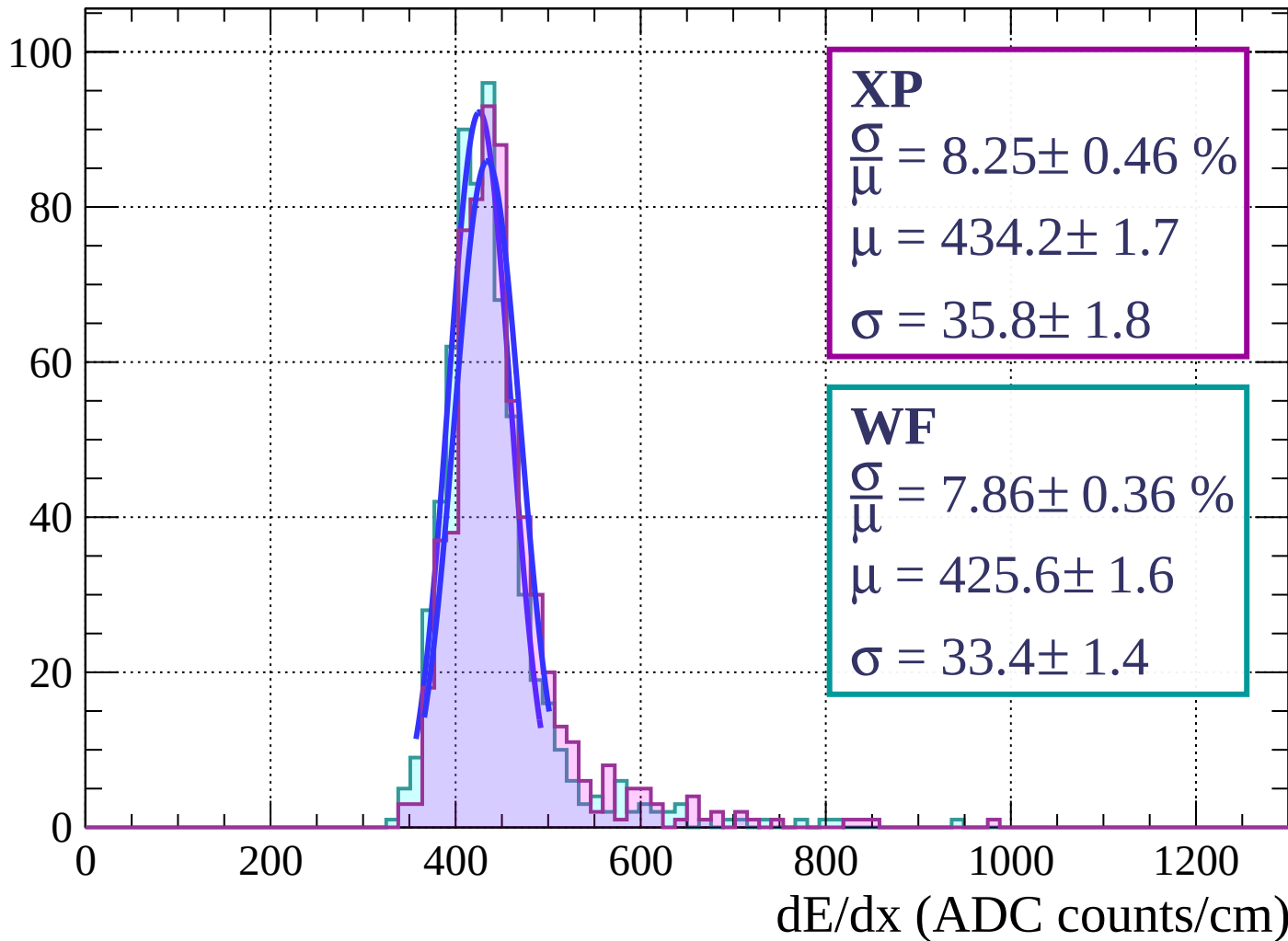
0



dE/dx (ADC counts/cm)

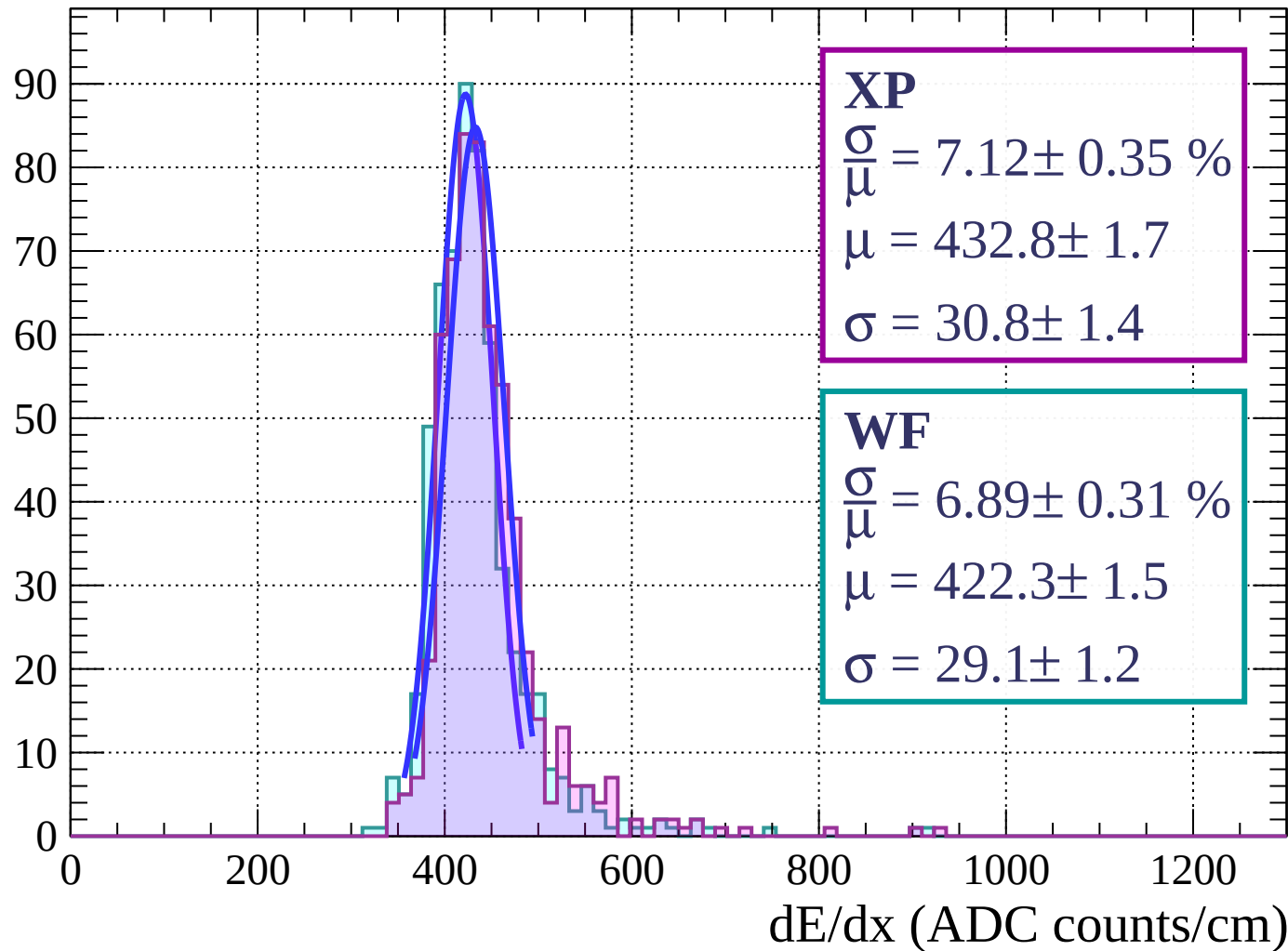
Energy loss | $15 < \theta < 18$

Count



Energy loss | $18 < \theta < 21$

Count



Energy loss | $21 < \theta < 24$

Count

XP

$$\frac{\sigma}{\mu} = 8.15 \pm 0.46 \%$$

$$\mu = 432.8 \pm 1.9$$

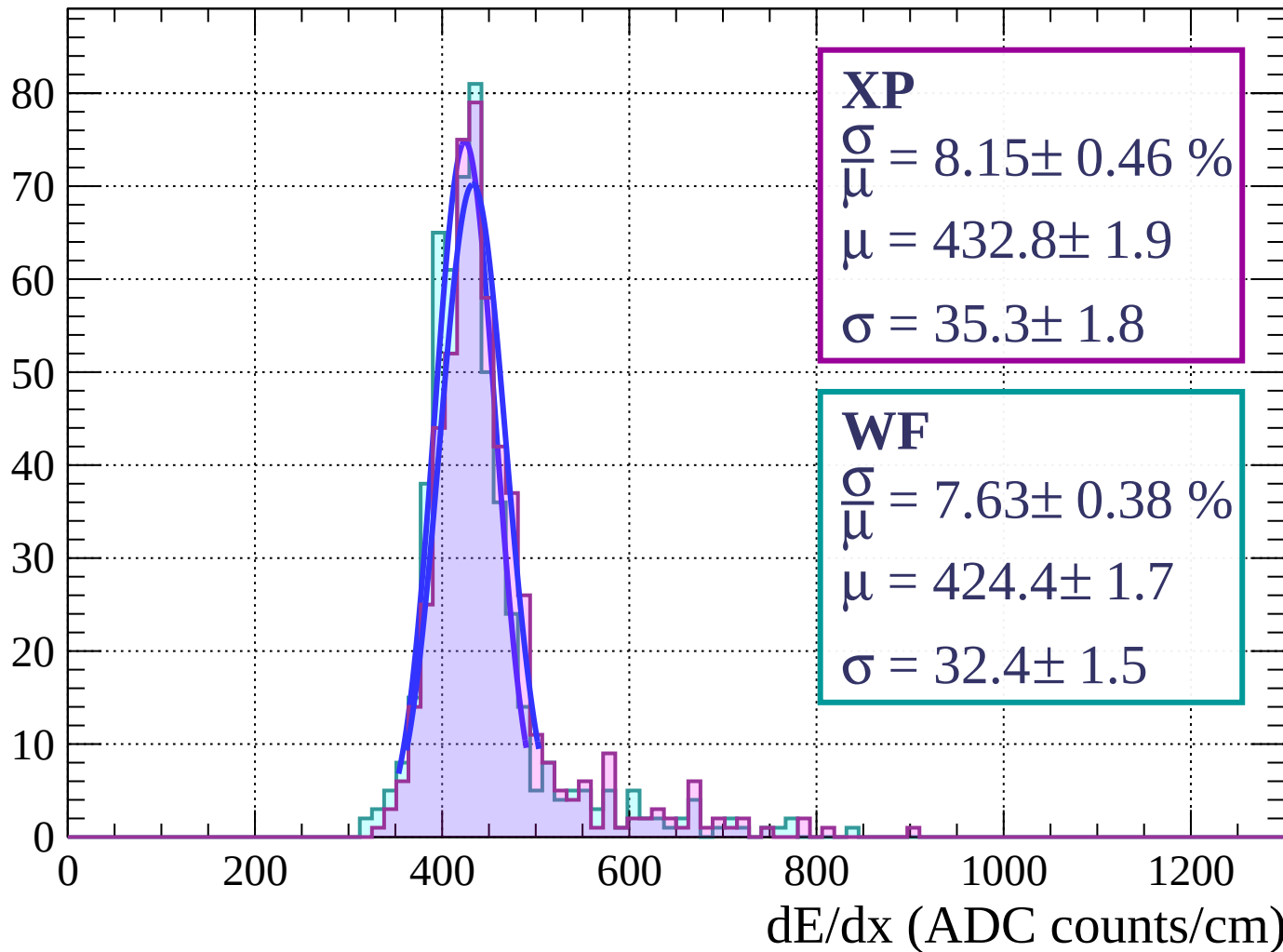
$$\sigma = 35.3 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.63 \pm 0.38 \%$$

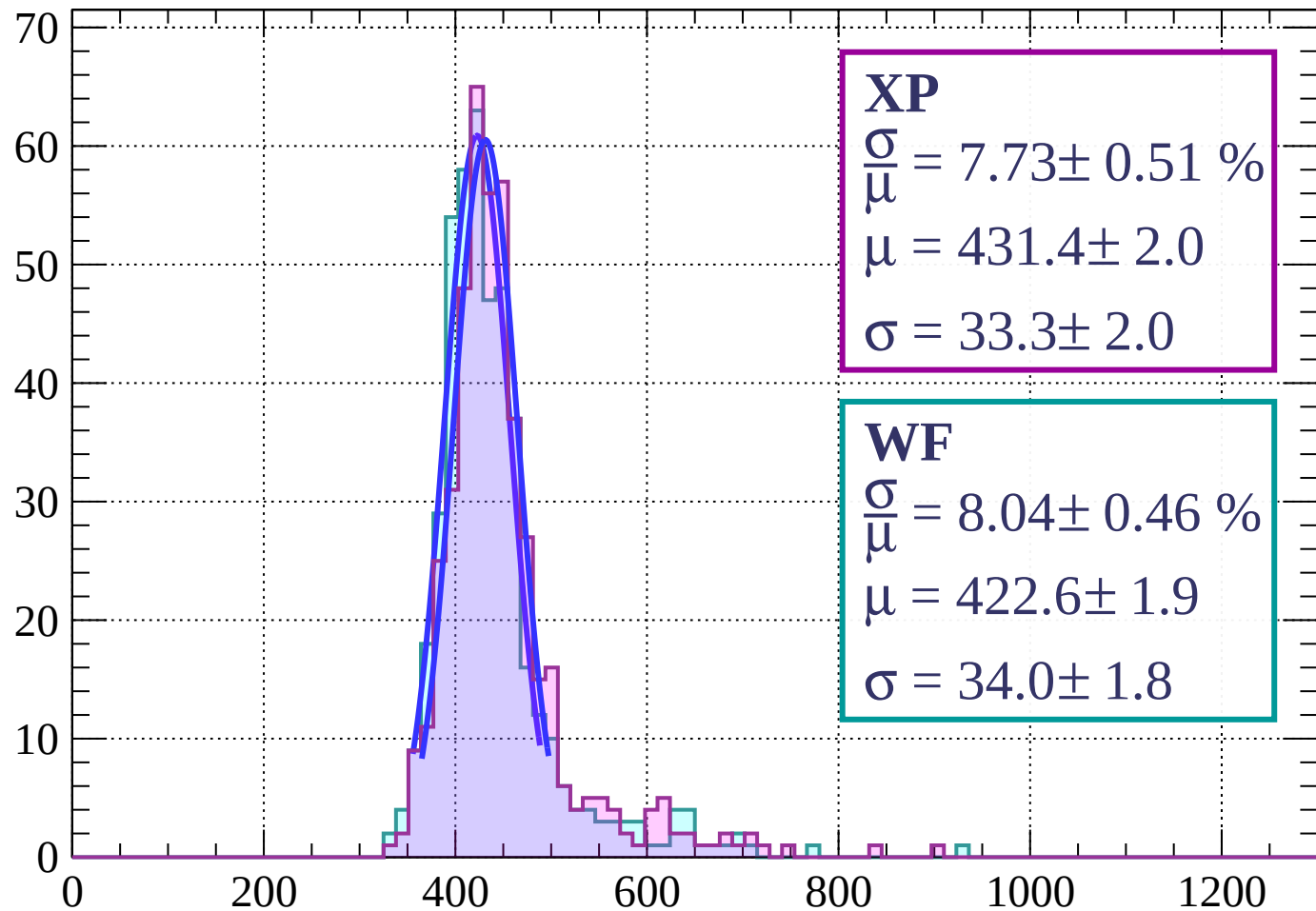
$$\mu = 424.4 \pm 1.7$$

$$\sigma = 32.4 \pm 1.5$$



Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 7.73 \pm 0.51 \%$$

$$\mu = 431.4 \pm 2.0$$

$$\sigma = 33.3 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.04 \pm 0.46 \%$$

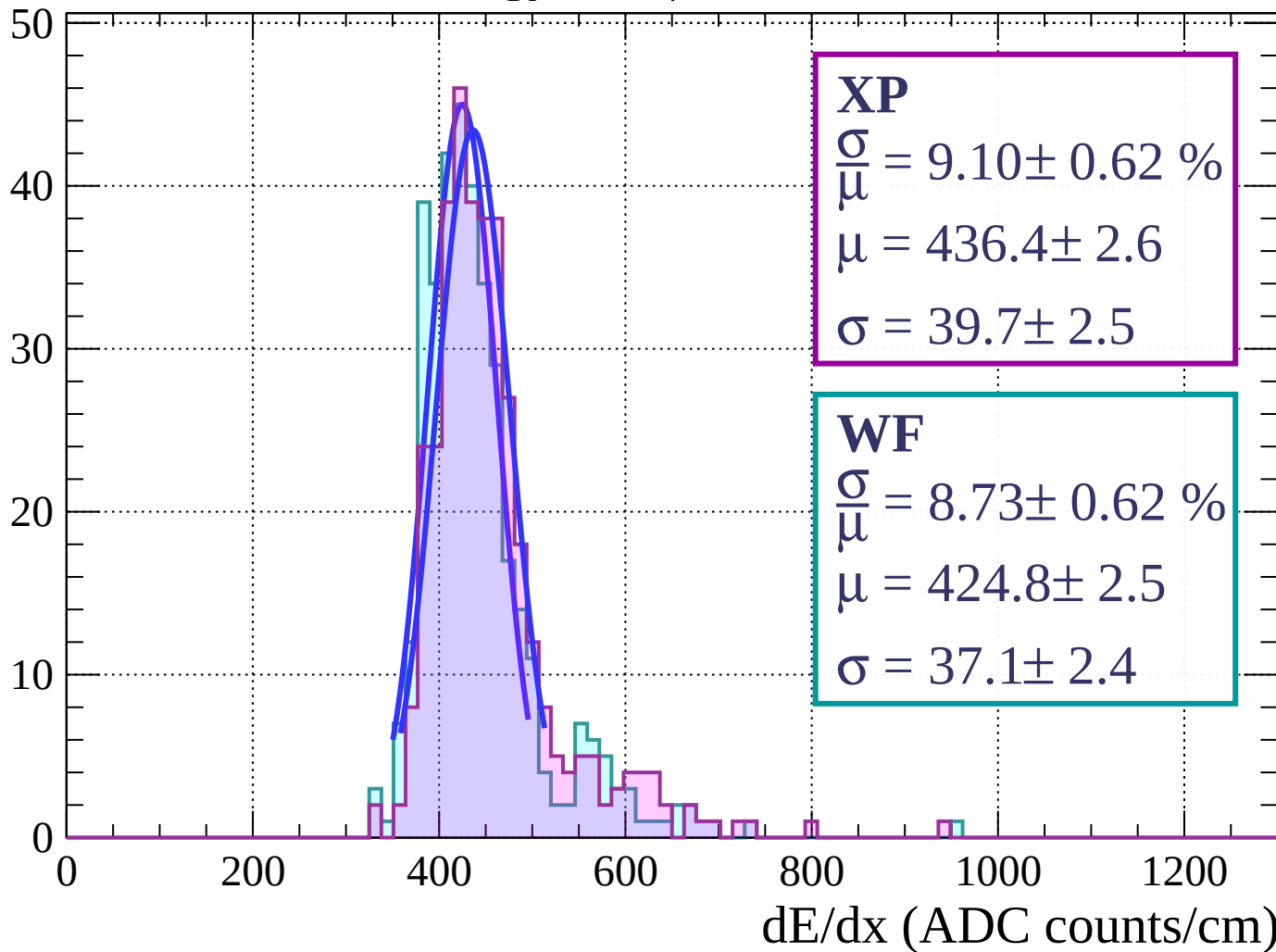
$$\mu = 422.6 \pm 1.9$$

$$\sigma = 34.0 \pm 1.8$$

dE/dx (ADC counts/cm)

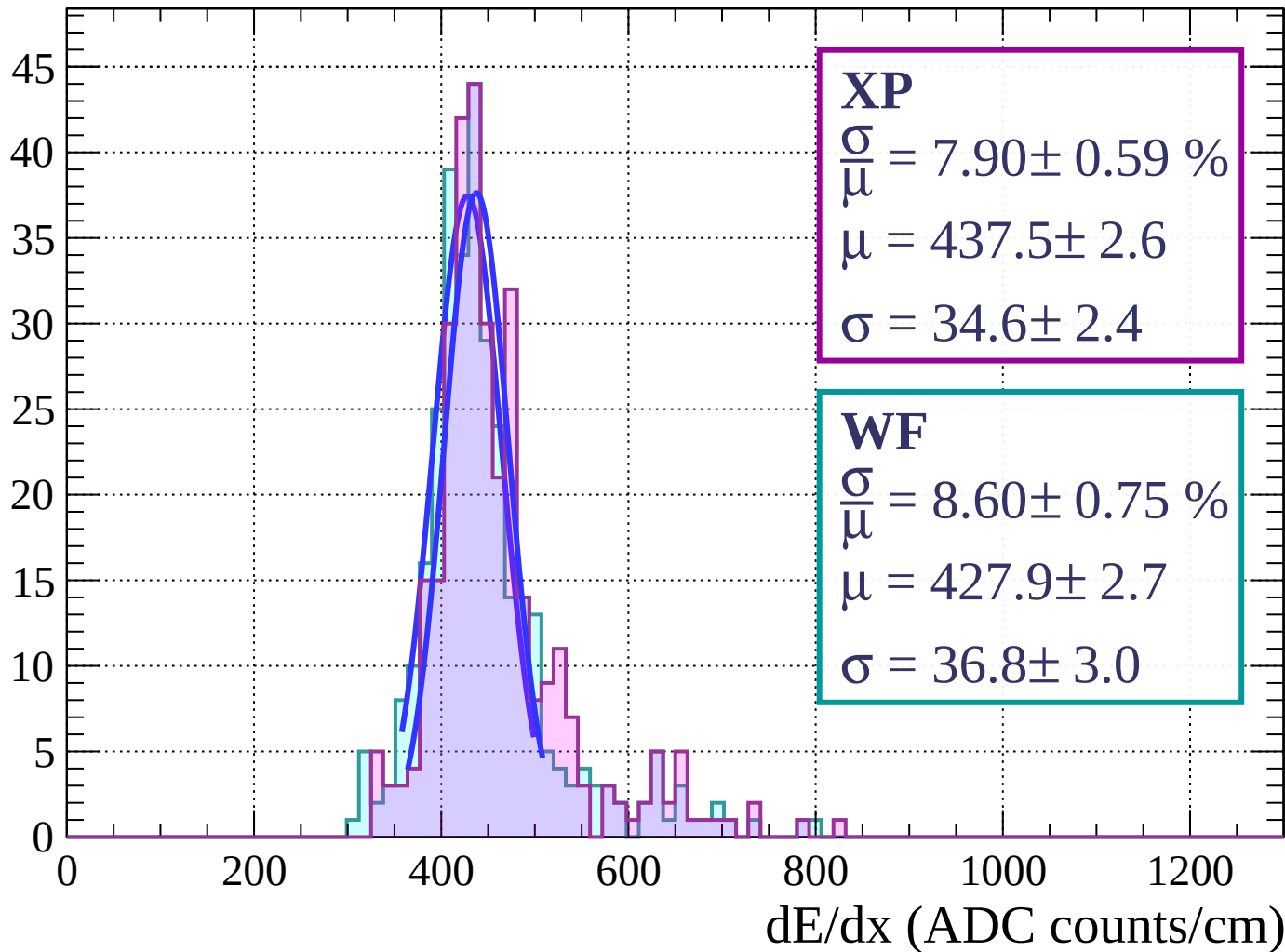
Energy loss | $27 < \theta < 30$

Count



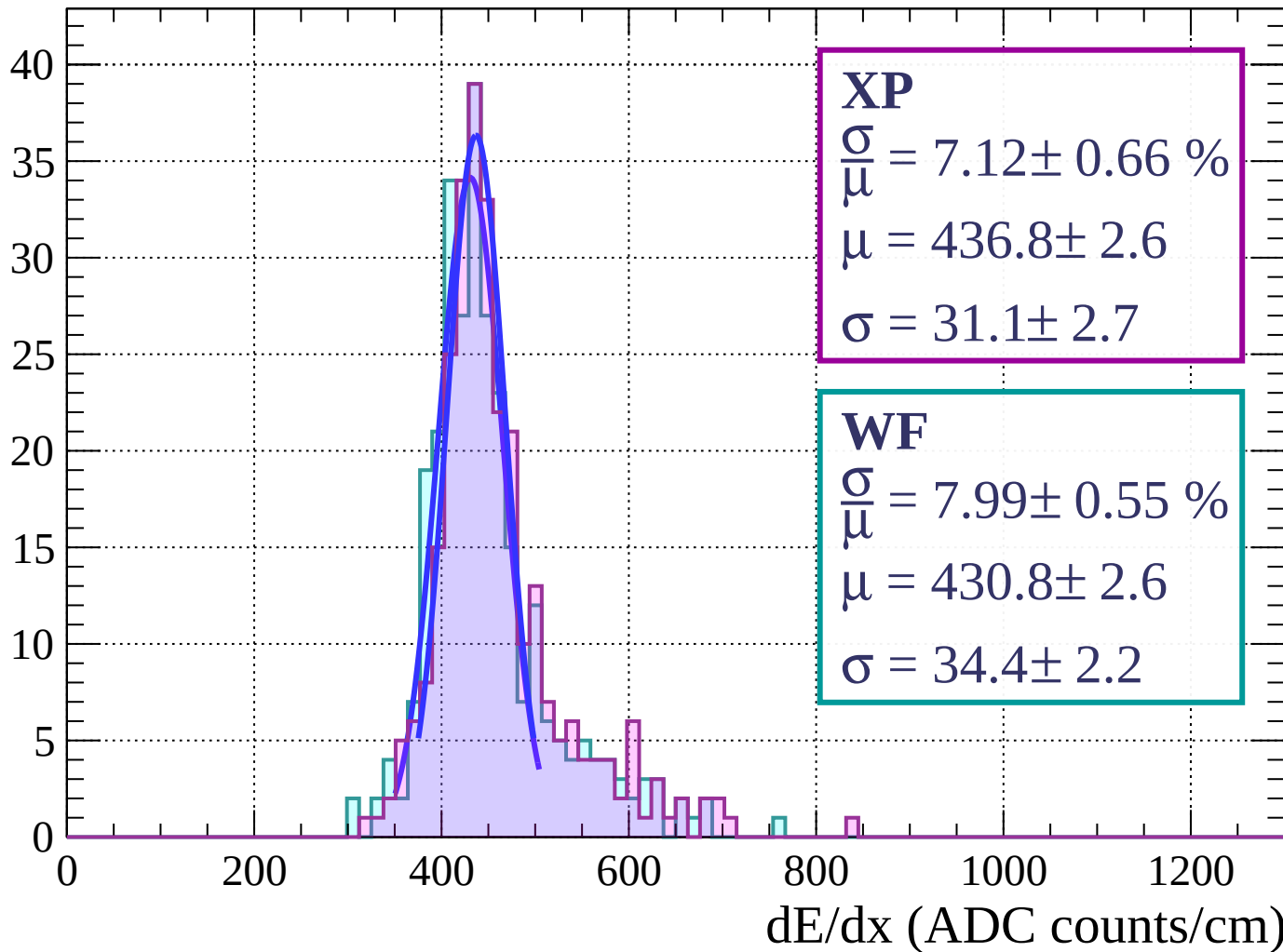
Energy loss | $30 < \theta < 33$

Count



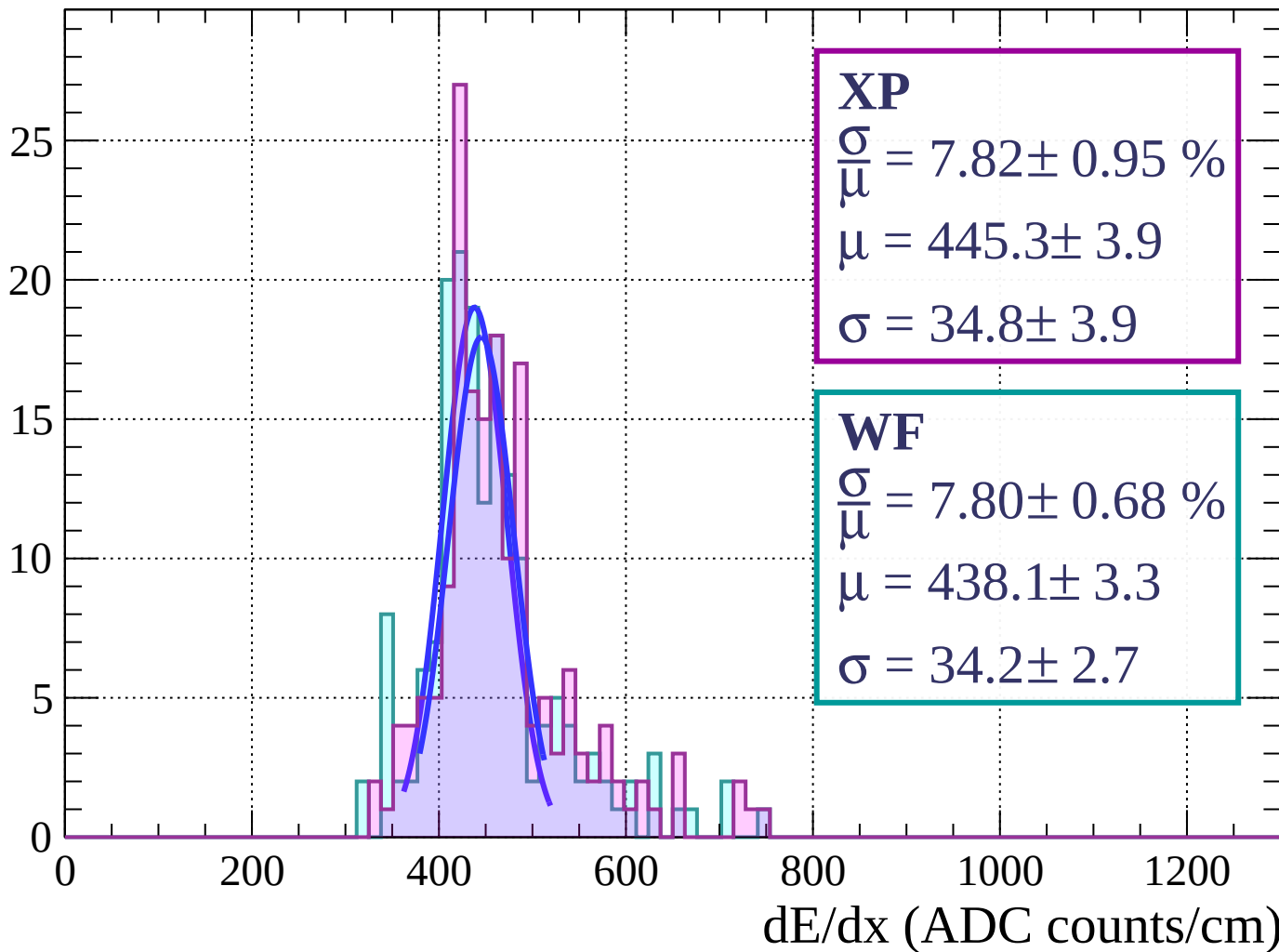
Energy loss | $33 < \theta < 36$

Count



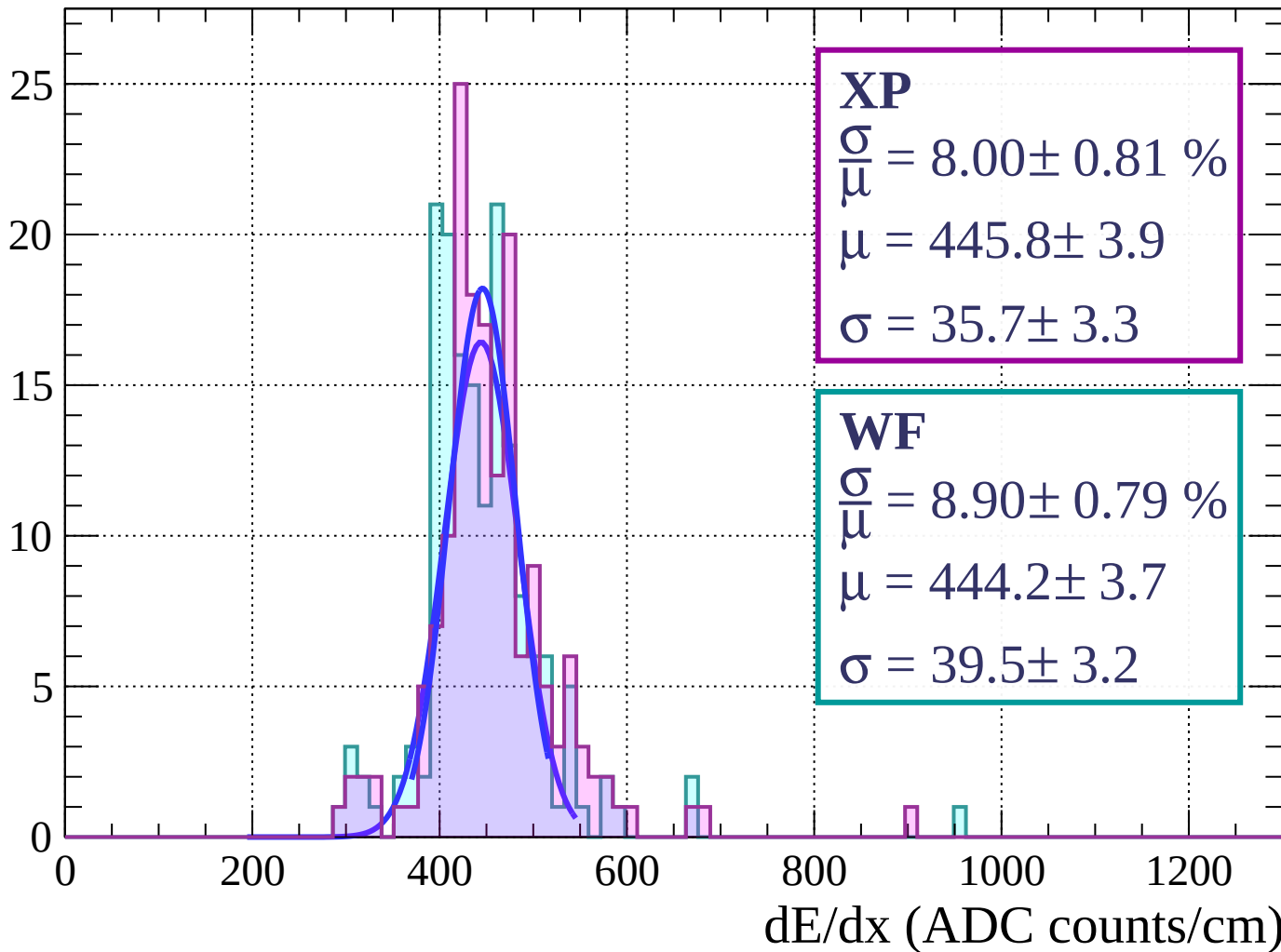
Energy loss | $36 < \theta < 39$

Count



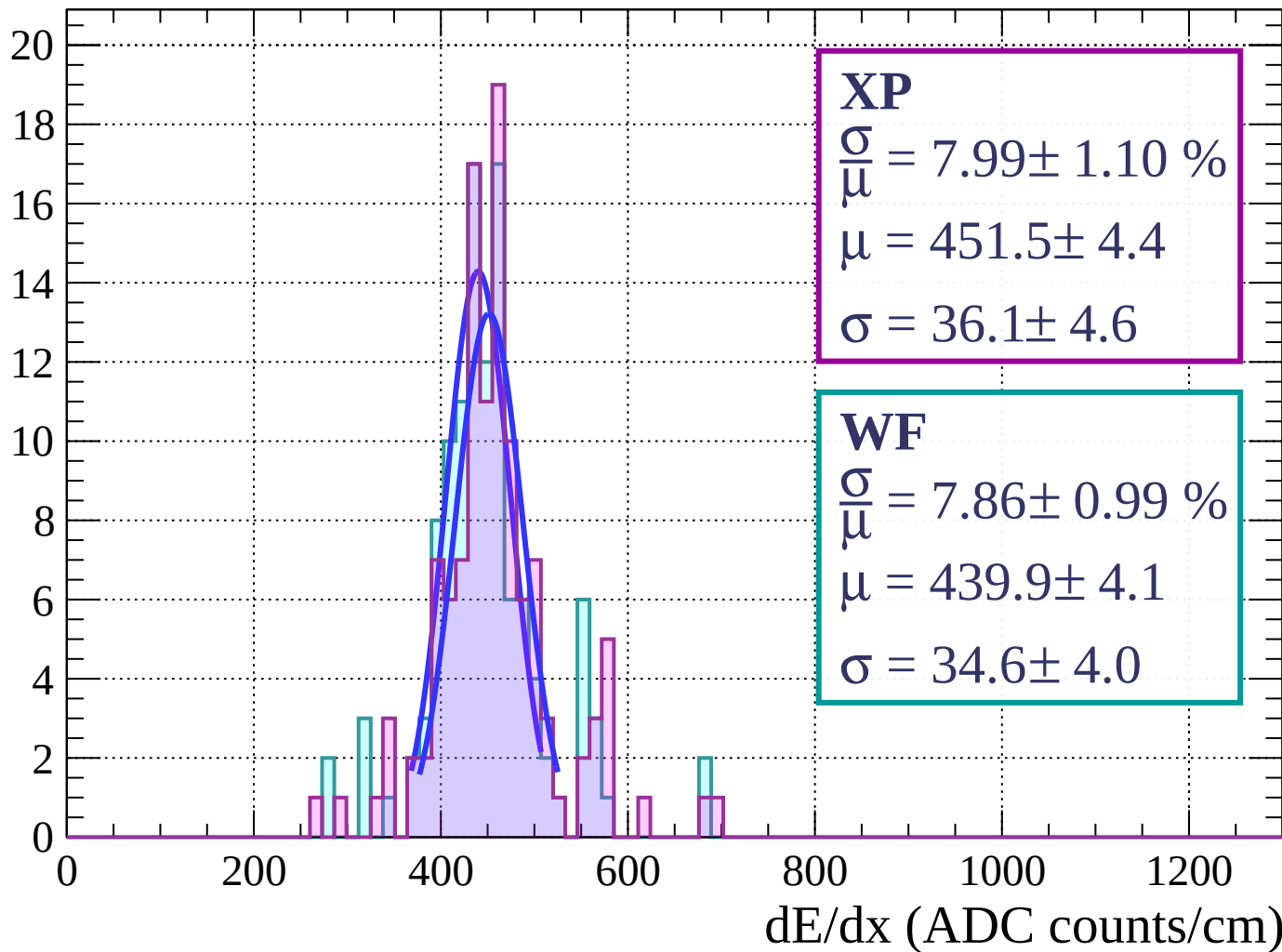
Energy loss | $39 < \theta < 42$

Count



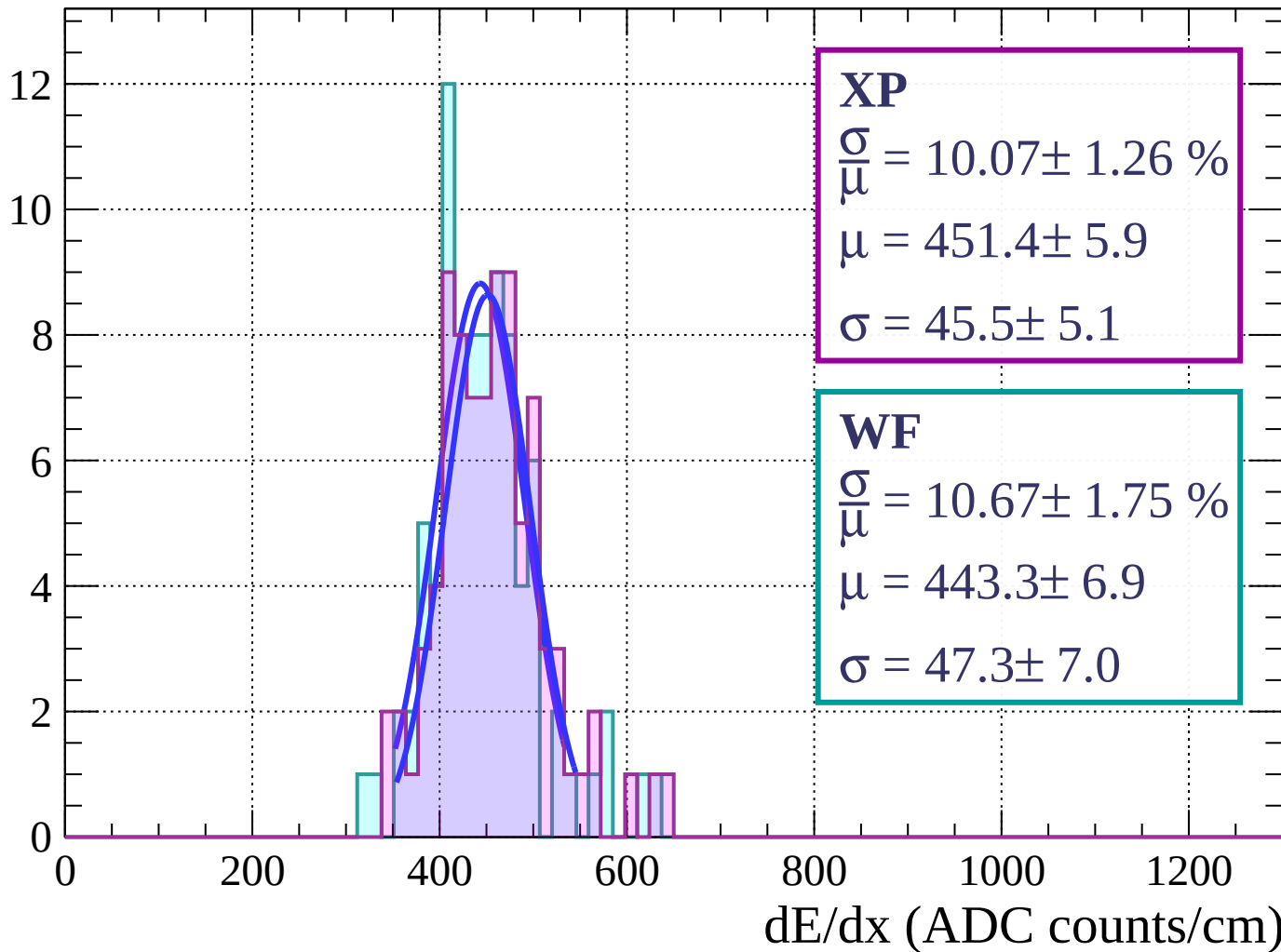
Energy loss | $42 < \theta < 45$

Count



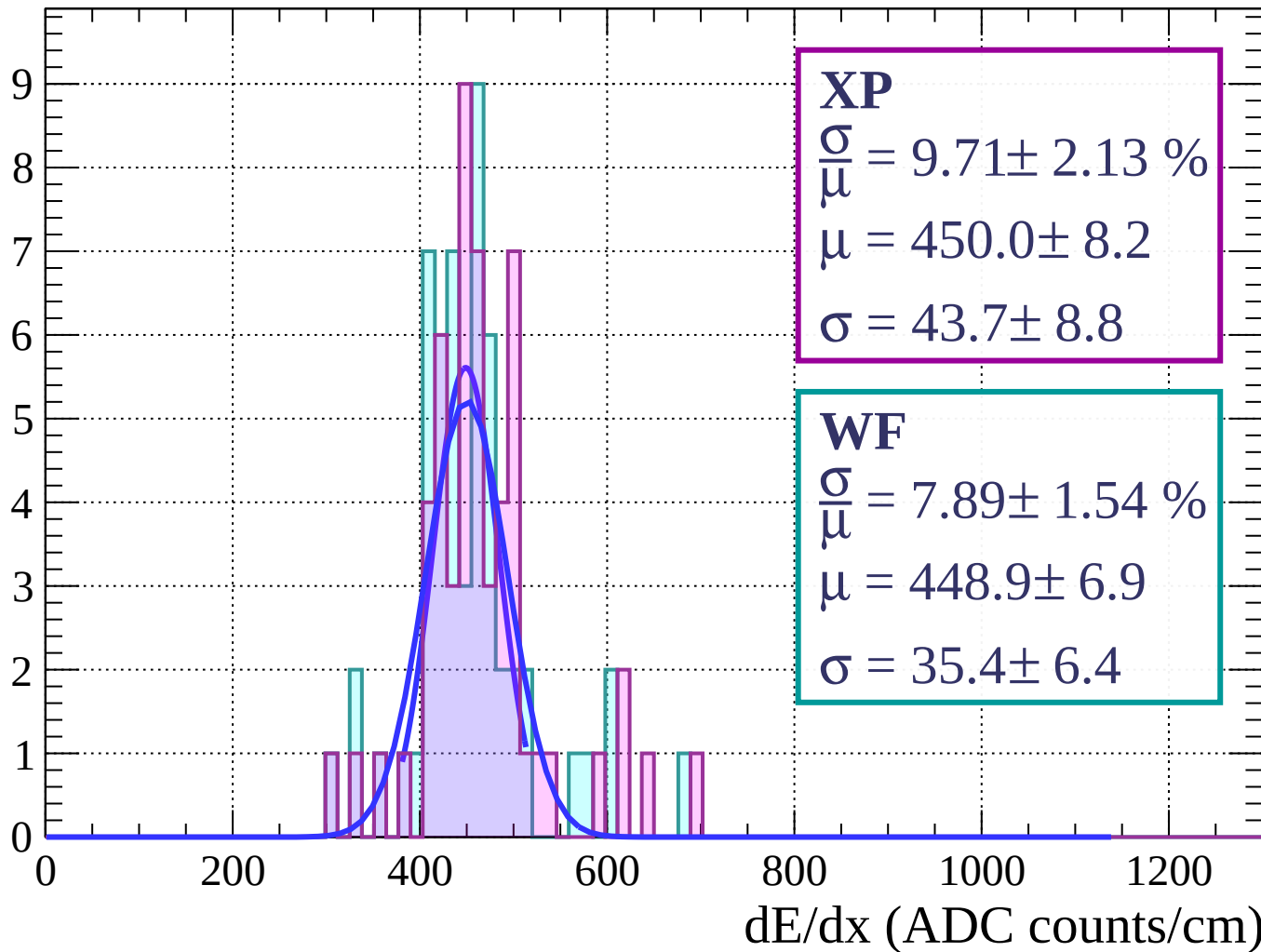
Energy loss | $45 < \theta < 48$

Count



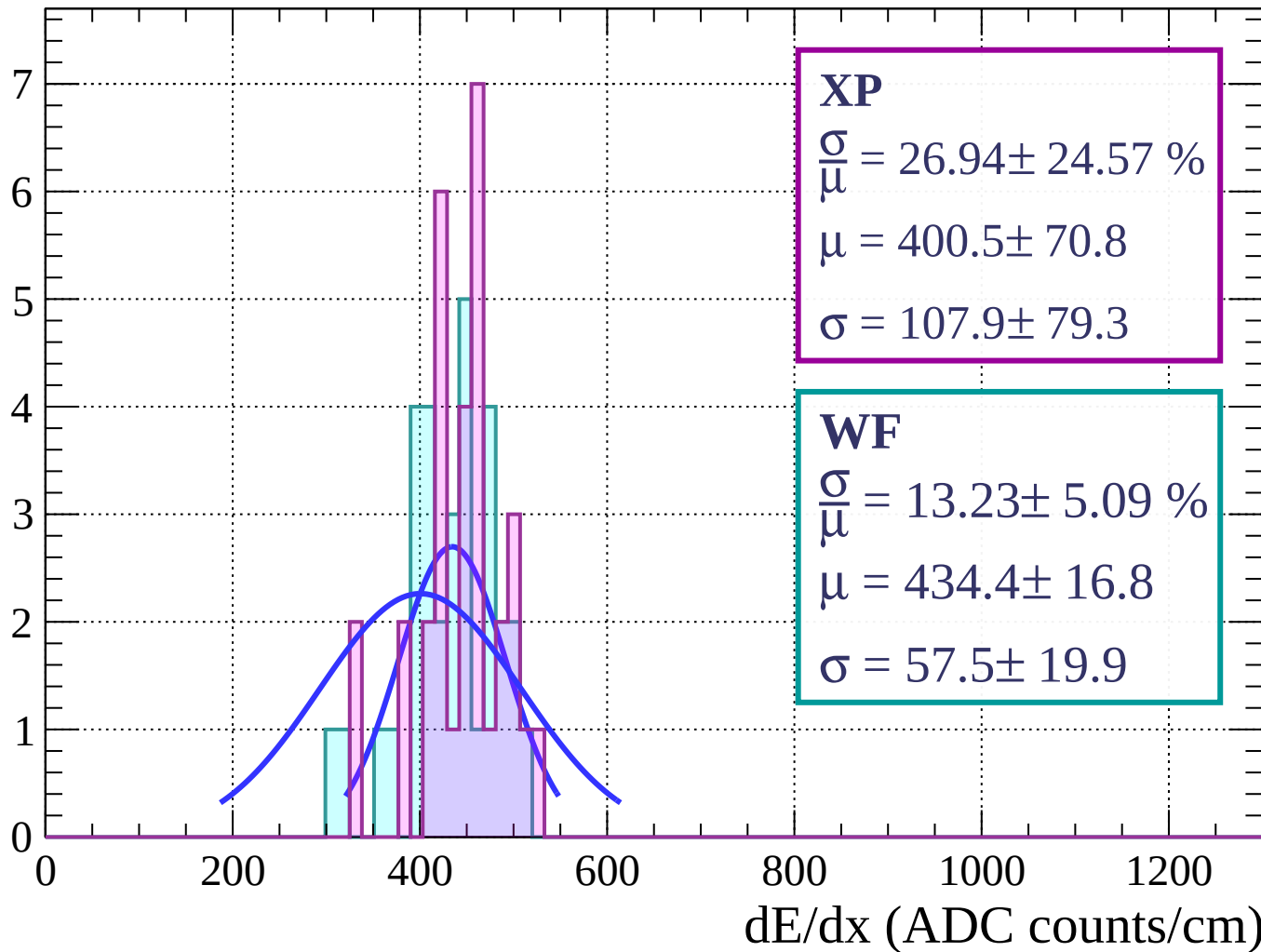
Energy loss | $48 < \theta < 51$

Count



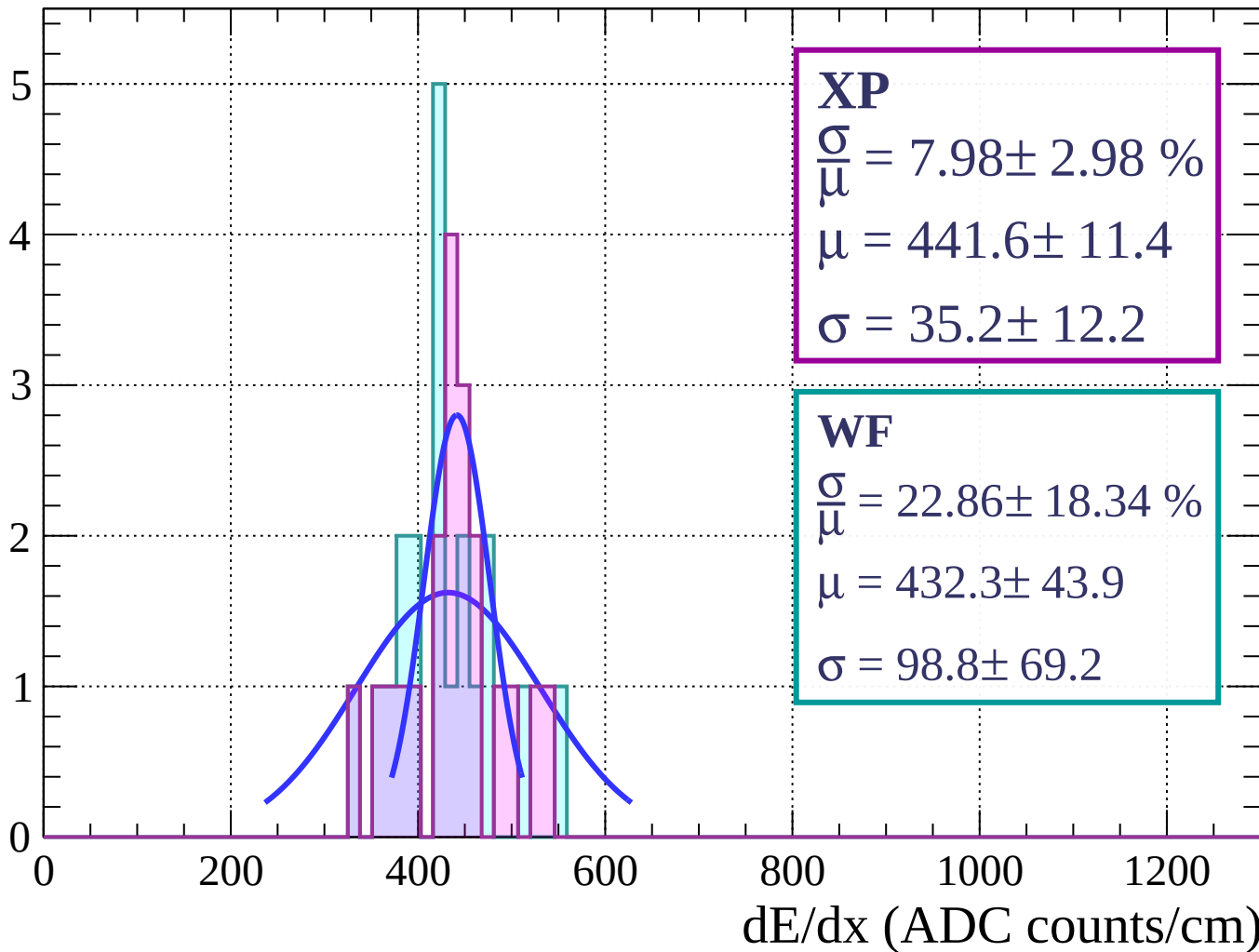
Energy loss | $51 < \theta < 54$

Count



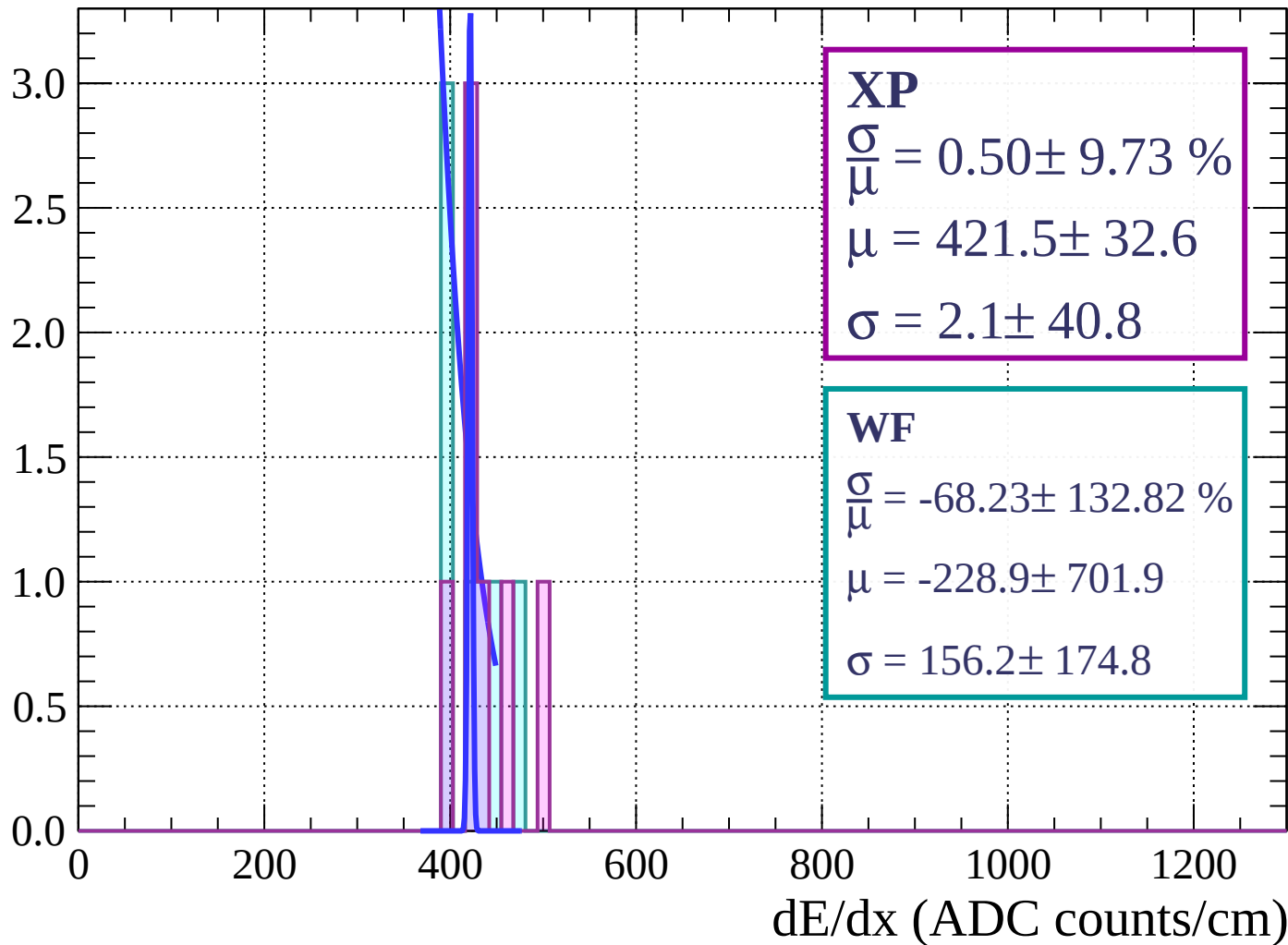
Energy loss | $54 < \theta < 57$

Count



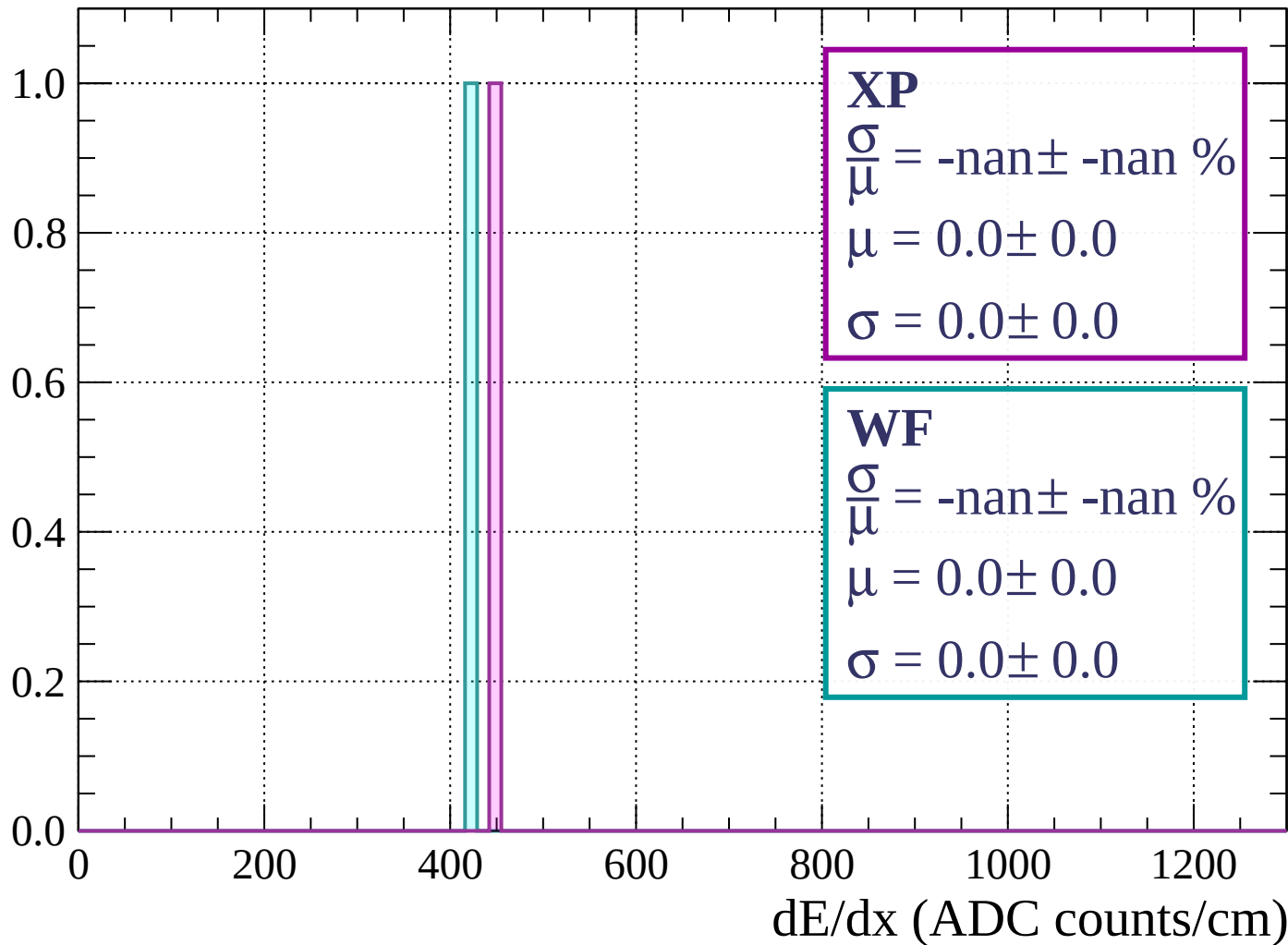
Energy loss | $57 < \theta < 60$

Count



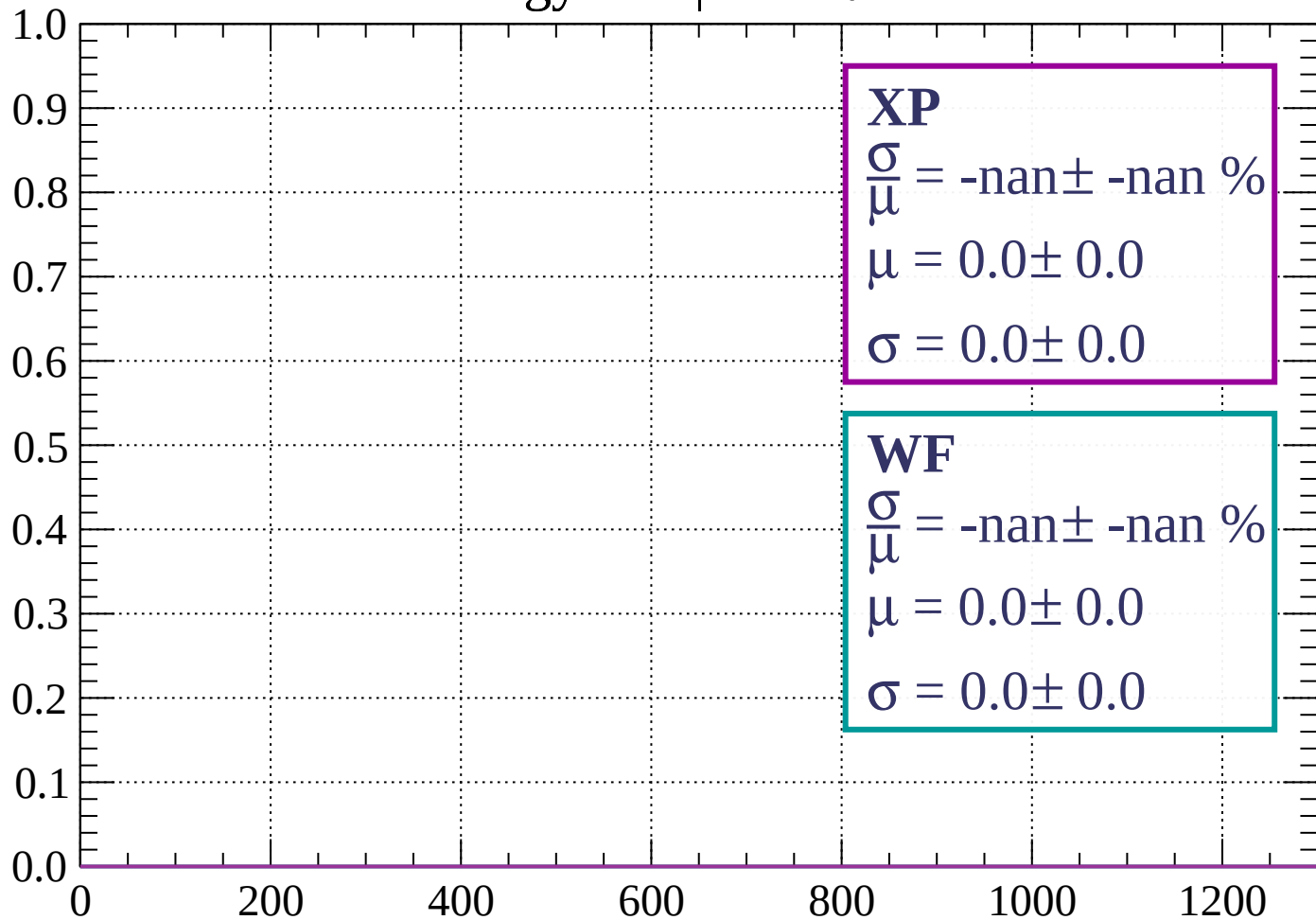
Energy loss | $60 < \theta < 63$

Count



Energy loss | $63 < \theta < 66$

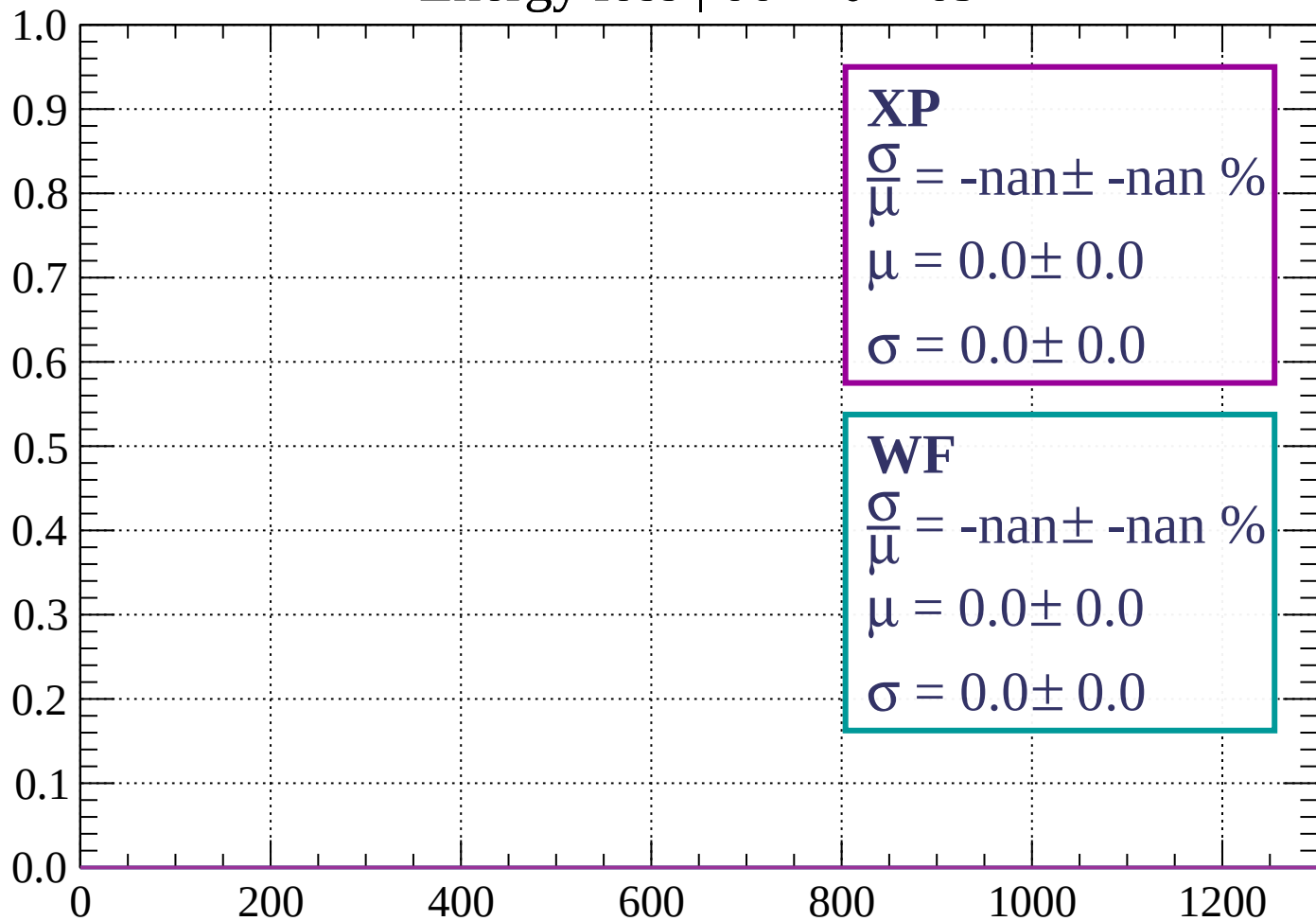
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

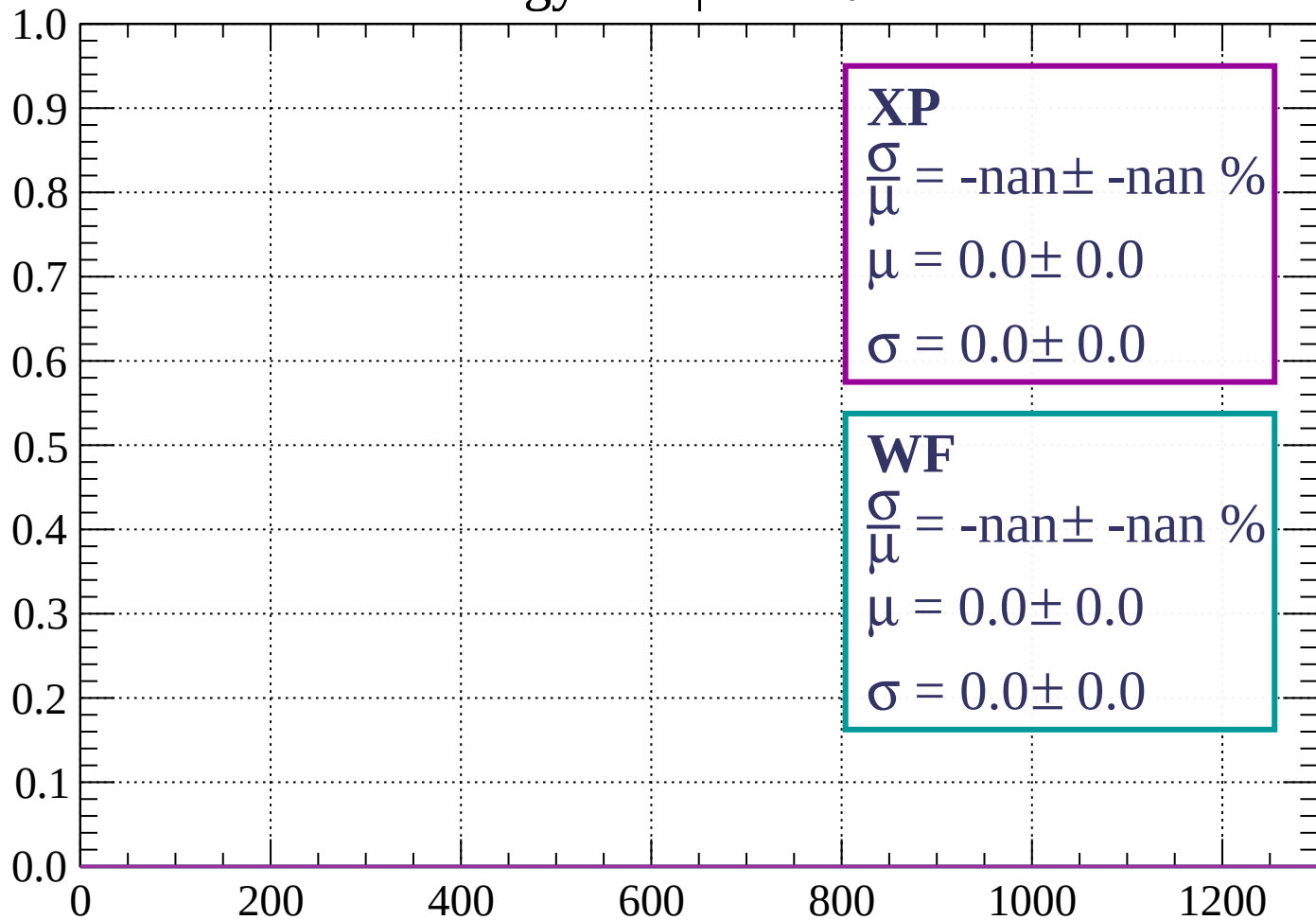
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

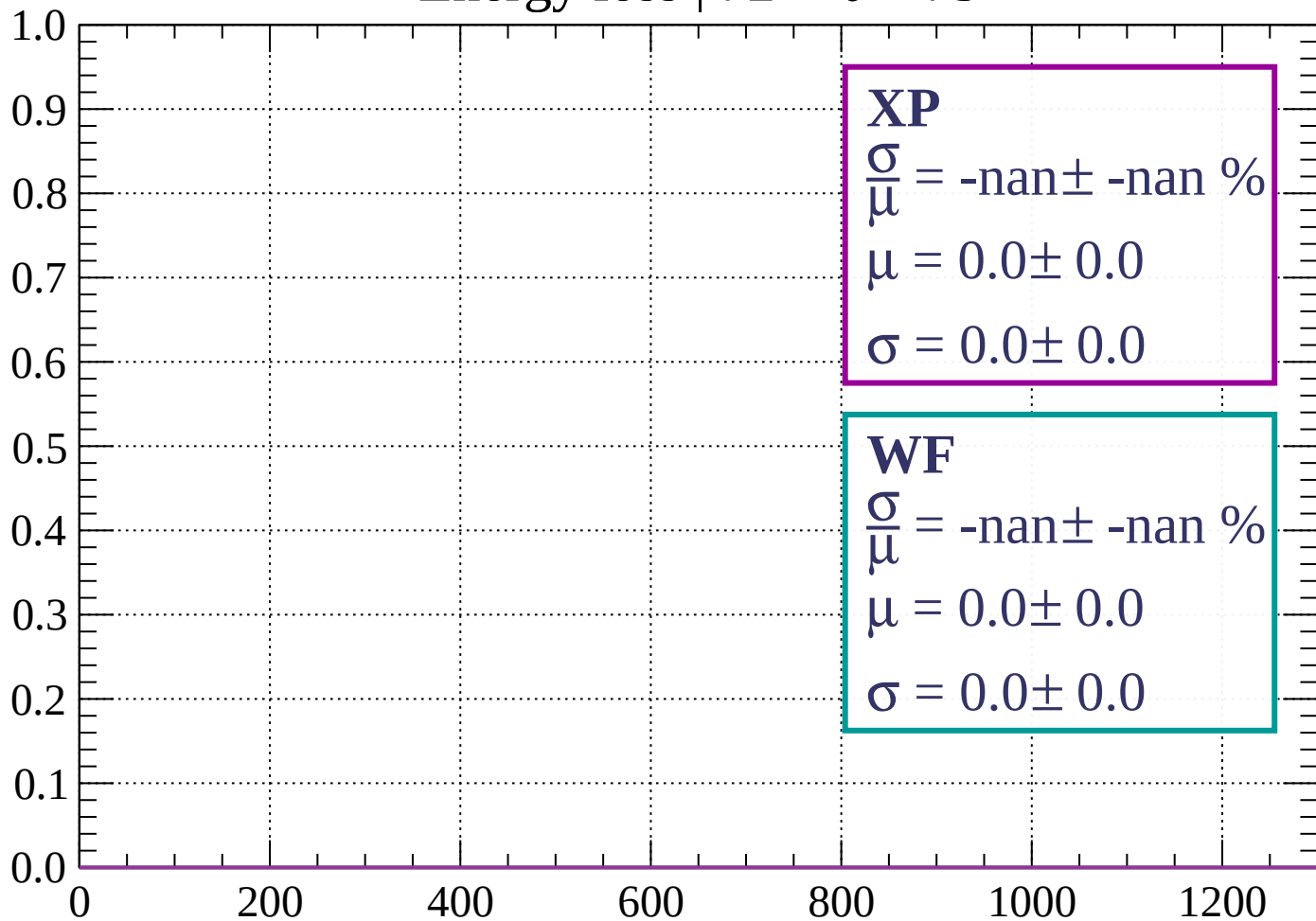
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

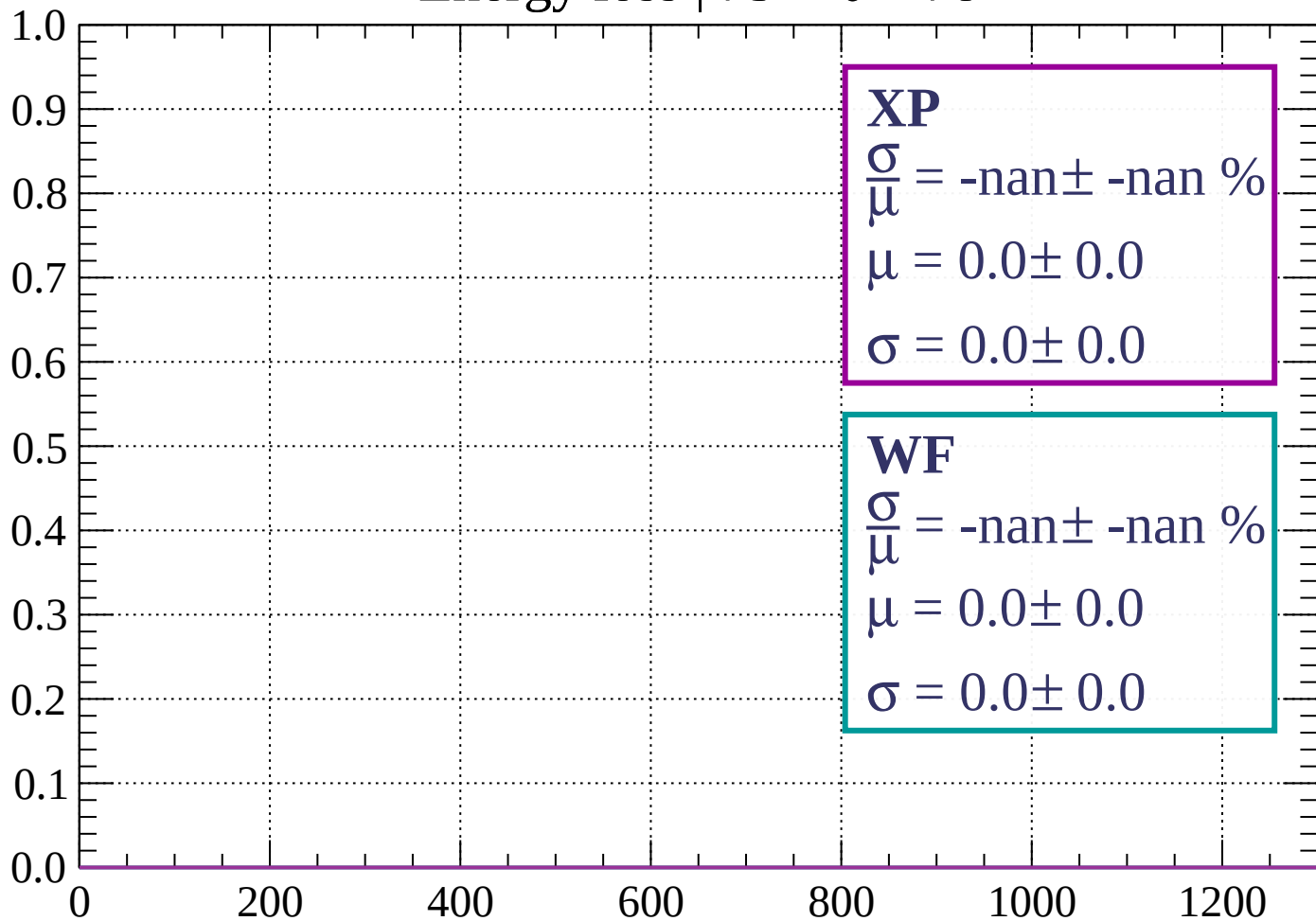
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

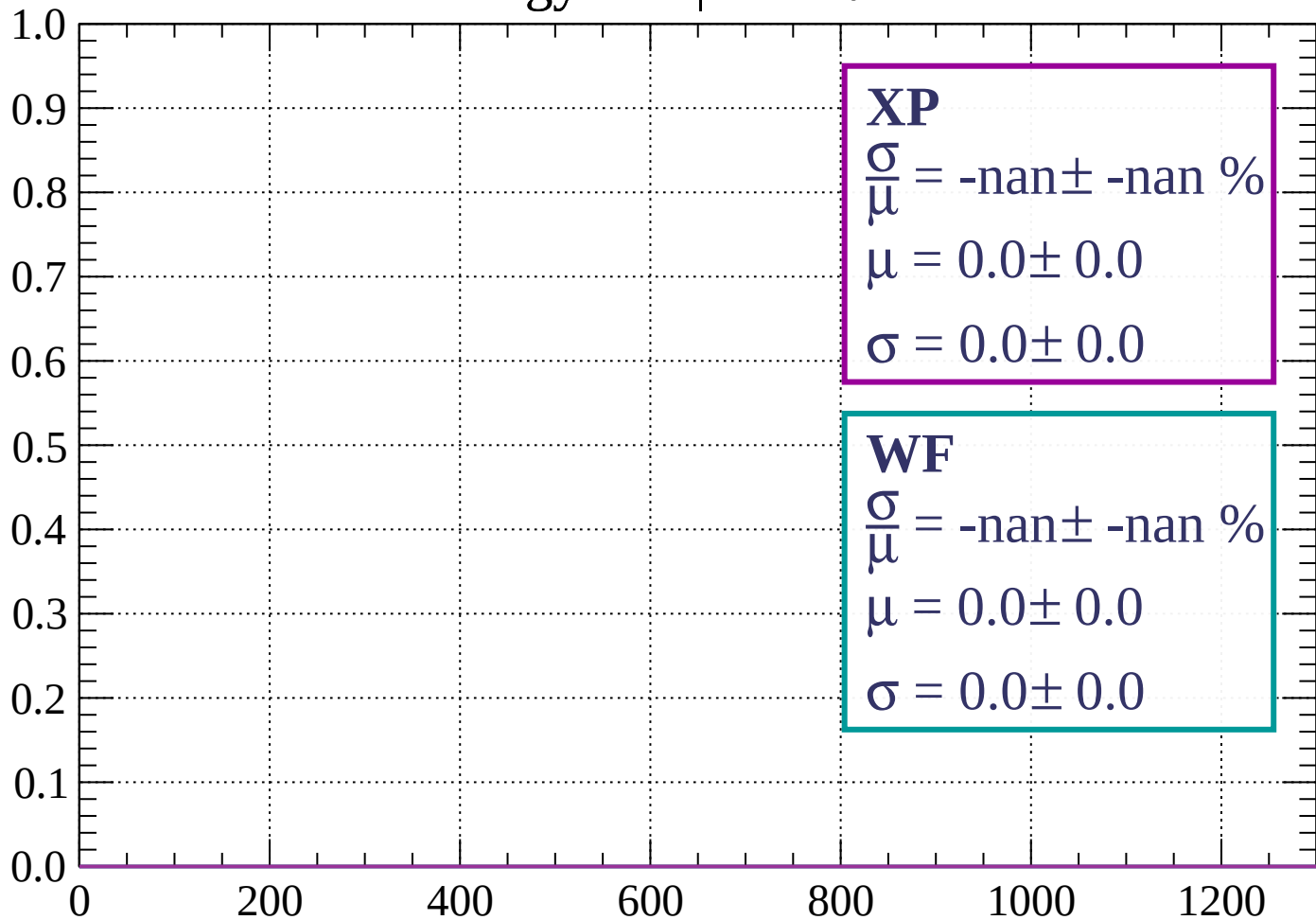
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

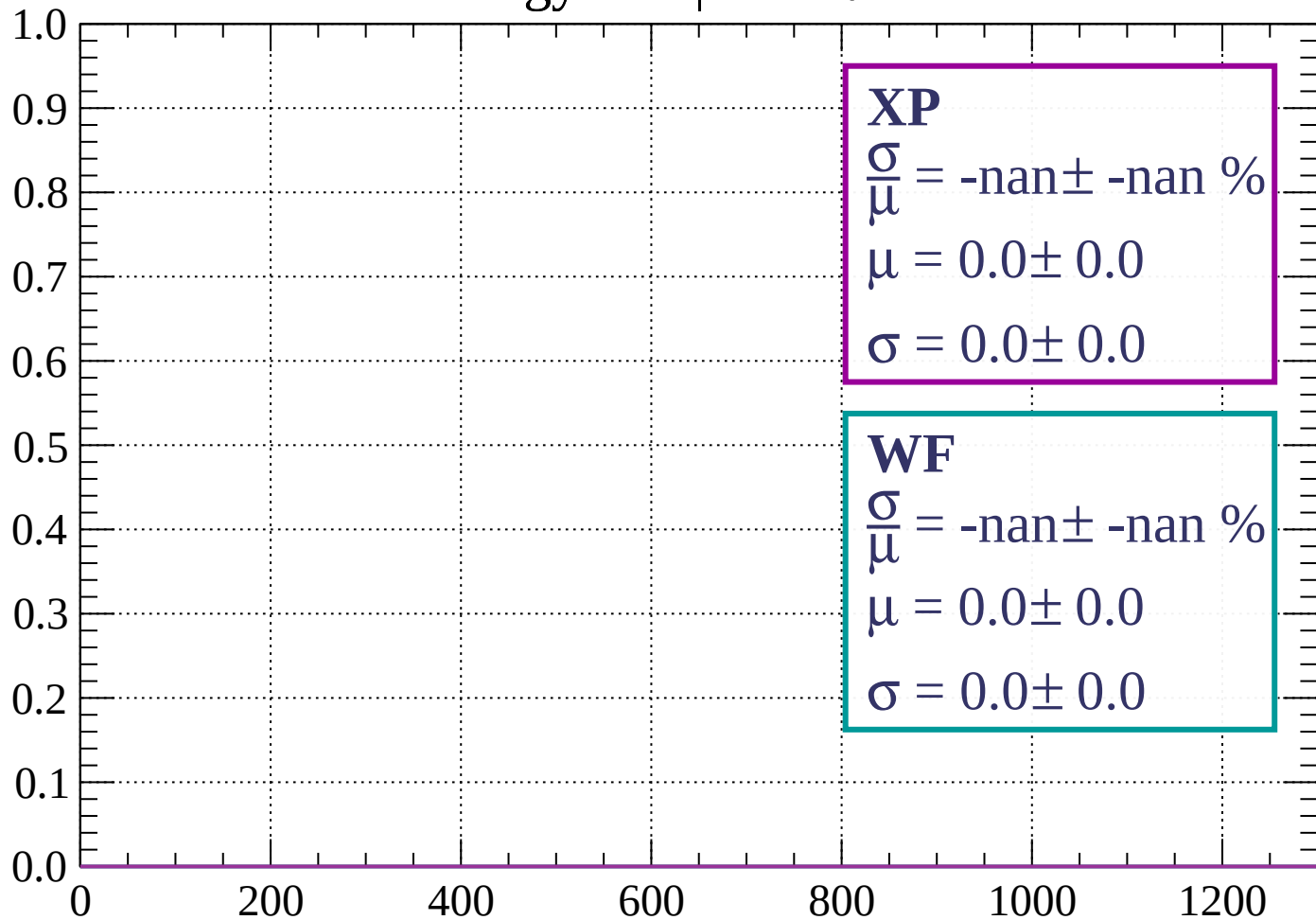
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

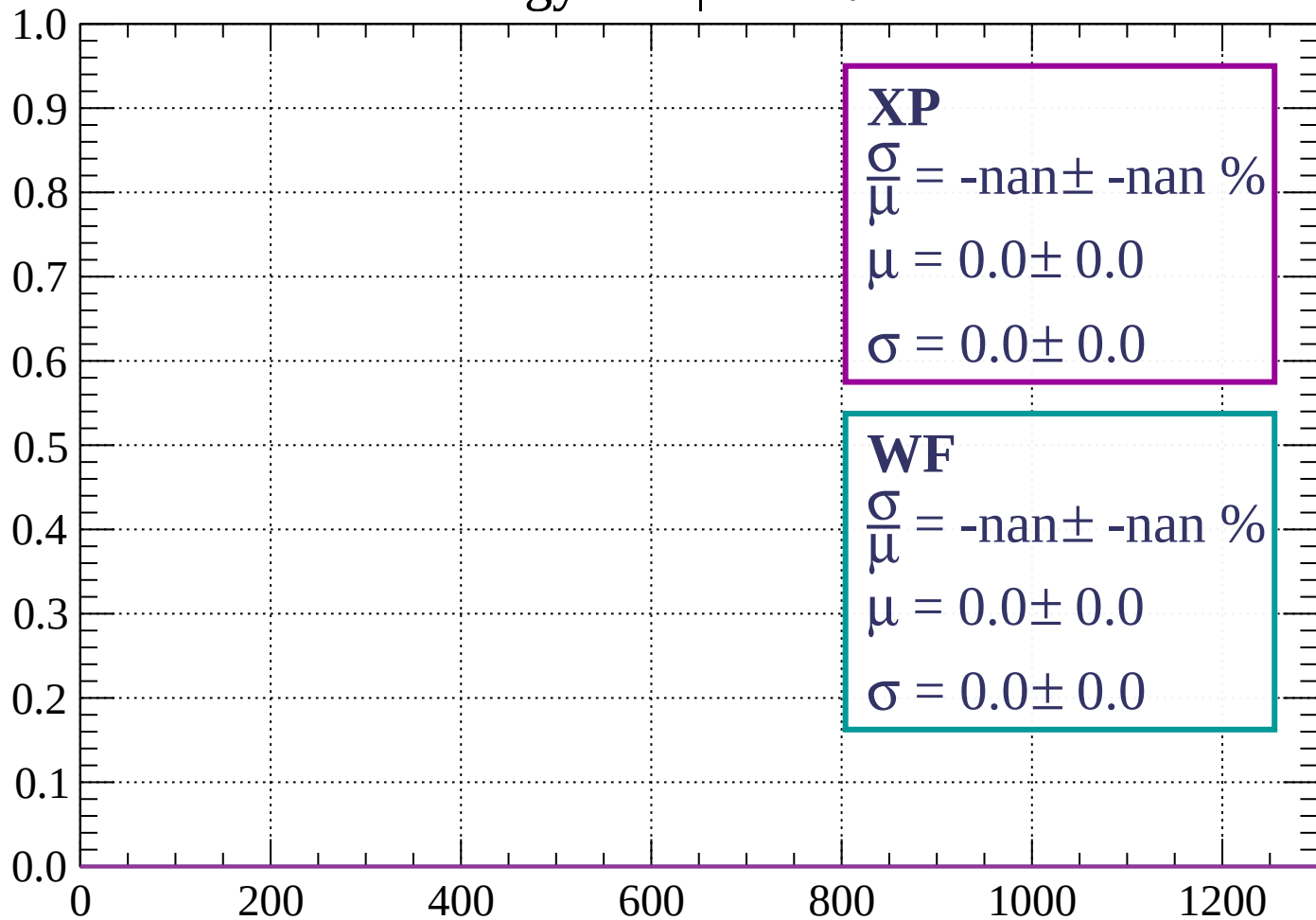
Count



dE/dx (ADC counts/cm)

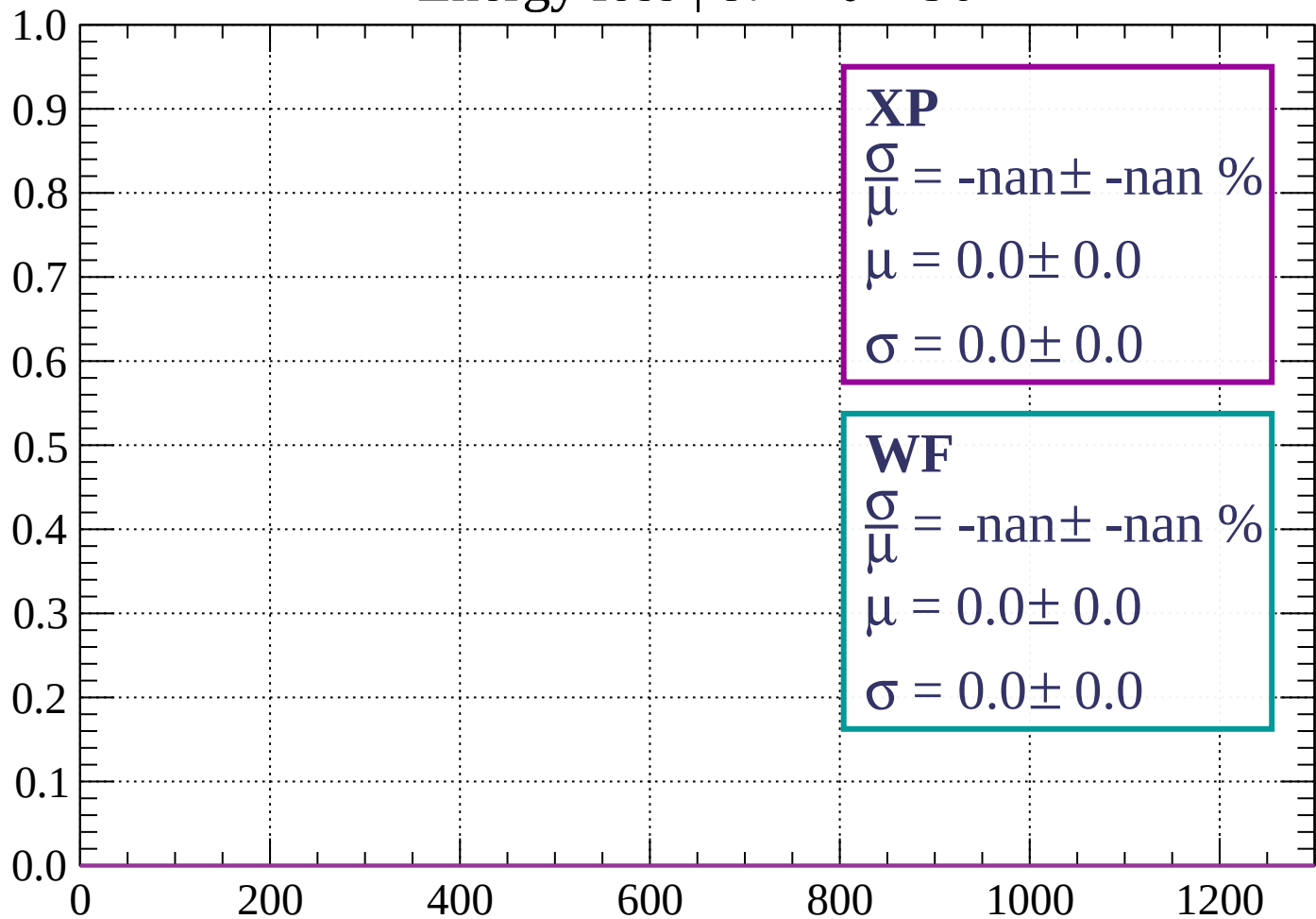
Energy loss | $84 < \theta < 87$

Count



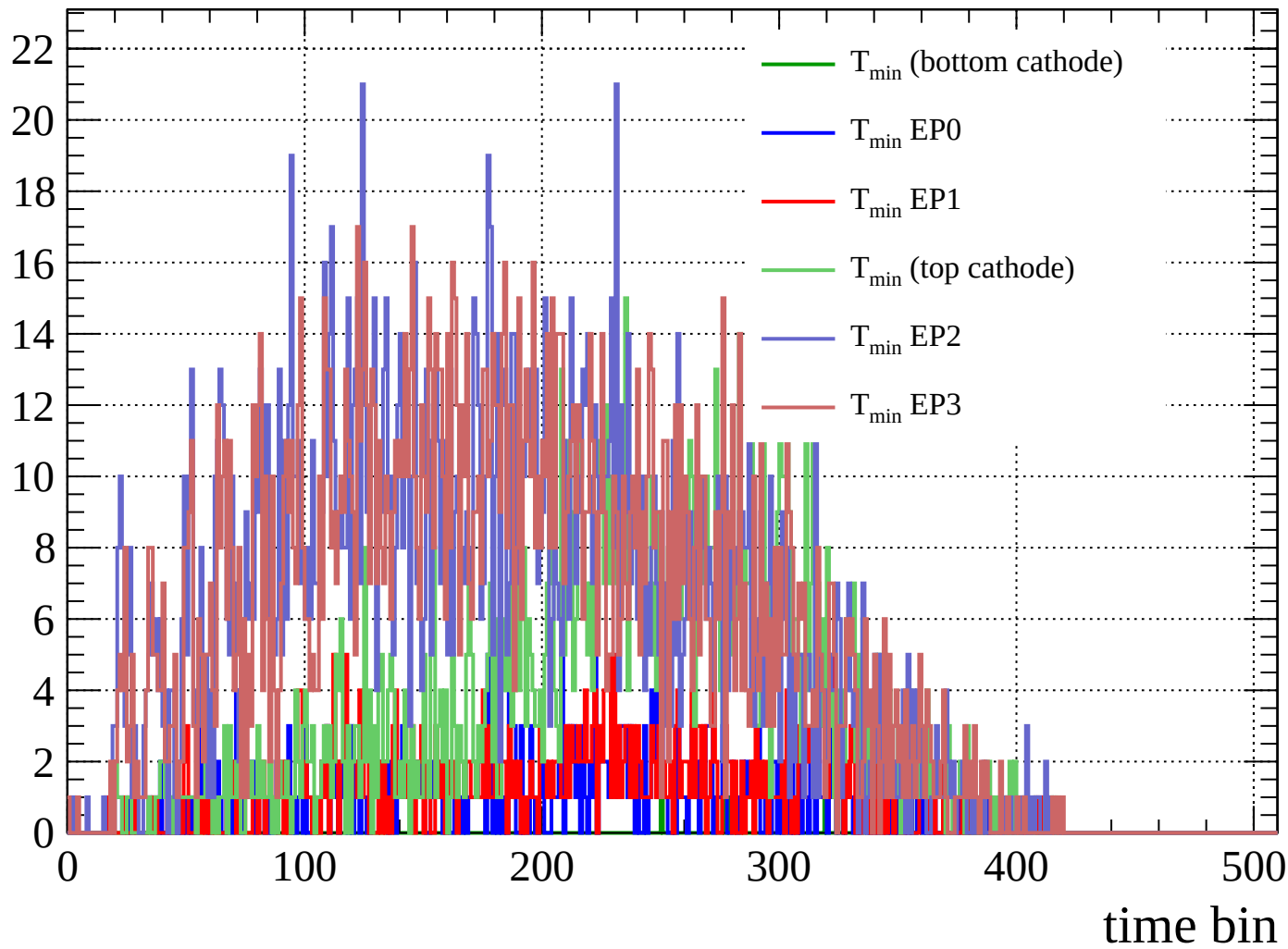
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

